Centre-of-Expansion Shell Cosmology: Collected Draft Papers

Chris Alameddin

October 2026

DRAFT — work in progress

Abstract

This collection asks whether non-accelerating expansion in a finite, off-centre geometry can look accelerated to an observer who is not at the Centre of Expansion (CoE). The first paper builds a CoE-plus-thick-shell geometry from observables, derives its angular morphology before comparing it with the independent Migkas et al. cluster anisotropy $A/A_{\rm all}$, and finds that the pure geometry overshoots the $\sim 10^{-5}$ CMB isotropy level at last scattering. Special-relativistic Doppler plus central-mass gravitational redshift cancel 95.9% of the geometric RMS while keeping the low-z morphology; residual multipoles, a CatWISE/Secrest matter dipole ($D \approx 0.0155$) and redshift-dependent cancellation $A(z, \theta)$ are examined. In this draft, apparent acceleration is interpreted as an off-centre sampling effect. A supplement repeats the anisotropy calculation exactly in a Lemaître–Tolman–Bondi dust model. With no mass it returns the Milne value $H_0 = 70.00$ in every direction, and in the classical limit it matches the classical values to about 10^{-3}. The 0° bump grows with central mass and with closeness to the CoE, the minimum moves from about 120° toward 105°, and the 180° rise reaches 1.02–1.14. The sky mean never reaches Planck or SH0ES, but at $R_{\rm obs} = 3\,$Gpc individual directions span both when $2GM_c/c^2(0.3\text{–}1)\,D$.

The second paper treats the leftover CMB residual (237 times the 10^{-5} level) as a fingerprint and asks which arrangements of masses and voids cancel it. Outside the visible sphere 11 bodies (6 masses, 5 voids) reach the target and inside it 4–8 suffice. Editing 12% of a balanced random web brings the residual from 789× to 1.00× without forming a ring. The third paper classifies twenty spontaneous patterns into five classes by how their scale is selected, giving each its forcing function, threshold and selected scale. It then applies the thin-shell (Vishniac) instability to the CoE onion, where wrinkled shells seed clumps and voids. A closing note argues that every observed physical system has a geometric, mass or momentum centre, and that proposed counter-examples have so far been mathematical abstractions. All results are preliminary.

Full table of contents on the next page. Page and figure numbers run continuously through the whole collection.

Contents

1. CoE cosmology: Apparent acceleration from an off-centre finite geometry 3

Introduction: the inverse problem of apparent acceleration 3
Off-centre geometric solution and prior angular predictions 5
Migkas comparison — out-of-sample angular morphology 6
CMB null test — the purely geometric model 8
Physical completion — motion, gravity, and photon propagation 11
Residual multipoles and limits 21
Independent signed matter-dipole constraint 25
Redshift evolution of $A(z, \theta)$ 27
Discussion and conclusions 28
Appendix figures 31
Supplement: Exact relativistic (LTB) calculation of $A/A_{\rm all}$ (inserted) 34

2. Reading the CMB fingerprint of the CoE shell model 36

Introduction 36
The fingerprint 36
Method 37
Results 39
Comparison with observed structure 53
Editing filled webs to reach the CMB level 55
Mesh, area and energy of an edited web 55
Crystal analogy and the compensation rule 58
A compensated random web reaches the CMB level 58
Pair energies and forces 62
Concentric shell models 62
Caveats 76
Conclusions 76

3. Pattern formation (v5) 77

Introduction 77
A common framework 77
Classification 79
Reference case: Chladni figures (forced mode, D) 79
Class A: bulk finite-wavelength instabilities 80
Class B: equilibrium modulated phases 84
Class C: coarsening 85
Class D: interfacial instabilities 86
Class E: fronts and waves 88
Self-similarity 90
Chaos 90
Summary: what the patterns share 91
Wrinkled thin shells: supernova remnants and the CoE onion 92
Inserted figures: Michaut et al. (2012) and wedge storyboard 93
Summary Table 107

4. Does everything have a centre? A philosophical note 109

References 111

Fig. 114: CoE on the northern celestial hemisphere 115

Apparent acceleration from an off-centre finite geometry: morphology, CMB null test, and redshift-dependent cancellation

Chris Alameddin

Draft preprint — October 7, 2026

Abstract

Type Ia supernovae established apparent late-time acceleration, yet the distance modulus $\mu = 5\log_{10}(D_L/10\,\mathrm{pc})$ alone does not specify the underlying geometry. We pose the inverse problem: can non-accelerating expansion in a finite, off-centre geometry appear accelerated to an observer who is not at the Centre of Expansion (CoE)? From observables we construct a candidate CoE-plus-thick-shell geometry, derive prior angular morphology (amplitude and $\varphi(\theta)$ rings), and only then compare to the independent Migkas et al. cluster anisotropy—morphology out-of-sample, amplitude partly construction-motivated. Extrapolating the same pure geometry to last scattering ($z_\star \approx 1100$) overshoots the $\sim 10^{-5}$ CMB isotropy level, so geometry alone is incomplete. Special-relativistic Doppler plus central-mass gravitational redshift cancel 95.9% of the geometric RMS while preserving low-z morphology (Fix Point 2): the CMB is addressed qualitatively by redshift-dependent cancellation, leaving a new inverse problem in the higher multipoles. Residual multipoles ($m = 3$–5 after SR; $m = 6$ and $m = 1$ after a four-clump sketch), an independent CatWISE/Secrest matter dipole ($\mathcal{D} \approx 0.0155$; CoE end $\sim 195°$ vs anti-CoE end $\ell_0 = 15.2°$), and redshift-dependent cancellation $A(z, \theta)$ complete the chronological programme. Apparent acceleration remains an off-centre sampling artefact—no dark energy required.

1 Introduction: the inverse problem of apparent acceleration

1.1 Motivating observation and the boxed question

In 1998–99, Type Ia supernovae established that the late universe appears to accelerate [1, 2]. The operational observable is minimal: the distance modulus

$$\mu = 5\log_{10}\big(D_L/10\,\mathrm{pc}\big) \tag{1}$$

depends only on luminosity distance. It does not, by itself, specify whether the physical expansion is accelerating, whether dark energy is required, or where the observer sits relative to any preferred centre.

> **Question.** Can non-accelerating expansion in a finite geometry appear accelerated to an off-centre observer?

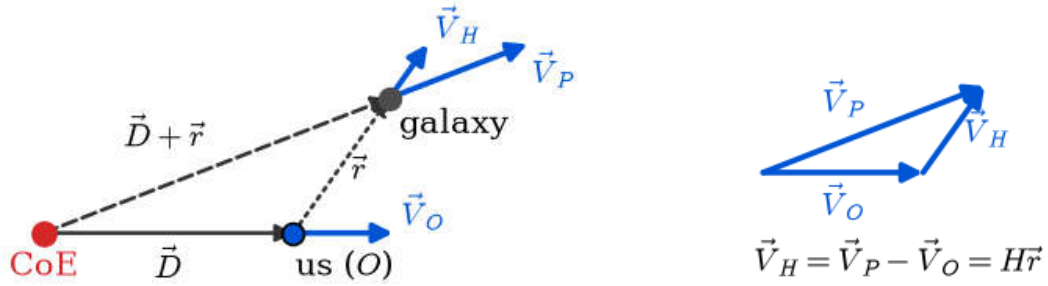


Figure 1: Hubble velocity vectors: Centre of Expansion (CoE), observer, and galaxy; $\vec{V}_H = \vec{V}_g - \vec{V}_O$.

We distinguish *physical* expansion (the true radial motion of matter relative to a Centre of Expansion) from *inferred* expansion (what an off-centre observer reconstructs from μ and redshifts under homogeneous–isotropic assumptions). Figure 1 shows the kinematic picture: local Hubble velocity as free-fall relative to the CoE minus the observer's own motion.

1.2 Forward versus inverse

The standard route is forward: assume FLRW (or ΛCDM), predict $D_L(z)$, and fit H_0, Ω_m, w. We take the inverse route instead [3]: start from observables, propose a candidate finite geometry consistent with them, then generate new observables that the same geometry must confront. Ellis and Stoeger framed this as the cosmological fitting problem; here the candidate is an off-centre CoE plus thick ordinary-matter shell (the donut-universe configuration of Fig. 2), not a new dark-energy parameter.

1.3 Degrees of freedom

The finite-visibility, off-centre setup has a controlled set of degrees of freedom: shell distance scale D, observer offset d from the CoE (with $d = D$ when the observer sits mid-shell), enclosed volume $V(D)$, total mass $M_{\rm tot}$, radial density $\rho(r)$ or an equivalent mass distribution, and sky angle θ from the CoE–anti-CoE axis. Lines of sight sample radii in the annulus

$$D - d \le r \le D + d. \tag{2}$$

Symmetry is axial about the CoE axis, not spherical about the observer. The vision radius $R \approx 14.3\,\mathrm{Gpc}$ is fixed at the Migkas sample depth and is not fitted; $A/A_{\rm all}$ depends only on D/R; the shell rule with that vision radius requires $D \gtrsim 14.7\,\mathrm{Gpc}$ so the observer remains mid-shell.

1.4 Parameter discipline and Fix Points

We name two Fix Points later: Fix Point 1 freezes the geometry (and its prior angular predictions) before the Migkas comparison; Fix Point 2 requires that special-relativistic and gravitational corrections cancel last-scattering anisotropy without destroying the low-z morphology. No silent geometric degrees of freedom are introduced between those points.

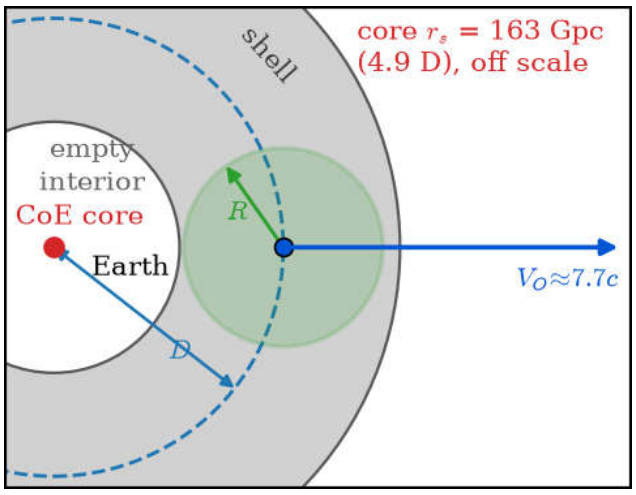


Figure 2: Hubble-bubble scale: observer inside a thick ordinary-matter shell at distance D from the CoE, with vision sphere of radius R.

1.5 Roadmap

The chronology is deliberate: apparent acceleration as inverse problem → off-centre geometric solution and prior rings → out-of-sample Migkas morphology → CMB null test of pure geometry → physical completion (SR + central mass) → residual multipoles → independent matter dipole → redshift evolution of $A(z, \theta)$.

2 Off-centre geometric solution and prior angular predictions

2.1 Ray geometry and local Hubble velocity

Local Hubble velocity is free-fall toward the CoE minus our own motion,

$$\vec{V}_H = \vec{V}_g - \vec{V}_O. \tag{3}$$

The observer sits inside a thick shell at comoving distance D from the CoE. Sky anisotropy is an off-centre sampling of radial structure, not a direction-dependent scale factor $a(t, \theta)$.

2.2 Amplitude $A(\theta)/A_{\rm all}$ and morphology

Along a line of sight at angle θ from the CoE axis, the sampled expansion rate (or equivalently the normalisation A of an L_X–T scaling when distances enter through D_L) differs from the all-sky mean. Writing $A(\theta)/A_{\rm all}$ for that ratio, the geometry predicts both an overall amplitude and an angular morphology. Amplitude is partly tied to how the construction is normalised (density – distance degeneracy); morphology—where maxima, minima, and rings sit on the sky—is fixed by the ray geometry once D, d, and the radial mass profile are chosen.

2.3 Phase rings $\varphi(\theta)$

Off-centre placement produces closed curves of constant phase on the celestial sphere: the $\varphi(\theta)$ rings. At low redshift those rings appear as a preferred locus of nearby sources—a "blue" ring when coloured by z.

2.4 Prior low-z rings (before Migkas maps)

Figure 3 illustrates that prior morphology on a cluster sky (Galactic coordinates, redshift colour-bar). In this narrative the ring and the longitude dependence of $A/A_{\rm all}$ are geometric predictions of the off-centre shell, not discoveries extracted from a newly released catalogue.

> **Fix Point 1.** The CoE-plus-shell geometry—and with it the predicted amplitude shape and $\varphi(\theta)$ rings—is fixed before the Migkas comparison. Migkas et al. do not redefine the geometric degrees of freedom.

3 Migkas comparison — out-of-sample angular morphology

3.1 The Migkas dataset

Migkas et al. [5] test sky isotropy with the X-ray luminosity–temperature (L_X–T) scaling of galaxy clusters. Measured L_X depends on luminosity distance; T does not. Directional variation in the fitted normalisation A of $L_X \propto A\,T^B$ therefore traces an apparent directional Hubble / distance scale. They report A in sliding sky cones relative to the all-sky fit $A_{\rm all}$: the plotted quantity is $A/A_{\rm all}$ (their Fig. 6; our Figs. 4–6). Under the usual conversion with $A \times H_0^2$ held fixed, $H/H_{\rm ref} = 1/\sqrt{A/A_{\rm all}}$, so troughs in $A/A_{\rm all}$ are high-H_0 / short-D_L directions—provided the variation is cosmological rather than cluster or X-ray systematics.

3.2 Coordinate placement and fits

We place the already-fixed geometry on the Galactic sky and compare to Migkas $A/A_{\rm all}$ versus Galactic longitude ℓ. Fits are realised in Newtonian gravity and in relativistic LTB about the CoE. The LTB origin is the CoE, not the observer: we sit at radial coordinate $r = D$, sampling a radial mass-multiplier $\beta(r)$.

Figure 4 shows Migkas $A/A_{\rm all}$ versus Galactic longitude with the Newtonian and relativistic shell-plus-core curves (top) and Migkas–A_F residuals as connect-the-dots with cumulative multipole curves $M \leq 5$–7 (bottom).

Figure 5 shows the same $A(\ell)$ pattern in three redshift slices (0.03–0.07, $z > 0.07$, and all-z). Fit parameters are held fixed; the model is recomputed at each slice's median sampling distance, so that the $A(\ell)$ maxima and minima move with that distance. The D–core-mass degeneracy is shown explicitly.

3.3 Einstein and Friedmann scaffolding

Einstein's equation is $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}$. For concentric dust shells about the CoE the Lemaître–Tolman–Bondi line element [8, 9, 10] is

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{(R')^2}{1+2E(r)}\,\mathrm{d}r^2 + R(t,r)^2\,\mathrm{d}\Omega^2, \tag{4}$$

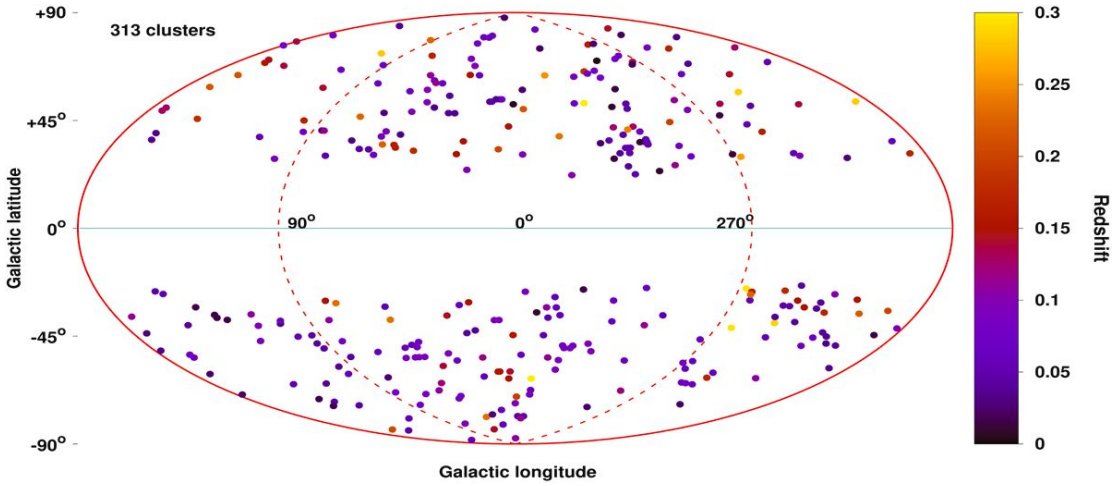


Figure 3: Sky distribution of the 313 clusters in Galactic coordinates, with the colorbar indicating their redshift (blue = low z). The low-redshift "blue" ring on the celestial sphere, together with the longitude dependence of $A/A_{\rm all}$, is the observable geometry pinned before Migkas et al. published. There are two clusters at $z > 0.3$; the scale is set so the colour contrast is optimal.

with areal radius $R(t, r)$. Each shell obeys

$$\dot{R}^2 = 2E(r) + \frac{2GM(r)}{R}, \tag{5}$$

and the interior mass uses a background density scale ν and radial mass multiplier $\beta(r)$,

$$M(r) = \frac{4\pi}{3} R^3 \nu \beta(r), \tag{6}$$

so that $\beta = 1$ recovers critical-density normalisation. Including radiation on the same scaffolding,

$$H_a^2 \propto \frac{\nu_m \beta_m(r)}{a^3} + \frac{\nu_r \beta_r(r)}{a^4} - \frac{k(r)}{a^2}. \tag{7}$$

Curvature $k(r)$ follows from $E(r)$ and introduces no angular dependence into the metric coefficients.

For the CatWISE-locked fits to Migkas $A(\ell)$ (Table 1),

$$\begin{aligned} D &\approx 58.2\,\text{Gpc}, \quad D-d \approx 43.9\,\text{Gpc}, \\ D+d &\approx 72.5\,\text{Gpc}, \quad \ell_0 = 15.2^\circ. \end{aligned} \tag{8}$$

The free-L exponential gives $D \approx 27.95\,\text{Gpc}$, $L \approx 6.07\,\text{Gpc}$, $\ell_0 \approx 18.6^\circ$; an earlier, superseded LTB shell fit gave $D \approx 32.9\,\text{Gpc}$. In all cases D lies above the shell-rule bound $D \gtrsim 14.7\,\text{Gpc}$ (vision radius $R \approx 14.3\,\text{Gpc}$), and the observer sits mid-shell.

Figure 6 confirms that both the Newtonian (earlier shell fit, $\Omega_s = 0.049$) and relativistic LTB curves follow the Migkas longitude dependence; the bottom panel shows Migkas–A_F residual bars with cumulative multipole curves $M \leq 5$–7.

Table 1: Local vs global σ (Migkas $A/A_{\rm all}$, $\chi^2_{\rm null} = 45.14$).

Model	R^2	Local σ	Global σ
Point mass at CoE	0.521	4.4	4.1
Exponential, L free	0.759	5.3	5.1
Exponential, CatWISE 3% lock	0.716	5.4	5.1
Soft core $1/(1+(r/L)^{0.3})$, lock	0.743	5.4	5.1
Exponential (lock) + rotation & phase	0.845	5.5	5.3
Soft core + rotation & phase	0.870	5.4	5.1

Local: axis angle ℓ_0 taken as given. Global: ℓ_0 counted as one fitted parameter; for the free-L row the global σ counts both L and ℓ_0 (four parameters). R^2 is unweighted.

3.4 Sky orientation

Starting from an untilted, no-roll configuration, we apply a $+15°$ tilt, then a radial gradient, then an $+80.8°$ roll about the CoE axis.

Figure 7 shows that sequence; Figure 8 overlays the predicted MAX/MIN bands on Migkas Fig. 8.

3.5 Epistemic status

Amplitude is partly construction-motivated (density–distance trade in the Migkas normalisation). Morphology—ring locus, MAX/MIN placement, longitude shape across redshift slices—is out-of-sample relative to Fix Point 1: no new morphological degree of freedom was opened to absorb the Migkas maps. The geometric R^2 versus Migkas (geometry only, no sine) is 0.521 (4.1σ global) for the point-mass row of Table 1; with rotation and the one-cycle phase factor, R^2 rises to 0.85–0.87 (0.845 exponential locked, 0.870 soft core). The point mass with no rotation gives weighted $R^2 = 0.534$ (unweighted 0.521).

4 CMB null test — the purely geometric model

With Fix Point 1 held, we extrapolate the *same* geometry to the last-scattering surface at $z_\star \approx 1100$. Off-centre placement implies a geometric multipole on the CMB sky of several percent—orders of magnitude above the Planck isotropy floor of $\sim 10^{-5}$ [11].

In the donut-universe picture the CoE is fixed while thick shells expand; the sphere of events visible to us grows at c. Recombination is a spherical surface of fixed physical radius around the CoE. Figure 9 shows the time sequence (frames A–D). Figure 10 shows the two-emitter baseline: if opposite plates share the observer's bulk velocity, the kinematic sky is flat before gravitational multipoles are included.

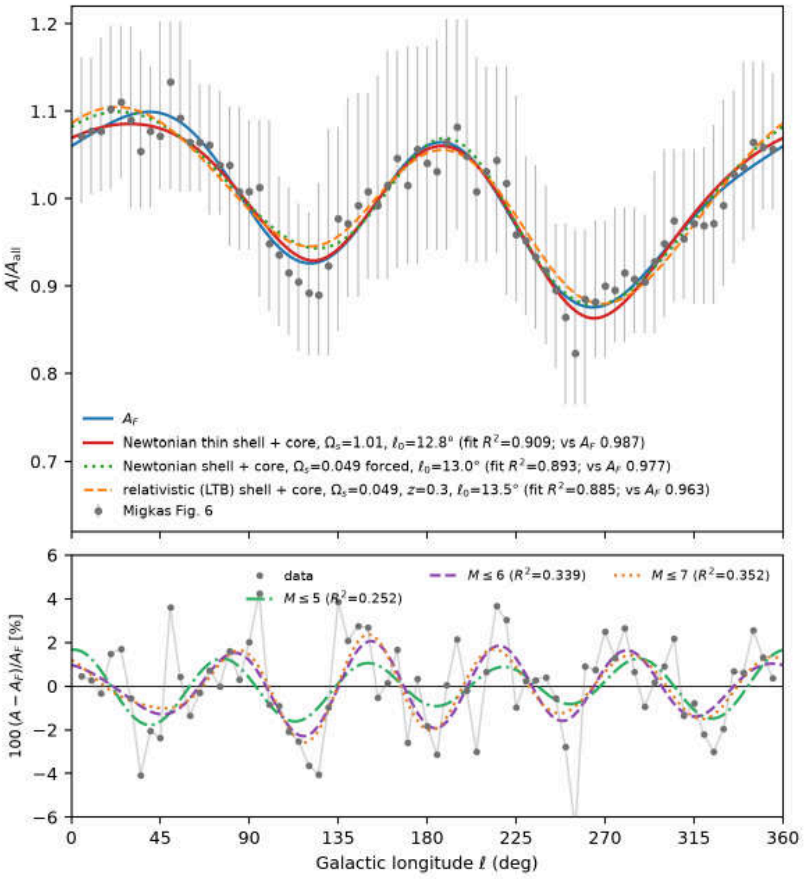


Figure 4: Migkas $A/A_{\rm all}$ versus Galactic longitude ℓ: top, Migkas points with A_F and Newtonian/relativistic CoE+shell curves; bottom, Migkas–A_F residuals (connect-the-dots) with cumulative Fourier multipole curves $M \leq 5$–7 and R^2 in the legend (not model–A_F).

4.1 Gravitational multipole from the geometry

On the emission sphere the Newtonian today's-field residual splits into a kinematic Doppler piece and a static gravitational piece,

$$\left(\frac{\Delta T}{T}\right)_{\rm Doppler} = -\,\hat{\mathbf{n}}\cdot\frac{\mathbf{V}_{\rm emit}-\mathbf{V}_{\rm obs}}{c}, \qquad \left(\frac{\Delta T}{T}\right)_{\rm static} = \frac{\Phi_{\rm emit}-\Phi_{\rm obs}}{c^2}. \tag{9}$$

The Migkas fit is carried by the shear in $\mathbf{V}$. For the Sedov–Taylor / shell-plus-core flow [6, 7] written $\vec{V} = h(q)\,\vec{q}$ with $q = s/D$, $\beta \equiv h'/h|_{q=1}$ and $\gamma \equiv h''/h|_{q=1}$, the line-of-sight amplitude a_f expands (with $\varepsilon = R/D$, $\mu = \cos\psi$, $\psi = \ell - \ell_0$) as

$$\frac{a_f}{\langle a_f\rangle} = 1 + a_1\cos\psi + a_2\cos 2\psi + a_3\cos 3\psi + \cdots, \tag{10}$$

where even harmonics are fixed by β and odd harmonics by the εQ tidal corrections. Pure geometry contributes only $\cos m\psi$ terms; the fitted sine rotation $A = a_f[1 - \mathrm{Amp}\sin(\psi+\varphi)]$ then mixes sine parts into the odd multipoles. Axisymmetric Legendre coefficients about the CoE axis are

$$a_\ell = \frac{2\ell+1}{2}\int_{-1}^{1} f(\mu)\,P_\ell(\mu)\,\mathrm{d}\mu, \qquad \mu = \cos\theta. \tag{11}$$

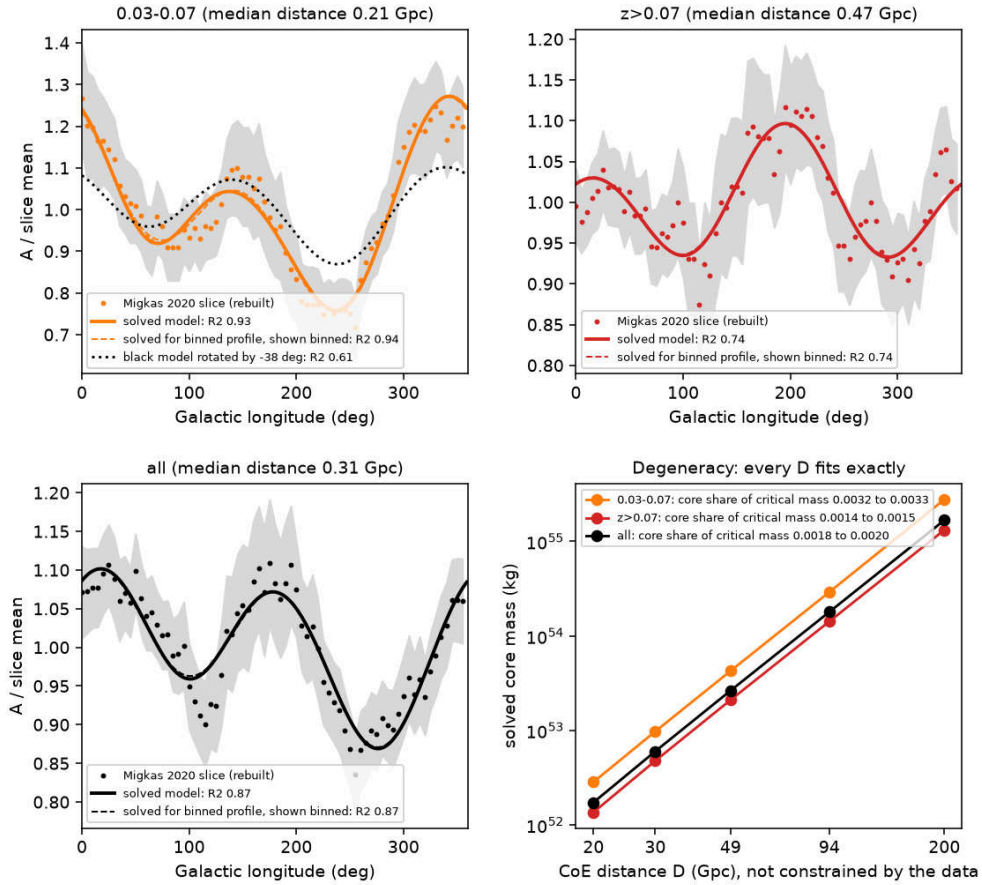


Figure 5: Migkas $A(\ell)$ in redshift slices, with the CoE model recomputed at each slice's median sampling distance (fit parameters held fixed). The D–core-mass degeneracy is shown explicitly.

4.2 Concentric-cell electromagnetic multipoles (even)

For concentric shells about the CoE with equal rest-frame temperature and shared bulk velocity, the kinematic sky is isotropic; anisotropy appears only from relative motion. The first-order electromagnetic Doppler from a hit on the emitting surface is

$$\frac{\Delta T}{T} \approx -\,\hat{\mathbf{n}} \cdot (\boldsymbol{\beta}_{\rm em} - \boldsymbol{\beta}_{\rm obs}), \qquad \boldsymbol{\beta}_{\rm em} = \beta_{\rm rec}\,\frac{r_{\rm hit}}{R_{\rm out}}\,\hat{\mathbf{e}}_r. \tag{12}$$

Axisymmetry about the CoE–anti-CoE axis makes the demeaned electromagnetic pattern even under $\phi \to 360^\circ - \phi$, selecting even multipoles once the monopole is removed. Odd electromagnetic multipoles require breaking that reflection symmetry.

> **Null-test verdict.** The purely geometric CoE-plus-shell model, extrapolated to $z_\star \approx 1100$ without additional photon-propagation physics, produces a last-scattering anisotropy far above $\sim 10^{-5}$. Geometry alone is therefore incomplete rather than final. Section 5 supplies the physical completion, which addresses the CMB qualitatively through redshift-dependent cancellation and leaves the higher multipoles as an open inverse problem.

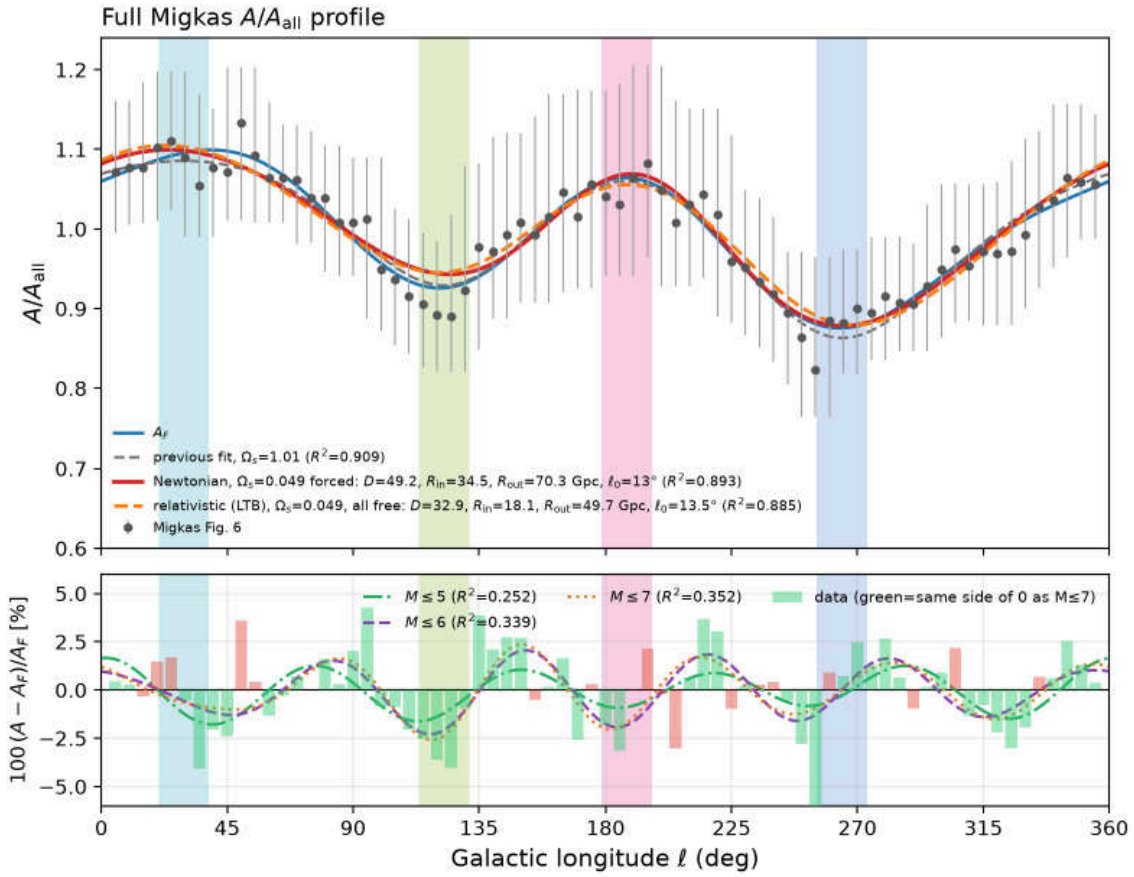


Figure 6: Full Migkas $A/A_{\rm all}$ amplitude profile (top) with Newtonian and relativistic fits; bottom: Migkas–A_F residual bars (green: same sign as $M \leq 7$; red: opposite; $\alpha \approx 0.45$) with cumulative multipole curves $M \leq 5$–7.

5 Physical completion — motion, gravity, and photon propagation

5.1 What must be added

The Migkas $A/A_{\rm all}$ sky pattern is fitted by an off-centre CoE plus thick shell. That geometry alone overshoots the CMB isotropy floor. We ask whether two standard effects—special-relativistic Doppler between a moving emitter and a moving observer, plus gravitational redshift from a central mass at the preferred origin—can cancel most of that geometric anisotropy without inventing a free "kill function." We build a multiplicative correction $S_{\rm model}$ from those effects, apply it to the geometric pattern G, measure the residual R, and read leftover multipoles as a fingerprint. Numbers are from existing fits; no new optimisation is reported here.

Distinguish the physical three-velocity $\beta = V/c$ with $|\beta| < 1$ from any effective Newtonian shell speed β_E that may formally exceed unity when Newtonian kinematics are extrapolated carelessly: only subluminal β enters the SR formulae below.

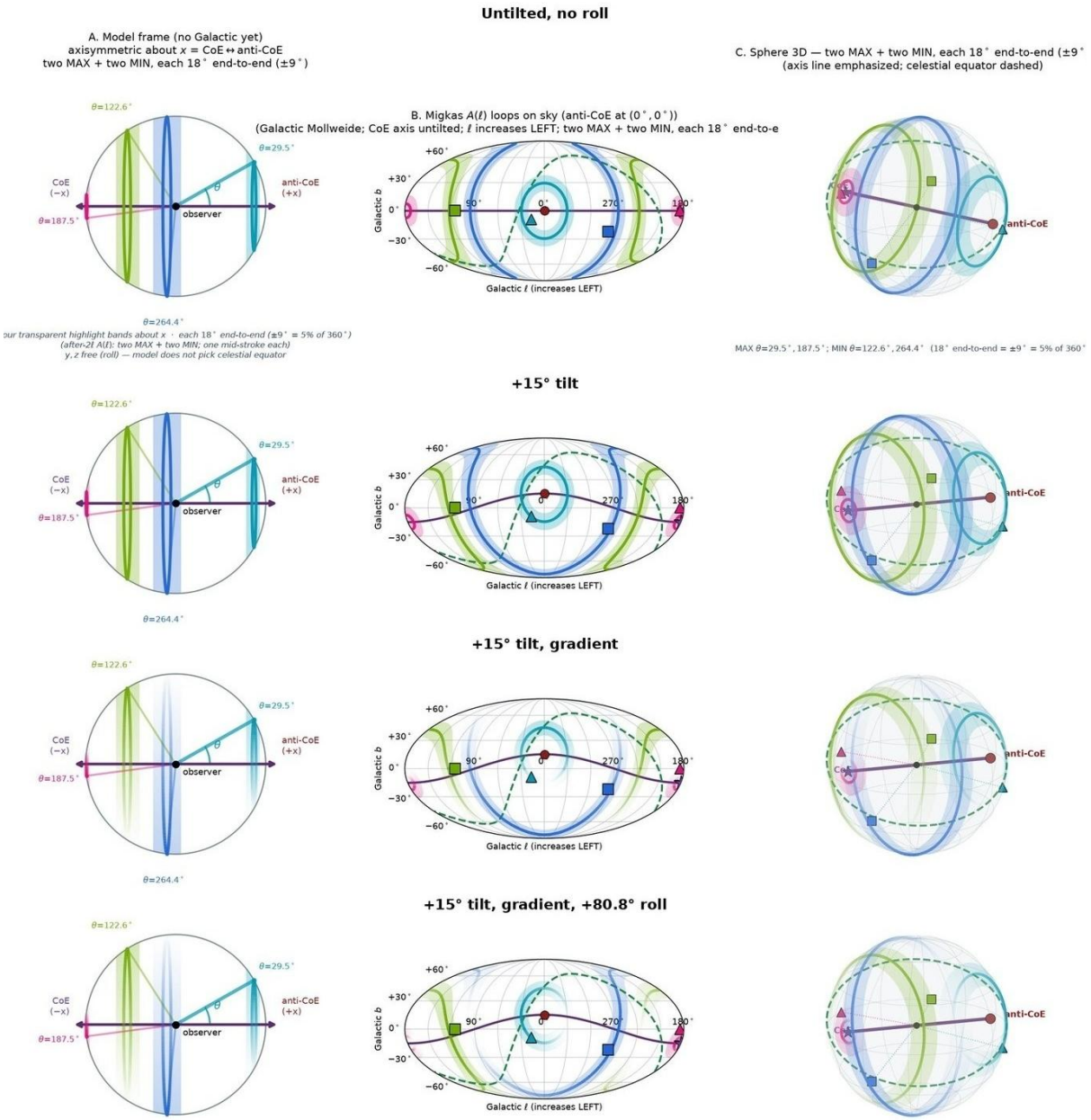


Figure 7: Sky-orientation sequence: untilted configuration through +15° tilt, radial gradient, and +80.8° roll about the CoE axis.

5.2 Composite anisotropy and the exact redshift

A useful bookkeeping split is

$$\Delta_{\rm obs} \approx \Delta_{\rm geom} + \Delta_{\rm SR} + \Delta_{\rm grav}, \tag{13}$$

but this linear decomposition is only approximate. The exact photon redshift between emitter and observer four-velocities and null tangent is

$$1 + z = \frac{(u \cdot k)_e}{(u \cdot k)_o}. \tag{14}$$

Morphology at low z must survive the high-z cancellation (Fix Point 2).

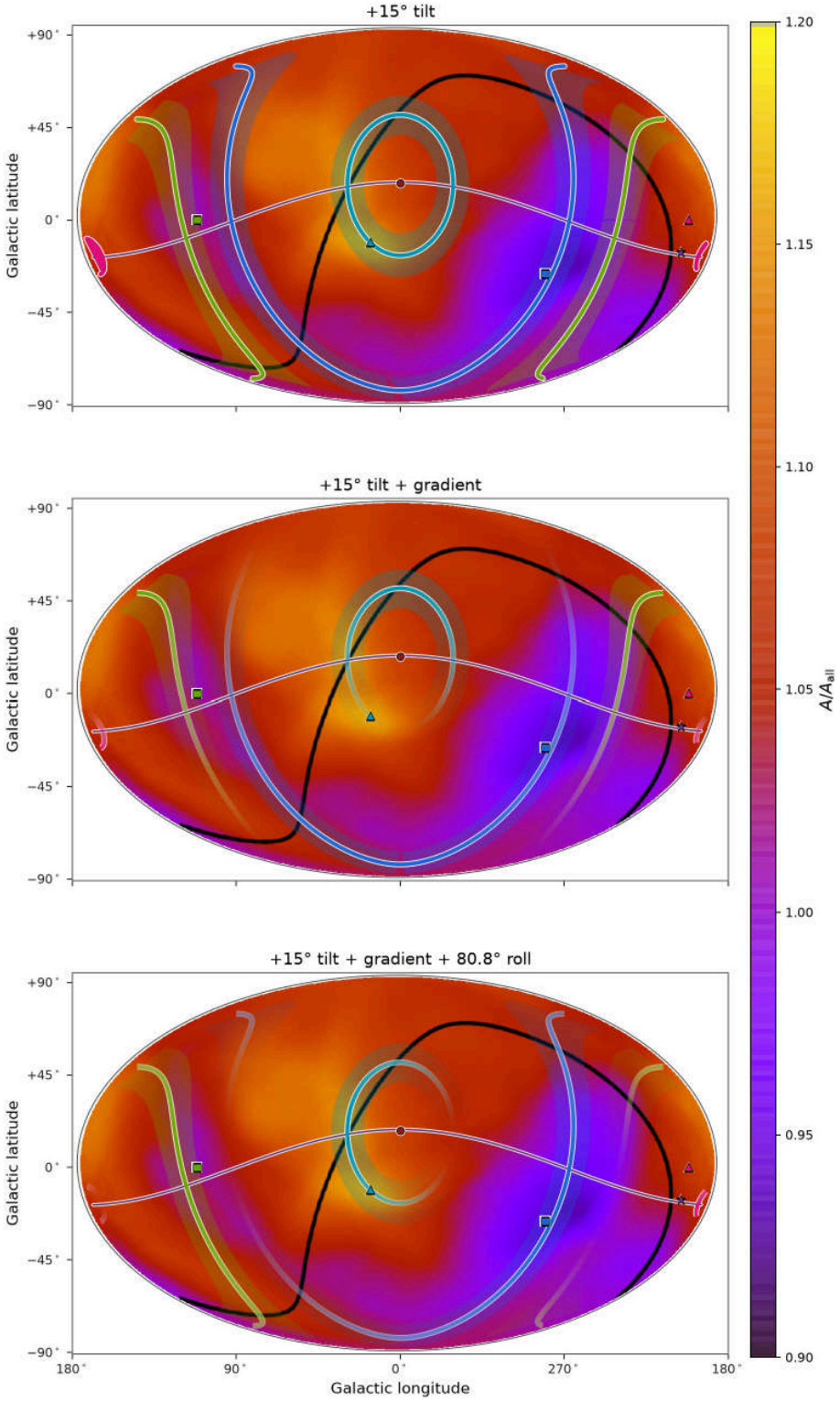


Figure 8: Predicted MAX/MIN bands overlaid on Migkas Fig. 8 for +15° tilt alone, tilt with gradient, and tilt with gradient plus +80.8° roll. Each highlight band is 18° end-to-end (±9° = 5% of 360°): two MAX and two MIN (cyan / green / magenta / blue). Migkas MAX markers include $(17°, -9°)$ and $(185°, 0°)$; the CoE axis is tilted +15° with anti-CoE near $(0°, +15°)$.

5.3 Kinematics and light travel

We work in units with $c = 1$, and set the observer's radial offset from the preferred centre O in the reference configuration to $d_1 = 1$. Figure 11 shows the exact photon intercept between two

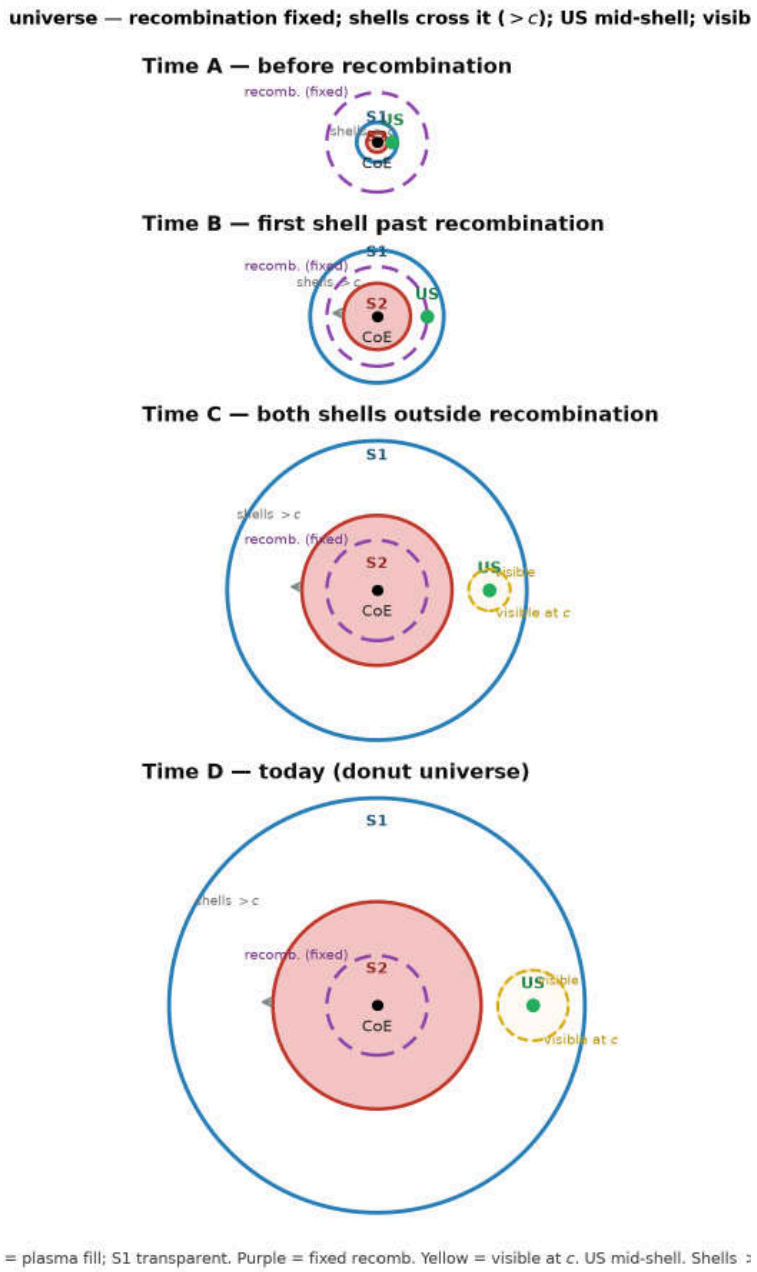


Figure 9: Geometric extrapolation of the same CoE+shell to last scattering (frames A–D): shells expand while the visible sphere grows at c, with recombination as a spherical surface about the CoE. Used here as the setup for the CMB null test (Sec. 4), not as a completed cosmology.

radially moving objects; the light-travel root t_f and arrival direction $\hat{\mathbf{n}}$ enter D_{motion} below.

Angles are measured from the fitted shell axis,

$$\theta = \ell - \phi, \tag{15}$$

with ϕ free in the fit (it returns to ℓ_0). The central mass sits at O; the observer is at $(d_1, 0) = (1, 0)$. The emitter (a last-scattering parcel in direction θ) sits at reference distance q from the observer:

$$\mathbf{x}_{e0} = (1 + q\cos\theta,\ q\sin\theta), \qquad r_{e0} = \sqrt{1 + q^2 + 2q\cos\theta}. \tag{16}$$

Thus $\theta = 0$ points away from O and $\theta = 180°$ toward O. A common expansion scale s

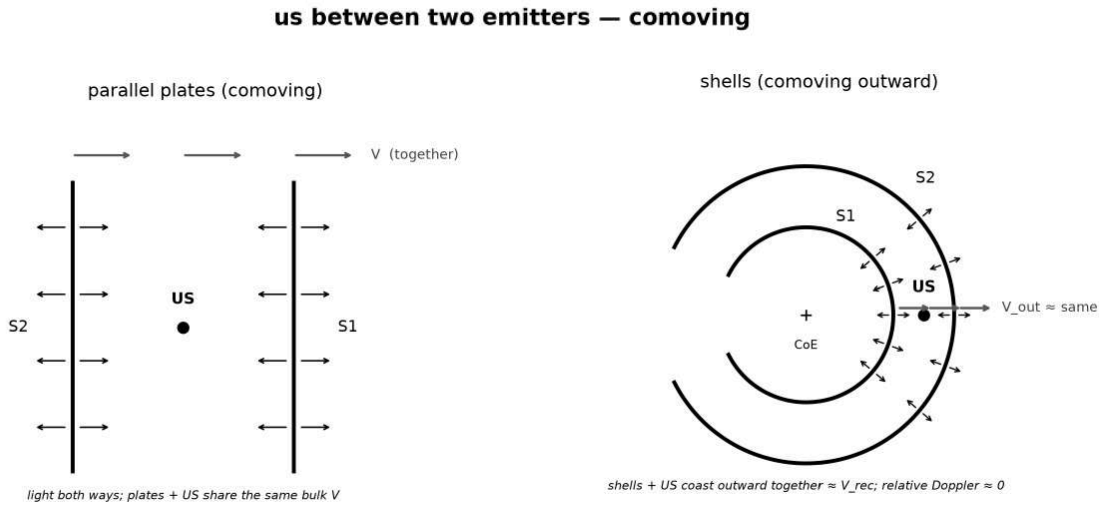


Figure 10: Two-emitter geometric baseline: opposite plates or shells sharing the observer's bulk velocity yield a flat kinematic sky before gravitational multipoles. Part of the pure-geometry extrapolation, which overshoots the CMB floor until SR and central-mass corrections are added (Sec. 5).

maps reference radii to present radii, $r_o = s$ and $r_e = s\, r_{e0}$. Velocities are radial from O (outward positive). Their magnitudes follow energy conservation in the central mass $\mu \equiv GM/c^2$ (measured in units of d_1):

$$v^2 = \beta_1^2 r_0^2 + 2\mu\Big(\frac{1}{r} - \frac{1}{r_0}\Big), \tag{17}$$

applied separately at the observer ($r_0 = 1$, $r = r_o = s$) and at the emitter ($r_0 = r_{e0}$, $r = r_e = s\, r_{e0}$). The corresponding velocity vectors are

$$\boldsymbol{\beta}_o = v_o\, \hat{\mathbf{x}}, \qquad \boldsymbol{\beta}_e = v_e\, \mathbf{x}_e / r_e. \tag{18}$$

With $\mathbf{d} = \mathbf{x}_o - \mathbf{x}_e$, the positive light-travel root t_f of

$$(v_o^2 - 1)\, t_f^2 + 2\, d_x\, v_o\, t_f + |\mathbf{d}|^2 = 0 \tag{19}$$

defines the photon arrival direction

$$\hat{\mathbf{n}} = \frac{\mathbf{d} + v_o t_f\, \hat{\mathbf{x}}}{t_f}. \tag{20}$$

5.4 Temperature pattern from the same kinematics

Figure 12 illustrates how the fractional blackbody-temperature anisotropy grows with expansion speed for the same intercept geometry, at the three speeds drawn (not the run 2 values). The shallow local minimum near 180° (inward) appears at every speed shown.

The special-relativistic Doppler factor between emitter and observer is

$$D_{\text{motion}} = \frac{\gamma_o\big(1 - \boldsymbol{\beta}_o \cdot \hat{\mathbf{n}}\big)}{\gamma_e\big(1 - \boldsymbol{\beta}_e \cdot \hat{\mathbf{n}}\big)}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2}}, \tag{21}$$

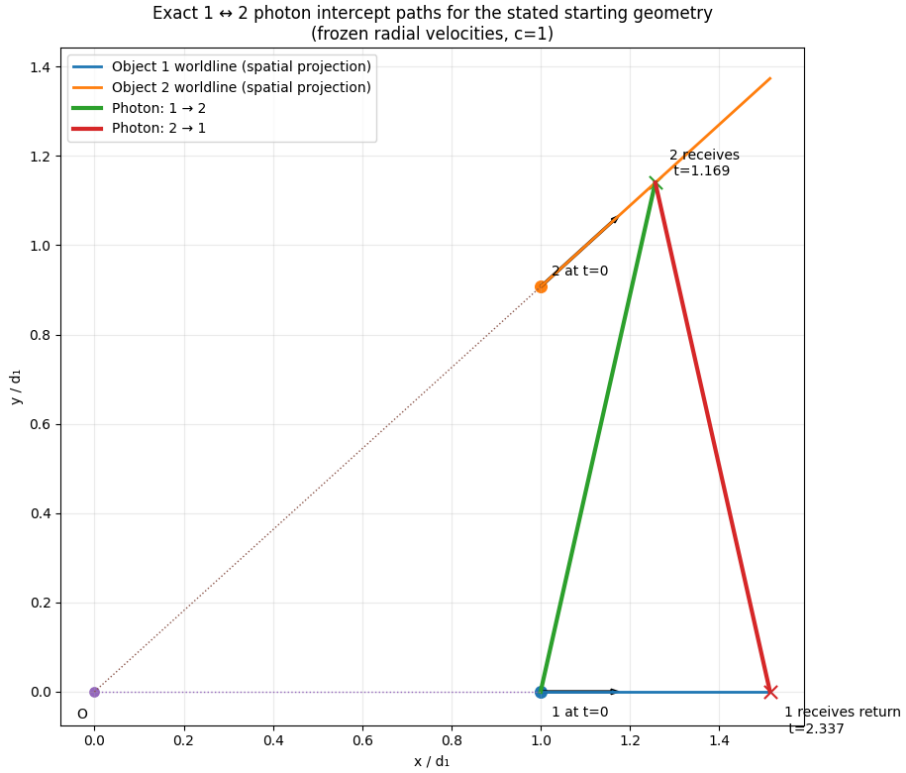


Figure 11: Illustration of exact photon intercept paths between two objects moving radially away from the centre O at frozen speeds ($c = 1$), for the speeds drawn in the figure—not the run 2 fit values. A photon sent from object 1 reaches object 2 at $t = 1.169$; the return reaches object 1 at $t = 2.337$. The light-travel root t_f and arrival direction $\hat{\mathbf{n}}$ enter D_{motion} [Eqs. (19)–(21)]. In the fit, the observer is at $0.469\,c$ and emitters span $0.205\,c$–$0.648\,c$.

and the gravitational redshift between the same radii is

$$D_{\text{grav}} = \sqrt{\frac{1 - 2\mu/r_e}{1 - 2\mu/r_o}}. \tag{22}$$

Their product is the model correction,

$$S_{\text{model}} = D_{\text{motion}}\, D_{\text{grav}}. \tag{23}$$

Applying it to the geometry and re-normalising gives the residual

$$R(\ell) = \frac{G\, S_{\text{model}}}{\langle G\, S_{\text{model}} \rangle} - 1. \tag{24}$$

(Here $R(\ell)$ is the residual, not the vision radius.) The objective minimised is RMS(R). Parameter points with $v^2 \notin (0, 1)$ or $2\mu/r \geq 1$ anywhere on the sky are rejected.

5.5 Geometric input

The target is the normalised Migkas shell curve from the weighted least-squares (WLS) fit: $D = 58.2\,\text{Gpc}$, fixed vision radius $R = 14.3\,\text{Gpc}$, shell axis $\ell_0 = 15.16°$ ($\approx 15.2°$), on 71 bins

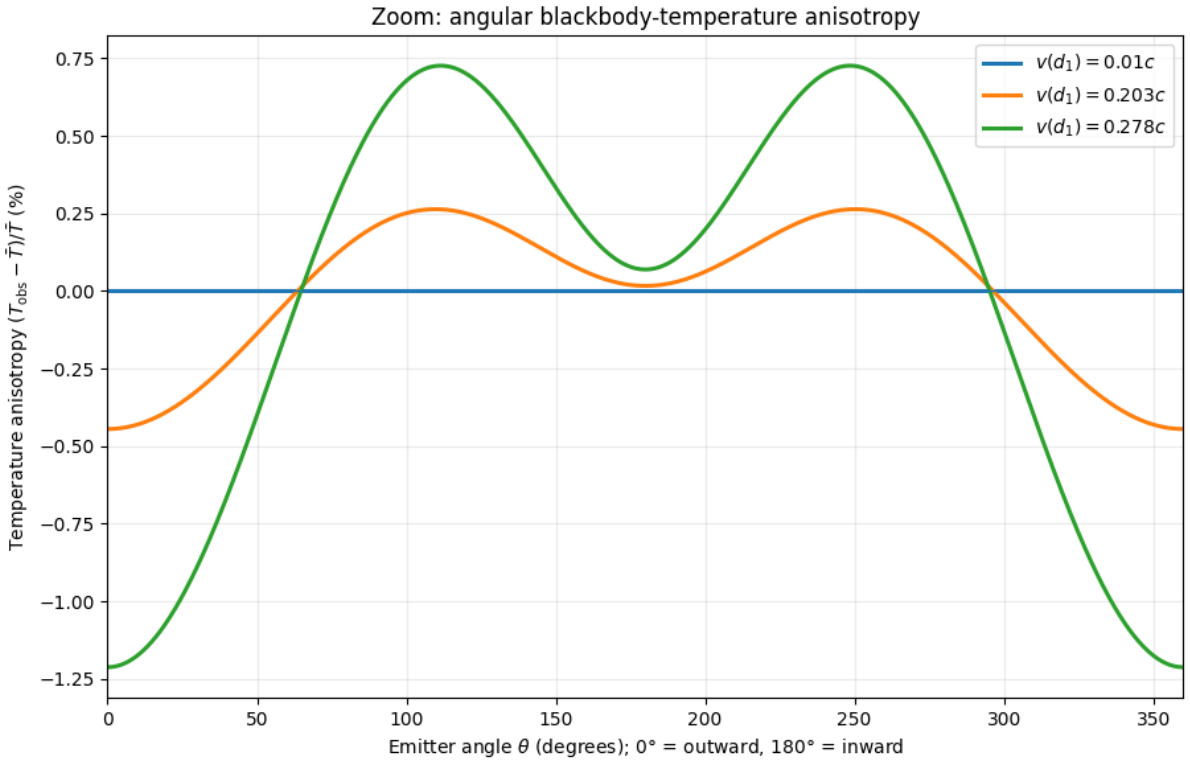


Figure 12: Illustration of fractional blackbody-temperature anisotropy $(T_{\rm obs} - \bar{T})/\bar{T}$ (percent) against emitter angle θ (0° outward, 180° inward), for the pedagogical speeds $v(d_1) = 0.01c$, $0.203c$, and $0.278c$ shown—not the run 2 fit. The SR sky pattern grows with expansion speed; a shallow local minimum near 180° appears in every case. In the fit the observer is at $0.469\,c$ and emitters span $0.205\,c$–$0.648\,c$, so the percent-level swings here are not the fitted anisotropy.

(Table 1). Let $A/A_{\rm all}(\ell)$ be that full-precision WLS curve on a 1° grid $\ell = 0, \ldots, 359$. Define

$$G(\ell) = \frac{A/A_{\rm all}(\ell)}{\langle A/A_{\rm all}\rangle}, \qquad \mathrm{RMS}(G-1) = 5.75 \times 10^{-2}, \tag{25}$$

where $\langle\cdot\rangle$ denotes the mean over the 360 longitude samples. A factor that would cancel G exactly (up to an overall scale) is

$$S_{\rm req}(\ell) = \frac{1/G(\ell)}{\langle 1/G\rangle}. \tag{26}$$

Figure 13 shows that geometric input before any SR or mass correction. The remainder of this section constructs a physical $S_{\rm model}$ and compares it to $S_{\rm req}$.

5.6 Required inverse versus physical correction

Figure 14 compares $S_{\rm req}$ to the SR-plus-mass model $S = D_{\rm motion} D_{\rm grav}$, with the Doppler and gravitational factors shown separately. The Doppler term carries most of the longitude structure; the gravitational factor is smoother and smaller.

5.7 Fitted parameters (run with $1 \le s \le 3$)

The run used below is the existing run 2 of the cancellation pipeline:

$$\beta_1 = 0.5866, \quad q = 0.2303, \quad \mu = 0.09332, \quad s = 3.000 \text{ (upper bound)}, \quad \phi = 15.2^\circ. \tag{27}$$

Here β_1 is the free-expansion speed at unit reference radius, q is the emitter distance from the observer in units of d_1, μ is the central-mass length, s is the common expansion scale, and ϕ coincides with the shell axis ℓ_0. At this point the present observer speed is $v_o = 0.469\,c$; emitter speeds range from $0.205\,c$ to $0.648\,c$; the largest $2\mu/r$ on the sky is 0.081. Starting from $\mathrm{RMS}(G-1) = 5.75 \times 10^{-2}$, one obtains

$$\mathrm{RMS}(R) = 2.37 \times 10^{-3} \qquad \big(\max|R| = 3.76 \times 10^{-3}\big), \tag{28}$$

which removes 95.9% of the starting RMS by amplitude and 99.83% of the variance. Equivalently, the starting anisotropy is about 5,750 times the CMB 10^{-5} level; after SR plus mass it is about 237 times that level. Parameter s sits on its imposed ceiling (a run with $s \le 50$ reaches $s = 50$ with a slightly lower RMS of 2.15×10^{-3}). Because the intercept geometry is scale-free, s is only weakly constrained; the physical content is carried by β_1, q, μ, and ϕ. Figure 15 shows the residual R after that correction, with the cosine series through $m = 12$ overlaid.

> **Fix Point 2.** Special-relativistic Doppler and central-mass gravitational redshift cancel most of the geometric last-scattering anisotropy while the low-redshift Migkas morphology (Fix Point 1) remains intact. No new geometric degree of freedom is opened to force the cancellation.

5.8 Fourier analysis of the residual

The residual is expanded about the shell axis $\phi = 15.2^\circ$:

$$R(\ell) = a_0 + \sum_{m=1}^{M} \Big[c_m \cos\big(m(\ell-\phi)\big) + s_m \sin\big(m(\ell-\phi)\big) \Big]. \tag{29}$$

Every sine coefficient satisfies $|s_m| \sim 10^{-8}$–10^{-12}, so R is a pure cosine series about ℓ_0. Through $M = 12$,

$$\mathrm{RMS}\Big(R - \sum_{m=1}^{12} c_m \cos\big(m(\ell-\phi)\big)\Big) = 1.83 \times 10^{-6} \tag{30}$$

on the full-precision geometry curve (below the CMB 10^{-5} level). This Fourier step diagnoses the residual after the physical SR-plus-mass correction; it does not replace that correction. Modes $m = 1$ and $m = 2$ of $\ln G$ are matched almost exactly by $\ln(1/S_{\mathrm{model}})$, which is why SR plus mass removes 95.9% of the starting RMS. Modes $m = 3$, 4, and 5 together carry about 92% of the leftover variance: the shell geometry's multipole spectrum falls faster with m than the relativistic Doppler pattern of a point-mass-plus-Hubble flow. Modes $m \ge 7$ are almost absent from $\ln G$ and are the geometric high-m Doppler tail. Figure 16 shows the longitude harmonics $m = 1$–12 of that residual.

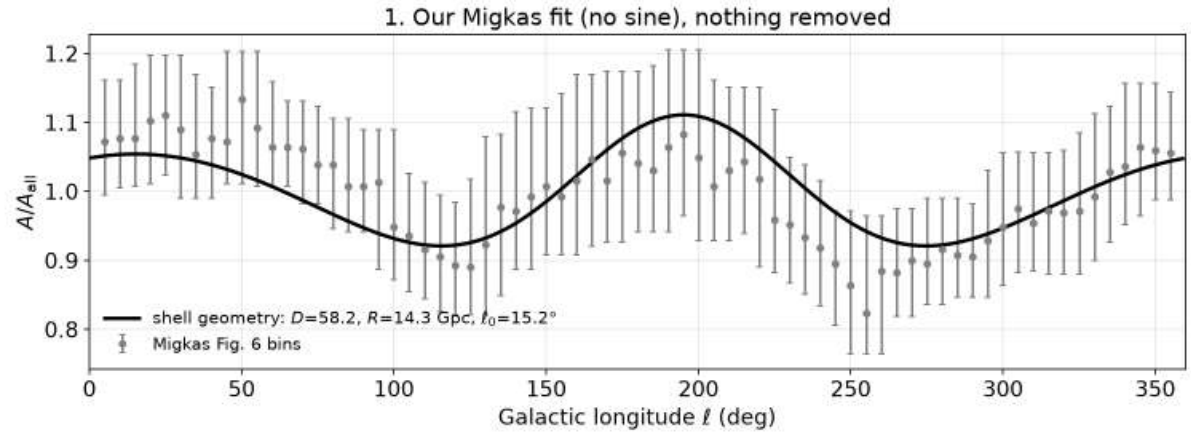


Figure 13: Migkas $A/A_{\rm all}$ bins and the WLS shell geometry ($D = 58.2\,$Gpc, fixed $R = 14.3\,$Gpc, $\ell_0 = 15.2°$) before any SR or mass correction.

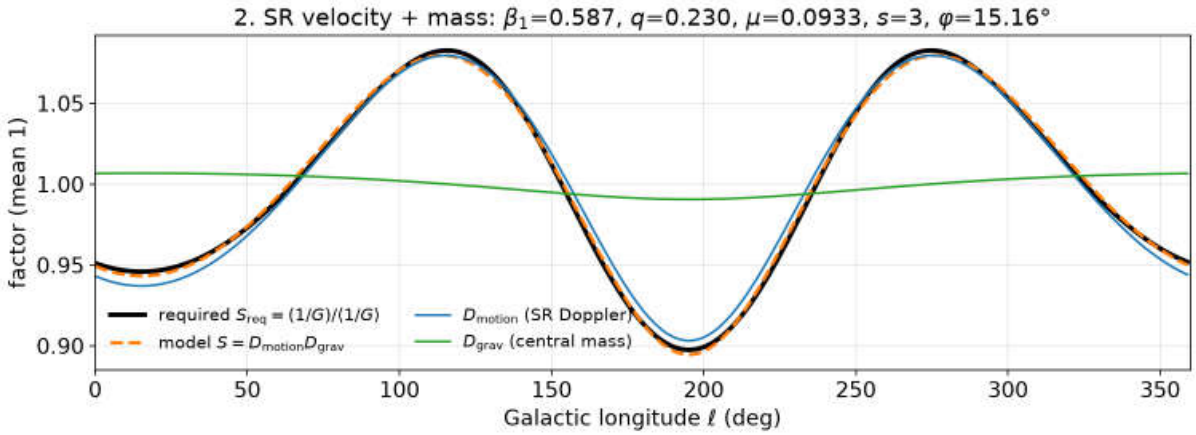


Figure 14: Required inverse $S_{\rm req}$ versus the SR-plus-mass model $S = D_{\rm motion} D_{\rm grav}$, with the separate Doppler and gravitational factors.

5.9 Leftover after removing $m = 1$–12

Figure 17 shows what remains after subtracting the $m = 1$–12 cosine series on the full-precision geometry curve (RMS 1.83×10^{-6}), compared with $\pm 10^{-5}$. Reaching the CMB amplitude still requires accounting for the $m = 3$–5 mismatch by additional physics, not by the SR-plus-mass model alone.

Figure 18 repeats the same removal on the four-decimal Migkas CSV: the leftover is then dominated by rounding noise (RMS 2.79×10^{-5}).

5.10 Error reduction in CMB units

Figure 19 summarises the RMS drop after each analysis step in units of the CMB 10^{-5} floor, and the linear share of the starting 5,751× removed by SR plus mass versus the Fourier diagnostic.

5.11 Exploratory four-clump sketch

As an existence check that off-axis mass can source the leftover multipoles, consider four equal clumps at distance $D\,R_{\rm vis}$ with $R_{\rm vis} = 14.3\,$Gpc, longitudes α_i, and masses M_i. With the observer at the origin and a dust direction $\hat{\mathbf{n}} = (\cos\ell, \sin\ell)$, the radial tidal acceleration (observer

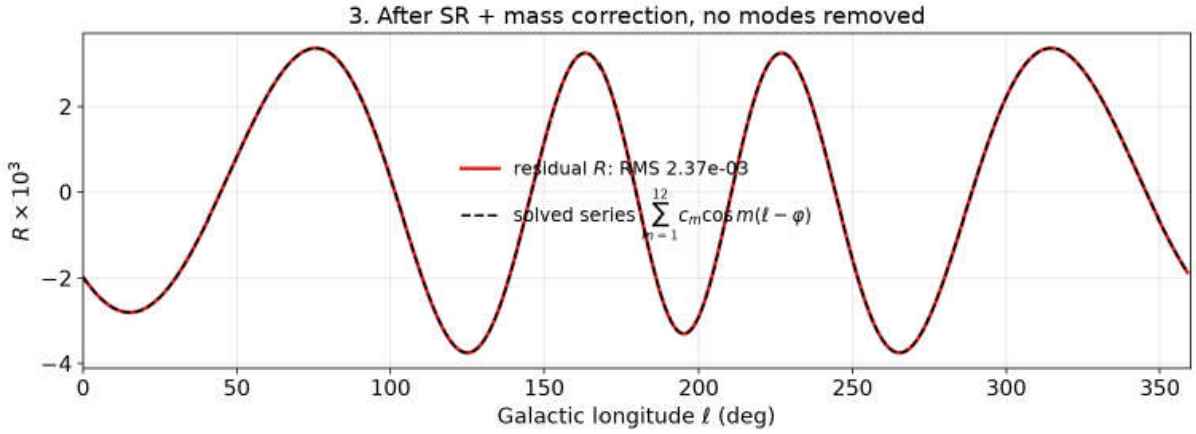


Figure 15: Residual R after SR plus central mass [Eq. (28)], with the cosine series through $m = 12$ overlaid.

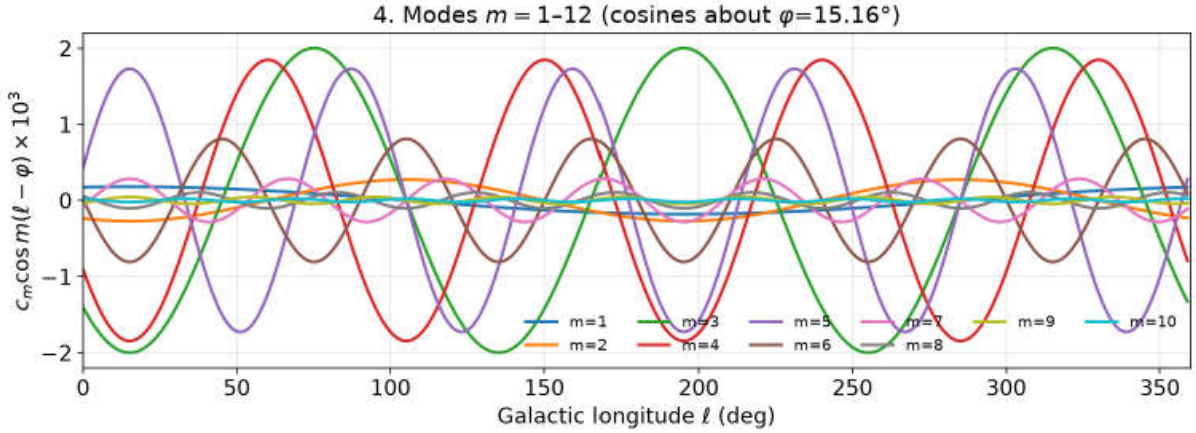


Figure 16: Longitude harmonics $m = 1$–12 of the post-SR residual (cosines about $\phi = 15.2°$).

common mode subtracted) is

$$a_r(\ell) = \sum_{i=1}^{4} GM_i \left[\frac{(\mathbf{X}_i - \hat{\mathbf{n}}) \cdot \hat{\mathbf{n}}}{|\mathbf{X}_i - \hat{\mathbf{n}}|^3} - \frac{\mathbf{X}_i \cdot \hat{\mathbf{n}}}{|\mathbf{X}_i|^3} \right], \qquad \mathbf{X}_i = DR_{\rm vis}(\cos\alpha_i, \sin\alpha_i), \tag{31}$$

and the corresponding Doppler imprint over a cosmic age $t = 13.8\,\mathrm{Gyr}$ is

$$\frac{\Delta T}{T}(\ell) \approx \frac{a_r(\ell)\, t}{c}. \tag{32}$$

Two mirror pairs about $\phi = 15.2°$ give the best-fit longitudes $\{75°, 162°, 228°, 315°\}$ at $D = 1.715$ ($r = 24.5\,\mathrm{Gpc}$) with equal masses $M = 7.7 \times 10^{51}\,\mathrm{kg}$. Figure 20 shows where those masses sit relative to the post-SR residual. It is exploratory only and is not part of the SR treatment proper.

5.12 Residual left after the four clumps

Figure 21 shows only the leftover after subtracting the four-clump model from the post-SR residual. Starting geometry RMS is 5.75×10^{-2}; after SR plus central mass it is 2.37×10^{-3} (95.9% of the starting RMS removed). After the four clumps it falls to 8.34×10^{-4}—a further

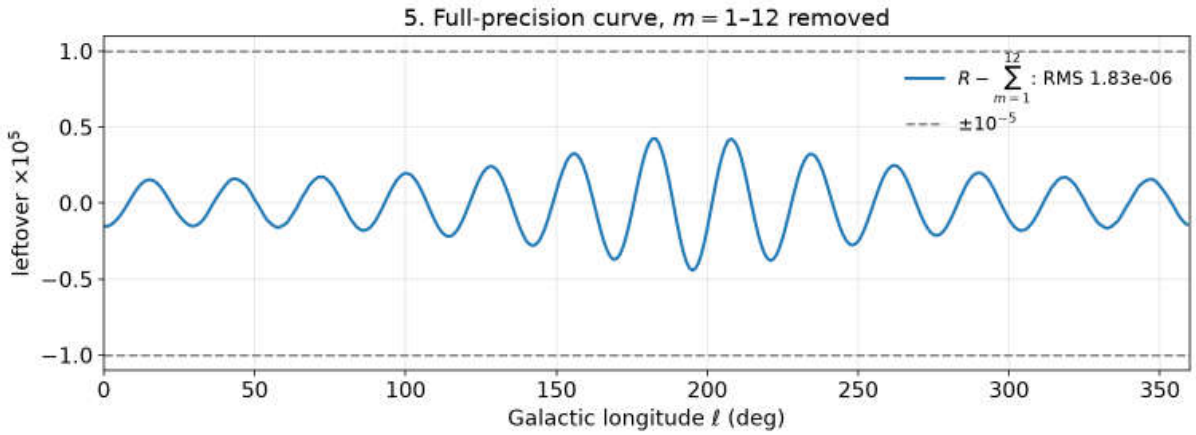


Figure 17: Leftover after removing $m = 1$–12 on the full-precision geometry curve (RMS 1.83×10^{-6}), compared with $\pm 10^{-5}$.

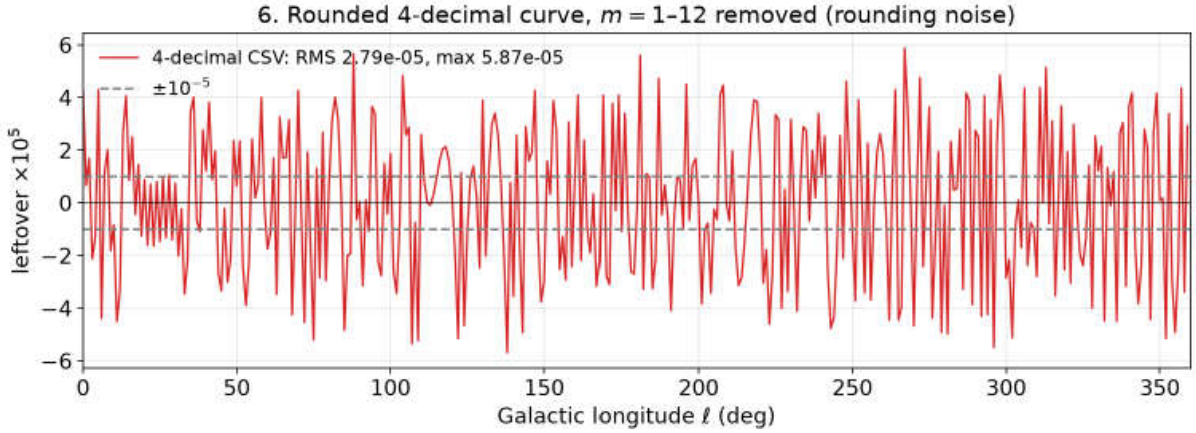


Figure 18: Same removal on the four-decimal Migkas CSV: leftover is dominated by rounding noise (RMS 2.79×10^{-5}).

2.7% of the starting RMS, 98.5% removed in total (99.98% by variance), and 88% of the post-SR variance. The maximum absolute residual is 2.0×10^{-3} near $\ell \approx 195°$, about 83 times the CMB 10^{-5} level. That leftover is mostly $m = 6$ ($\sim 53\%$ of its variance) plus an $m = 1$ dipole ($\sim 32\%$), with smaller $m = 5$ and $m = 9$. The clumps remove $m = 3$ and $m = 4$ almost completely but raise $m = 1$ from about 1.8×10^{-4} to 6.7×10^{-4}, so they trade low-order power for a larger dipole.

6 Residual multipoles and limits

6.1 What remains after SR plus central mass

The SR-plus-central-mass treatment describes radial expansion slowed by the central mass. Real matter is not a smooth shell: voids, filaments, and clumps, plus shear and rotation, sit on top of the primary flow. The multipole residuals of Sec. 5 are the fingerprint those secondary motions would have to reproduce. No new fits are reported in this section.

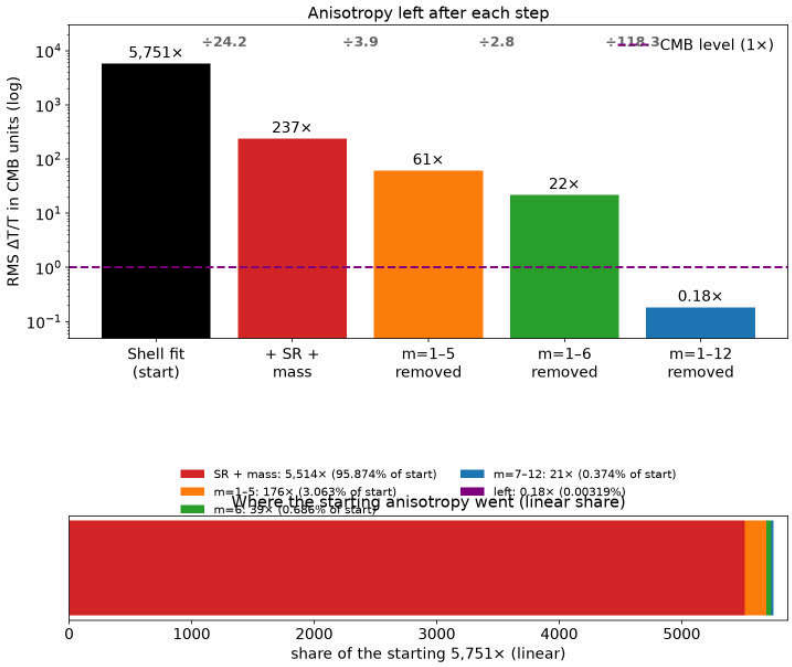


Figure 19: Anisotropy left after each analysis step in CMB units, and the linear share of the starting 5,751× removed by SR plus mass versus Fourier modes.

6.2 Decomposing the motion

At any point, the velocity field $\mathbf{v}(\mathbf{x})$ can be split into its primary expansion and a perturbation,

$$\mathbf{v} = \mathbf{v}_0(r)\,\hat{\mathbf{r}} + \delta\mathbf{v}(\mathbf{x}), \tag{33}$$

where $\mathbf{v}_0$ is the radial CoE flow used in the SR fit. The local velocity gradient separates uniquely into three parts:

$$\frac{\partial v_i}{\partial x_j} = \frac{1}{3}\,\Theta\,\delta_{ij} + \sigma_{ij} + \omega_{ij}. \tag{34}$$

Here $\Theta = \nabla \cdot \mathbf{v}$ is the expansion rate, σ_{ij} is the trace-free symmetric shear, and ω_{ij} is the antisymmetric vorticity. The primary CoE flow carries only Θ. Voids, threads and clumps are made by σ_{ij} together with spatial changes in Θ; rotation is carried by ω_{ij}.

6.3 The medium is dust, not an elastic solid

If the matter shell behaved like a constant-density elastic solid at the mean cosmic density, $\rho \approx 8.5 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$, the residual harmonics would be standing normal modes. Their frequencies would be

$$f_\ell \simeq \frac{\sqrt{\ell(\ell+1)}}{2\pi R}\sqrt{\frac{K}{\rho}}, \tag{35}$$

where K is the bulk modulus. Even with a radiation-fluid sound speed $c/\sqrt{3}$, the $m = 3$ mode on the $R = 14.3\,\mathrm{Gpc}$ scale would have a period of roughly 50 to 100 billion years. After recombination the matter is pressureless dust with $K \to 0$, so there are no restoring oscillations. The appropriate picture is a velocity ripple: matter flows slowly under gravity, and such gravitational flows grow monotonically rather than oscillating.

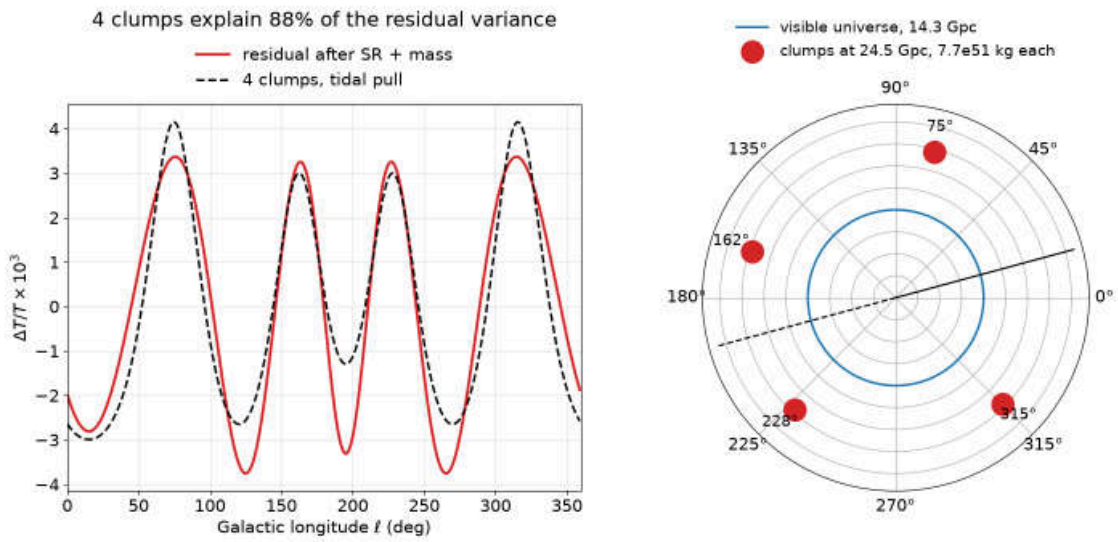


Figure 20: Exploratory four-clump placement [Eqs. (31)–(32)]: post-SR residual compared with the tidal pull of four equal masses of 7.7×10^{51} kg at 24.5 Gpc, at $\ell = 75°$, 162°, 228°, and 315°.

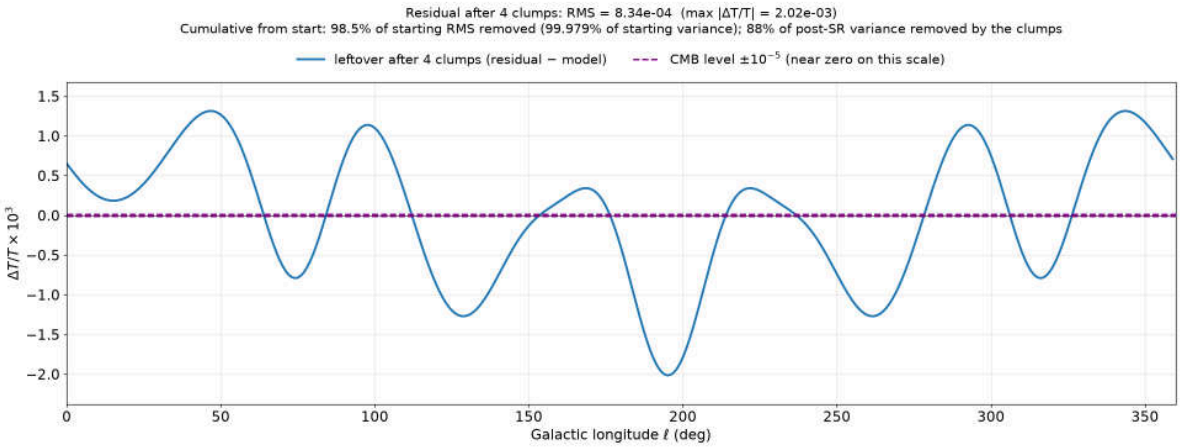


Figure 21: Residual after the four clumps only (no repeat of Fig. 20): RMS 8.34×10^{-4}, max $|\Delta T/T| = 2.0 \times 10^{-3}$ near 195°, $\sim 83\times$ the CMB 10^{-5} floor; 98.5% of starting RMS removed in total (99.98% by variance), 88% of post-SR variance. Leftover multipoles ($m = 6$ dominant, then $m = 1$, with smaller $m = 5$ and $m = 9$) are the fingerprint still to be explained.

6.4 Voids, threads, clumps, and rotation

Through the Doppler factor $D_{\rm motion}$, peculiar velocities change the observed temperature by $\Delta T/T \approx \delta\mathbf{v} \cdot \hat{\mathbf{n}}/c$. The dominant residual modes $m = 3$ to 5 have angular wavelengths $\lambda_m = 2\pi R/m \approx 18$–30 Gpc at $R = 14.3$ Gpc. Treated as velocity ripples in a shell holding an assumed shell fraction $\Omega_s = 0.049$ (earlier shell fit) of the critical mass, they need about 7×10^{64} J, roughly 0.002% of the shell's kinetic energy. The clumps are not proposed as real objects. Four positive point masses (about 7.5–7.8×10^{51} kg each at 24.5 Gpc; Figs. 20 and 21), each standing for the centre of gravity of accumulated matter in a chaotic web of filaments and voids, reduce the post-SR residual to 8.34×10^{-4} without rotation or rheology. Nothing requires them to be symmetric. Adding further positive points either drives D or the masses to unphysical values or rearranges them into near-uniform rings that shift power to higher multipoles; none reaches 10^{-5}. The remaining residual is therefore posed as a statistical problem for the cosmic web

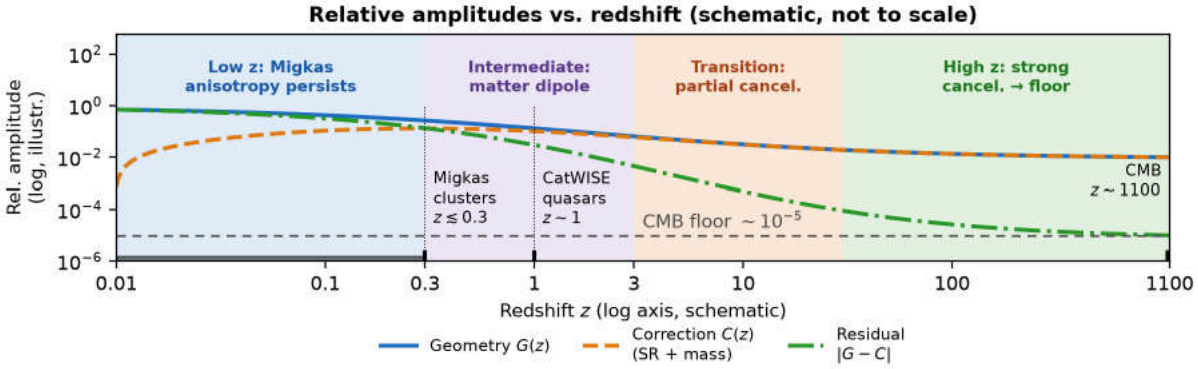


Figure 22: Schematic of relative amplitudes versus redshift in four regions: low z, where the Migkas anisotropy persists; intermediate z, the matter-dipole regime; a transition with partial cancellation; and high z, strong cancellation down to the 10^{-5} CMB floor. The geometric multipole G falls with z while the SR-plus-mass correction C rises in importance relative to geometry (C/G rises), not in absolute amplitude; the residual is |geometry − correction|. Markers indicate the Migkas, CatWISE, and CMB regimes. Illustrative, not to scale; region boundaries schematic.

rather than a list of objects.

If the shell rotates, vorticity ω_{ij} is non-zero. In the Fourier analysis about $\phi = 15.2°$ [Eq. (29)], rotation shows up as non-zero sine coefficients s_m. In the present SR fit, $|s_m| \sim 10^{-8}$, so the residual is axisymmetric: the current model contains no rotation, and the residual itself does not require any. Adding the one-cycle phase factor of Migkas et al. raised the geometry fit from $R^2 = 0.521$ to 0.845–0.870 (Table 1). A rotating model refit to the Migkas data and run through the same SR pipeline would predict specific s_m that the residual could confirm or rule out.

Table 2 lists construction degrees of freedom versus the sky observables they set and the Migkas-qualified data used here. Secondary motions map onto multipoles in a simple way: a bulk flow gives $m = 1$, large-scale shear gives $m = 2$, horizon-scale voids, threads and clumps give $m = 3$–6, smaller structures give $m \geq 7$, and rotation gives sine terms s_m that break the axisymmetry. The polar shapes of the $m = 1$–12 harmonics about the shell axis are collected on their own appendix page (Fig. 35).

6.5 LTB as the eventual exact class

Phenomenological Newtonian-plus-SR bookkeeping is not itself an LTB solution. LTB (and related exact inhomogeneous dust cosmologies) remain the natural exact class in which to embed the same off-centre shell once residual multipoles are under control; we do not claim that the present phenomenological fit *is* LTB. Figure 25 (placed after the redshift-evolution figures to preserve float order) shows LTB shell-evolution branches.

6.6 Caveats

This is a qualitative map, not a solved model. The kinetic-energy estimate assumes velocity ripples in a uniform shell with an assumed shell fraction $\Omega_s = 0.049$ (earlier shell fit). The clump sketch assumes a steady tidal pull over 13.8 billion years, so its masses are uncertain by a factor of a few. Rotation has not been fitted. A full treatment would solve the continuity,

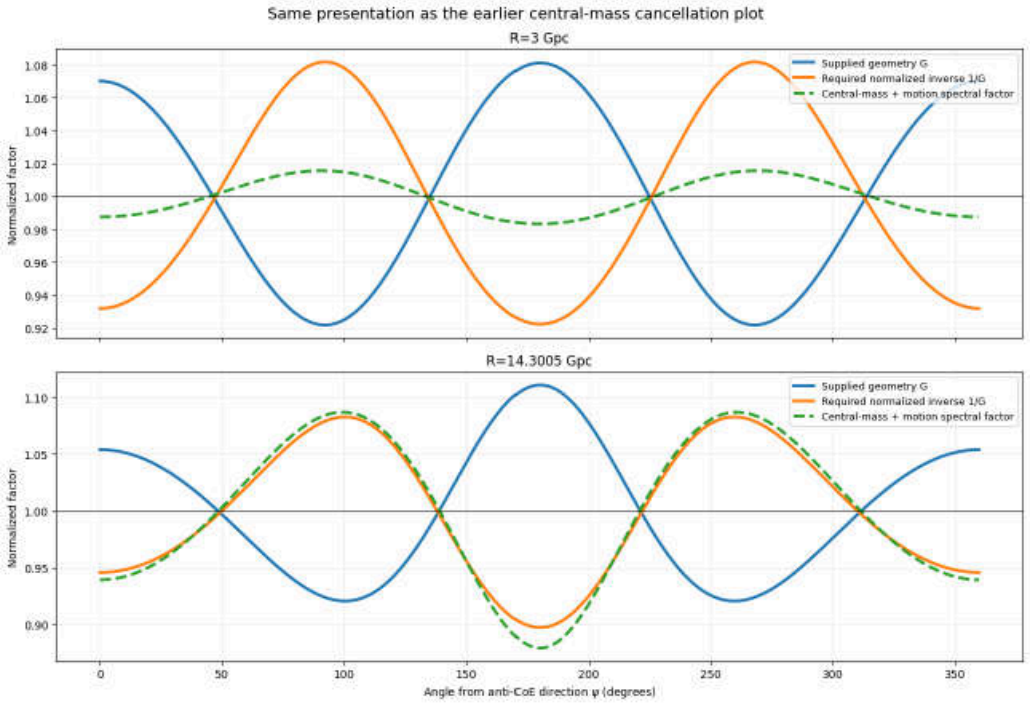


Figure 23: Same cancellation presentation as the earlier central-mass plus motion spectral factor, at near ($d_{\rm near} = 3\,$Gpc) and far ($d_{\rm far} = 14.3\,$Gpc) distances: supplied geometry G, required normalised inverse $1/G$, and the spectral factor. Cancellation is weak at low z and strong at $d_{\rm far}$ (illustrative). The green curve is that illustrative spectral factor, not $\gamma(1-\boldsymbol{\beta}\cdot\hat{\mathbf{n}})$ evaluated on Newtonian $|\beta_E| \approx 12.7$ shell speeds; true SR ratios are those of Sec. 5.

Table 2: Construction degrees of freedom, the sky observable each sets, and the Migkas-qualified datum used in this paper. Geometry is Fix Point 1 before Migkas; amplitude partly construction-motivated, morphology out-of-sample.

DoF	Observable (what that DoF sets on the sky)	Migkas-qualified observation
D	Overall distance/Hubble scale via D/R ($R = 14.3\,$Gpc fixed); annulus $D-d \ldots D+d$	CatWISE lock: $D \approx 58.2\,$Gpc; free-L: $D \approx 27.95\,$Gpc
d	Near/far sampling asymmetry along lines of sight	Redshift-slice $A(\ell)$ at each slice median distance (Fig. 5)
$V(D)$	Enclosed-volume / density scale for mass budget	CatWISE lock, Table 1
$M_{\rm tot}$	Overall amplitude of $A/A_{\rm all}$ (density–distance degeneracy)	R^2 and σ rows in Table 1
$\rho(r)$	Angular morphology and $\varphi(\theta)$ rings (Fix Point 1)	Sky MAX/MIN bands vs Migkas Fig. 8 (Fig. 8); cluster blue-ring morphology (Fig. 3)
θ	Directional $A(\theta)/A_{\rm all}$ shape along the axis	Migkas $A/A_{\rm all}$ vs ℓ (71 bins / sliding cones; Figs. 4, 6)
ℓ_0	Preferred axis on the Galactic sky	$\ell_0 \approx 15.2^\circ$ (CatWISE lock); 18.6° free-L

Euler and Poisson equations for pressureless matter on the CoE background and compare the resulting c_m and s_m with the residual fingerprint.

7 Independent signed matter-dipole constraint

An independent probe of large-scale anisotropy is the dipole in the number counts of distant quasars. Secrest et al. [12], using a CatWISE2020 sample of $\sim 1.36 \times 10^6$ quasars (with Sarkar

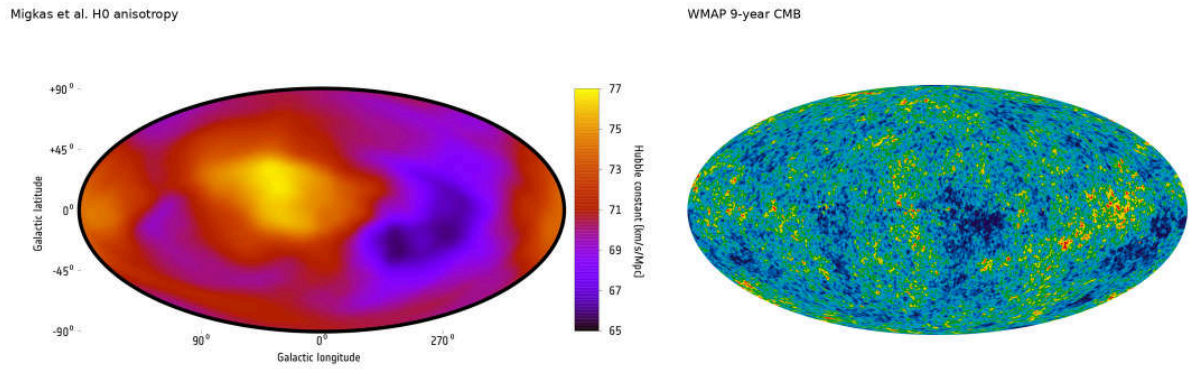


Figure 24: Side-by-side contrast of a low-z directional sky and a high-z microwave sky: Migkas et al. H_0 anisotropy map (left) and the WMAP 9-year CMB temperature map (right; NASA/WMAP Science Team). The pair illustrates the observational tension that the z-dependent cancellation is meant to address, not a quantitative prediction from the schematic above.

and collaborators), report a dipole amplitude

$$\mathcal{D} \approx 0.0155 \tag{36}$$

whose direction is broadly related to the CMB dipole but whose amplitude exceeds the purely kinematic expectation. We use this only as a signed axial constraint on the CoE geometry—not as a new fit target for $\mathcal{D}$.

7.1 Hemisphere counts and density contrast

Write $N_{\rm high}$ and $N_{\rm low}$ for source counts in opposite hemispheres along a candidate axis. For a pure dipole, the hemisphere-averaged $\langle \cos\theta \rangle$ factor is $1/2$, so the fractional count asymmetry relates to $\mathcal{D}$ but is not identical to a local density contrast:

$$\frac{\Delta N}{N} \neq \frac{\Delta \rho_m}{\rho_m}. \tag{37}$$

Number-count dipoles mix density, magnification, and selection; they must not be read as raw $\Delta\rho_m/\rho_m$ at the shell.

7.2 Signed axis: CoE versus antipode

The geometry's preferred axis has a CoE end and an anti-CoE end. The shell axis $\ell_0 = 15.2°$ (CatWISE lock; 18.6–19.9° in free-L/stretched fits) marks the anti-CoE end; the CoE end lies near 195°. The observational question is which end is the high-count hemisphere. Great-circle separations between the CatWISE dipole direction, the CMB dipole, and the CoE/anti-CoE ends diagnose whether the matter dipole prefers the CoE or the antipode.

The model density contrast between the near and far sides of the shell, $\rho(D-d)$ versus $\rho(D+d)$, is fixed by the already-chosen mass distribution (Fix Point 1). It is *not* adjusted to reproduce $\mathcal{D} = 0.0155$. Whether the high-count end coincides with the denser side is then a genuine prediction / consistency check. The original CoE spreadsheet already had $\rho(r)$ among

Table 3: Signed matter-dipole checklist (text-only constraint; not a fit to $\mathcal{D}$).

Quantity	Value / note
CatWISE/Secrest dipole $\mathcal{D}$	≈ 0.0155
Hemisphere factor $\langle \cos\theta \rangle$	1/2 for a pure dipole
$\Delta N/N$ vs $\Delta\rho_m/\rho_m$	not interchangeable
CoE-end longitude (Migkas-oriented)	$\sim 195^\circ$
Anti-CoE / ℓ_0 end	15.2°
Model $\rho(D-d)$ vs $\rho(D+d)$	predicted by geometry; *not* fitted to $\mathcal{D}$

its DoFs $(D, d, V(D), M_{\rm tot}, \rho(r), \theta)$; off-centre, non-uniform $\rho(r)$ implies a denser side of the sky, which CatWISE [12] measured at $\sim 3\%$. Existence and sign were predicted; the amplitude is from data (a 3% lock). With it ($D = 58.2$, $D \pm d = 43.9/72.5$ Gpc) four laws fit Migkas $A/A_{\rm all}$: soft core+tail $R^2 = 0.743$ (5.1σ), exponential 0.716 (5.4σ local, 5.1σ global), power law 0.72 (5.0σ; not in Table 1) at $\ell_0 = 15.2^\circ$; stretched exp. 0.745 (5.0σ, $\ell_0 = 19.9^\circ$). The morphology is insensitive to the law.

7.3 Observational caveat

The CatWISE dipole result is debated: later reassessments emphasise partial-sky mode coupling and higher multipoles that can inflate the apparent dipole significance. We therefore treat $\mathcal{D} \approx 0.0155$ and the signed-axis comparison as an independent external constraint with acknowledged systematic uncertainty, not as a definitive rejection or confirmation of the CoE picture.

8 Redshift evolution of $A(z, \theta)$

The SR-plus-central-mass cancellation of Sec. 5 is not a single number at one redshift. Both the geometric gravitational multipole and the relativistic (Doppler plus central-mass) correction change with distance, so their relative sizes change with z. Multipoles $a_\ell(z)$ therefore evolve: at low z the geometric anisotropy is large compared with the relativistic correction, so a Migkas-scale directional Hubble anisotropy can persist; at intermediate z a matter-dipole regime is natural; through a transition region the cancellation is partial; at high z, approaching last scattering, the relativistic correction grows in importance relative to the geometric term (C/G rises, not its absolute amplitude) and can cancel most of the geometric multipole, leaving a residual near the CMB 10^{-5} floor.

Figure 22 is a schematic of that hierarchy, with residual |geometry − correction| (illustrative, not to scale; region boundaries schematic). Figure 23 shows the same cancellation presentation used earlier for the central-mass plus motion spectral factor, now at a near distance ($d_{\rm near} = 3$ Gpc) and a far distance ($d_{\rm far} = 14.3$ Gpc, numerically equal to R): at low z the spectral factor is too weak to track $1/G$; at high z it closely follows $1/G$ (illustrative).

Caveat on the green spectral factor. The green curve in Fig. 23 is the same illustrative spectral factor as in the earlier central-mass plot, not a raw insertion of Newtonian shell speeds into the special-relativistic Doppler factor $\gamma(1 - \boldsymbol{\beta} \cdot \hat{\mathbf{n}})$. Absolute CoE-frame speeds from the Newtonian shell geometry reach $|\beta_E| \approx 12.7$, which cannot be fed unmodified into that formula.

The quantitative SR temperature ratios remain those of Sec. 5, with subluminal β from the fitted kinematics. Likewise, schematic comparisons of G versus $1/G$ must not be confused with a claim that the physical correction equals $1/G$ at every z.

Figure 24 places the low-z Migkas H_0 anisotropy map beside a WMAP 9-year CMB temperature map to contrast a strongly directional late-time sky with the nearly isotropic high-z microwave sky. This section does not introduce a new fit; it frames the coexistence problem already quantified in Sec. 5.

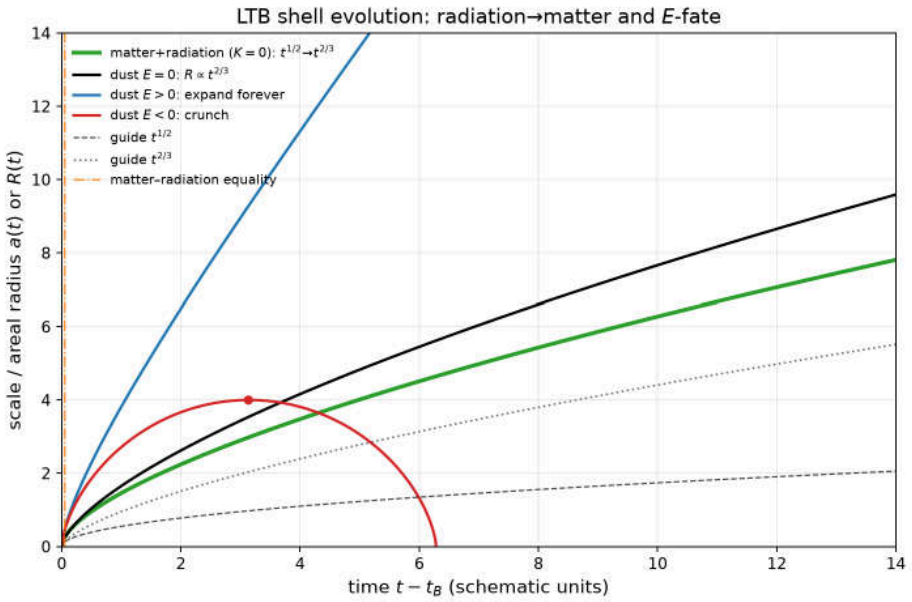


Figure 25: LTB shell-evolution solutions: flat matter-plus-radiation track and dust branches at $E = 0$, $E > 0$, and $E < 0$.

9 Discussion and conclusions

9.1 Chronology recapitulated

We began from apparent SN Ia acceleration as an inverse problem, constructed an off-centre finite geometry with prior angular morphology (Fix Point 1), compared morphology out-of-sample to Migkas et al., showed that pure geometry alone overshoots the CMB floor, completed the photon physics with SR Doppler and central mass (Fix Point 2; 95.9% RMS cancellation), mapped residual multipoles, confronted an independent signed matter dipole, and framed the redshift hierarchy $A(z, \theta)$. Apparent late-time acceleration remains an off-centre sampling artefact: neither dark energy nor outward shell acceleration is required. A contracting CoE geometry can share the average appearance of acceleration while differing in angular signature.

9.2 Fitted versus predicted

The CatWISE-locked fits give $D \approx 58.2\,\mathrm{Gpc}$, $D \pm d \approx 43.9/72.5\,\mathrm{Gpc}$, $\ell_0 = 15.2°$; the free-L exponential gives $D \approx 27.95\,\mathrm{Gpc}$; an earlier, superseded LTB shell fit gave $D \approx 32.9\,\mathrm{Gpc}$. Special-relativistic Doppler plus central-mass gravitational redshift cancel 95.9% of the geomet-

Table 4: Prediction ledger: each claim, where it came from, how it is tested, and the outcome.

Claim	Origin	Test	Outcome
Migkas $A(\ell)$ curve	Shape a priori (Fix Point 1); orientation and scale fitted	Migkas et al. cluster anisotropy	Fit, not an out-of-sample pass: $R^2 \approx 0.52$–0.87, 4.1–5.3σ global
$\varphi(\theta)$ rings	Prior geometry	Celestial-sphere rings (Fig. 3)	Observed
Density gradient sign	$\rho(r)$ DoF, before CatWISE	Secrest et al. dipole, longitude only, axisymmetry assumed	Sign check passed at hemisphere level ($\Delta\ell \approx 43°$; chance ~ 1 in 4); competes with kinematic reading
CMB isotropy	SR + central mass (Fix Point 2)	Planck $\sim 10^{-5}$ floor	Addressed qualitatively: 95.9% cancelled, residual 2.37×10^{-3}; four-clump sketch 8.34×10^{-4} ($\sim 83\times$ floor). Open inverse problem
$A(z, \theta)$ hierarchy	Redshift-dependent cancellation	Binned anisotropy vs z	Open

ric RMS (residual 2.37×10^{-3}); leftover $m = 3$–5 multipoles after SR ($m = 6$ and $m = 1$ after the four-clump sketch) are the fingerprint of what remains. The open problem is whether those leftovers can be brought to the 10^{-5} level by additional physics without spoiling the Migkas anisotropy. We frame it as a ladder, each rung scored by a Bayes factor $\ln B$ against a ΛCDM random-fluctuation baseline. Rung 0 is the off-centre bound of Alnes and Amarzguioui and of Foreman et al.: an observer ~ 58 Gpc from the CoE needs an explicit dipole budget, including the extra $m = 1$ the clumps introduce. Rung 1, mass points, must predict number counts, lensing and flows. Rung 2, rotation, needs a spin-handedness or CMB-parity test. Rung 3, rheology, must fit $A(z, \theta)$ with one law. The end goal is $\sim 10^{-5}$ on every ring plus one new dataset predicted in advance; the next step is to find further multipole signals to fit against the geometry. The goal is coexistence of both datasets under one off-centre shell geometry—a logical alternative to the homogeneous–isotropic cosmological principle, not a new concordance constant.

The model universe at a glance.

Distances	CoE–observer distance $D \approx 58.2$ Gpc (CatWISE lock; free-L: 27.95 Gpc); lines of sight sample the annulus $D-d \ldots D+d \approx 43.9 \ldots 72.5$ Gpc; vision radius $R = 14.3$ Gpc fixed (shell rule $D \gtrsim 14.7$ Gpc, observer mid-shell).
Masses	$M_{\rm tot}$ sets the amplitude of $A/A_{\rm all}$ through the density–distance degeneracy and is not quoted separately; $\rho(r)$ sets the morphology (exponential, $L \approx 6.07$ Gpc free-L, or soft core). The CMB sketch adds four equal clumps of 7.7×10^{51} kg at 24.5 Gpc, $\ell = 75°, 162°, 228°, 315°$.
$V(D)$, expansion	$V(D)$ is the enclosed-volume/density scale of the mass budget (CatWISE lock); expansion is radial from the CoE with a common scale s, velocities from energy conservation in the central mass.
Viewing angle	$\theta = \ell - \phi$ from the CoE axis, axis at $\ell_0 \approx 15.2°$ (18.6° free-L); $\theta = 0$ points away from the CoE.
Age	$t = 13.8$ Gyr for the clump Doppler imprint.

References

[1] A. G. Riess et al., *Astron. J.* **116**, 1009 (1998).

[2] S. Perlmutter et al., *Astrophys. J.* **517**, 565 (1999).

[3] G. F. R. Ellis and W. R. Stoeger, *Class. Quantum Grav.* **4**, 1697 (1987).

[4] E. Hubble, *Proc. Natl. Acad. Sci. USA* **15**, 168 (1929).

[5] K. Migkas et al., *Astron. Astrophys.* **636**, A15 (2020); arXiv:2004.03305.

[6] L. I. Sedov, *J. Appl. Math. Mech.* **10**, 241 (1946).

[7] G. I. Taylor, *Proc. R. Soc. Lond. A* **201**, 159 (1950).

[8] G. Lemaître, *Ann. Soc. Sci. Bruxelles* **A53**, 51 (1933).

[9] R. C. Tolman, *Proc. Natl. Acad. Sci. USA* **20**, 169 (1934).

[10] H. Bondi, *Mon. Not. R. Astron. Soc.* **107**, 410 (1947).

[11] Planck Collaboration, *Astron. Astrophys.* **641**, A7 (2020).

[12] N. J. Secrest, S. von Hausegger, M. Rameez, R. Mohayaee, S. Sarkar, and J. Colin, *Astrophys. J. Lett.* **908**, L51 (2021); arXiv:2009.14826.

Appendix

Appendix captions use the earlier spreadsheet parametrisation (D/L, d/D, α, κ, Amp, φ), superseded by Table 1. **Appendix contents.** Supporting material for the chronological body (Secs. 1–9); the fit-path figures and the heatmap, level-set and Amp$\times\varphi$ sweeps are combined into multi-panel figures. Figure 26 gathers the $A/A_{\rm all}$ and $B/B_{\rm all}$ fit path: sine rotation, a multiplicative 2ℓ correction, then the coupled A/B solutions; Migkas reports both the normalisation A and the slope B of L_X–T versus Galactic longitude, with $A/A_{\rm all}$ the main-text directional Hubble proxy and $B/B_{\rm all}$ the second feature on the same bins. Figure 27 gathers the sheet-parameter R^2 heatmaps; Figure 28 gathers the matching one-parameter level sets (D/L, α, κ, d/D). Figure 29 shows the Amp$\times\varphi$ heatmaps; Figure 30 shows the Amp$\times\varphi$ level-set families on Migkas axes. Figure 35 shows the $m = 1$–12 harmonics. Figure 114 (last page) places the CoE on the NCP-centered sky.

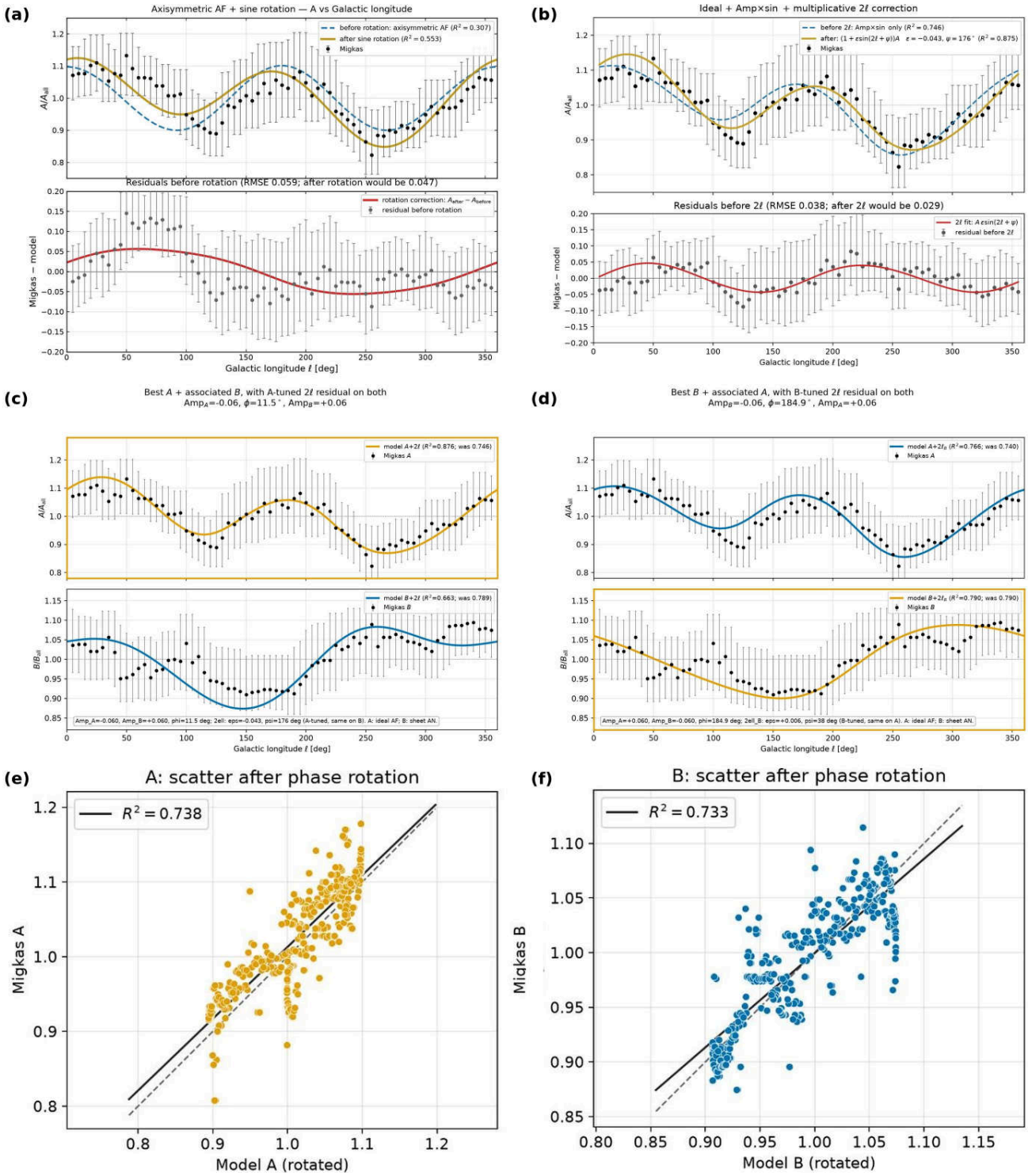


Figure 26: A/A_{all} and B/B_{all} fit path versus Migkas (fit over residual where stacked). (a) Axisymmetric AF before/after sine rotation: R^2 from 0.307 to 0.553 (earlier AF model); bottom residual is the rotation correction itself. (b) Amp× sin before/after a multiplicative 2ℓ correction ($\varepsilon = -0.043$, $\psi = 176°$): R^2 from 0.746 to 0.875; bottom isolates the 2ℓ residual. (c) Best A + associated B with A-tuned 2ℓ on both ($\mathrm{Amp}_A = -0.06$, $\varphi = 11.5°$, $\mathrm{Amp}_B = +0.06$): A improves ($R^2 = 0.876$); the same 2ℓ hurts B ($0.789 \to 0.663$). (d) Best B + associated A with B-tuned 2ℓ on both ($\mathrm{Amp}_B = -0.06$, $\varphi = 184.9°$, $\mathrm{Amp}_A = +0.06$): B holds $R^2 \approx 0.79$; A gets only a mild lift. (e, f) Sheet sine-rotation scatter after phase $\varphi \approx 0.35$ rad: Model A versus Migkas A ($R^2 \approx 0.74$) and Model B versus Migkas B ($R^2 \approx 0.73$).

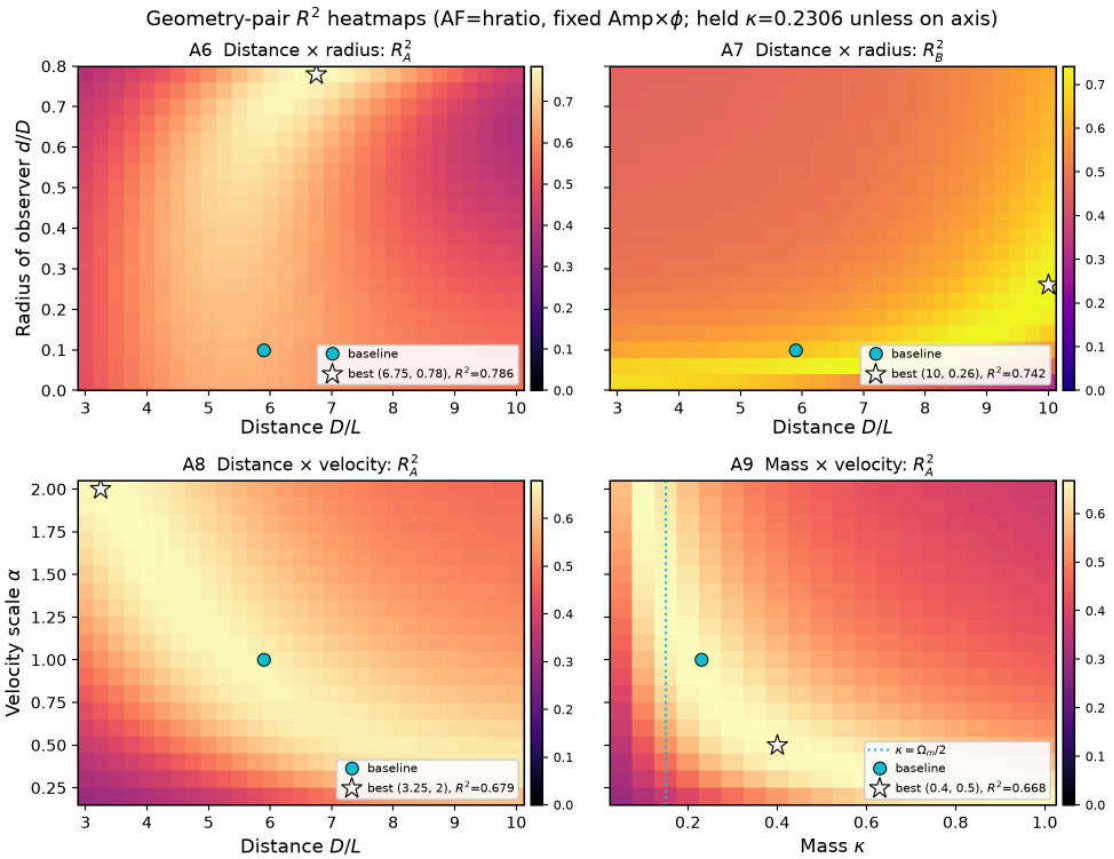


Figure 27: R^2 heatmaps (circle baseline, star best). (a) $D/L \times d/D$, R_A^2: best (6.75, 0.78), 0.786. (b) Same, R_B^2: (10.0, 0.26), 0.742. (c) $D/L \times \alpha$, R_A^2: (3.25, 2.0), 0.679. (d) $\kappa \times \alpha$, R_A^2: (0.4, 0.5), 0.668. Best fits in (a)–(c) lie on the grid edge.

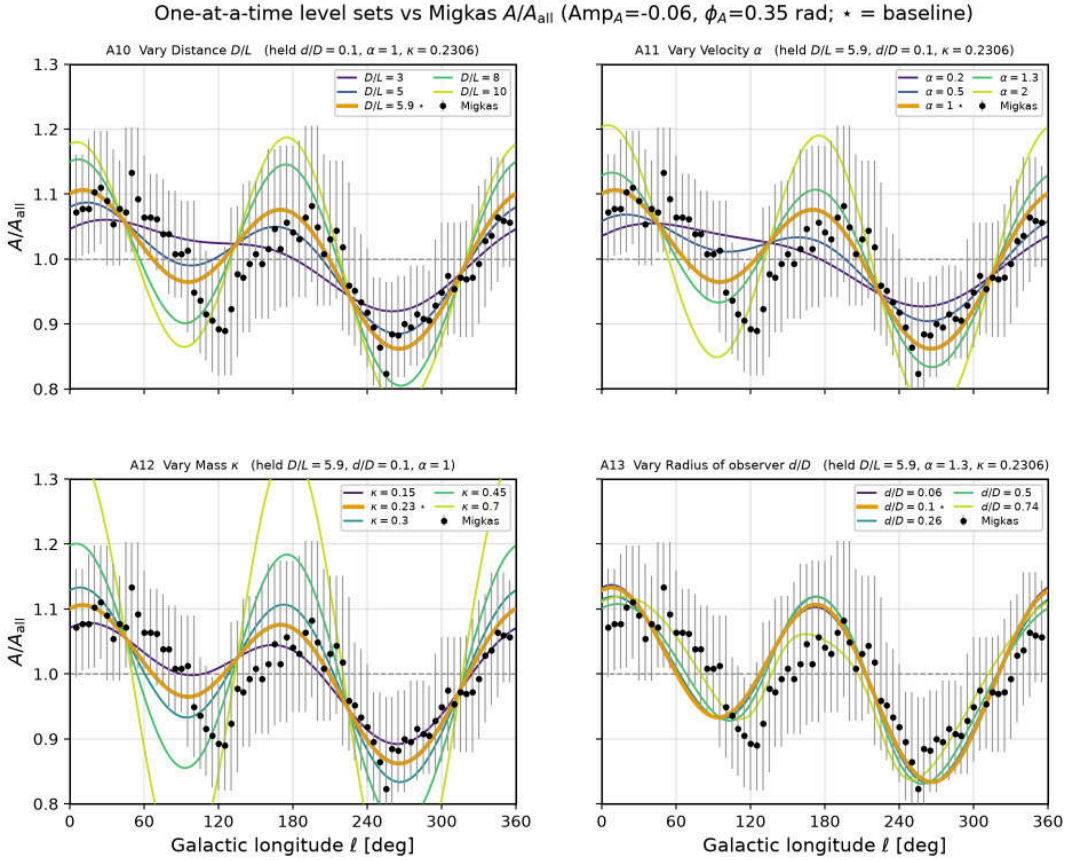


Figure 28: One-at-a-time level sets of $A/A_{\rm all}$ vs. Migkas, others at baseline ($D/L = 5.9$, $d/D = 0.1$, $\alpha = 1$, $\kappa = 0.2306$): vary (a) D/L, (b) α, (c) κ, (d) d/D with α held at 1.3.

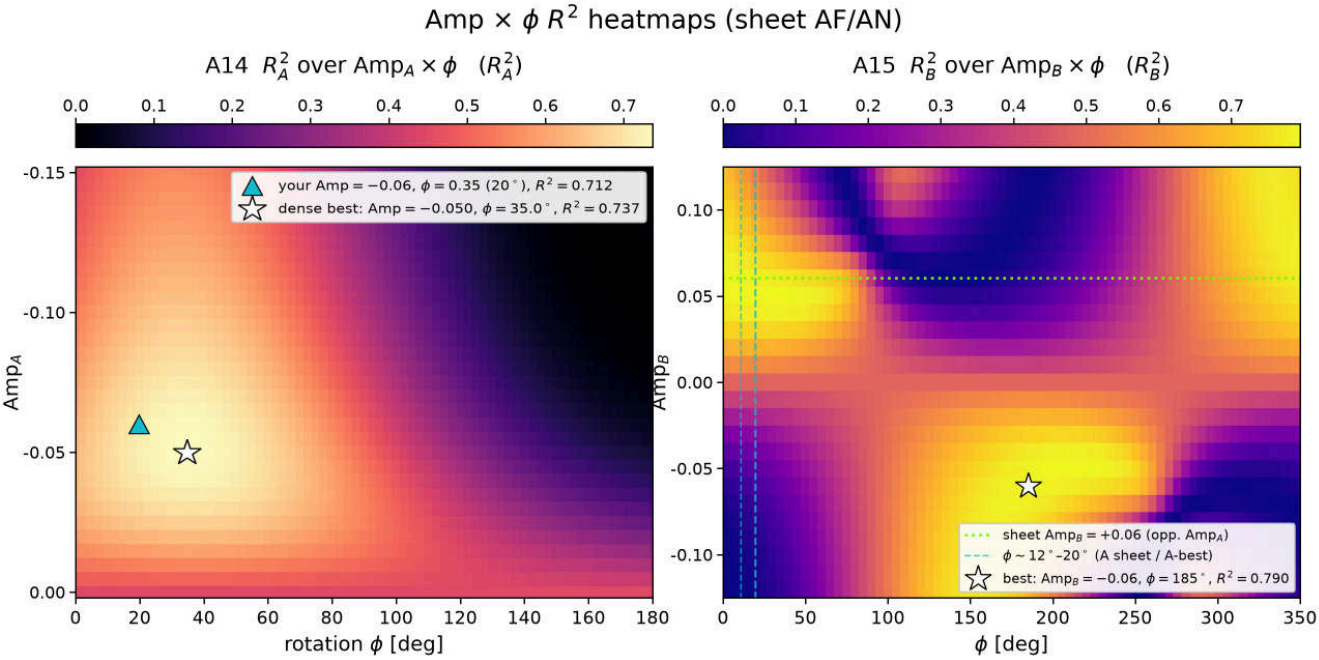


Figure 29: Amp×φ R^2 heatmaps (sheet AF/AN). (a) $A/A_{\rm all}$: your Amp= -0.06, $\varphi = 0.35$ rad (20°), $R^2 = 0.712$; dense best Amp= -0.050, $\varphi = 35.0°$, $R^2 = 0.737$. (b) $B/B_{\rm all}$ (sheet AN×Amp$_B$ × sin): best Amp$_B$ = -0.06, $\varphi = 185°$, $R^2 = 0.790$; reference Amp$_B$ = +0.06 (opp. Amp$_A$) and $\varphi \sim 12°$–$20°$ (A sheet).

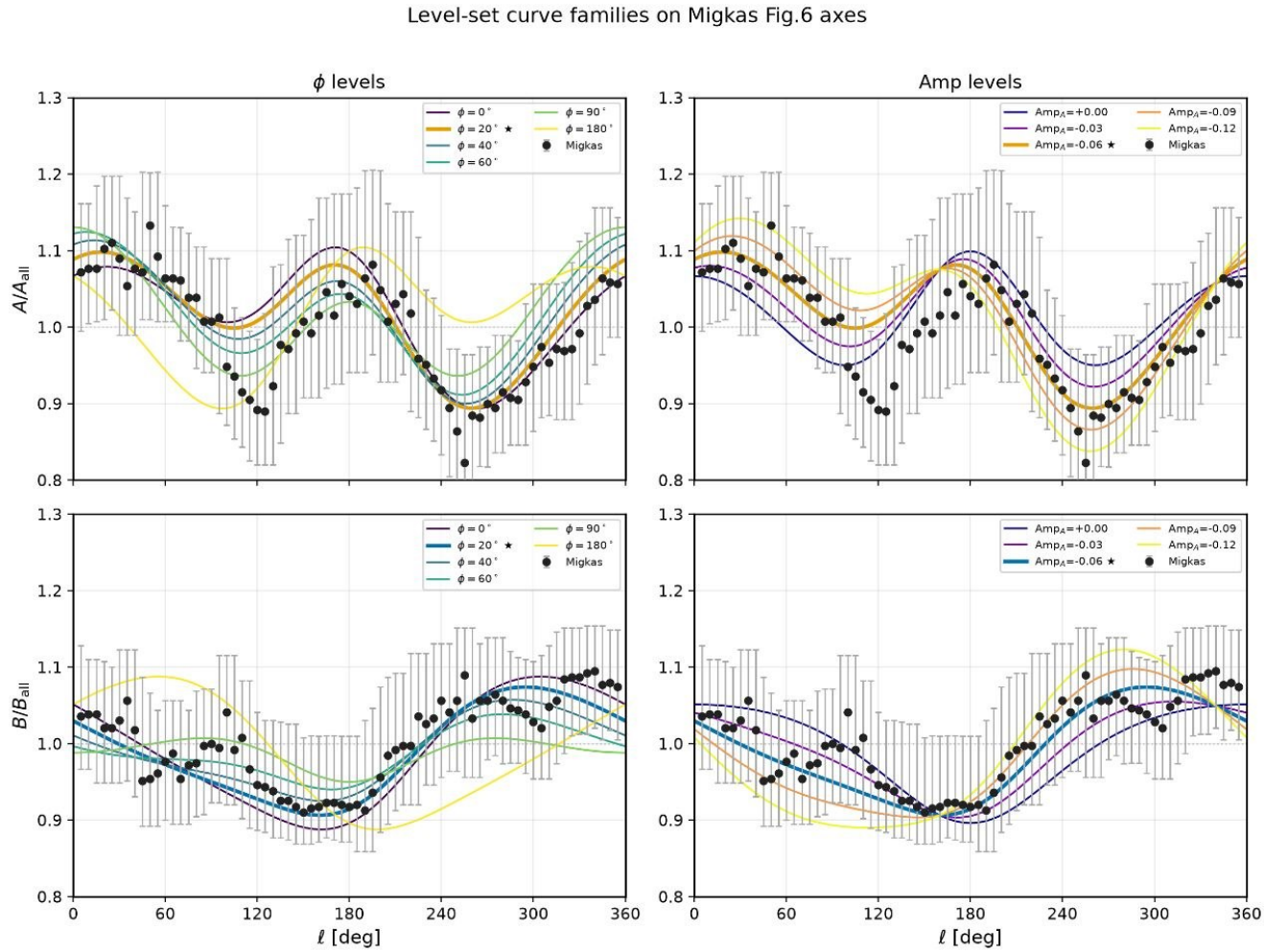


Figure 30: Level-set curve families on Migkas Fig. 6 axes: φ and Amp$_A$ sweeps for $A/A_{\rm all}$ (top) and $B/B_{\rm all}$ (bottom). Starred curves mark Amp$_A = -0.06$, $\varphi = 20°$.

Supplement: Exact relativistic (LTB) calculation of $A/A_{\rm all}$

This supplement accompanies the Migkas comparison section of the paper and replaces its Newtonian anisotropy curves with an exact general-relativistic calculation. Only statements are given; derivations are omitted.

Model. All shells start at the CoE in a simultaneous bang at $t = 0$. We are an off-centre observer on the shell with label $r = D$ in a Lemaître–Tolman–Bondi (LTB) dust universe,

$$ds^2 = -c^2 dt^2 + \frac{R'^2}{1 + 2E(r)/c^2}\, dr^2 + R^2\, d\Omega^2, \tag{S1}$$

where r labels each shell, $R(t, r)$ is its areal (proper) radius from the CoE, $R' = \partial R/\partial r$, $E(r)$ is the shell's energy per unit mass and $d\Omega$ is the solid-angle element. Each shell obeys

$$\dot{R}^2 = \frac{2GM(r)}{R} + 2E(r), \qquad R(0, r) = 0, \tag{S2}$$

with $M(r)$ the mass enclosed. The central mass is smoothed over a core $r_c = 0.1\,\mathrm{Gpc}$, $M(r) = M_c\, r^3/(r^2 + r_c^2)^{3/2}$. It is quoted as $2GM_c/c^2$ and is not scaled with D; its Schwarzschild radius may exceed D. $E(r)$ is tuned so that with no mass the model is Milne with $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and age $t_0 = 13.97\,\mathrm{Gyr}$. Bulk recession of our shell faster than c is allowed; only local relative speeds stay below c.

Light rays and redshift. Null geodesics are traced backward from our shell, in our frame, with conserved angular momentum J about the CoE. Along each ray

$$\frac{d\ln(1+z)}{dt} = H_r \cos^2\psi + H_\perp \sin^2\psi, \qquad H_r = \frac{\dot{R}'}{R'}, \quad H_\perp = \frac{\dot{R}}{R}, \tag{S3}$$

where ψ is the angle between the ray and the local radial direction. The direction-dependent Hubble rate is

$$H(\theta) = \frac{c\,\ln(1+z)}{R_{\rm obs}}, \tag{S4}$$

with $R_{\rm obs}$ today's proper distance along the ray and θ measured from the direction pointing away from the CoE (0°). Rays that reach the CoE or the bang are rejected (horizon): our visible region starts at zero size and grows, and we never see light from the far side of the CoE. Finally

$$\frac{A}{A_{\rm all}}(\theta) = \frac{H(\theta)}{\langle H\rangle}, \qquad \langle H\rangle = \frac{\int_0^\pi H(\theta)\,|\sin\theta|\,d\theta}{\int_0^\pi |\sin\theta|\,d\theta}, \tag{S5}$$

the raw per-angle value over the full-sky $|\sin\theta|$-weighted mean, with no cone averaging.

Validation. (i) With no mass the code returns $H = 70.00$ at every angle. (ii) In the classical limit ($D = 1\,\mathrm{Gpc}$, $R_{\rm obs} = 0.025\,\mathrm{Gpc}$) the exact values at 0°/90°/180° are 72.919/67.677/73.074 against the classical 72.924/67.675/73.076. (iii) Locally the result reduces to $H(\theta) = H_r\cos^2\theta + H_\perp\sin^2\theta$. Figure 31(a) compares exact and classical curves for one case.

Results. At $R_{\rm obs} = 3\,\mathrm{Gpc}$ (Fig. 31b–e) the curves share one shape. The bump at 0° grows with mass and as we move closer to the CoE (smaller D): for $2GM_c/c^2 = 100\,\mathrm{Gpc}$ it is 1.17 at $D = 20$ and 1.02 at $D = 60$. The minimum sits near 120° for weak mass and deepens and moves toward $\sim 105°$ as mass grows or D falls. The rise at 180° is usually the largest feature, about 1.02–1.14. At $R_{\rm obs} = 14.3\,\mathrm{Gpc}$ the curves are nearly flat until the edge of the region from which light can reach us, then dip and spike.

The sky-mean H never reaches Planck (67.4 ± 0.5) or SH0ES (73.04 ± 1.04). At 3 Gpc, however, individual directions span both values when $2GM_c/c^2 \gtrsim (0.3\text{–}1)\,D$: 90° reads like Planck and 0°, 180° like SH0ES (Fig. 31f). By 14.3 Gpc this spread is gone.

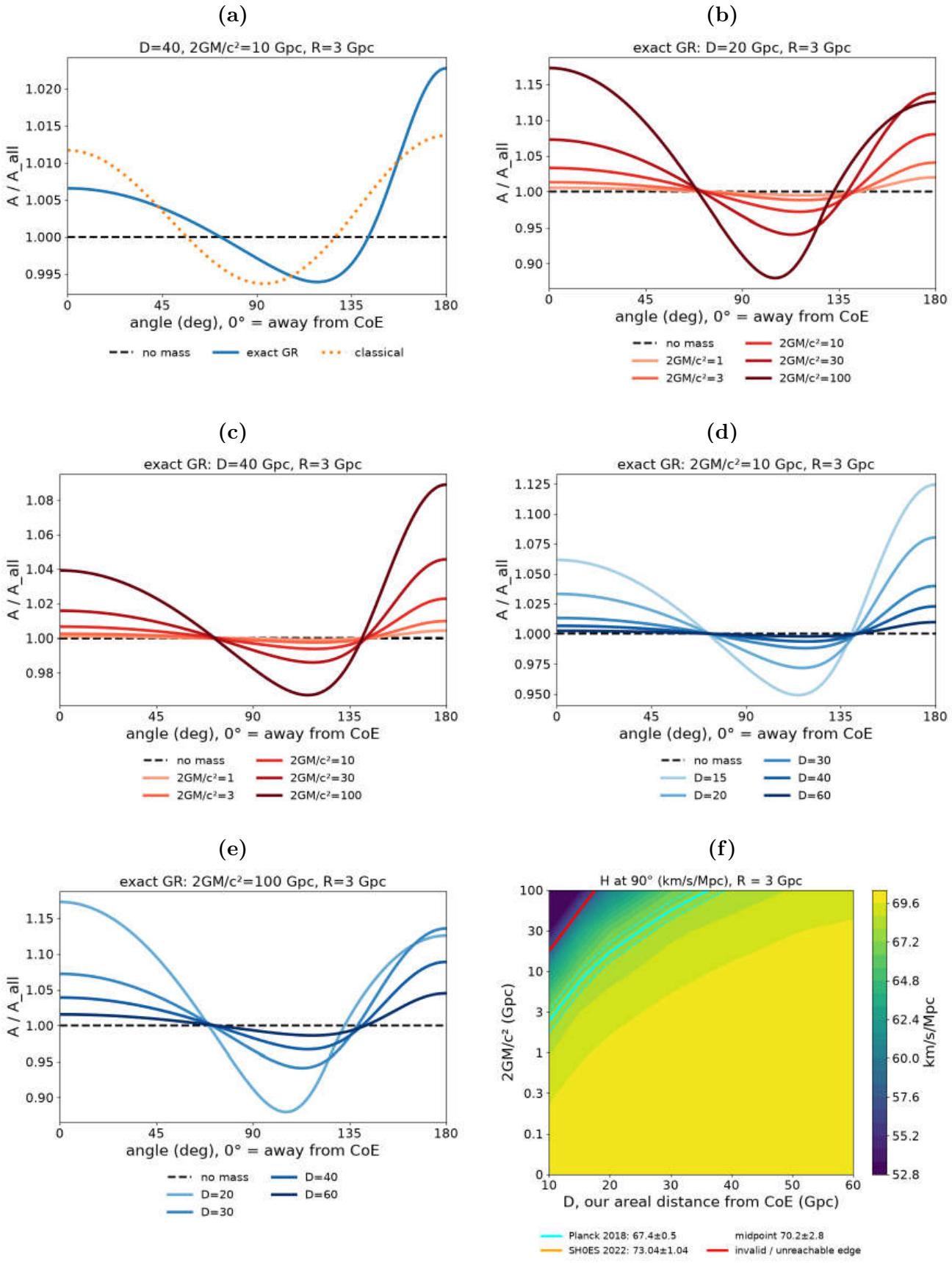


Figure 31: Exact GR (LTB) results at $R_{\rm obs} = 3\,$Gpc. (a) Exact versus classical $A/A_{\rm all}(\theta)$ for $D = 40$, $2GM_c/c^2 = 10\,$Gpc; black dashed: no mass. (b) $D = 20$ fixed, mass varied. (c) $D = 40$ fixed, mass varied. (d) $2GM_c/c^2 = 10\,$Gpc fixed, D varied. (e) $2GM_c/c^2 = 100\,$Gpc fixed, D varied. (f) H at 90° over D and mass, with Hubble-tension contours (Planck cyan, SH0ES orange, dashed $\pm1\sigma$; midpoint white dotted).

Reading the CMB fingerprint of the CoE shell model: what arrangements of masses and voids cancel the leftover residual?

Chris Alameddin
(Dated: October 11, 2026)

In the center-of-explosion (CoE) shell model, the shell's gravity imprints a pattern on the cosmic microwave background around the galactic ring that is 5,751 times the 10^{-5} level at which the CMB is flat. Special relativity plus the central mass cancel 95.9% of it, leaving a residual of 237 times that level. We treat this residual as a fingerprint and pose an inverse problem: which arrangements of added masses and voids cancel it? Using a Newtonian tidal and velocity kernel, we fit compact masses, deep voids, giant partner regions and smooth density fields under several sets of placement rules. Outside our visible sphere, 11 bodies (6 masses and 5 voids) reach the flat target; inside it, 4 to 8 bodies suffice; removing all placement walls brings this to 8. Opening the CoE core completely brings the outside count to 10, but the fits use only its near edge. Sequential, one-at-a-time growth never gets below 27×, but editing a filled web does reach the target: in a random web that obeys published void and cluster size laws and keeps every void balanced by its own clusters, moving or resizing 12% of the bodies, with no deletions and no decrease in any of seven disorder measures, brings the leftover from 789× to 1.00× without forming a ring. In every edited web that reaches the target, the large-scale pattern (modes $m = 1$–12) is cancelled to 0.02–0.1×, and what is left is chaotic, asymmetric fine-scale ripple with an rms at the 10^{-5} limit, statistically like Gaussian noise. A smooth field reaches 1× with only about 1% rms density contrast, but at gigaparsec scales this is roughly 8σ even against the most generous extrapolation of observed structure. We summarize which solutions exist, how unique they are, and what they would require of the real universe.

I. INTRODUCTION

In the CoE shell model the matter of the universe sits in a thick shell, 43.9–72.5 Gpc from a center of explosion (CoE). We are inside the shell, 58.2 Gpc from the CoE, and our visible universe is a sphere of radius 14.3 Gpc around us (Fig. 32). The visible sphere exactly spans the shell: its near edge touches the inner edge of the shell in the direction of the CoE ($\ell \approx 195°$) and its far edge touches the outer edge in the opposite direction ($\ell \approx 15°$). The us–CoE line is the symmetry axis of the problem. The cosmic microwave background (CMB) we analyze is the ring of sightlines in the galactic plane at the edge of the visible sphere.

The logic of this brief starts from Fig. 33. On its own, the shell's gravity imprints a strong direction-dependent pattern on the ring: 5,751 times the 10^{-5} level at which the CMB is observed to be flat. Our motion relative to the CoE (special relativity: Doppler shift and aberration) and the redshift from the central mass produce a pattern of almost the same shape and opposite sign. Fitting these (speed parameter $\beta_1 = 0.587$, distance ratio $q = 0.230$, mass parameter $\mu = 0.0933$, with the axis free; it settles at $\ell = 15.16°$) removes 95.9% of the starting anisotropy, a reduction by a factor of 24. What is left has an rms of 237 times the floor.

We call this leftover the *fingerprint*. It is a specific shape around the sky, and any structure the model adds must cancel that shape rather than merely reduce its size. The question of this brief is an inverse problem: *what arrangement of masses and voids, placed where, cancels the fingerprint down to the flat target (1×)?* We report every scenario we have run, how many bodies each needs, what the best layouts look like, and how the required structure compares with what is observed.

II. THE FINGERPRINT

Gravity against special relativity. Both large patterns can be broken into waves around the ring. We solve for the residual with modes $m = 1$–12, where m counts the number of waves in one full turn and the fingerprint is $R(\ell) = \sum_m c_m \cos m(\ell - \varphi)$ with $\varphi = 15.16°$. Figure 34 compares, mode by mode, the gravitational multipoles of the shell, $(\ln G)_m$, with the multipoles of special relativity plus the central mass, $(\ln 1/S)_m$. For $m = 1$ and 2 they agree to within a few percent and cancel; these modes are where the fit is anchored. For $m = 3$–6 they no longer match: at $m = 3$ the shell's pattern is about 10% larger, and from $m = 4$ on it falls off faster than the Doppler pattern. This mismatch is what remains. For $m \geq 7$ the shell contributes almost nothing and only the tail of the Doppler pattern is left. The difference $c_m = (\ln G)_m - (\ln 1/S)_m$ is the fingerprint (Fig. 34b, Table I).

The fingerprint mode by mode. Figure 35 draws each mode around the sky. Modes $m = 3$, 4 and 5, with amplitudes of 200, 185 and 173 times the floor, carry 92% of the fingerprint's variance; $m = 6$ (81×) carries most of the rest. Their sum, a three-lobed shape with an rms of 2.373×10^{-3} (237×), is the pattern every scenario below tries to cancel. Modes above 12 are negligible in the fingerprint itself, but they become important once a fit has removed the large ones (Sec. IV B).

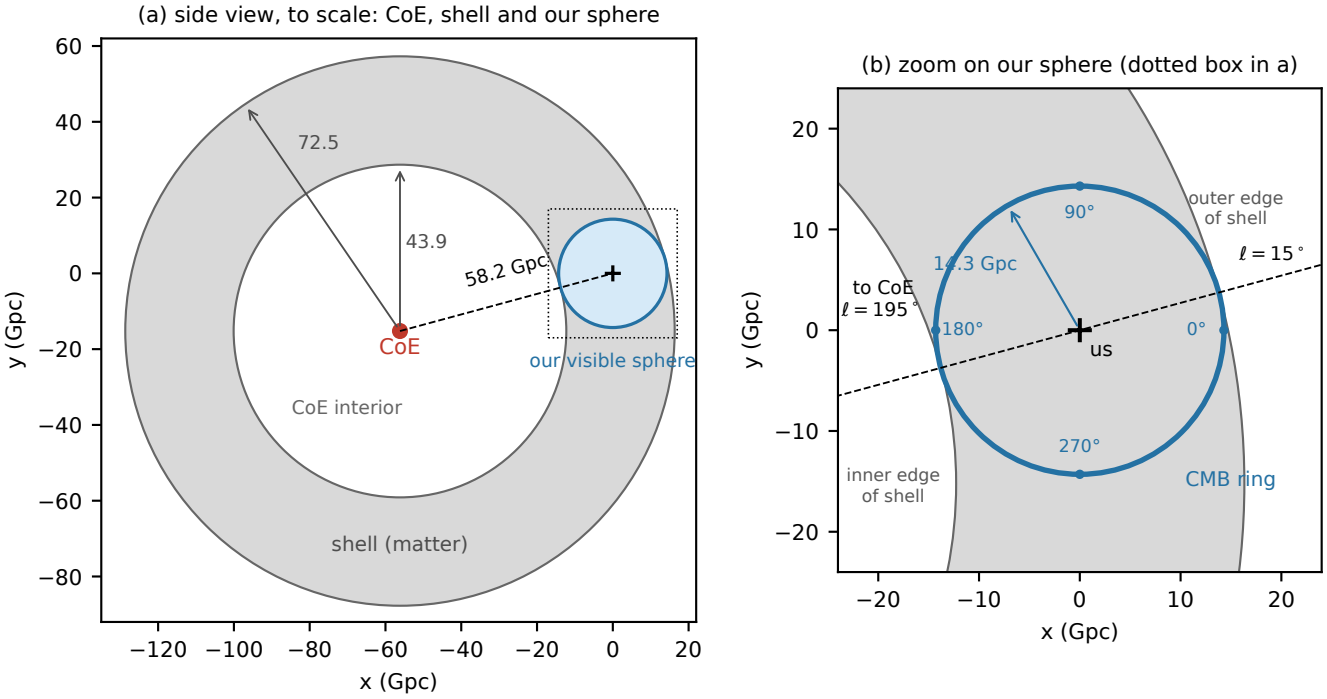


FIG. 32: Geometry of the model, side view (looking face-on at the plane of the CMB ring). (a) To scale: the CoE (red), the matter shell (grey, 43.9–72.5 Gpc from the CoE), and our visible sphere (blue, radius 14.3 Gpc) 58.2 Gpc from the CoE. (b) Zoom on our sphere: us (+), the CMB ring with galactic longitudes, and the us–CoE axis ($\ell = 195°$ toward the CoE, 15° away). Our sphere just touches both edges of the shell.

TABLE I: The modes of the fingerprint. c_m: amplitude (absolute and in units of the 10^{-5} floor); peak: longitude of the first maximum; % of var.: share of the fingerprint's variance; $(\ln G)_m$ and $(\ln 1/S)_m$: the shell-gravity and special-relativity plus central-mass multipoles whose difference is c_m.

m	period (°)	c_m	$\lvert c_m\rvert$ (×)	peak (°)	% of var.	$(\ln G)_m$ shell	$(\ln 1/S)_m$ SR + mass	origin
1	360.0	$+1.80 \times 10^{-4}$	18	15.16	0.29	-5.68×10^{-3}	-5.86×10^{-3}	fit matches (cancels)
2	180.0	-2.73×10^{-4}	27.3	105.16	0.66	$+7.81 \times 10^{-2}$	$+7.83 \times 10^{-2}$	fit matches (cancels)
3	120.0	-2.00×10^{-3}	200	75.16	35.56	-2.08×10^{-2}	-1.88×10^{-2}	shell vs Doppler mismatch
4	90.0	-1.85×10^{-3}	185	60.16	30.29	$+2.42 \times 10^{-3}$	$+4.27 \times 10^{-3}$	shell vs Doppler mismatch
5	72.0	$+1.73 \times 10^{-3}$	173	15.16	26.56	$+1.96 \times 10^{-4}$	-1.53×10^{-3}	shell vs Doppler mismatch
6	60.0	-8.08×10^{-4}	80.8	45.16	5.80	-1.13×10^{-4}	$+6.97 \times 10^{-4}$	shell vs Doppler mismatch
7	51.4	$+2.84 \times 10^{-4}$	28.4	15.16	0.71	$+1.56 \times 10^{-6}$	-2.80×10^{-4}	Doppler tail only
8	45.0	-1.04×10^{-4}	10.4	37.66	0.10	$+9.37 \times 10^{-6}$	$+1.13 \times 10^{-4}$	Doppler tail only
9	40.0	$+4.63 \times 10^{-5}$	4.63	15.16	0.02	-3.05×10^{-6}	-5.02×10^{-5}	Doppler tail only
10	36.0	-2.18×10^{-5}	2.18	33.16	0.00	$+3.10 \times 10^{-7}$	$+2.34 \times 10^{-5}$	Doppler tail only
11	32.7	$+1.01 \times 10^{-5}$	1.01	15.16	0.00	$+7.87 \times 10^{-8}$	-1.09×10^{-5}	Doppler tail only
12	30.0	-4.74×10^{-6}	0.474	0.16	0.00	-3.20×10^{-8}	$+5.17 \times 10^{-6}$	Doppler tail only

III. METHOD

Geometry and density. We are at the origin. The CMB ring is the set of 360 in-plane sightlines at radius $R_{\rm vis} = 14.3\,$Gpc, one per degree of galactic longitude ℓ. The CoE lies at $(-56.17, -15.22, 0)\,$Gpc, 58.2 Gpc from us, in the direction $\ell \approx 195°$. The shell spans 43.9–72.5 Gpc from the CoE. The matter density depends on the distance r from the CoE as

$$\rho(r) = 2.9 \times 10^{-27}\,\mathrm{kg\,m^{-3}}\,\exp\left[-\frac{r - 58.2\,\mathrm{Gpc}}{922.65\,\mathrm{Gpc}}\right]. \quad (1)$$

Kernel. A body of mass M at position $\mathbf{x}$ changes the signal at ring point $\hat{n}$ by the line-of-sight difference between its pull at the ring point and its pull on us,

$$\Delta(\hat{n}) \propto [\mathbf{g}_M(R_{\rm vis}\hat{n}) - \mathbf{g}_M(\mathbf{0})] \cdot \hat{n}, \quad (2)$$

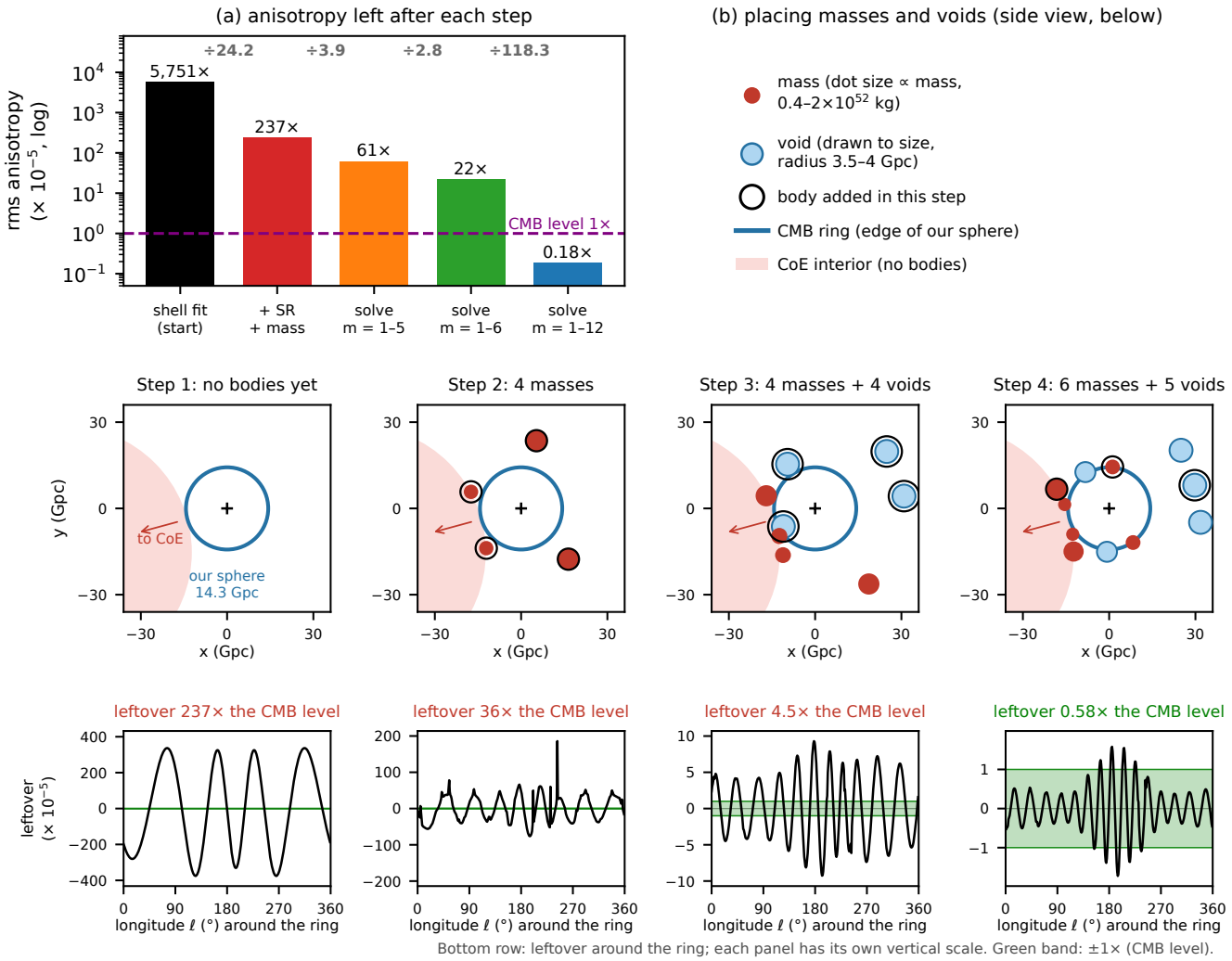


FIG. 33: The starting logic, step by step. (a) rms anisotropy around the ring after each step, in units of the 10^{-5} CMB level (purple dashed line). The shell alone predicts 5,751×; adding special relativity and the central mass brings this to 237×, the fingerprint. Solving for its modes $m = 1$–5, 1–6 and 1–12 would leave 61×, 22× and 0.18×. (b) How placing masses and voids does this, using real fitted layouts outside our sphere (side view: the plane of the CMB ring, seen face-on; bodies are drawn at their position projected onto that plane). Top: the layout; red dots are masses (size grows with mass), blue circles are voids drawn to size, black rings mark the bodies added in that step, and the pink area is the CoE interior, where no body may sit. Bottom: the leftover around the ring, each panel on its own vertical scale; the green band is ±1×. Step 1: no bodies, 237×. Step 2: the best 4 masses, 36×. Step 3: the best 4 masses plus 4 voids, 4.5× (the masses are refitted, so they move). Step 4: 2 masses and 1 void added to step 3 one at a time, refitting all bodies each time (3.0×, 1.3×, then 0.58×), below the CMB level. What remains at step 4 is short waves beyond the solved range, mostly $m = 13$–19 (97% of it).

with the ring mean removed. This tidal-plus-velocity kernel is plain Newtonian; it is valid for sources inside and outside the ring (checked against an independent calculation for 200 interior positions, correlation ≥ 0.99999999). The *leftover* of a layout is $\mathrm{rms}_\ell(R - \Delta)/10^{-5}$, where R is the starting residual; the start is 237× and the target is 1×.

Bodies. *Compact masses* are point masses of 10^{49}–2×10^{52} kg with a $\delta = +100$ core. *Deep voids* are uncompensated spheres with $\delta = -0.7$ and radius 0.5–4 Gpc. *Shallow voids* follow a compensated profile (mean $\delta \approx -0.35$ inside, compensation out to three radii, net zero) [1]. *Giant partners* are uniform spheres of radius 1–15 Gpc and δ between −0.9 and +2, paired with a uniform contrast δ_{us} of our own sphere.

Placement rules. "Outside" bodies lie between our sphere and 43 Gpc from us and may not enter the CoE interior (the "wall"). "Inside" bodies lie wholly within our sphere, at least 0.1 Gpc from the ring and clear of us. In the *no-wall* runs the CoE-interior exclusion is removed; its *smooth-density limit* replaces the wall by a mass cap that scales with the local density, $M \leq$

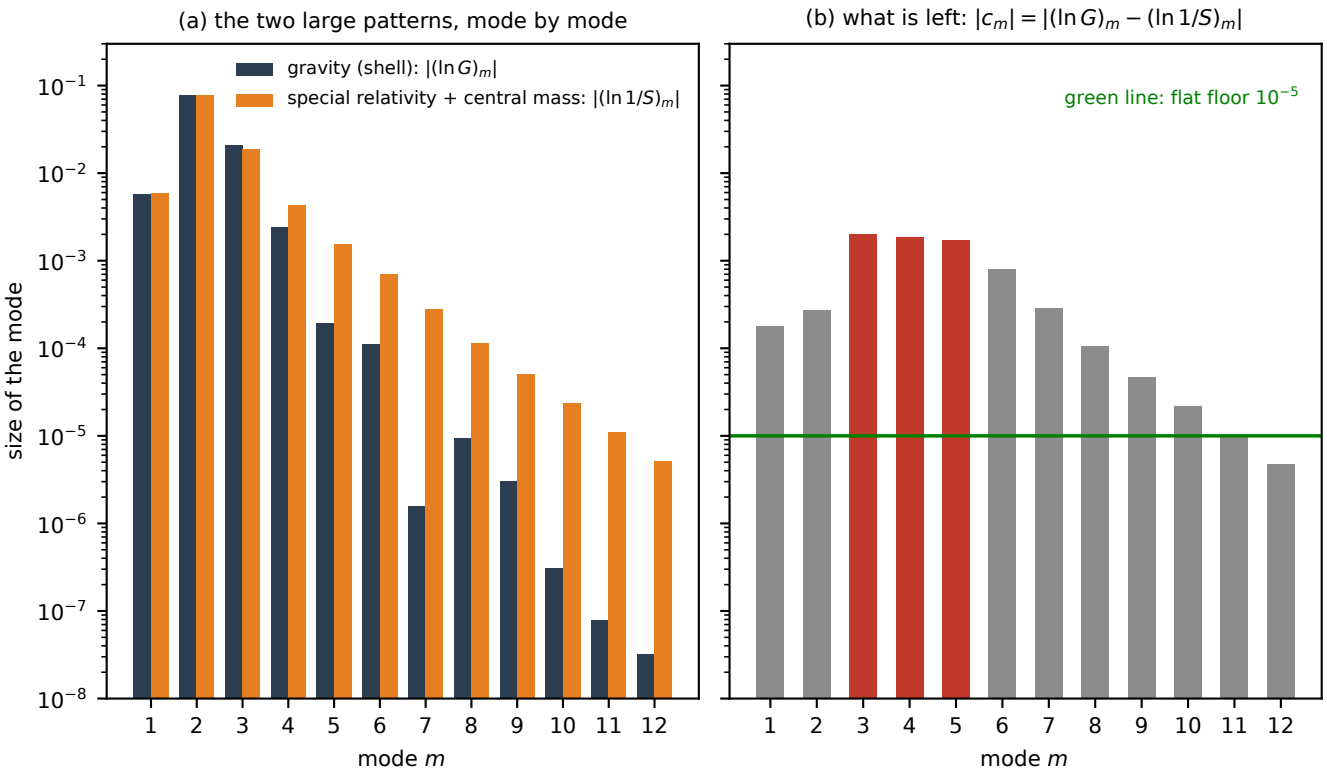


FIG. 34: Where the fingerprint comes from. (a) The size of each mode of the shell's gravitational pattern, $|(\ln G)_m|$ (dark), and of the special-relativity plus central-mass pattern, $|(\ln 1/S)_m|$ (orange), on a log scale. They nearly cancel at $m = 1$–2 and part ways from $m = 3$ on. (b) What is left, $|c_m|$: the fingerprint, with the big modes $m = 3$–5 in red. The green line is the 10^{-5} flat level.

$2 \times 10^{52}\,\mathrm{kg}\,\rho(\mathbf{x})/\rho(58.2\,\mathrm{Gpc})$, and requires each mass to draw its material from a non-overlapping supply sphere $V = M/\rho(\mathbf{x})$. The *all-open* runs also remove the visible-sphere exclusion: bodies may sit anywhere 0.5–43 Gpc from us but may not straddle the ring. Cores never overlap.

Net mass. Every fit has zero net mass. Outside fits remove or add the net mass smoothly over the surrounding shell region; inside and all-open fits remove it smoothly over our sphere with the same density shape (whose uniform part produces no signal). We also test no compensation, compensation drawn from outside, and, for partners, strict balance by the partners themselves.

No-mirror rule. The problem is nearly symmetric about the plane through us, the CoE and the galactic pole ($\ell \to 2 \times 15.16° - \ell$), and the best fits use mirror pairs. In the no-mirror runs two bodies of the same kind are forbidden if one lies within 20° of the other's mirror image *and* their distances agree within 15% *and* their sizes within 25%; no body may lie within 10° of the mirror plane.

Fitting. Each body has four free numbers (distance, longitude, latitude, and mass or radius). All are fitted together by bounded least squares (trust-region reflective) from many random starts: 40–100 per body class outside, 12–40 inside, plus warm starts that add one body to the best $N - 1$ layout. Sequential runs instead place one body at a time and never refit earlier ones. Field solutions (Sec. IV E) solve a regularized linear inverse problem for a continuous $\delta(\mathbf{x})$.

IV. RESULTS

A. Leftover versus number of bodies

Figure 36 collects the best leftover against the number of added bodies N for every point-body scenario, and Fig. 37 splits them into families. Table II lists the first N that reaches the flat target and the best value reached.

Three facts stand out. First, masses alone never reach the target from outside (they stall near 4× at $N = 20$), and voids alone stall near 85×; the outside solutions need both, and reach 1× at $N = 11$ (6 masses and 5 voids). Second, bodies inside our sphere are far more effective: 7 masses or 4 masses plus 4 voids suffice, and with a looser mass cap of 10^{53} kg only 4 masses are needed. Third, each wall that is removed saves bodies: without the CoE-interior wall the target is reached at $N = 10$, and with everything open at $N = 8$, where all 8 bodies move in-

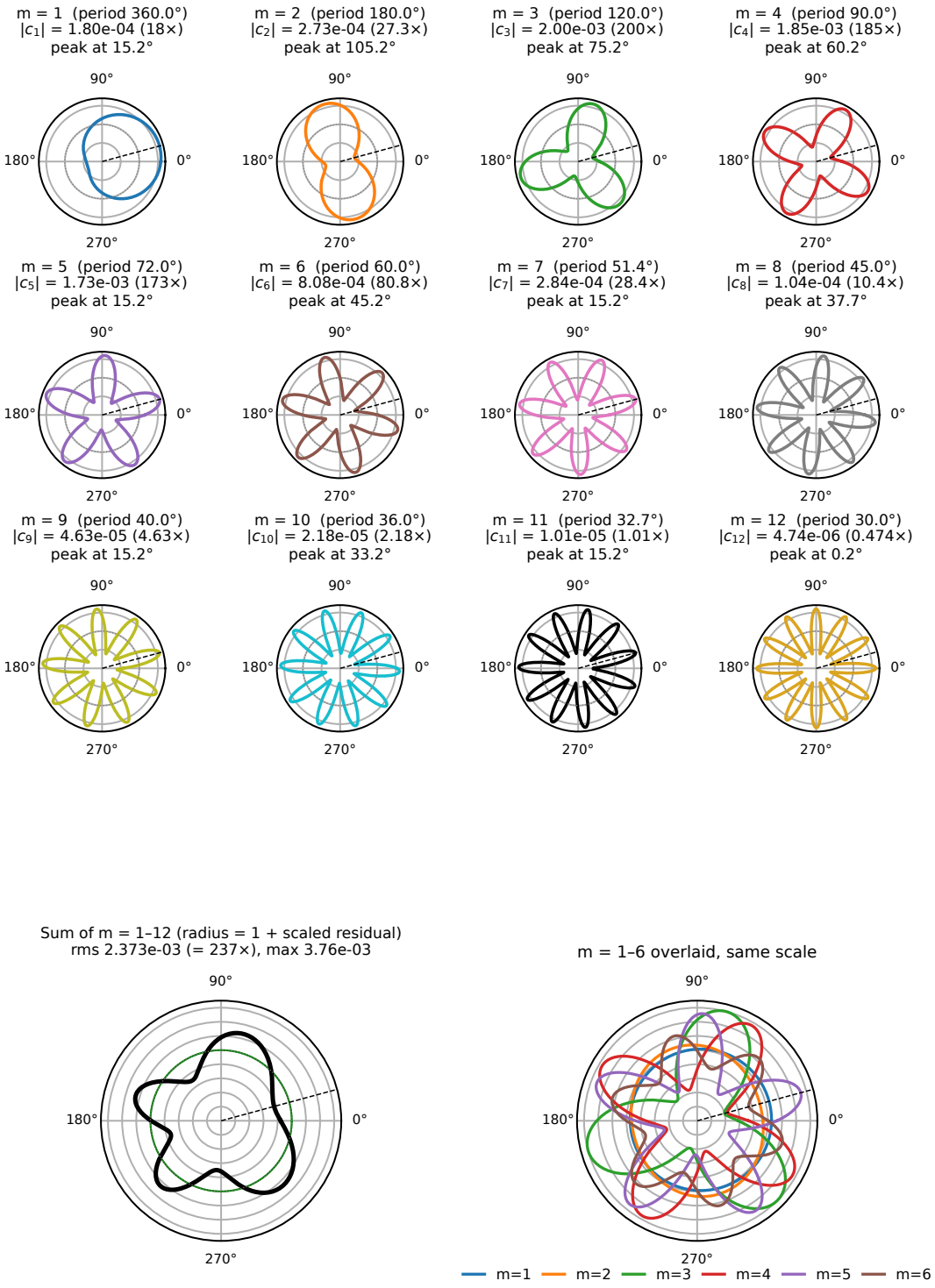


FIG. 35: The fingerprint: the modes we must solve for. Each small panel draws one mode $c_m \cos m(\ell - \varphi)$ around the sky as radius = 1+ (scaled mode); the dotted circle is zero and the dashed line is the symmetry axis at $\ell = 15.16°$. Each small panel has its own scale; the numbers give the size, in absolute units and in units of the 10^{-5} floor. Bottom left: the sum of $m = 1$–12, the full fingerprint; the green circle is the $\pm 1\times$ CMB band, thinner than the line at this scale. Bottom right: $m = 1$–6 overlaid on one common scale, showing how $m = 3$–5 dominate.

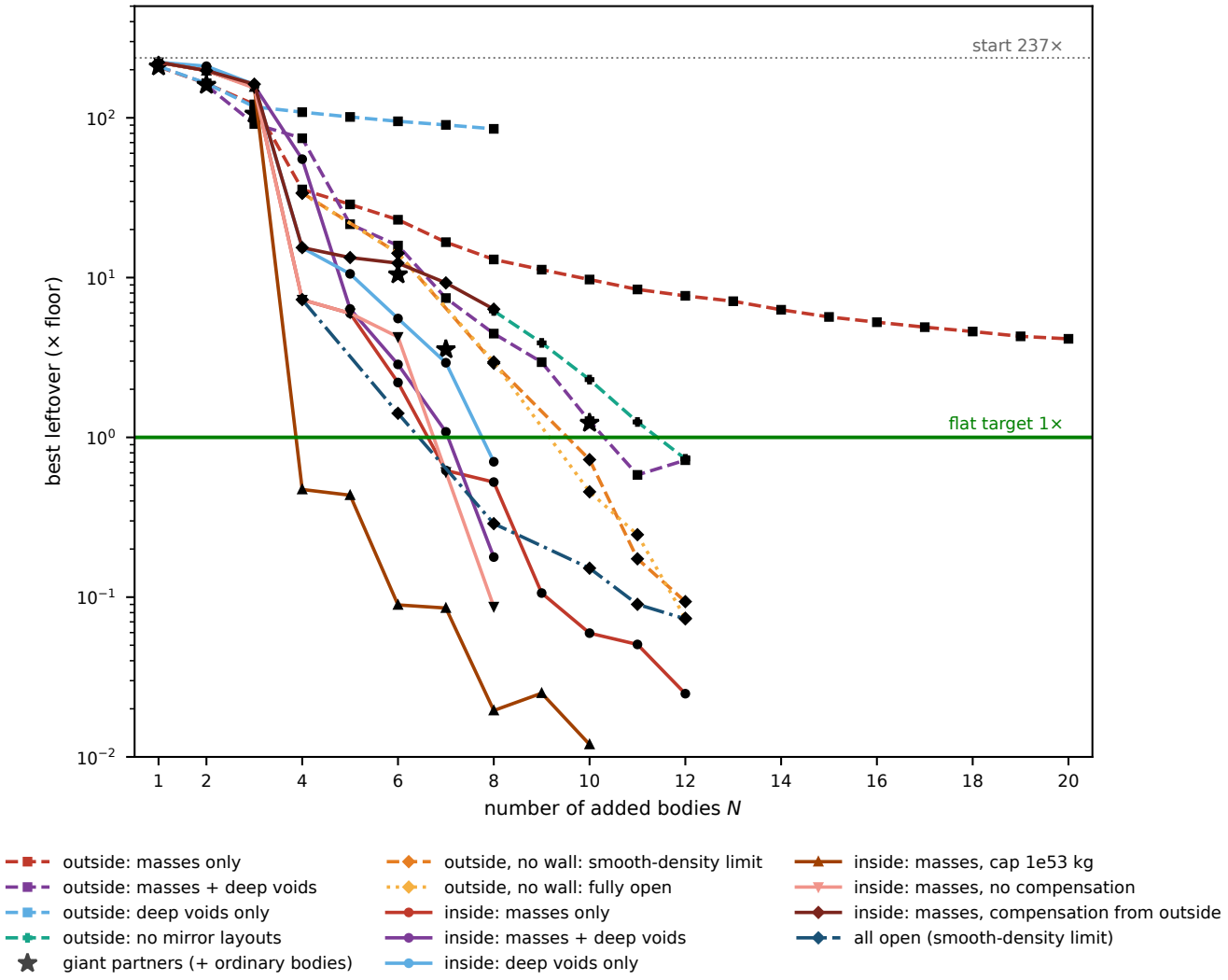


FIG. 36: Best leftover (in units of the 10^{-5} floor, log scale) against the number of added bodies, for all point-body scenarios. Solid lines: inside our sphere; dashed: outside; dash-dot: all open; stars: giant partners. Green: the flat target (1×). Dotted: the start (237×).

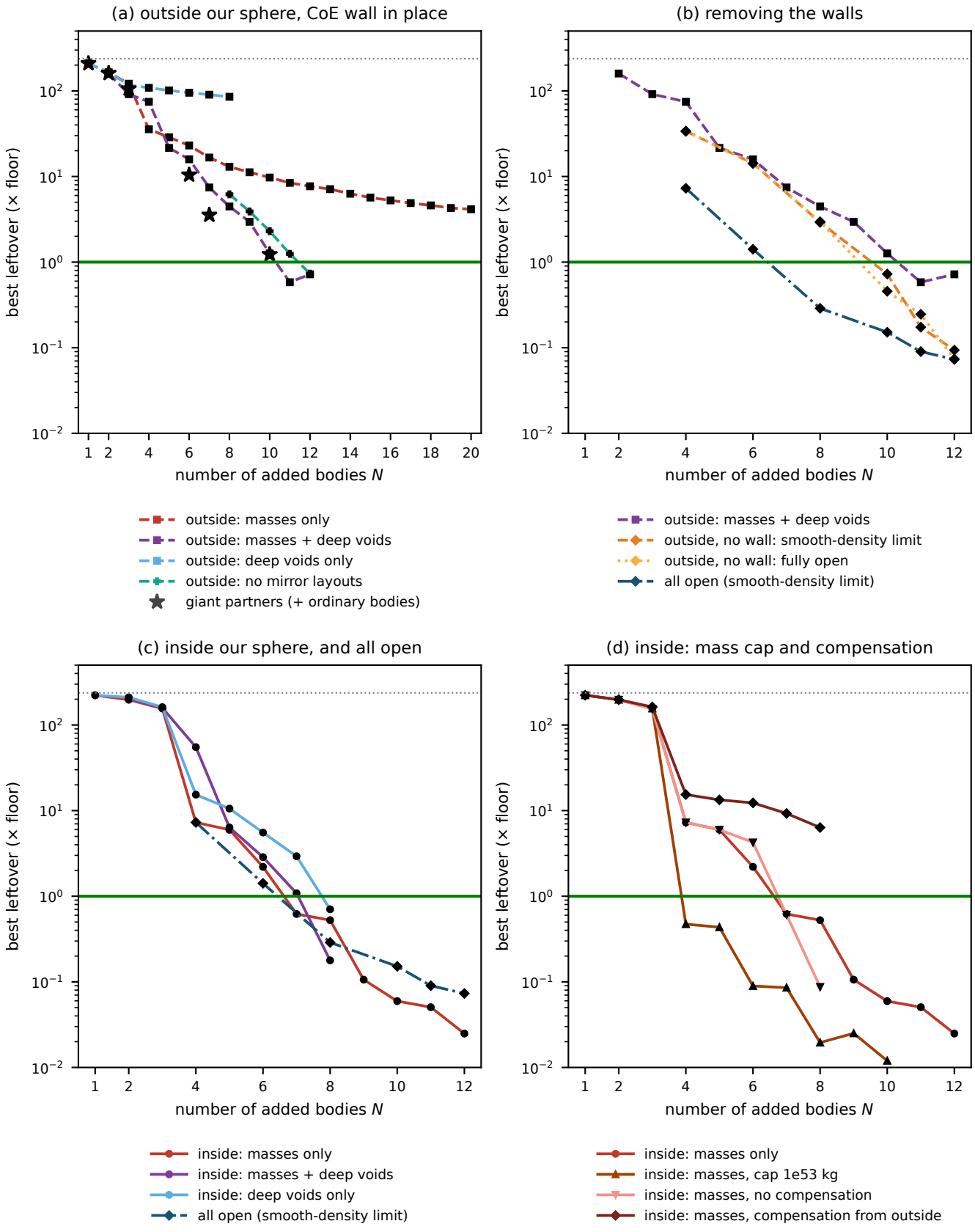


FIG. 37: The same curves by family. (a) Outside our sphere with the CoE wall in place. (b) The effect of removing the CoE wall and then the visible-sphere exclusion. (c) Inside our sphere, compared with all open. (d) Inside variants: a larger mass cap helps most; compensating from outside hurts most.

TABLE II: Fewest bodies that reach the flat target, and the best value reached (in units of the floor).

Scenario	first $N < 1\times$	best (at N)
Outside: masses only	–	4.14 (20)
Outside: masses + deep voids	11 (0.58)	0.58 (11)
Outside: deep voids only	–	85.3 (8)
Outside: no-mirror layouts	12 (0.73)	0.73 (12)
Giant partners + bodies	–	1.23 (10)
No wall: smooth-density limit	10 (0.73)	0.094 (12)
No wall: fully open	10 (0.46)	0.074 (12)
Core open: masses + deep voids	10 (0.46)	0.069 (12)
Core open: no-mirror layouts	11 (0.37)	0.34 (12)
Core open: giant partners + bodies	10 (0.17)	0.080 (11)
Core open: all open	8 (0.18)	0.18 (8)
All bodies in the core	–	143 (12)
Inside: masses only	7 (0.62)	0.025 (12)
Inside: masses + deep voids	8 (0.18)	0.18 (8)
Inside: deep voids only	8 (0.70)	0.70 (8)
Inside: masses, cap 10^{53} kg	4 (0.47)	0.012 (10)
Inside: no compensation	7 (0.61)	0.087 (8)
Inside: compensation from outside	–	6.35 (8)
All open (smooth limit)	8 (0.29)	0.073 (12)
Seq. A: deep voids + clusters	–	27.4 (48)
Seq. A: shallow voids	–	135 (30)
Seq. B: CMB-aware growth	–	37.2
Seq. B: foam rule	–	222
Seq. B: random	–	213
Seq. B: shallow voids	–	163

side our sphere. Sequential placement (Fig. 38), which never revisits earlier choices, never reaches the target (final universes in Figs. 39 and 40); editing a few percent of such grown webs afterwards does (Secs. VI and IX).

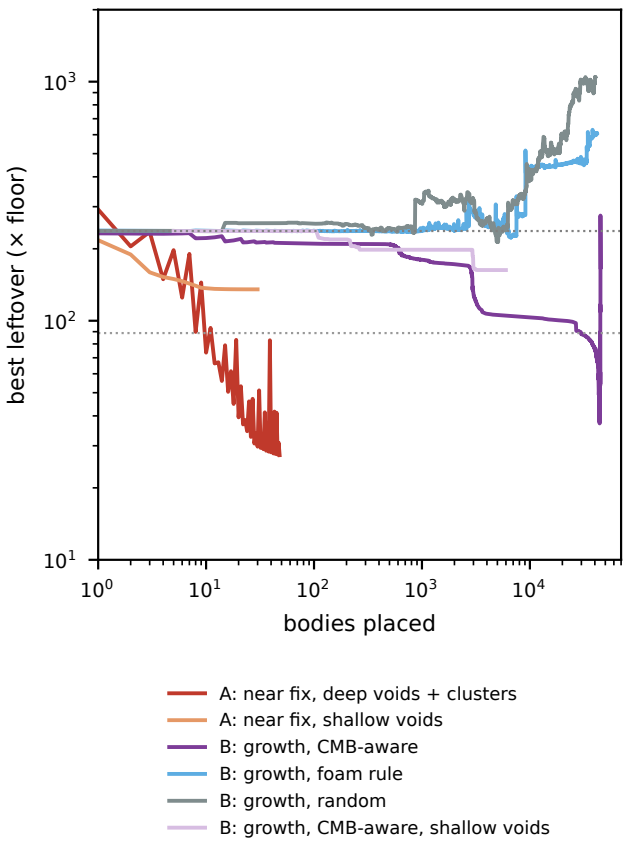


FIG. 38: Sequential placement, one body at a time with no refitting (log scale in bodies placed). A: the near region fixed pair by pair. B: growth from empty until the region is full, under three placement rules. None reaches the flat target (green); the dotted lines mark the start (237×) and the four-clump result (89×).

B. Example universes and what is left

Figures 41–43 show one representative layout per scenario, nine in all, numbered 1–9 from the largest to the smallest leftover; Fig. 44 continues with examples 10–13, the core-open reruns (Sec. IV D), again from largest to smallest. Each is shown at the first N that reaches $1\times$ (or the best N otherwise), with the mode-by-mode leftover and the residual that remains around the ring. The starting residual is not repeated; it is in Fig. 35. Two patterns recur. The successful fits remove $m = 3$–5 almost completely (by factors of 10^4), and what remains is spread over $m \gtrsim 8$, often dominated by modes above 12. The remaining residual is also concentrated near $\ell \approx 180°$–$240°$, around the CoE direction.

C. Symmetry, walls and partners

The best unrestricted fits are close to mirror images across the plane through us and the CoE. The old $0.58\times$ outside layout has five exact mirror pairs and one body on the axis. Forbidding mirror layouts costs one body: the target is reached at $N = 12$ (7 masses and 5 voids, $0.73\times$). A stricter rule that forbids mirror angles alone, regardless of distance and size, stalls at $4.5\times$.

Removing the CoE-interior wall gains one body ($N = 10$ instead of 11) and lowers the floor at $N = 12$ to about 0.07–$0.09\times$. Opening our visible sphere as well gains two more ($N = 8$). With the no-mirror rule applied at those N, the open solutions rise to $0.54\times$ ($N = 11$, no wall) and $3.7\times$ ($N = 8$, all open).

If our own sphere were on average over- or underdense, its net mass would have to sit in giant partner regions. A uniform contrast of our sphere mostly produces an $m = 1$ pattern ($515\times$ per unit $\delta_{\rm us}$). The best partner fit (2 partners plus 4 masses and 4 voids, $\delta_{\rm us} = -0.025$) reaches $1.23\times$; a layout with a giant 9.2 Gpc partner void and $\delta_{\rm us} = +0.21$ reaches $1.57\times$. Partners help, but do not beat ordinary bodies.

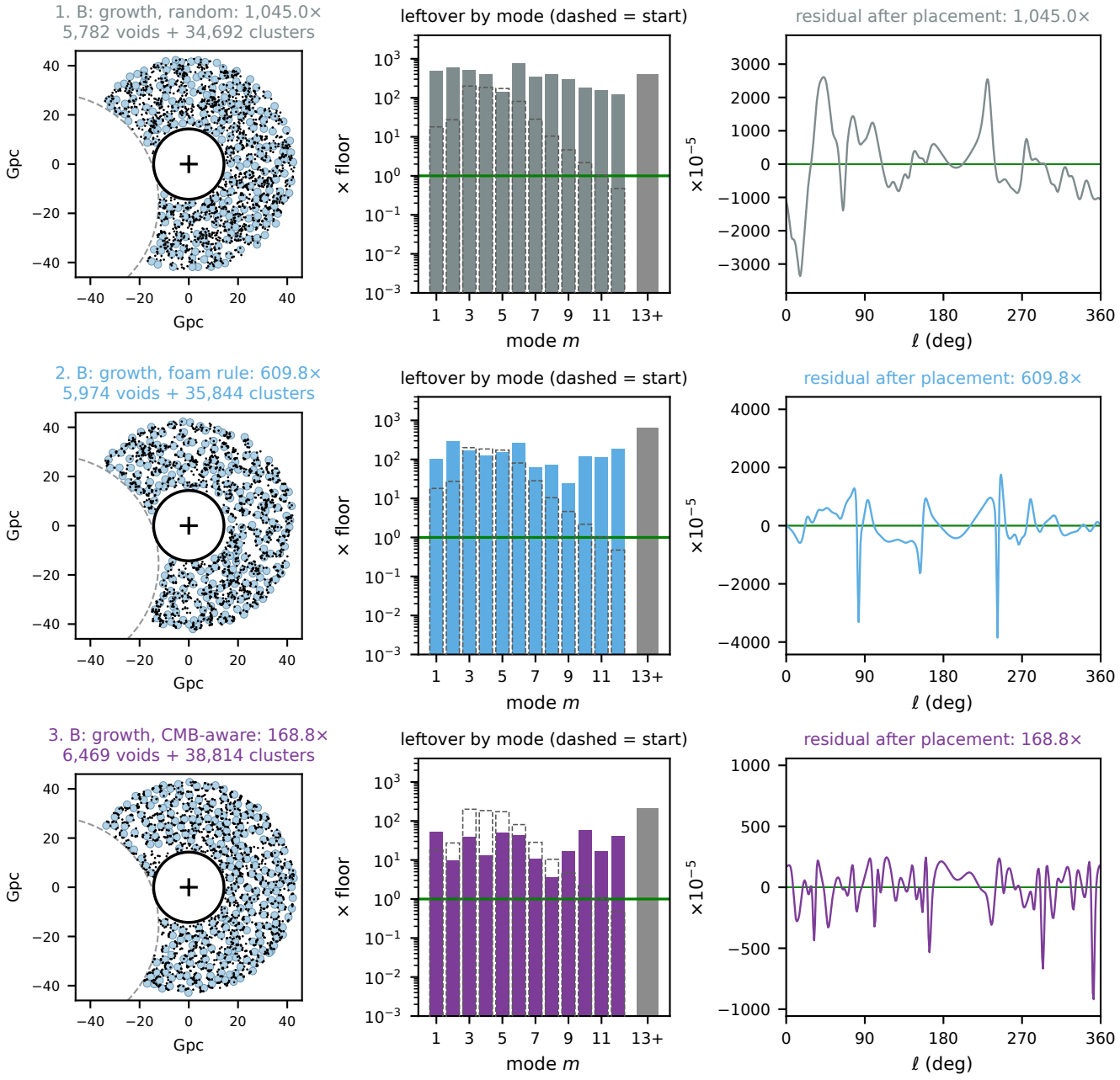


FIG. 39: Sequential-placement universes 1–3: the final universe of each curve of Fig. 38. Figures 39 and 40 are one sequence of six runs, numbered 1–6 from the largest to the smallest final leftover; each row's titles, bars and residual curve have the colour of its curve in Fig. 38. Left: side view, looking face-on at the plane of the CMB ring (us: +; black circle: our 14.3 Gpc sphere and the CMB ring; grey dashed: the CoE-interior wall; blue: voids at their true radius; red dots: clusters). The A runs fill only the near region and all their bodies are shown, projected onto the plane; the B runs fill the whole region out to 42.9 Gpc in three dimensions, so only bodies centred within 1.5 Gpc of the ring plane are drawn. Middle: leftover by mode (dashed outline: start; grey bar: all modes above 12 combined; green: 1×). Right: residual around the ring after placement, each on its own vertical scale; the green band is ±1×. The B layouts were not saved by the original runs and were regenerated with the same code and random seed; they reproduce the saved final leftovers. This figure: 1, B growth with random placement (5,782 voids and 34,692 clusters, 1,045×); 2, B growth with the foam rule (5,974 voids and 35,844 clusters, 610×); 3, B growth, CMB-aware (6,469 voids and 38,814 clusters, 169×).

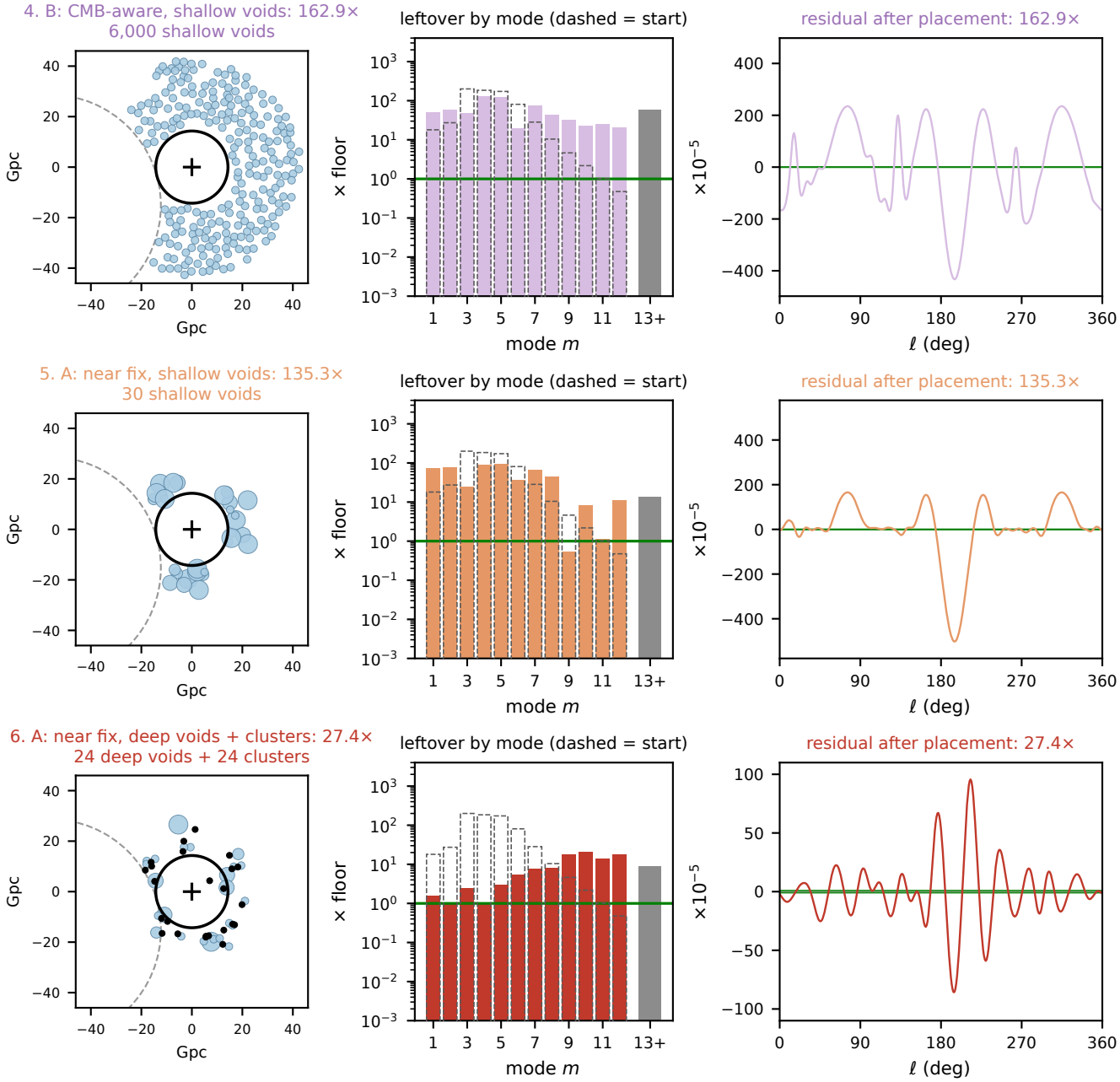


FIG. 40: Sequential-placement universes 4–6 of the sequence begun in Fig. 39 (largest to smallest final leftover; panels as in Fig. 39): 4, B growth, CMB-aware, with shallow voids (6,000 voids, 163×); 5, A near region fixed first, shallow voids (30 voids, 135×); 6, A near region fixed first, deep voids and clusters (24 of each, 27.4×; the same layout as example 1 in Fig. 41).

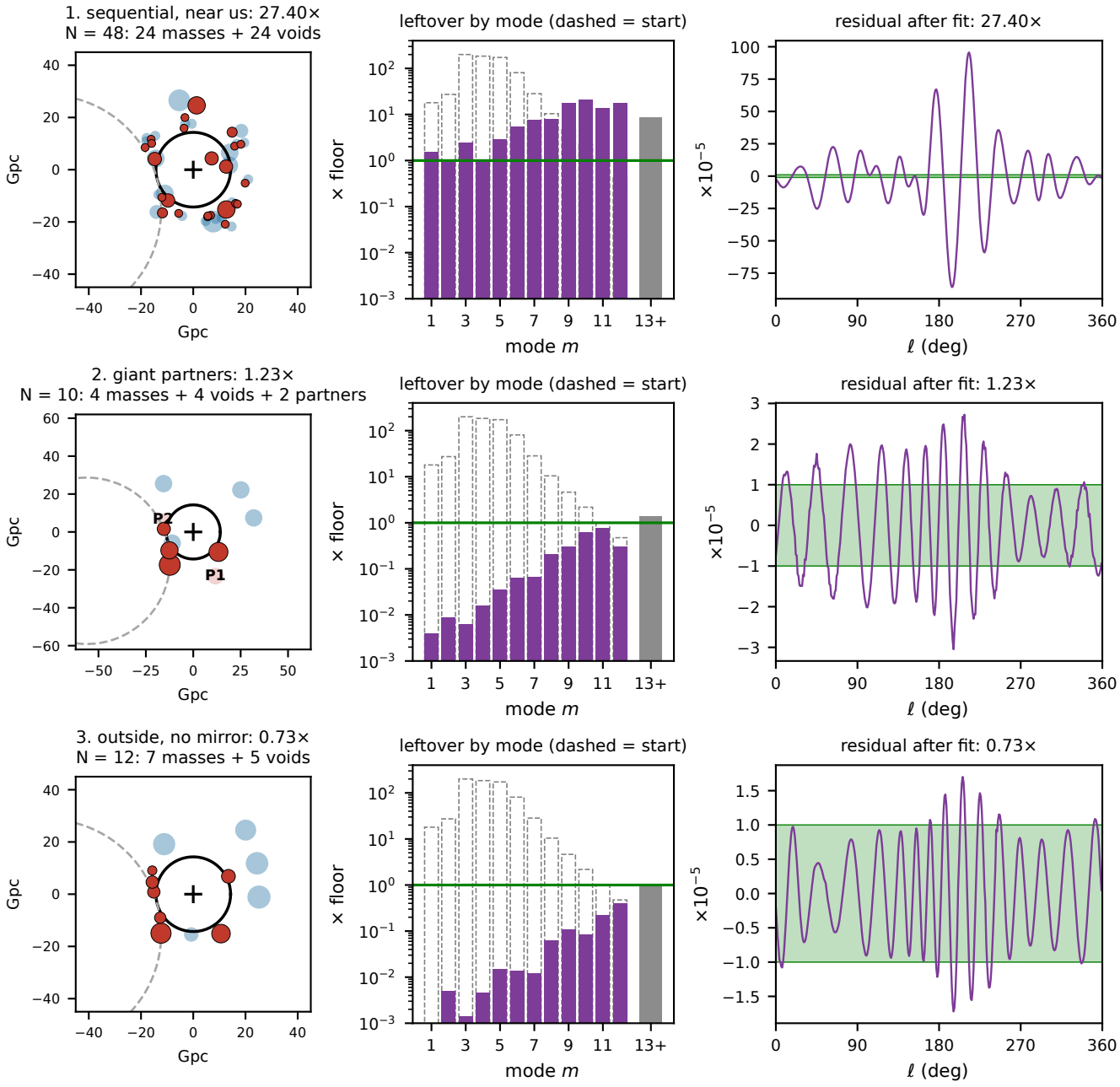


FIG. 41: Example universes 1–3. Left: side view, looking face-on at the plane of the CMB ring (the galactic plane) (us: +; black circle: our 14.3 Gpc sphere; grey dashed: the CoE-interior wall, where it applies; red: masses, sized by mass; blue: voids at their true radius; P1, P2: giant partners; bodies off the plane are projected). Middle: leftover by mode (dashed outline: start; grey bar: all modes above 12 combined; green: 1×). Right: residual around the ring after the fit; the green band is ±1×. Figures 41–43 are one sequence of nine example universes, numbered 1–9 from the largest to the smallest remaining leftover; Fig. 44 continues with the core-open examples 10–13. This figure: 1, sequential placement in the near region (48 bodies: 24 masses and 24 voids, 27.40×), where large residuals remain in every mode; 2, giant partners plus ordinary bodies (10 bodies, 1.23×); 3, outside, masses and deep voids with mirror layouts forbidden (12 bodies, 0.73×).

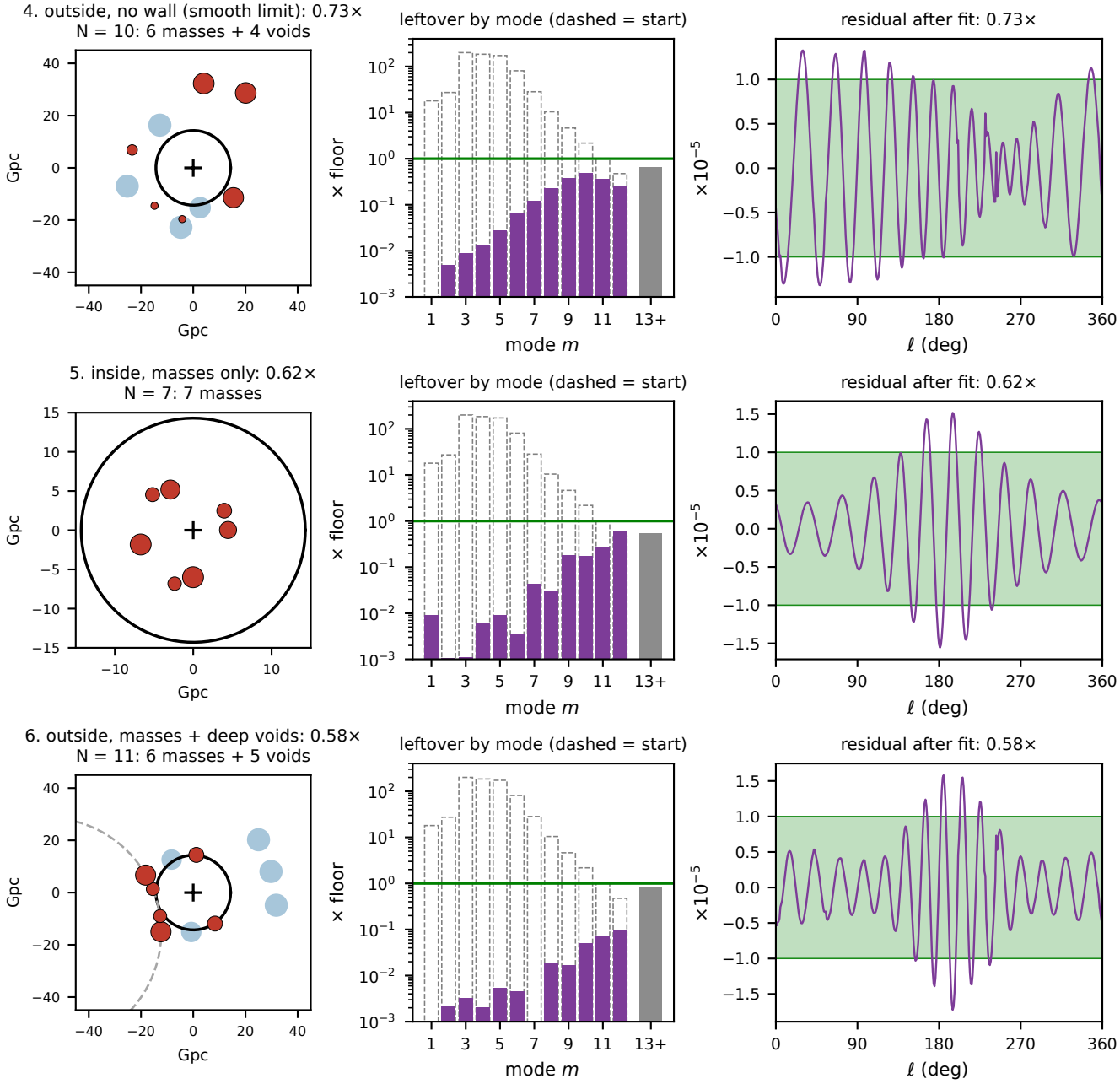


FIG. 42: Examples 4–6 of the sequence begun in Fig. 41 (largest to smallest leftover; panels as in Fig. 41): 4, outside with no CoE wall and the smooth-density limit (10 bodies, 0.73×; 0.726 against 0.734 for example 3); 5, inside our sphere, 7 masses (0.62×); 6, outside, masses and deep voids (11 bodies, 0.58×).

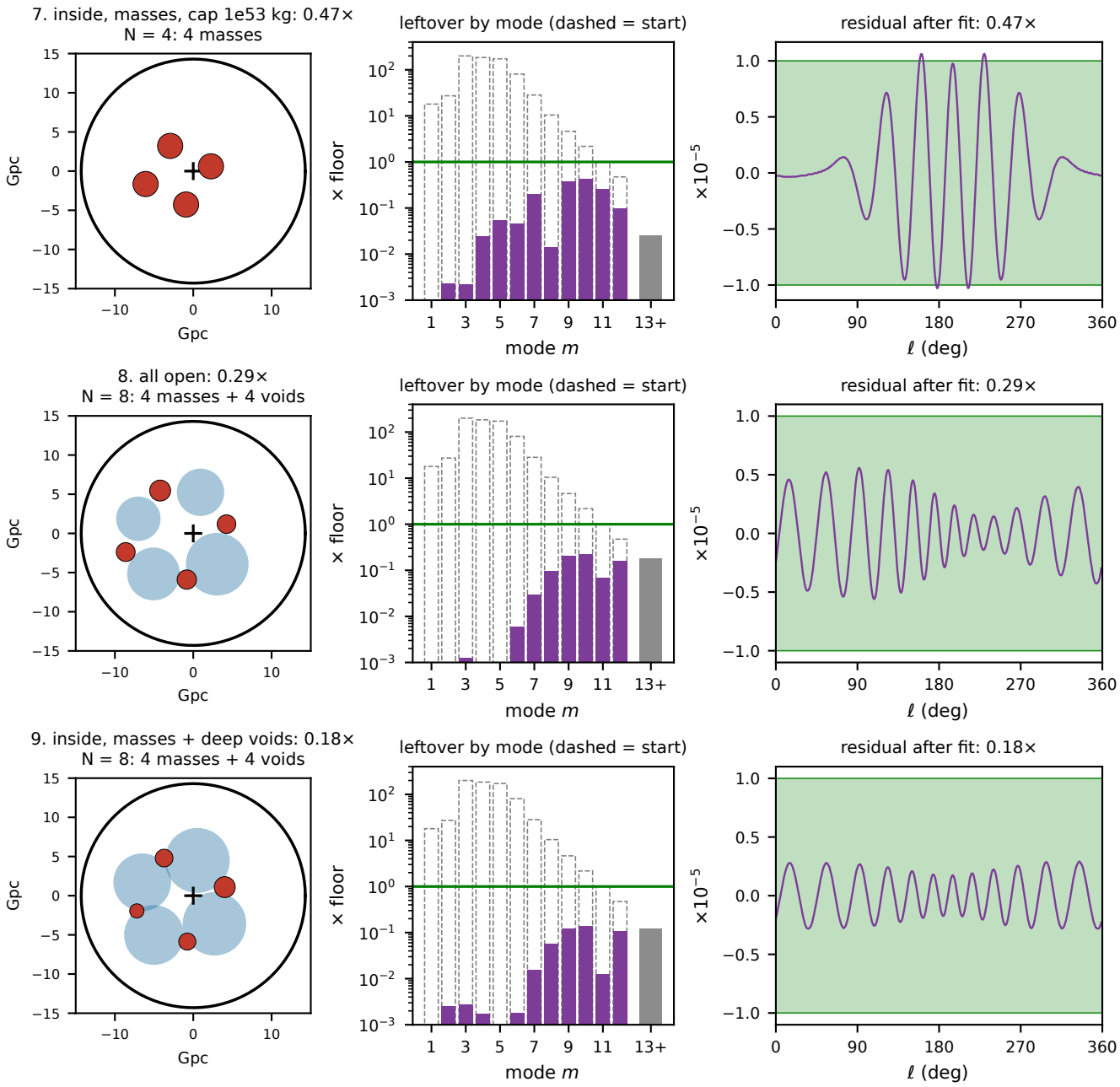


FIG. 43: Examples 7–9 of the sequence begun in Fig. 41 (largest to smallest leftover; panels as in Fig. 41): 7, inside our sphere, 4 masses with a 10^{53} kg cap (0.47×); 8, all open, where the eight bodies move inside our sphere (0.29×); 9, inside our sphere, 4 masses and 4 voids (0.18×).

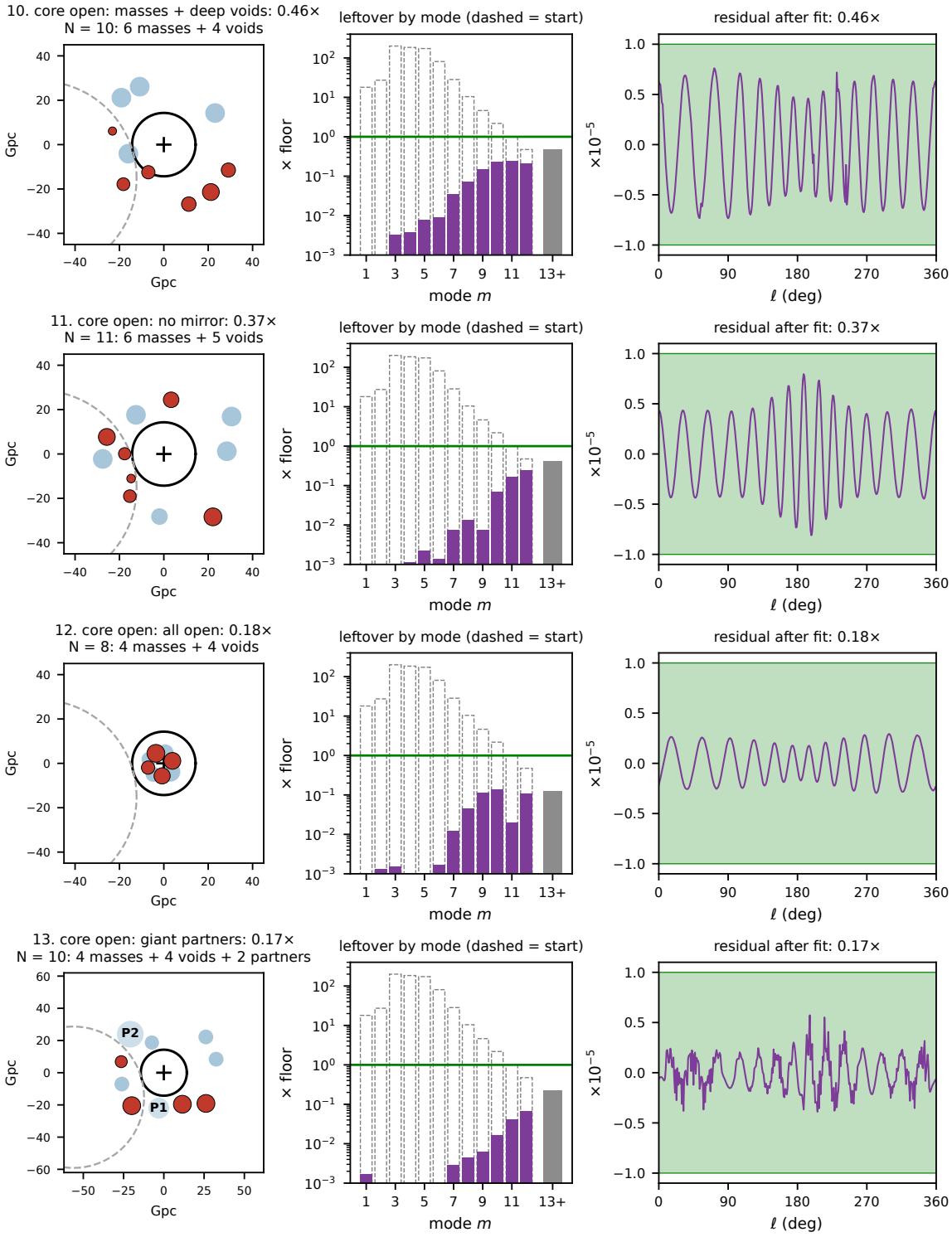


FIG. 44: Examples 10–13: the same scenarios rerun with the CoE core fully open (Sec. IV D), at the first N that reaches 1×, ordered from the largest to the smallest leftover; panels as in Fig. 41, and the grey dashed circle is now the edge of the open core rather than a wall. 10, masses and deep voids (10 bodies, 0.46×; 3 bodies 1–6 Gpc inside the core edge); 11, no mirror layouts (11 bodies, 0.37×; 5 in the core, 4 of them within 6 Gpc of the edge); 12, all open (8 bodies, 0.18×; none in the core; all eight sit inside our sphere, as in examples 8 and 9); 13, giant partners plus ordinary bodies (10 bodies, 0.17×; 3 in the core, 7–13 Gpc deep).

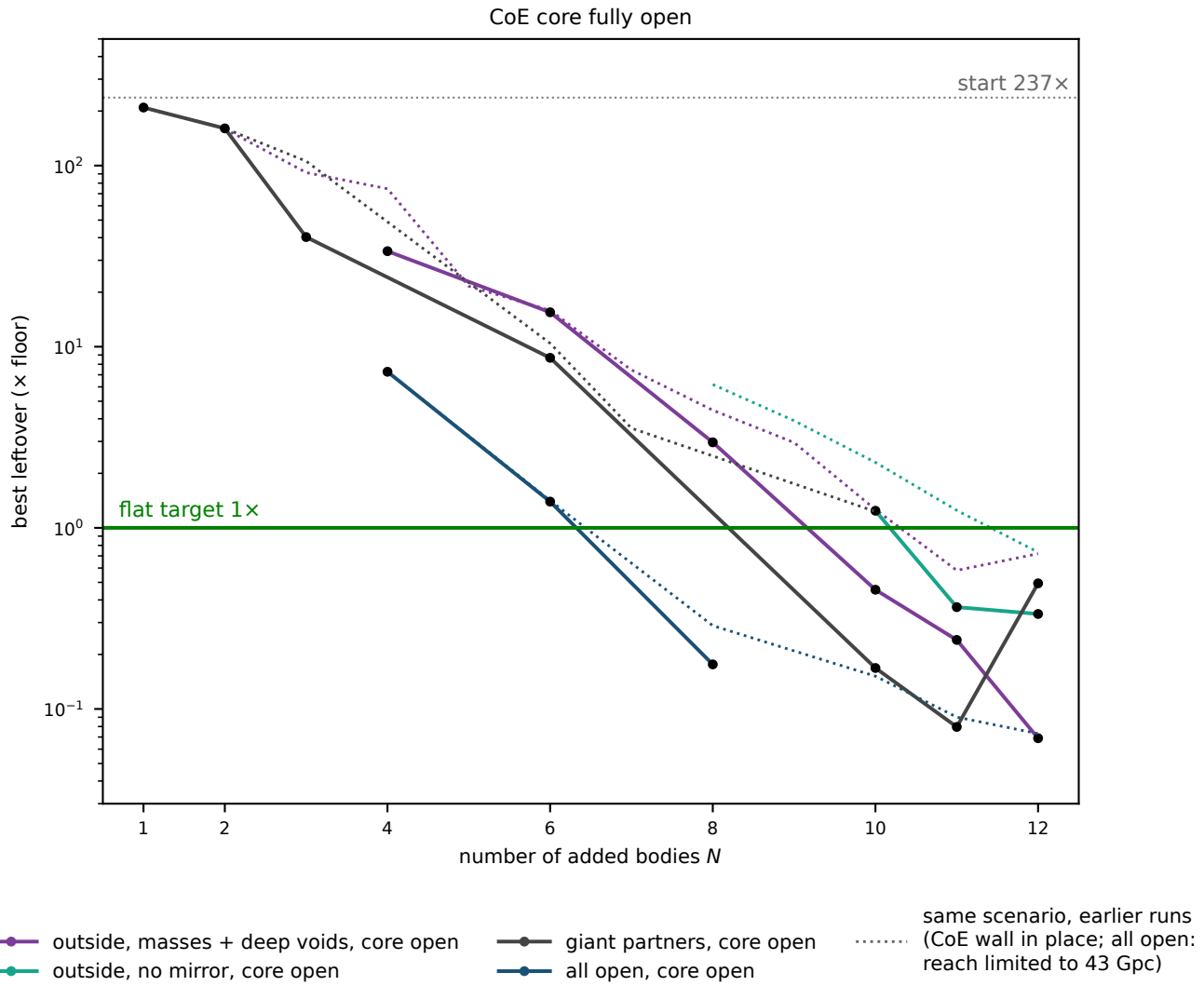
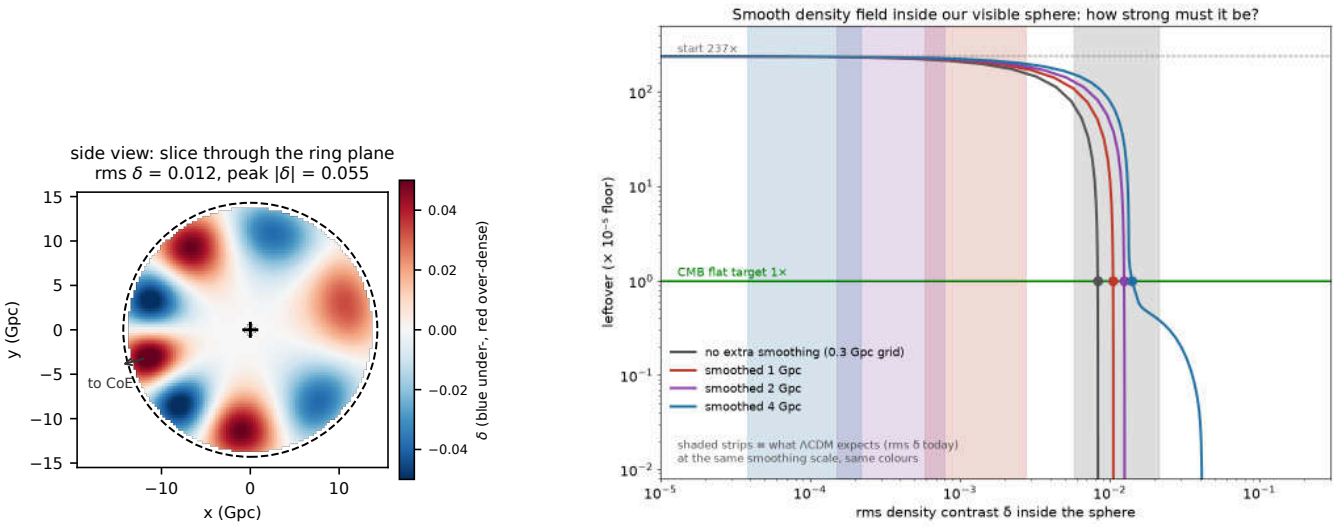


FIG. 45: Best leftover against the number of added bodies with the CoE core fully open (solid), compared with the same scenarios in the earlier runs (dotted: CoE wall in place; for all open, reach limited to 43 Gpc). Green: the flat target (1×); dotted grey: the start (237×).

D. CoE core fully open

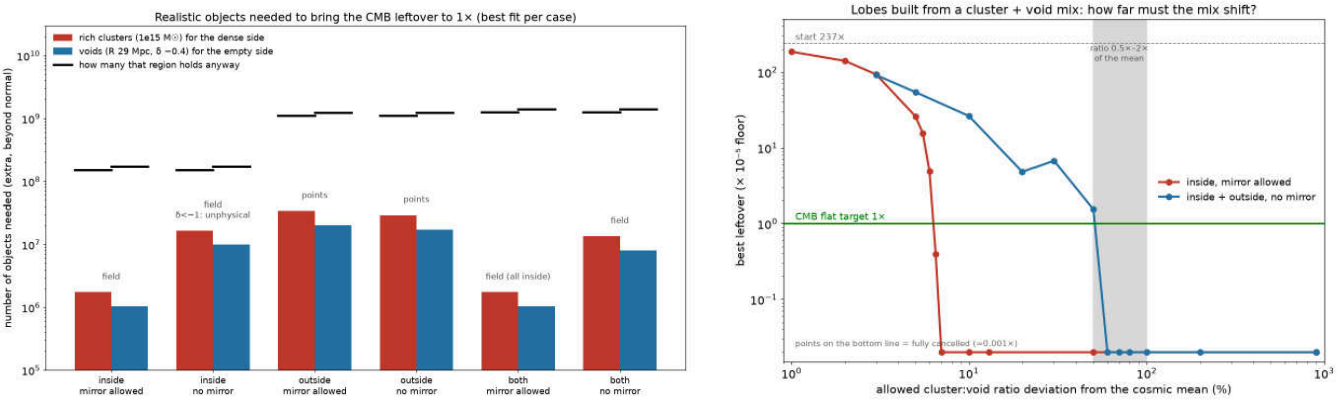
We reran the outside, no-mirror, giant-partner and all-open fits with the CoE interior fully open: no wall and no smooth-density mass cap, with the reach extended along every direction to the far side of the core, up to 102 Gpc from us toward $\ell = 195°$. The mass and radius bounds are unchanged, and of the 214 fits, many started with a third, half or all of their bodies inside the core (Fig. 45, examples 10–13 in Fig. 44). Outside our sphere the target is now reached at $N = 10$ (0.46×) and the leftover falls to 0.069× at $N = 12$; the no-mirror fits reach it at $N = 11$ (0.37×), and giant partners plus ordinary bodies reach it for the first time, at $N = 10$ (0.17×). The all-open result is unchanged: 0.18× with 8 bodies, as in the earlier all-open run without the density limit, and none of them in the core. Almost all of this gain was already present with the wall removed at the old 43 Gpc reach (0.46, 0.25 and 0.074× at $N = 10$, 11 and 12).

The fits use the core, but only its near edge. Bodies settle a median 3–6 Gpc inside the edge, about 25 Gpc from us around $\ell \approx 195°$; core-seeded starts move their bodies back out to the edge or leave them parked where they do nothing. With every body forced into the core, the leftover never drops below 143× (12 bodies). The reason is how fast the needed pull fades with distance. The ring responds to how much a body's pull changes across it, so a distant body pulls almost uniformly and leaves mainly a weak two-lobed ($m = 2$) stretch. Table III gives the pattern of one 2×10^{52} kg mass on the us–CoE line: from 20 Gpc to the CoE at 60 Gpc the $m = 3$–5 modes, which must cancel 200, 185 and 173×, weaken by a factor of 100–1000.



(a) (b)

FIG. 46: Smooth-field solution. (a) The weakest field that reaches 1× at 2 Gpc smoothing, side view: a slice through the ring plane (red: overdense, blue: underdense; dashed: our 14.3 Gpc sphere). (b) Leftover against the rms density contrast of the field for several smoothing scales; shaded strips mark what standard cosmology expects at the same scales.



(a) (b)

FIG. 47: (a) Number of extra rich clusters and voids that each 1× field corresponds to, compared with how many each region holds anyway. (b) Leftover against how far the local cluster-to-void ratio may differ from the cosmic mean; inside our sphere about ±6.5% suffices. The no-mirror curve is from a stalled solver and is superseded (Sec. IV F).

TABLE III: Ring pattern of one 2×10^{52} kg mass on the us–CoE line, by mode (units of the floor). The fingerprint's $m = 3$, 4, 5 amplitudes are 200, 185 and 173×.

distance from us	$m = 3$	$m = 4$	$m = 5$
20 Gpc (near edge of the core)	862	691	541
60 Gpc (at the CoE)	6.4	1.8	0.47
100 Gpc (far edge of the core)	0.81	0.13	0.02

E. Field solutions

Instead of individual bodies, one can solve directly for a smooth density contrast $\delta(\mathbf{x})$ inside our sphere with zero net mass (Fig. 46). The smallest such field that reaches 1× has an rms contrast of 1.0–1.4% and peaks of 5–7%, depending on the smoothing scale (1–4 Gpc); it moves only 2×10^{51}–5×10^{51} kg. It sits just inside the ring and alternates in eight lobes, the same $m = 3$–5 pattern turned inside out: an inside deficit acts like an outside excess. The leftover falls from 237× to 1× over a narrow range of field strength (Fig. 46b).

Translated into real objects (Fig. 47a), these fields cor-

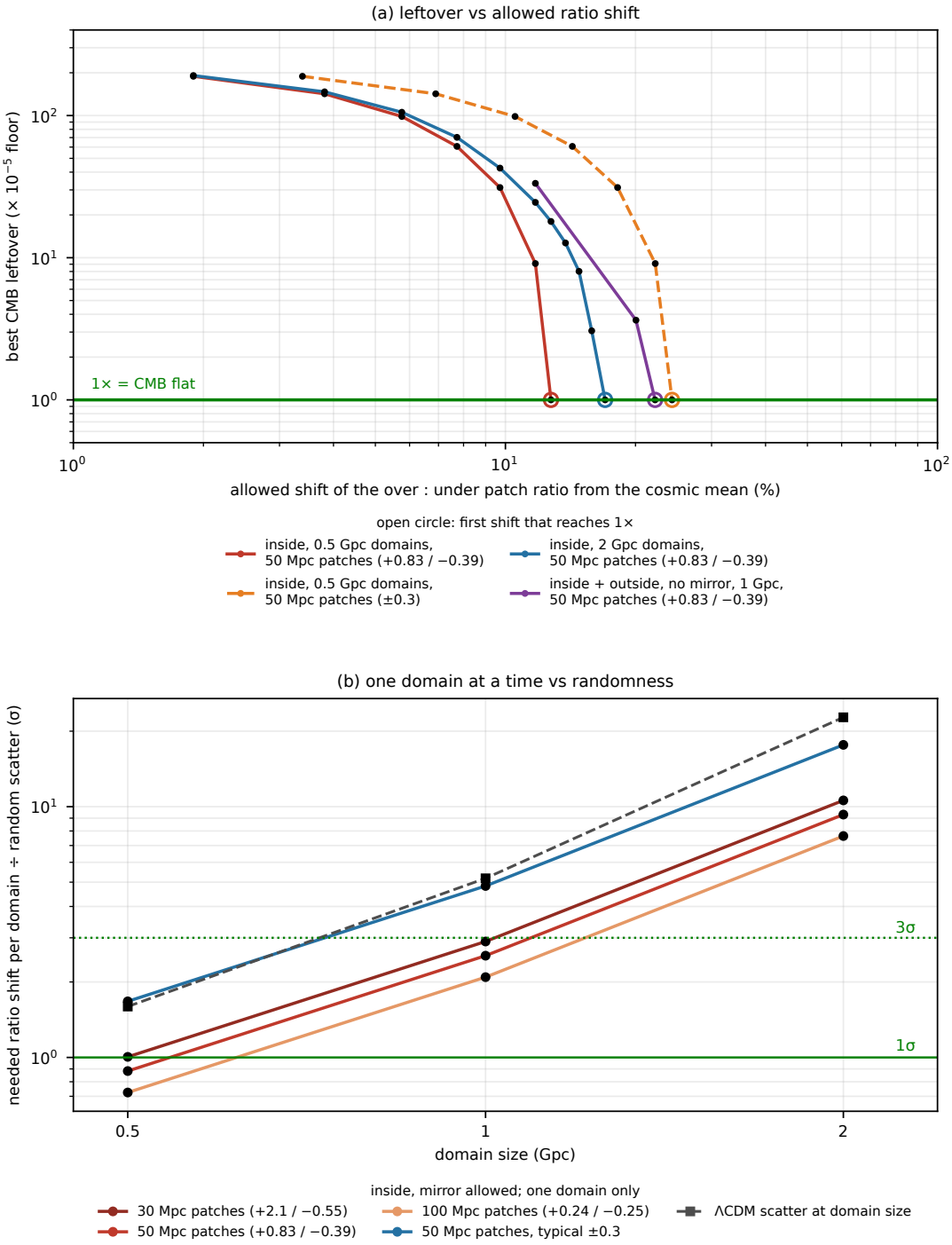


FIG. 48: Domains built from a fine mix of over- and underdense patches. (a, top) Best leftover against how far the over:under patch ratio of each domain may move from the cosmic mean. Open circles mark the first shift that reaches 1×: +12.7% / −11.8% for 0.5 Gpc domains and observed-extreme 50 Mpc patches, +17% / −15% for 2 Gpc domains, about +24% for typical ±0.3 patches, and +22% / −19% with mirror layouts forbidden (1 Gpc domains). (b, bottom) The required shift in one domain divided by the random scatter of that domain's patch count, inside our sphere with mirror layouts allowed. This counts one domain only; the coherence over thousands of domains is not included.

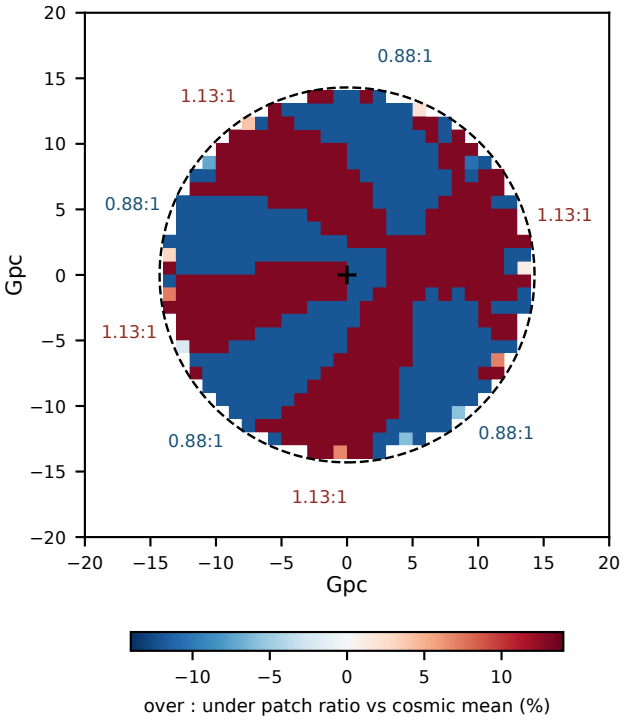


FIG. 49: The required over:under patch ratio, side view (the plane of the CMB ring, seen face-on; dashed: the ring), for 1 Gpc domains and 50 Mpc patches, as a percentage of the cosmic mean. Labels give each lobe's mean ratio relative to the mean. Every domain sits at the allowed limit, in the same eight lobes as the smooth field.

respond to a rearrangement of very many ordinary clusters and voids rather than a few giant bodies. If every lobe must be built from a mix of clusters and voids, the local cluster-to-void ratio must differ from the cosmic mean by about ±6.5% inside our sphere (Fig. 47b). Forbidding mirror layouts raises the required contrast to about 1.6 times the inside value (Sec. IV F; this supersedes the stalled no-mirror curve in Fig. 47b).

F. Over:under patch ratio

Real domains are not purely over- or underdense; each is a fine mix of small overdense and underdense patches. We therefore let every domain (0.5–2 Gpc) consist of patches of fixed contrast and vary only the volume fraction of overdense patches, so a domain's net contrast is set by its over:under patch ratio. With 50 Mpc patches at the observed extreme contrasts (+0.83 and −0.39), the cosmic mean ratio is 0.47:1. Solving for the residual with modes $m = 1$–12 (the box-constrained problem solved exactly through its dual), each 0.5–1 Gpc domain needs only about a ±12–13% shift in its over:under ratio to reach 1× (Fig. 48a); typical ±0.3 patches need about twice that, and forbidding mirror layouts raises it to about ±20% (1.6 times the inside contrast).

Taken one domain at a time, that shift is ordinary: it is about 1–2σ of the random scatter in the number of patches per domain (0.9σ for 0.5 Gpc domains, 2.6σ for 1 Gpc domains) (Fig. 48b). The difficulty is coherence. Thousands of domains must all shift the same way, in the same eight lobes, with the $m = 3$–5 pattern of the smooth field (Fig. 49): ratios of 1.13:1 and 0.88:1 relative to the mean, alternating around the ring and organised over about 5 Gpc. Random patch placement, propagated through the same kernel, produces only 1/6 to 1/13 of the needed CMB pattern, and a ΛCDM density field about 1/20. The needed pattern is therefore far beyond chance (formally 40–130σ; these numbers mean "not by chance" rather than a precise probability).

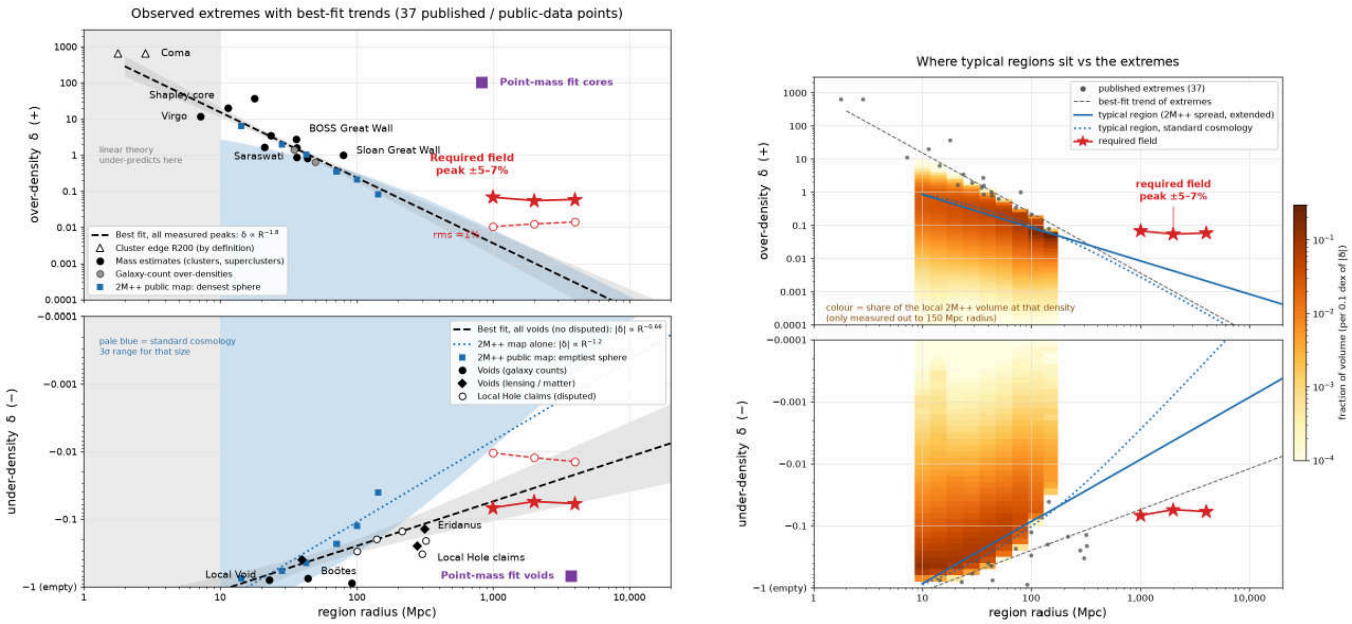
TABLE IV: Rarity of the required peak contrast $|\delta|$ for one region of radius R. "2M++ trend" extrapolates the observed spread; ΛCDM uses linear theory at $z = 0$.

R	peak $\|\delta\|$	width used	σ	peak in σ
1 Gpc	0.068	2M++ trend	0.0085	8.0 (6.0–10.5)
1 Gpc	0.068	ΛCDM	0.0028	24.5
2 Gpc	0.055	2M++ trend	0.0043	12.9
2 Gpc	0.055	ΛCDM	0.00079	69
4 Gpc	0.059	2M++ trend	0.0021	28
4 Gpc	0.059	ΛCDM	0.00022	267

V. COMPARISON WITH OBSERVED STRUCTURE

How unusual would the required structure be? Figure 50a compares the density contrast of the most extreme observed over- and underdense regions with their size: rich clusters [2–4], superclusters [5], voids [1, 6, 7], the Eridanus supervoid behind the Cold Spot [8, 9], the disputed local hole [10–12], and the densest and emptiest spheres in the 2M++ reconstruction [13]. Overdense extremes fall roughly as $|\delta| \propto R^{-1.8}$; at 1 Gpc the required peak contrast is about 19 times the fitted trend of observed extremes. Typical regions (Fig. 50) are far narrower than the extremes: the 2M++ spread falls as $\sigma \approx 8.6\,(R/\mathrm{Mpc})^{-1.0}$ over 20–80 Mpc, and nothing in any catalog reaches 1 Gpc. The homogeneity scale is about 100 Mpc [14]; the quasar dipole [15] and the cluster anisotropy toward $\ell \approx 280°$ [16] are the strongest observed hints of Gpc-scale variation.

Table IV gives the rarity of the required peak contrast if Gaussian statistics hold at Gpc scales. Even with the generous width extrapolated from 2M++, the 1 Gpc field is an 8σ event; against linear ΛCDM [17] it is 25σ or more.



(a) (b)

FIG. 50: (a) Density contrast of the most extreme observed regions against their radius, with power-law fits, the standard-cosmology band and the required model markers. (b) The same axes with typical regions (2M++) as a heat map under the extremes.

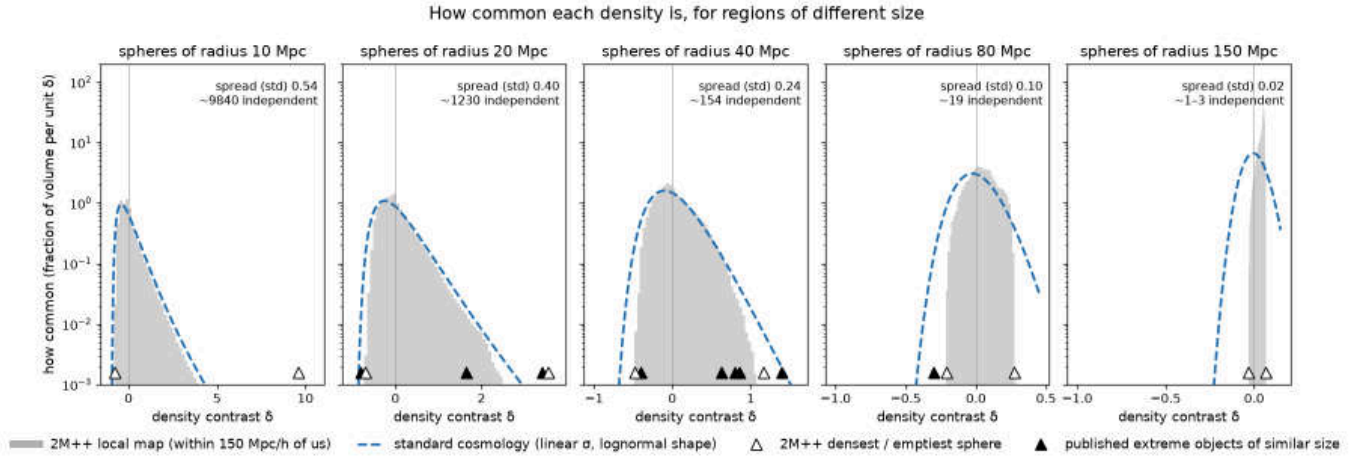


FIG. 51: How common each density contrast is, for 2M++ spheres of 10 to 150 Mpc. The distribution narrows and becomes symmetric as the spheres grow.

VI. EDITING FILLED WEBS TO REACH THE CMB LEVEL

The fits above place a handful of bodies in otherwise empty space. Here we pose the inverse problem the other way round: start from a universe already filled with ordinary voids and clusters, and ask how few of them must be moved or resized for the leftover to reach 1×. This section summarizes the work of October 10–11.

Method. The modelled region is our 14.3 Gpc sphere plus the shell out to 30 Gpc from us, including the part of the CoE core inside 30 Gpc (about the depth to which bodies still shape the ring, Table III). The starting webs are the open-region versions of the sequential growth runs (B in Fig. 38) (random and foam rules, seeds 1–3, about 17,000 bodies each; voids $R = 1.5$ Gpc, $\delta = -0.70$, each paid back by six $\delta = +100$ clusters). No body may straddle the ring sphere, come within 0.5 Gpc of us or overlap another. Each edited body has four free numbers, its position and a mass factor a (radius scales as $a^{1/3}$; voids stay ≥ 0.5 Gpc). Each round we rank bodies by how much the leftover would drop if they were nudged (a linearized step, capped at 5 Gpc), fit a batch of them jointly by bounded least squares, and keep the round only if the leftover falls. A fine-scale pass handles bodies within 6 Gpc of the ring, which mainly set the short waves ($m \geq 13$), by trying starts pushed 2–8 Gpc away. Every five rounds the most recent edits are refitted together. Runs stop at 1× or when the last 300 edits gain less than 1%.

The ring rule. Left free, the fitter piles bodies onto the CMB ring, because a body right at the ring has enormous leverage on the short waves. Our first fix was a hard *backstop*: every body surface stays at least 3 Gpc from the ring. That is a judgment call, not physics, and it produces an artificial pile-up just outside 3 Gpc. We replaced it with an *entropy rule*: no edit may make the web more ordered than it started. We measure disorder seven ways, and none may decrease: three volume-weighted position entropies $H = -\sum_b p_b \ln(p_b/w_b)$ (radial shells around us, distance from the ring circle, and sky direction; w_b is the volume share of bin b, so $H = 0$ means uniform in volume) and four Shannon entropies $S = -\sum p \ln p$ of the nearest cluster–cluster, cluster–void and void–void distances and of the number of neighbours within 3 Gpc. This rules out forming a ring, a pile-up, a hole or a regular pattern.

Results. Allowing deletion reaches 0.87–1.00×, but only by deleting the bodies nearest the ring, so we forbid it from here on. With move and resize only, under the 3 Gpc backstop, two foam webs reach the target: foam seed 3 needs 489 edits (2.8% of the web; 2,164× to 0.995×, Fig. 52) and foam seed 1 needs 614 (3.6%; 0.95×). Bodies near the ring move a median of about 2 Gpc, the others under 1 Gpc, and most masses change by less than a factor of 2. Under the entropy rule none of the six grown webs quite reaches 1× (Table V). The best, foam seed 2, goes from 1,223× to 2.40× with 638 edits (3.8%, Fig. 53). Solving for the residual with modes $m = 1$–12 leaves only 0.01–0.1× in every run; everything left is short-wave ripple ($m \geq 13$) from bodies near the ring. In plain words, the smooth large-scale pattern is gone, and what remains is chaotic, asymmetric fine-scale texture, at the CMB limit in the two backstop runs and a few times above it in the entropy-rule runs (Sec. IX A). There is no pile-up at the ring: the edits partly refill the gap the growth rule left there, and the density within 3 Gpc of the ring stays well below the average (Fig. 54). These runs are superseded by the compensated web of Sec. IX, which reaches 1× under the same entropy rule.

TABLE V: Move and resize under the entropy rule, for the six grown webs. Edits as a share of the web; leftover in units of the floor; "after $m \leq 12$" is what remains after solving for the residual with modes $m = 1$–12. All seven entropy measures ended at or above their starting values.

web	edits	leftover	after $m \leq 12$
random, seed 1	825 (4.9%)	2,569 → 13.0	0.02
random, seed 2	897 (5.3%)	2,106 → 6.41	0.02
random, seed 3	964 (5.6%)	2,222 → 18.3	0.04
foam, seed 1	856 (5.0%)	794 → 6.29	0.02
foam, seed 2	638 (3.8%)	1,223 → 2.40	0.01
foam, seed 3	665 (3.8%)	2,164 → 6.05	0.01

VII. MESH, AREA AND ENERGY OF AN EDITED WEB

Method. To see what the edits do to the shape of the web, we join the bodies in the side-view slab ($|z| < 1.5$ Gpc, projected onto the ring plane) by their full Delaunay triangulation: every node is linked to its neighbours with non-crossing links until only triangles remain, so there are no gaps. Each triangle after the edits is coloured by its area minus the area of the before-triangle that contains its centre. We then compare histograms of triangle area (for the whole slab and for the band 0–3 Gpc from the ring), of connections per node and of link length. Area entropy is $S = -\sum_i p_i \ln p_i$ with $p_i = A_i / \sum A$, and the toy energy is $U = -G \sum_{\langle ij \rangle} m_i m_j / r_{ij}$ over the links, with clusters at $+M$ and voids at minus their missing mass, plus a wall term and an entropy term $T_{\text{eff}} \Delta S$ with $k_B T_{\text{eff}} = \langle \frac{1}{2} M v^2 \rangle$.

Results (foam seed 2). The triangles change almost only inside about 20 Gpc, around our sphere (Fig. 55). Over the whole slab the median triangle grows slightly, from 0.59 to 0.66 Gpc^2 (Fig. 56a). Near the ring most spots are compressed, because large gap triangles get split, but the population evens out: the median goes from 0.79 to 1.35 Gpc^2 and the area entropy rises from 4.70 to 5.03. Connections per node do not change (mean 5.94, as expected for a triangulation), and the median link grows from 1.77 to 1.87 Gpc (Fig. 56b); the spike near 1.7 Gpc is the foam rule, which puts each cluster on a void rim. The energy rises near the ring (about +23%) and falls

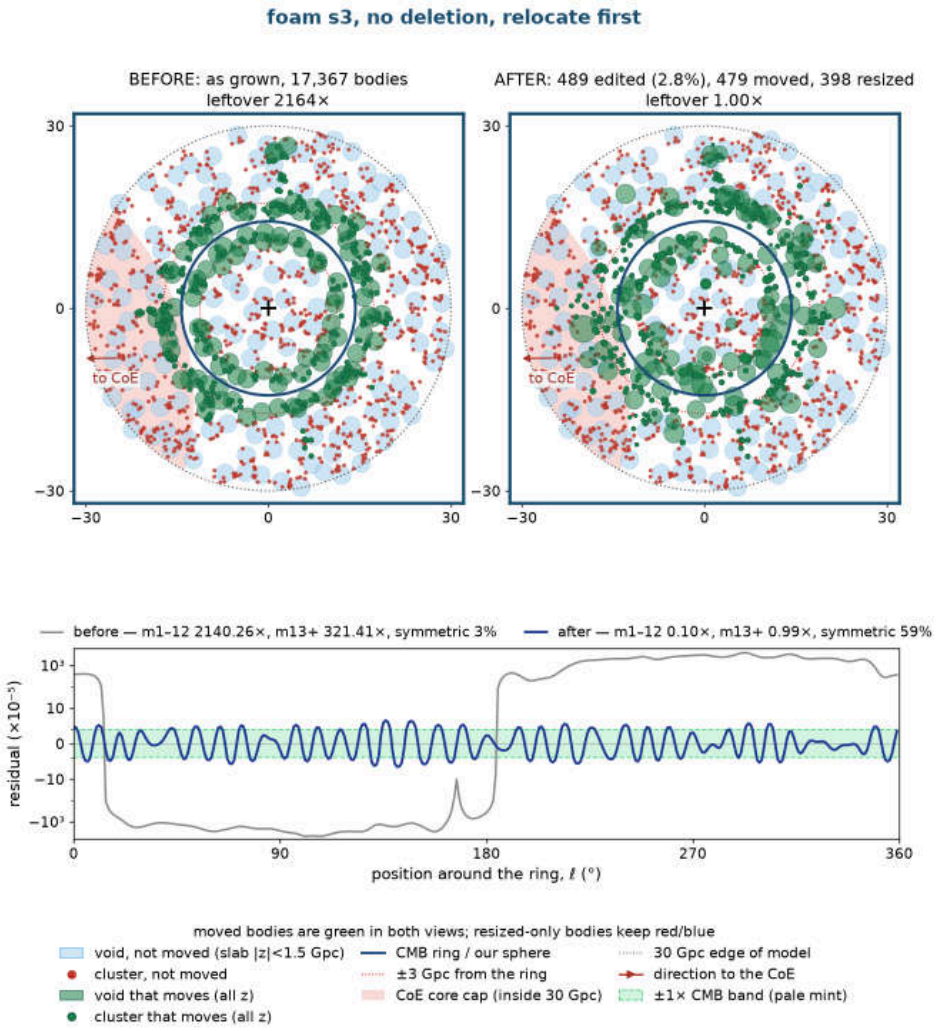


FIG. 52: Foam web, seed 3, edited by moving and resizing only (no deletion) under the 3 Gpc ring backstop. Top: side views before (as grown, 17,367 bodies, 2,164×) and after (489 bodies edited, 1.00×), slab $|z| < 1.5$ Gpc. Pale blue: voids; red dots: clusters; green: bodies that are moved (shown at their old place before and their new place after, all z); bodies that are only resized keep their red or blue colour. Dark blue circle: our sphere and the CMB ring; red dotted: ±3 Gpc from the ring; pink: the CoE core cap inside 30 Gpc; arrow: direction to the CoE; dotted grey: the 30 Gpc edge of the model. Bottom: residual around the ring before (grey) and after (dark blue), on a symmetric log scale; the pale mint band is ±1×. The leftover (dark blue) is no longer the smooth large-scale pattern: the $m = 1$–12 part is cancelled to about 0.1×, and what remains is chaotic, asymmetric fine-scale ripple at the CMB limit, the noise-like texture the model predicts (Fig. 71).

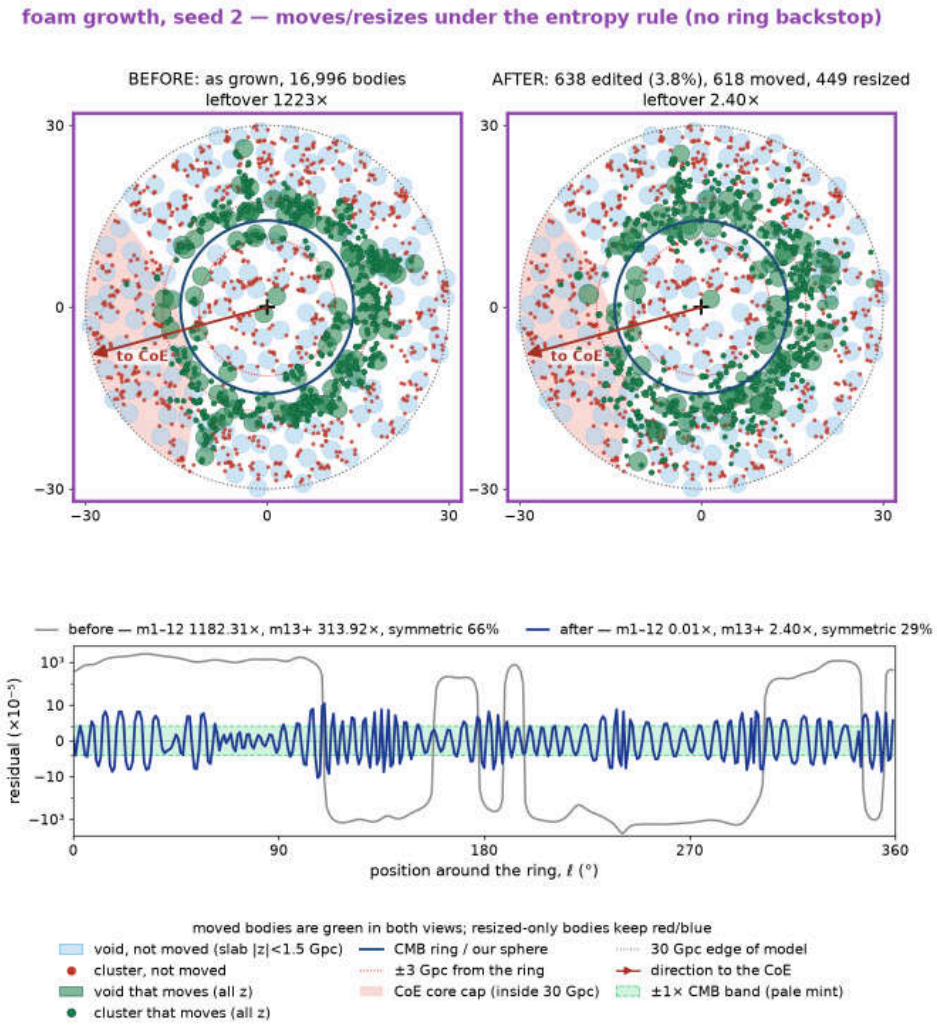


FIG. 53: Foam web, seed 2, edited by moving and resizing under the entropy rule instead of the ring backstop: 1,223× to 2.40× with 638 edits (3.8%). Panels and colours as in Fig. 52. Solving for the residual with modes $m = 1$–12 leaves 0.01×; what remains (dark blue) is short-wave ripple. The symmetric share of the leftover drops from 66% to 29%. The leftover is no longer the smooth large-scale pattern: the $m = 1$–12 part is cancelled to 0.01×, and what remains is chaotic, asymmetric fine-scale ripple, here still 2.4 times the CMB limit, the same noise-like texture as in the runs that reach it (Fig. 71).

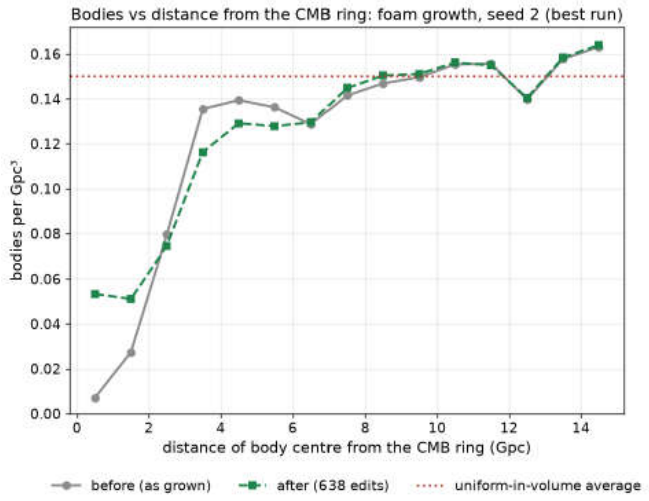


FIG. 54: Bodies per Gpc^3 against the distance of their centre from the CMB ring, foam seed 2, before (grey) and after the 638 entropy-rule edits (green, dashed). Red dotted: the uniform-in-volume average. The edits partly refill the gap left by the growth rule near the ring but stay far below average: no ring forms.

farther out (about −14% at 3–6 Gpc); overall the edited web is about 3% more bound. The entropy term is negligible next to gravity. (An earlier non-crossing mesh with a long-link cutoff left gaps; it gave the same trend near the ring, median cell 0.60 to 1.20 Gpc^2.)

VIII. CRYSTAL ANALOGY AND THE COMPENSATION RULE

Lattice reference. Einasto et al. found a roughly regular network of superclusters and voids with a step of about $120\,h^{-1}$ Mpc [18, 19], 171 Mpc for $h = 0.7$. As a perfectly ordered reference we put clusters on a face-centred cubic (FCC) lattice with that cell edge and voids in its tetrahedral holes, so each void sits inside a pyramid of four clusters (the fluorite arrangement; Fig. 57a). We then built a ladder from perfect to random: thermal jitter (5, 15 and 30% of the spacing), point defects (3% and 10% vacancies, interstitials and voids in the wrong holes), a polycrystal of ten randomly oriented grains with disordered boundaries, and fully random points. Each structure and the foam web are tetrahedralized in 3D and placed on a chart of disorder (Shannon entropy of the spread of tetrahedron volumes, 0.1-dex bins) against link energy per node, both scaled by each structure's own spacing.

Results. Disorder rises steadily from perfect through defects, jitter and polycrystal to random (Fig. 57b). The foam web lies *past* random: its cells are more uneven than random points, because the foam rule makes both very small and very large cells, and it is the least bound structure, because its clusters sit on void rims and cluster–void pairs repel when voids count as negative mass. The edits make it about 4% more bound and slightly more even, a small step toward the lattice. If the real web is a defective crystal, it should fall between the polycrystal and random points, which is a testable target for the generator.

Compensation rule. On a 2D triangular lattice with equal numbers of clusters and voids, the lowest-energy arrangement with voids as bare negative masses is full segregation, clusters on one side and voids on the other; this is an artifact of ignoring where a void's missing matter goes. We therefore balance every void with its own clusters: its missing mass $M_{\rm def} = |\delta_v|\rho\frac{4}{3}\pi R^3$ sits in clusters placed 1–1.3 void radii from its centre (the compensating ridge [1, 7]), so each void plus its wall has zero net mass. The number of clusters per void then grows in proportion to the missing mass (Fig. 58): $N = 0.92 + M_{\rm def}/2.10 \times 10^{50}\,\mathrm{kg}$ in the 2D test and $N = 0.49 + M_{\rm def}/1.48 \times 10^{50}\,\mathrm{kg}$ in the 3D web of Sec. IX, about 4.5 clusters per void. Annealed to its lowest energy, the balanced 2D layout looks like the cosmic web: clusters slide to the outer edge of their walls and bunch into knots where two or three voids meet, and the runaway collapse of unbalanced negative-mass pairs disappears.

IX. A COMPENSATED RANDOM WEB REACHES THE CMB LEVEL

Generator. We built a new random web that respects both the compensation rule and published size and abundance laws, using only their *shapes* rescaled to the Gpc body scale. Void radii follow the volume-conserving void size function of Jennings, Li and Hu [20], built on the two-barrier excursion set of Sheth and van de Weygaert [21], scaled to a median of 1.5 Gpc (1.36–2.14 Gpc for 5–95%). Void depths are shallow, δ_v uniform in $[-0.45, -0.25]$ [1]. Cluster masses follow the Tinker et al. mass function [2], scaled to a median of 1.40×10^{50} kg, with contrasts log-uniform in 30–300 (median radius 0.155 Gpc). Each void's clusters are placed in its wall until they hold exactly its missing mass. Voids go in uniformly in volume, largest first, until they fill 29% of the volume, under all the placement rules of Sec. VI. Each seed has about 9,000 bodies (seed 3: 1,645 voids and 7,417 clusters).

Edits that keep the balance exact. An edit never separates a void from its clusters. A *group* edit moves a void together with all its wall clusters and rescales them together (four numbers: shift and mass factor, with masses re-weighted by the local density), so the group's net mass stays exactly zero. A *cluster* edit slides one cluster within its own wall, between the void edge and 1.3 void radii, without changing its mass. The search and the entropy rule are as in Sec. VI, with moves capped at 5 Gpc for groups and 2 Gpc for clusters.

Results. Seed 3 reaches the CMB level: 789× to 0.998× with 1,099 bodies edited (12.1% of the web), no deletions and no ring (Table VI, Fig. 59). The edits are

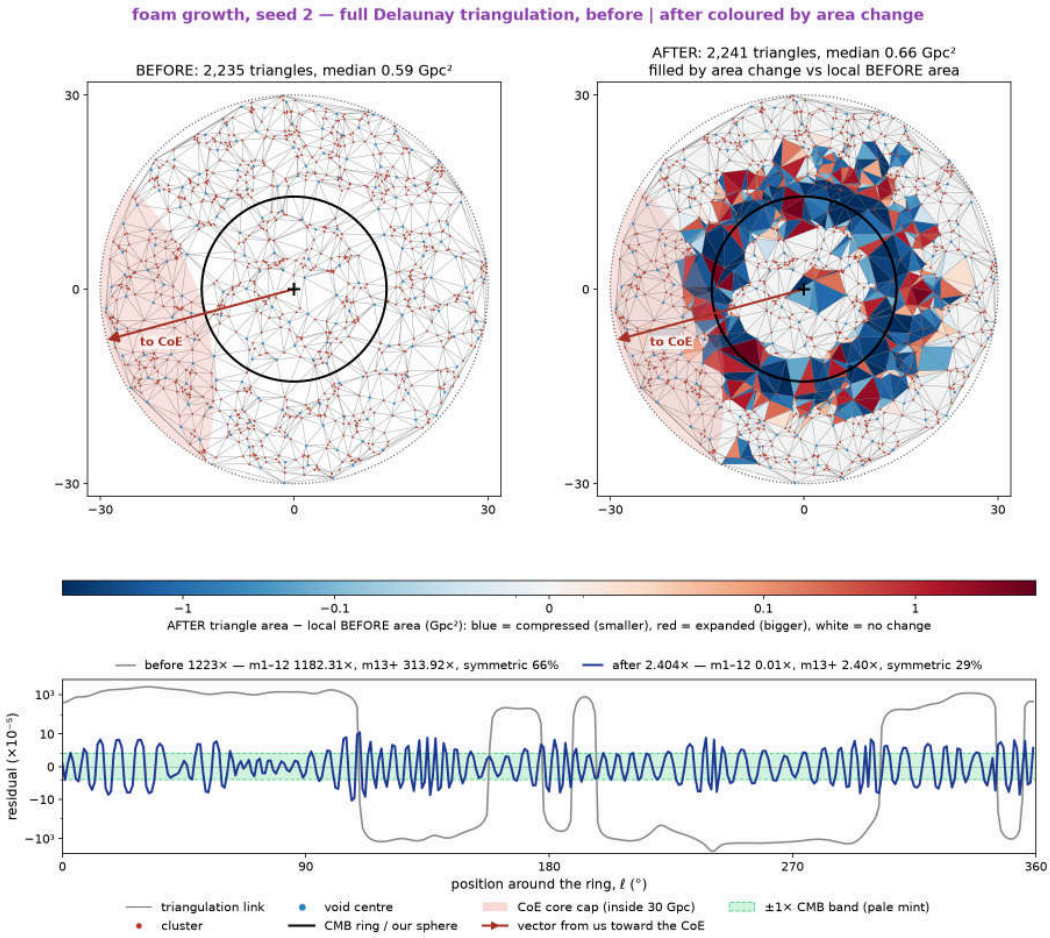
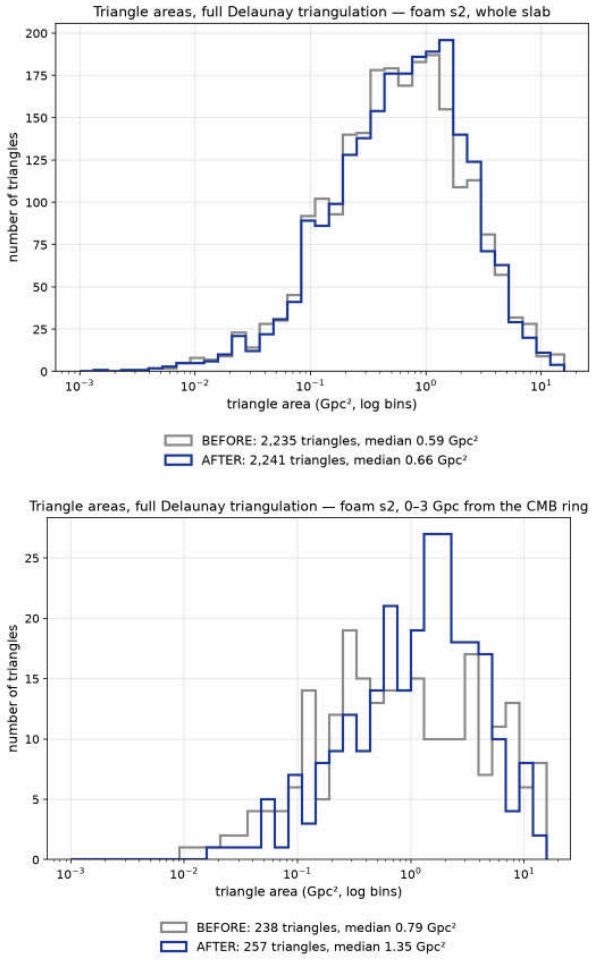
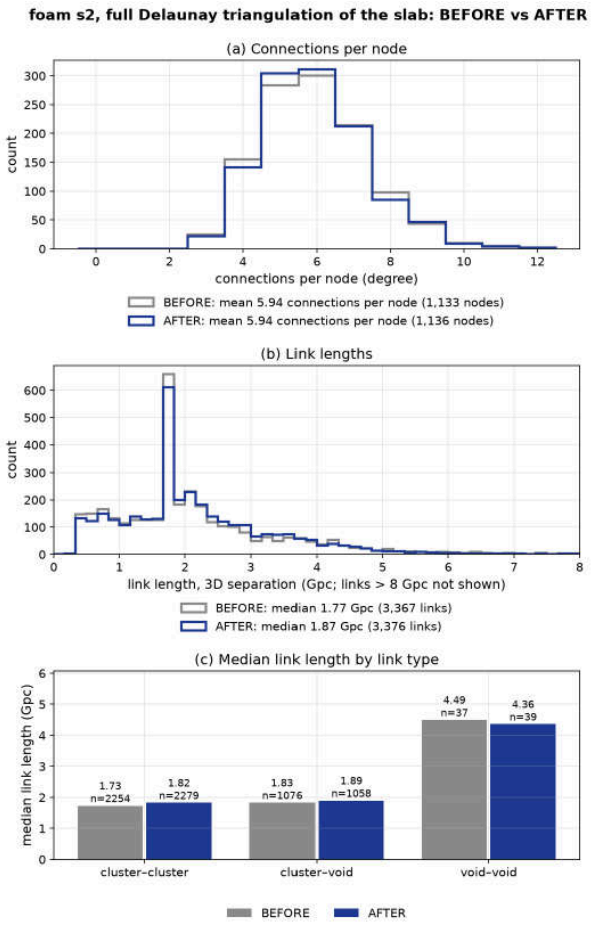


FIG. 55: Full Delaunay triangulation of the side-view slab, foam seed 2, before (left) and after the entropy-rule edits (right). After, each triangle is filled by its area change against the local before-area: blue, compressed (smaller); red, expanded (bigger); white, no change (symmetric log scale). Red dots: clusters; blue dots: void centres; grey: links; black circle: our sphere and the CMB ring; pink: CoE core cap; arrow: from us toward the CoE. Bottom: residual around the ring before (grey) and after (dark blue); pale mint band, ±1×. The leftover (dark blue) is no longer the smooth large-scale pattern: the $m = 1$–12 part is cancelled to 0.01×, and what remains is chaotic, asymmetric fine-scale ripple (2.4 times the CMB limit in this run).

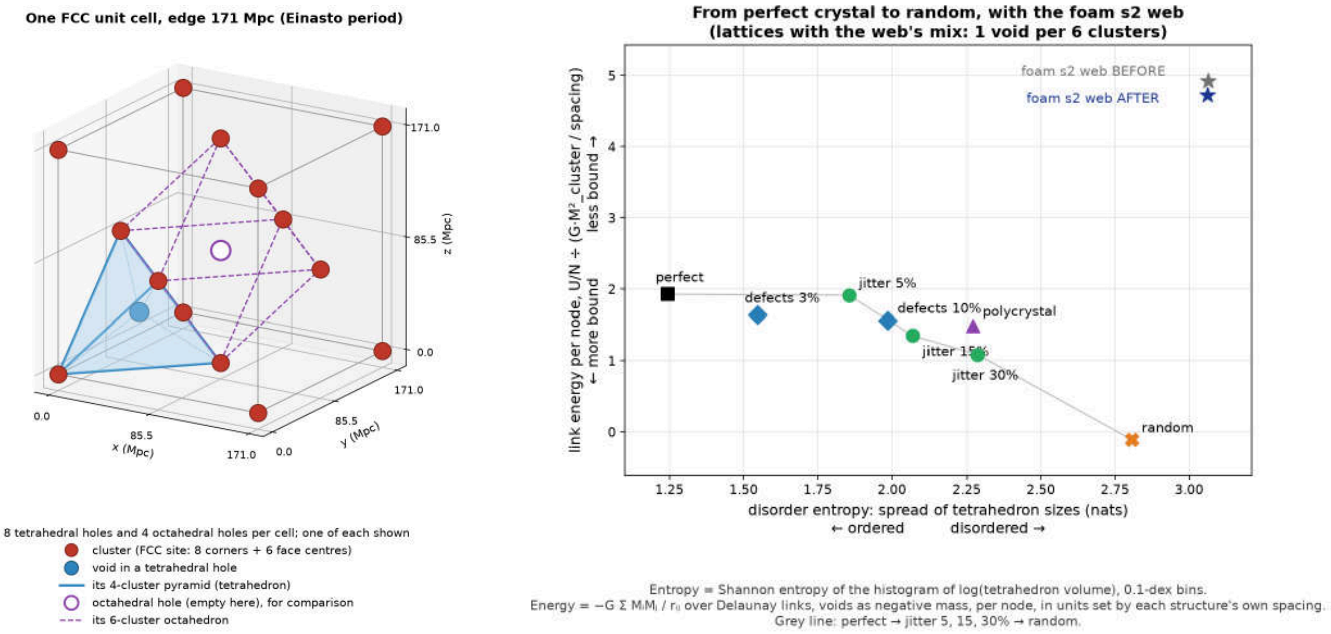


(a)



(b)

FIG. 56: Histograms of the triangulation of Fig. 55, before (grey) and after (dark blue). (a) Triangle areas on log bins, whole slab (top) and triangles centred 0–3 Gpc from the ring (bottom). (b) Connections per node, link lengths (3D separation), and median link length by link type.



(a) (b)

FIG. 57: (a) One FCC cell, 171 Mpc on a side (the Einasto period). Red: clusters at the corners and face centres; blue: a void in a tetrahedral hole, with its four-cluster pyramid; purple: an empty octahedral hole and its six-cluster octahedron, for comparison. (b) From perfect crystal to random, with the foam web (seed 2) before (grey star) and after the edits (blue star), for lattices with the web's mix of one void per six clusters. Horizontal: disorder (spread of tetrahedron sizes); vertical: link energy per node in units of GM_c^2 over the spacing (up is less bound). Black: perfect; blue diamonds: defects; green: jitter; purple: polycrystal; orange: random; grey line: perfect to jitter to random.

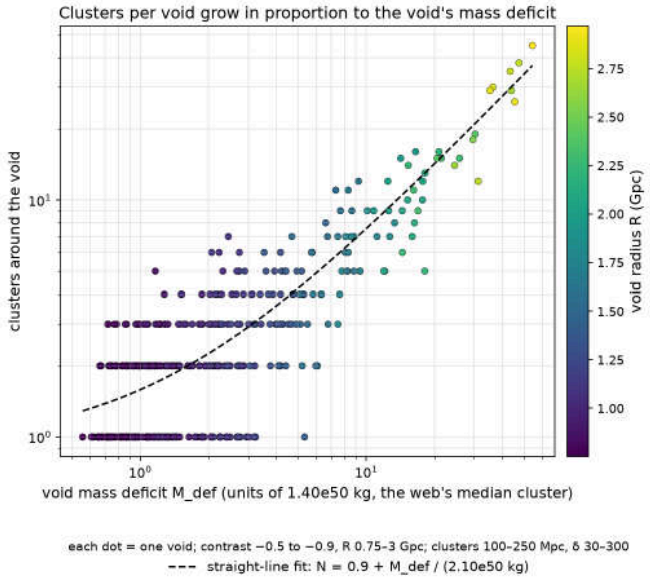


FIG. 58: Compensation rule in the 2D test: clusters around each void against the void's missing mass (in units of the median cluster, 1.40×10^{50} kg). Each dot is one void, coloured by its radius (purple, small, to yellow, large); dashed: the straight-line fit $N = 0.9 + M_{\rm def}/2.10 \times 10^{50}$ kg. The same rule in the 3D web gives $N = 0.49 + M_{\rm def}/1.48 \times 10^{50}$ kg.

191 groups moved or resized and 133 single clusters slid in their walls. The typical move is 0.6 Gpc (90% under 2.3 Gpc, largest 6.5 Gpc) and the median mass factor is 1.02. Solving for the residual with modes $m = 1$–12 now leaves 0.02× (from 640×), and the short waves fall from 461× to 1.00×. The symmetric share drops from 67% to 35%. Compensation stays exact (largest group net mass 4×10^{-16} of its deficit), and none of the seven entropy measures decreased. Bodies centred 0–3 Gpc from the ring go from 151 to 160 (0.060 to 0.063 Gpc^{-3}, against a web average of 0.080), so the region near the ring stays below average. The changes sit mostly inside about 20 Gpc, and the histograms barely move (Fig. 60): the median triangle goes from 1.45 to 1.38 Gpc^2 over the slab and from 1.84 to 1.59 Gpc^2 near the ring, connections per node stay at 6, and the median link goes from 2.09 to 2.05 Gpc. A modest rearrangement of a realistic, balanced web is therefore enough to flatten the CMB ring without making the web more ordered. Seeds 2 and 1 stalled at 1.18× and 7.40×. The random starts begin very high (789–1,260×) because some bodies land right next to the ring, and most of the edits go to clearing those.

TABLE VI: Compensated random web, move and resize under the entropy rule. Units: group (void plus its clusters) / single cluster.

seed	leftover	bodies edited	units	stop
3	789 → **0.998**	1,099 (12.1%)	191 / 133	reached 1×
2	1,077 → 1.18	1,927 (21.3%)	483 / 563	plateau
1	1,260 → 7.40	2,538 (27.7%)	1,122 total	plateau

A. What is left: chaotic, noise-like texture

The model predicts that once the large-scale pattern is cancelled, what is left of the residual is not a structured pattern but asymmetric, noise-like texture at the 10^{-5} limit. Figures 71 and 72 (placed after the shell figures so the earlier figures keep their numbers) show the leftover on its own, without the much larger 'before' curve, for the three web runs that reach the target (compensated web seed 3, 0.998×; foam seeds 3 and 1 edited without deletions, 0.995× and 0.95×) and, for contrast, compensated seed 2, which stalled just above at 1.18×. All numbers come from the saved final residuals; nothing was rerun.

The large scales are cancelled. In every example the modes $m = 1$–12, which carry the fingerprint, hold between 0.0003% and 1.1% of the leftover power (0.002–0.10×), and no low mode stands out (red bars in Fig. 71).

What is left is chaotic fine texture at the limit. The leftover rms is 1.00×, 0.995× and 0.95× (1.18× for the contrast case), spread over many fine modes: 10–90% of the power lies between $m = 46$ and 89 for compensated seed 3 and between $m = 50$ and 96 for seed 2. The foam webs are narrower, $m \approx 29$–41, so their ripple looks more regular, with a beating envelope. The largest single mode carries only 13%, 17%, 24% and 6% of the power, and neighbouring points stop being correlated within 2–3 degrees. Because the rms sits at the limit, 25–41% of the ring lies outside the ±1× band, with peaks of 2.4–3.5×; Gaussian noise with an rms of 1 would be outside 32% of the time.

There is no symmetric pattern, and the statistics look like noise. Of the leftover power, 43%, 43%, 33% and 41% is mirror-symmetric about the CoE axis. Rings of Gaussian noise with the same mode amplitudes and random phases give 31–70% (5–95% range), so every example sits inside or at the low edge of the noise range, never on the symmetric side. The residual values follow a Gaussian of the same rms (Kolmogorov–Smirnov $p = 0.86$, 0.32, 0.44 and 0.75; skewness within ±0.03); only foam seed 1 is slightly heavy-tailed (excess kurtosis 0.9; Fig. 72). In plain words: every example cancels the large scales and leaves an asymmetric, chaotic, noise-like texture at the CMB limit, as the model predicts. One caveat: this texture comes from bodies near the ring that the kernel treats as points, so its exact spectrum depends on that simplification (Sec. XII).

X. PAIR ENERGIES AND FORCES

Method. For every pair of 3D Delaunay neighbours in foam seed 2 we compute $U_{ij} = -Gm_im_j/r_{\rm eff}$ and the net force on each body $\mathbf{F}_i = \sum_j Gm_im_j(\mathbf{x}_j - \mathbf{x}_i)/(|\mathbf{x}_j - \mathbf{x}_i|\, r_{\rm eff}^2)$, with clusters at $+M$, voids at minus their missing mass, and $r_{\rm eff} = \max(r_{ij}, R_i + R_j)$ so that touching bodies do not blow up. Force statistics use bodies within 26 Gpc of us, away from the one-sided edge.

Results. Cluster–void repulsion dominates, about $+4.7 \times 10^{69}$ J in total, roughly twice the cluster–cluster attraction (-2.1×10^{69} J); void–void pairs are few but individually strong (-0.7×10^{69} J), so the net pair energy is positive, about $+1.9 \times 10^{69}$ J (Fig. 61). The edits change each total by only 1–3%. The two peaks in the cluster–void panel come from the foam rule, which puts clusters right on void rims. The moved bodies end about 20% closer to force balance (median net force 4.2×10^{39} to 3.4×10^{39} N), below that of the bodies that were not moved, and they are not pushed toward the ring: about half feel a push toward it and half away (Fig. 62). Voids feel larger forces than clusters because their missing mass is about six times a cluster's mass. This is a static snapshot of pairwise forces with no expansion, so it shows balance at one moment, not long-term stability.

XI. CONCENTRIC SHELL MODELS

In the CoE picture the matter left the centre as concentric shells. We built simple shell models to see how the spacing of the shells and the mass they carry shape the density profile, a wave tied to the shells, and the structure each shell grows.

Method. Twelve shells start at radius 3 (figure units), the first one 1 unit wide. Their spacing is constant (outer radius $R = 15$), widening by a factor $q = 1.097$ per shell ($R = 24$) or narrowing by $q = 0.954$ ($R = 12.37$). In the first set every shell holds the same mass; in the second, mass rises outward geometrically so that the outermost shell holds three times the innermost, with the same total. Each shell's density is its mass over its volume, shown relative to the innermost equal-mass shell on one logarithmic colour scale for all cases. A wave $\cos\phi$ puts exactly one cycle in each shell, so its crests sit on the shell boundaries. A smooth fit to the shell densities, $\rho_{\rm fit} = A\,x^{-n} + C$ with $x = r/R$ (with $C\,x^p$ in place of C when mass rises outward, so the curve can turn up again), is multiplied by the wave; the dashed curves are $\pm\rho_{\rm fit}$. For structure, the innermost shell is smooth and each shell outward gets slightly clumpier, with rms $\sigma_i = 0.8\,(i/11)\,(\rho_i/\rho_{i,\rm const})^{1/3}$ relative to the same shell in the constant case, so thinned shells soften and concentrated shells sharpen. Typical clump spacing is 0.6 of the shell width. Each shell's clumpy field is lognormal and rescaled so that its mean is exactly the shell density. Under each density step we draw the 16–84% and 2.5–97.5% ranges of the clumpy density and its full

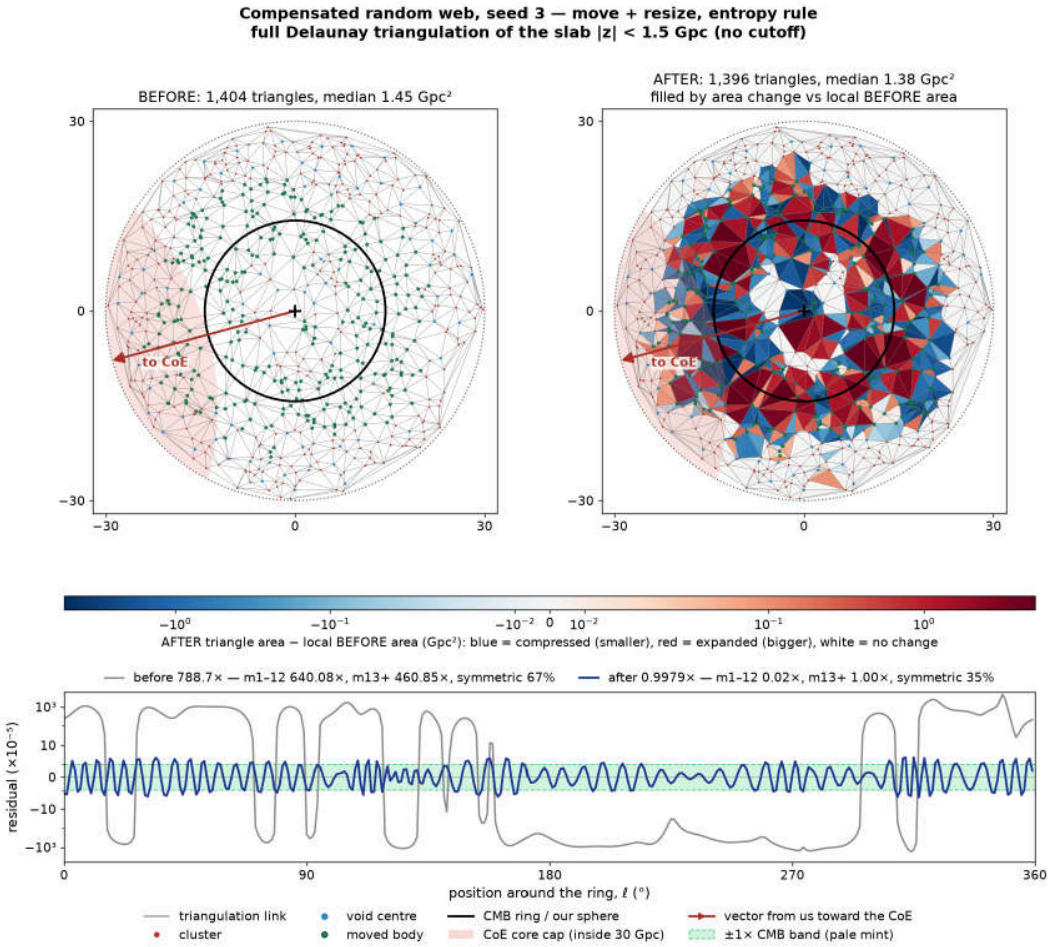


FIG. 59: Compensated random web, seed 3, before and after the move and resize edits under the entropy rule (789× to 0.998×). Full Delaunay triangulation of the slab $|z| < 1.5\,\mathrm{Gpc}$. Left: before; green dots are the bodies that will move. Right: after, each triangle filled by its area change against the local before-area: blue, compressed; red, expanded; white, no change (symmetric log scale). Red dots: clusters; blue dots: void centres; black circle: our sphere and the CMB ring; pink: CoE core cap; arrow: from us toward the CoE. Bottom: residual around the ring before (grey) and after (dark blue); pale mint band, ±1×. The leftover (dark blue) is no longer the smooth large-scale pattern: the $m = 1$–12 part is cancelled to 0.02×, and what remains is chaotic, asymmetric fine-scale ripple at the CMB limit, the noise-like texture the model predicts (shown enlarged in Fig. 71).

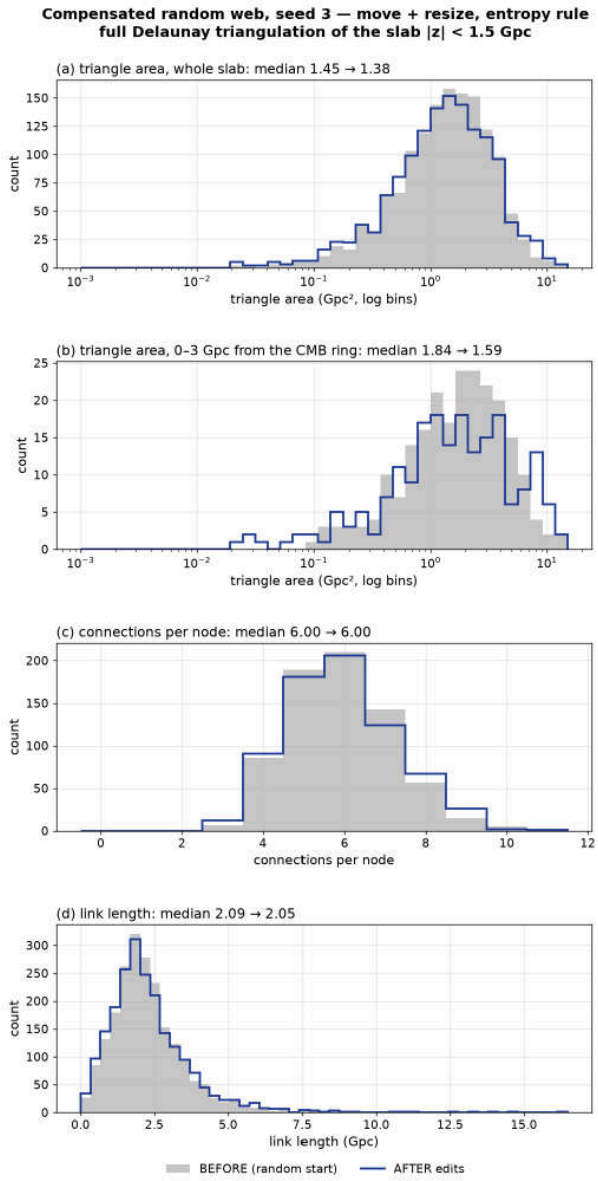


FIG. 60: Histograms of the triangulation of Fig. 59, before (grey fill) and after the edits (dark blue line): (a) triangle area, whole slab; (b) triangle area, 0–3 Gpc from the ring; (c) connections per node; (d) link length.

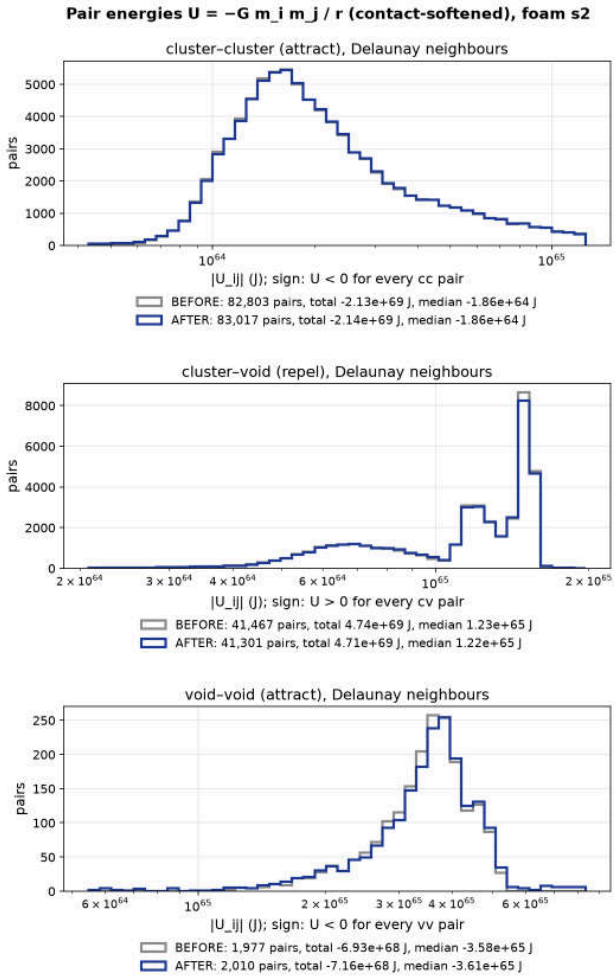


FIG. 61: Pair energies $|U_{ij}|$ by link type for foam seed 2 (3D Delaunay neighbours, contact-softened), before (grey) and after the edits (dark blue): cluster–cluster (attract), cluster–void (repel) and void–void (attract).

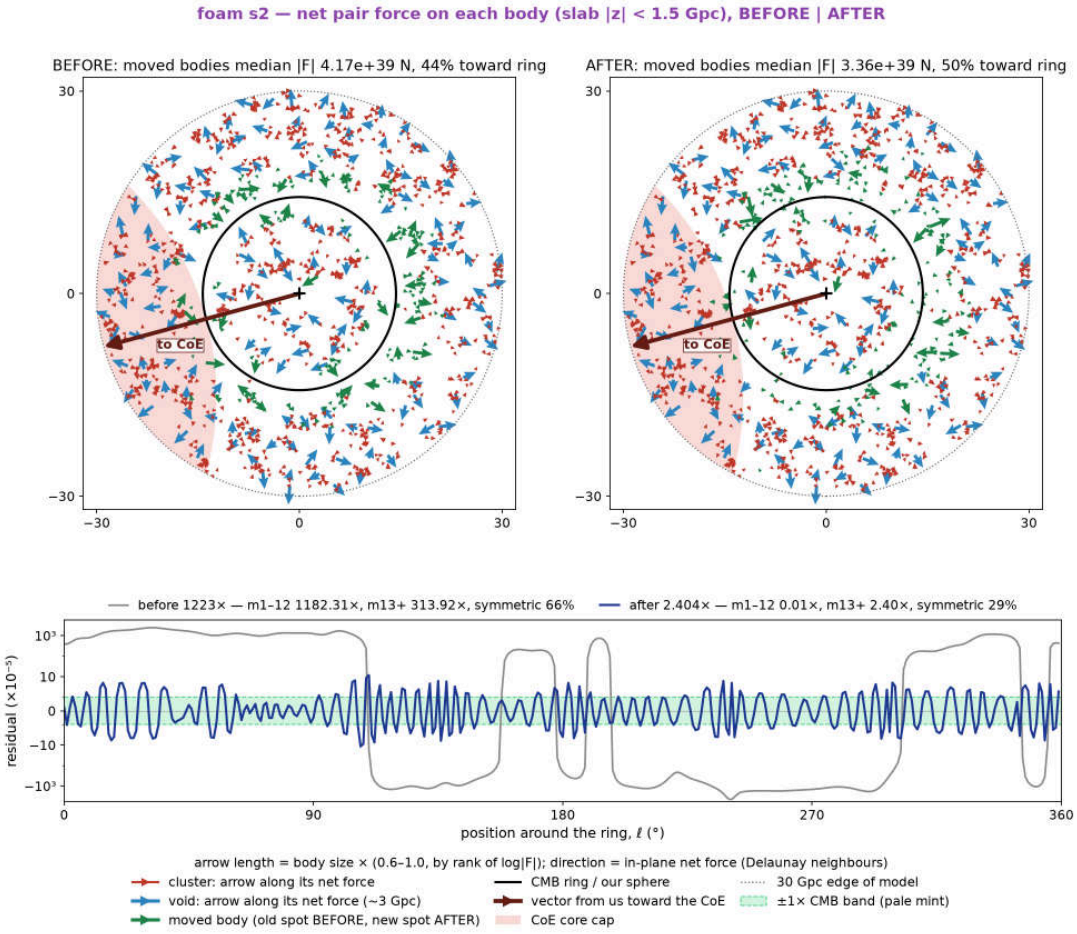
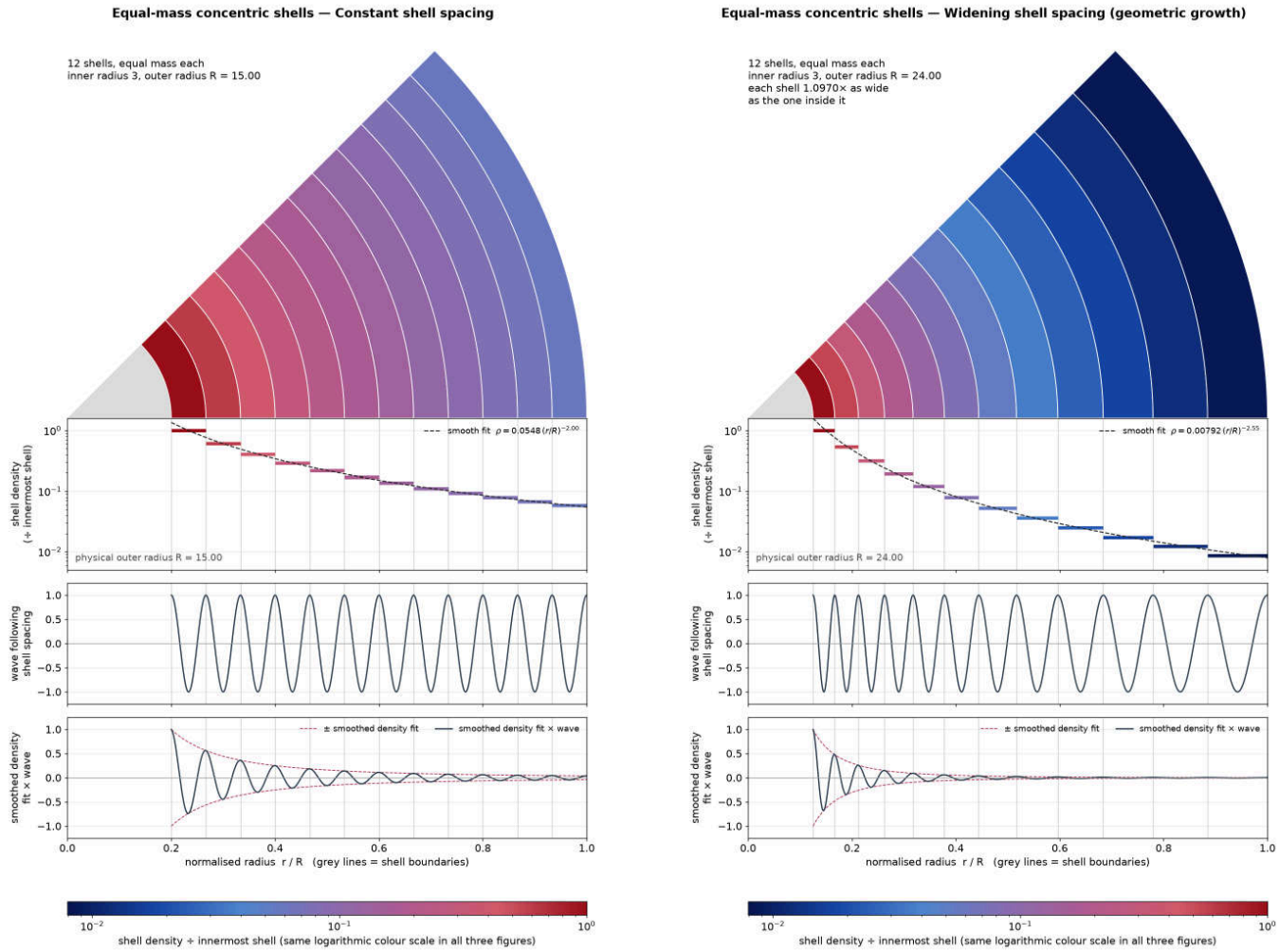


FIG. 62: Net pair force on each body in the slab, foam seed 2, before and after the edits. Each body is drawn as an arrow about its own size pointing along its in-plane net force (red: clusters; blue: voids, about 3 Gpc; green: moved bodies, at the old spot before and the new spot after). Black circle: our sphere and the ring; pink: CoE core cap; thick dark-red arrow: from us toward the CoE. Bottom: residual around the ring before (grey) and after (dark blue); pale mint band, ±1×. The leftover (dark blue) is no longer the smooth large-scale pattern: the $m = 1$–12 part is cancelled to 0.01×, and what remains is chaotic, asymmetric fine-scale ripple (2.4 times the CMB limit in this run).

minimum-to-maximum range. (An earlier version grew 10^{-5} ripples from recombination with linear growth set by each shell's density; it is replaced by this rule, which ties the colours to actual density.)

Results. With equal mass the fitted density falls as $r^{-2.00}$ for constant spacing ($C = 0$) and faster, $r^{-2.55}$, for widening spacing, so the density-times-wave envelope dies out by about $0.7\,R$ (Figs. 63, 64). Narrowing spacing levels off ($C = 0.058$), and the wave keeps a steady amplitude at large radius. With mass rising outward the constant-spacing profile flattens, the widening profile still falls steeply, only shifted up, because the extra outer mass cannot keep up with the growing shell volumes, and the narrowing profile bottoms out near $0.8\,R$ and rises toward the edge, so its envelope turns slightly upward (Figs. 67, 68). In the structure maps (Figs. 65, 66, 69, 70) clumpiness climbs steadily for constant spacing, widening shells spread their clumps into large faint blobs (outer-shell rms 0.43, densest spot only about 9 times the emptiest), and narrowing shells give the clumpiest outer shells with small sharp knots (rms 1.15, densest to emptiest about 1,400 times). With mass rising outward the outer shells are a little clumpier (rms 0.50, 0.89 and 1.38 for widening, constant and narrowing; extremes up to about 3,300 times), and clump sizes do not change because they follow the spacing.

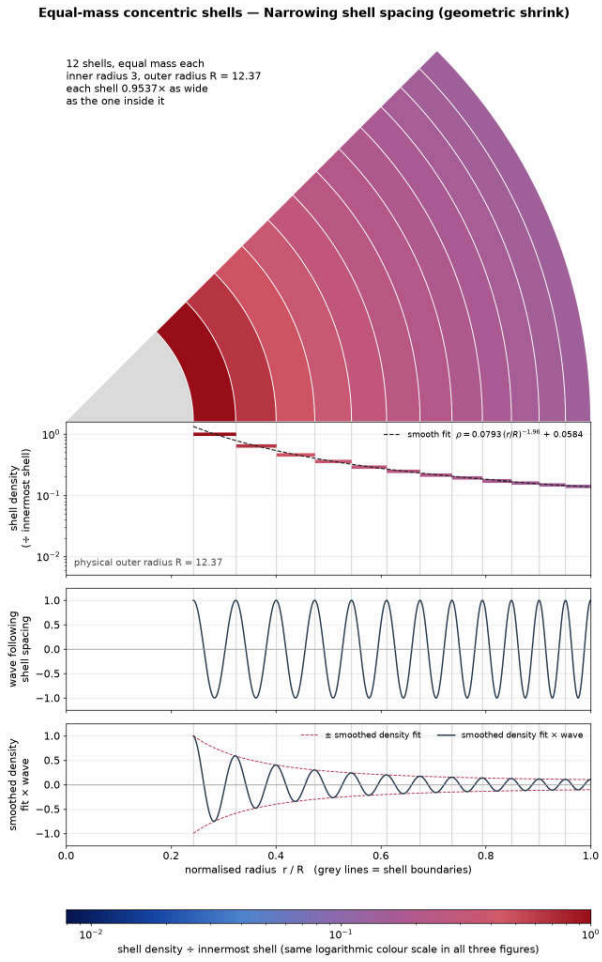
Interfaces between shells. Where density changes at a shell boundary the gravitational potential changes too, so there is a pull across the interface. Before recombination pressure pushed matter from the dense side to the thin side, much like a concentration gradient evening out. After recombination there is almost no pressure, and gravity runs the other way from osmosis: the denser side pulls matter toward itself and the contrast sharpens. Random motions of a few hundred km/s smear each boundary by only about 1–10 Mpc, tiny next to shells hundreds of Mpc to Gpc wide. We therefore expect each boundary to drain the thinner shell slowly into the denser one and leave a wall or filament at the interface rather than a blended zone; structure lined up along shell edges would be an observable. This is a qualitative expectation and was not modelled.



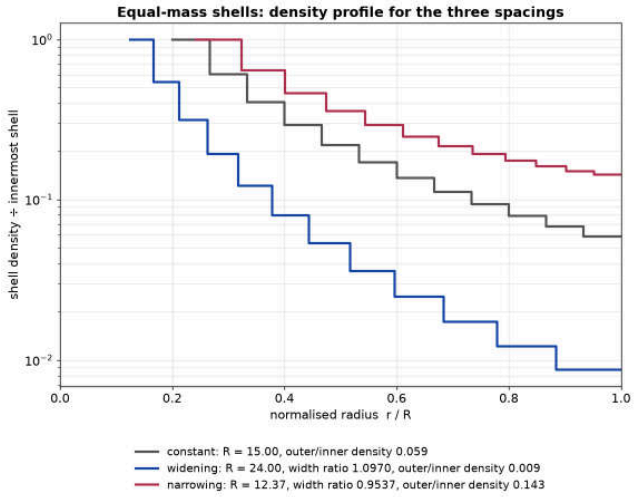
(a) constant spacing

(b) widening spacing

FIG. 63: Equal-mass concentric shells, (a) constant and (b) widening spacing. Top: the shell wedge, its flat edge on the radius axis of the panels below so each shell boundary (grey lines) drops straight into its density step; colour is shell density relative to the innermost shell (dark red, dense, to dark blue, thin; one log scale for every shell figure). Panels: shell density with the smooth fit (dashed); the wave with one cycle per shell; the smooth density fit times the wave, with ± the fit dashed.

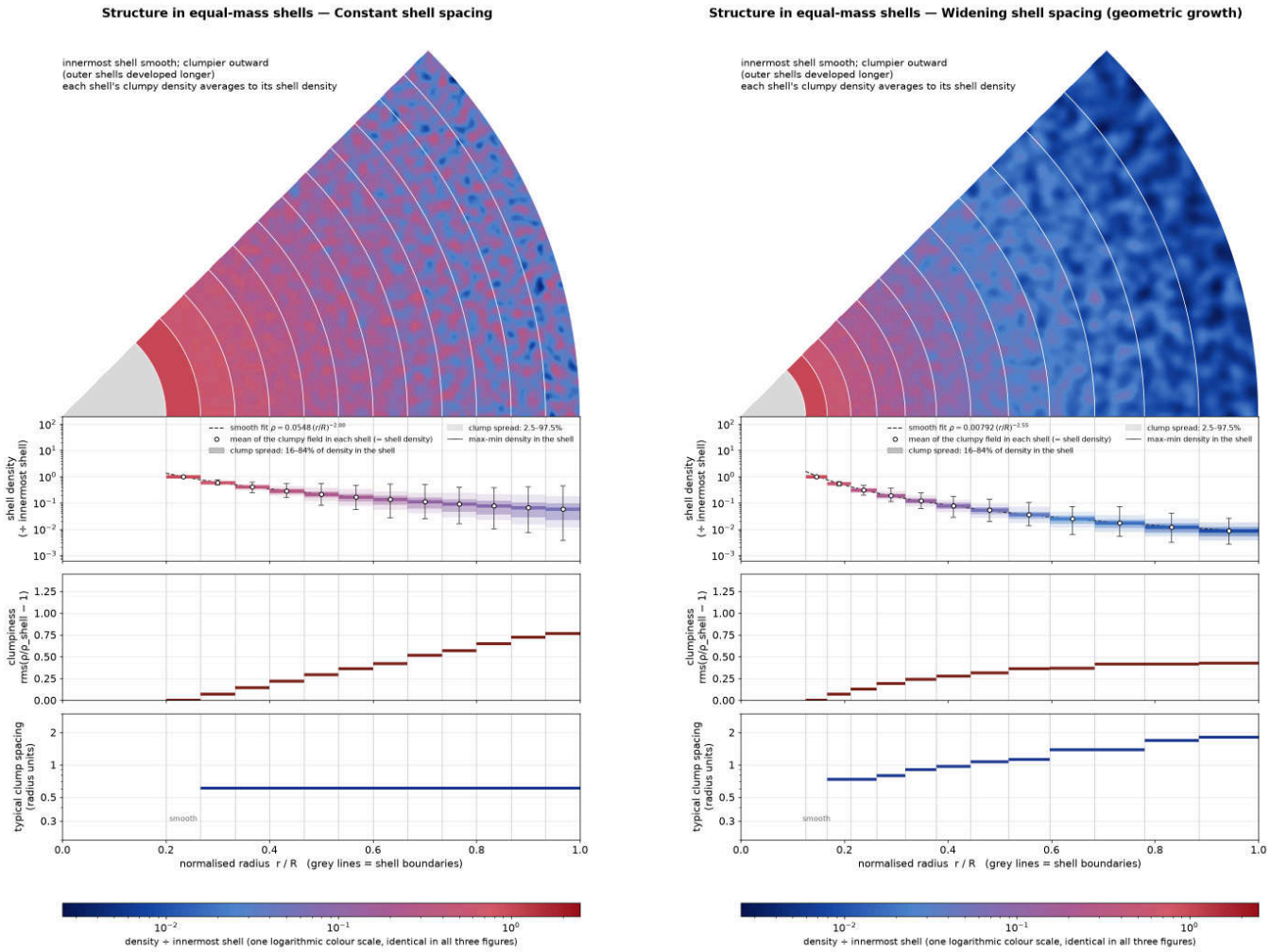


(a) narrowing spacing



(b) the three density profiles

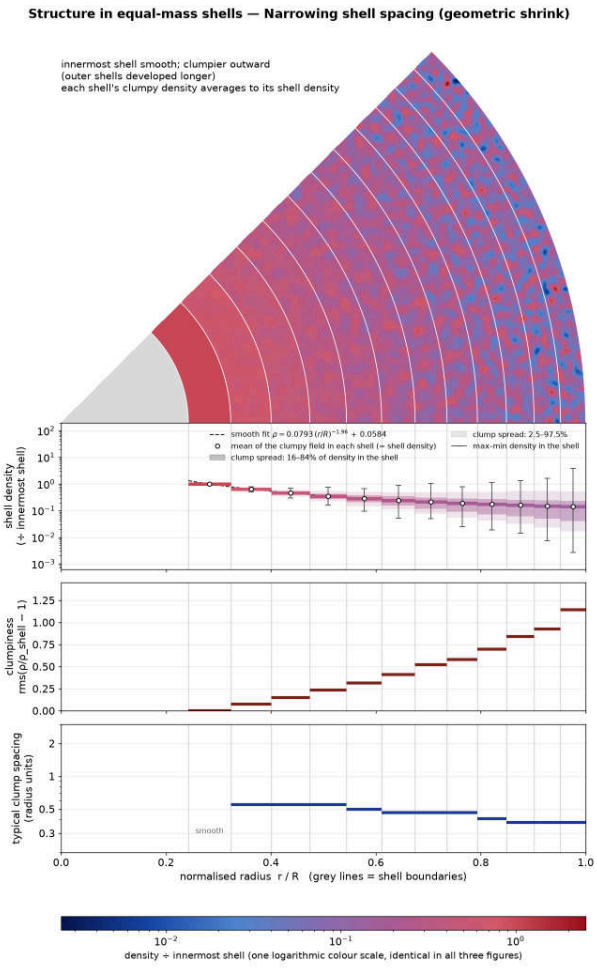
FIG. 64: (a) Equal-mass shells with narrowing spacing; panels and colours as in Fig. 63. The fit levels off, so the density-times-wave envelope keeps a steady amplitude at large radius. (b) Shell density against radius for the three spacings (grey: constant; blue: widening; red: narrowing).



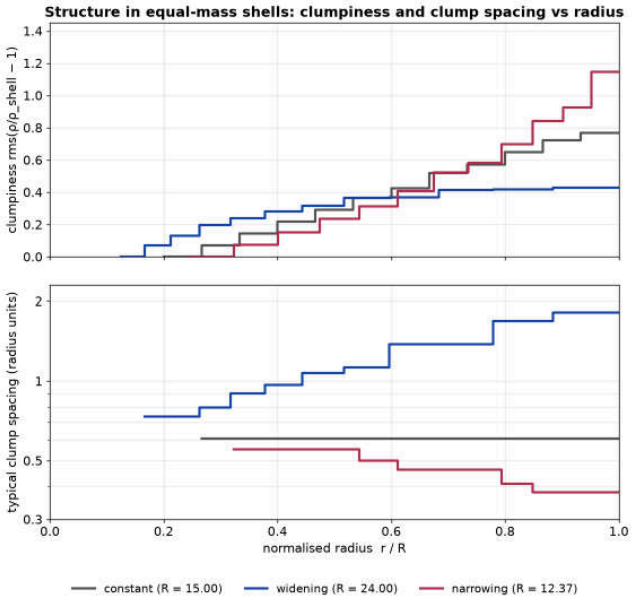
(a) constant spacing

(b) widening spacing

FIG. 65: Structure in equal-mass shells, (a) constant and (b) widening spacing. Top: the wedge filled with each shell's clumpy density (one log colour scale for all structure figures; red, dense; blue, empty); the innermost shell is smooth and each shell's clumps average exactly to its density. Panels: shell density (bar) with the mean of the clumpy field (white dot), the 16–84% (dark band) and 2.5–97.5% (light band) ranges and the minimum-to-maximum whisker; clumpiness, rms($\rho/\rho_{\rm shell} - 1$); typical clump spacing.

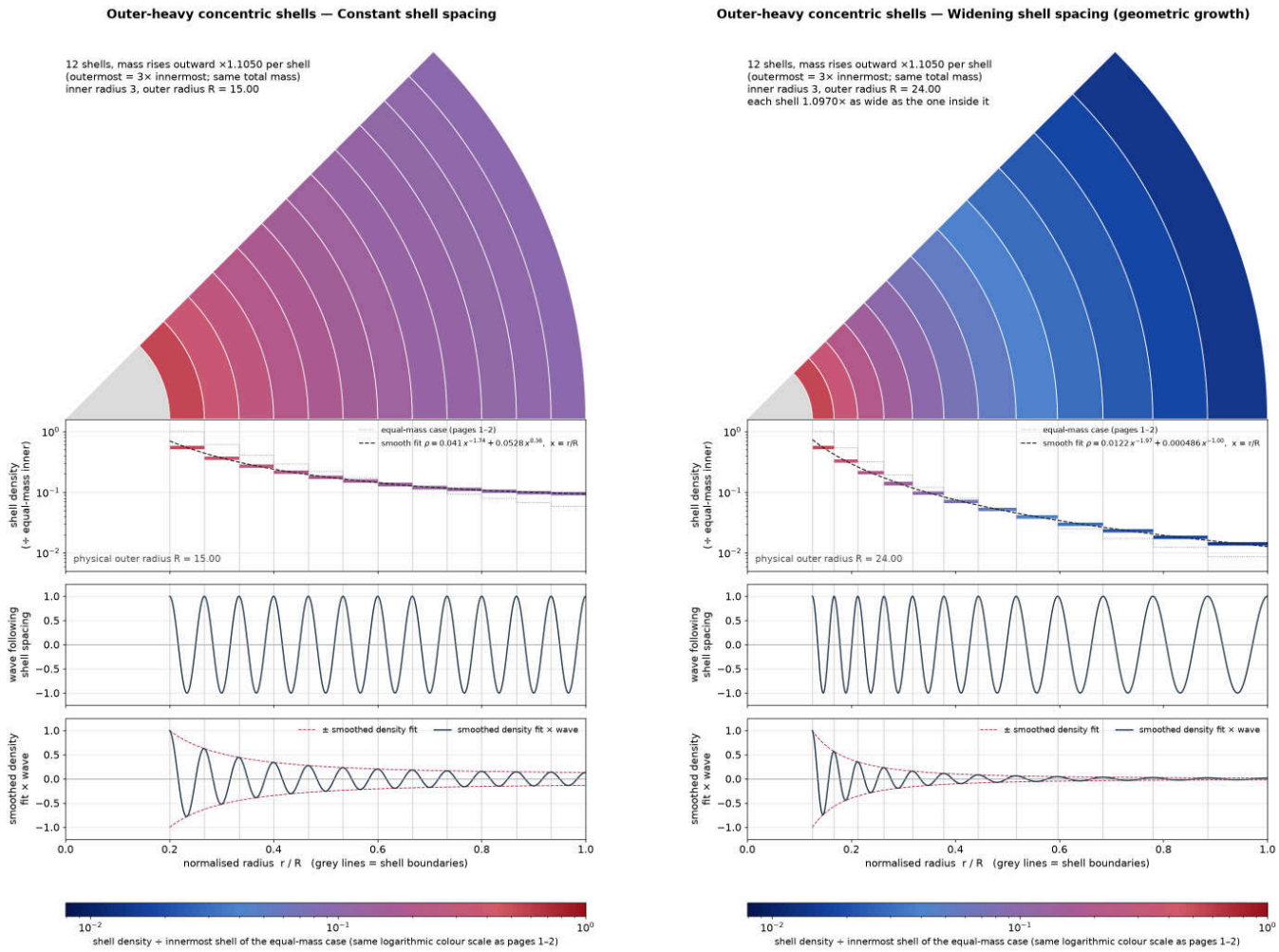


(a) narrowing spacing



(b) clumpiness and clump spacing compared

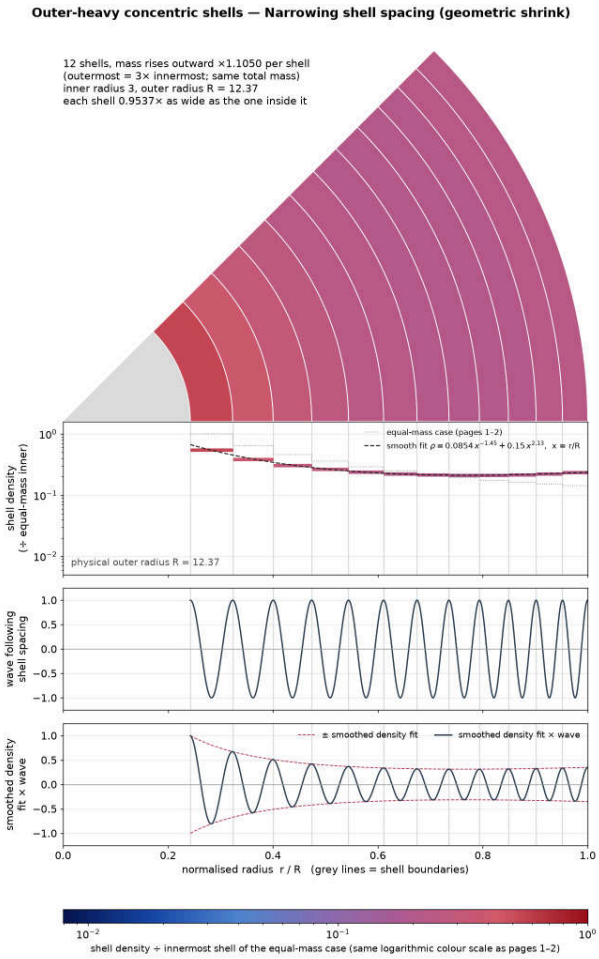
FIG. 66: (a) Structure in equal-mass shells with narrowing spacing; panels and colours as in Fig. 65. Concentration makes the outer shells the clumpiest, with small sharp knots. (b) Clumpiness and typical clump spacing against radius for the three spacings (grey: constant; blue: widening; red: narrowing).



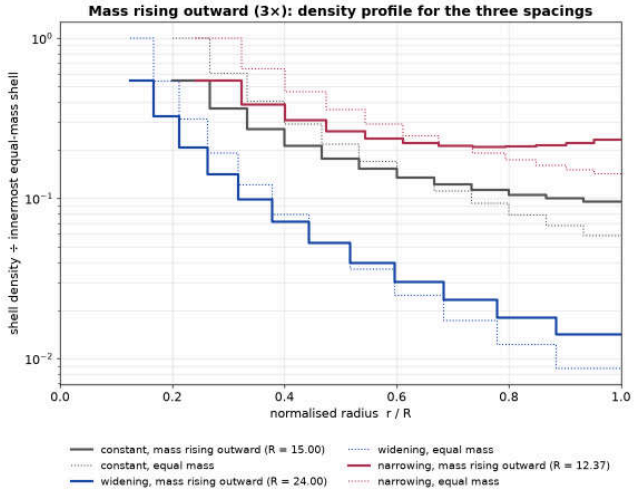
(a) constant spacing

(b) widening spacing

FIG. 67: Shells with mass rising outward (outermost shell three times the innermost, same total as the equal-mass case), (a) constant and (b) widening spacing. Panels and colour scale as in Fig. 63, so the colours compare directly; densities are relative to the innermost equal-mass shell.

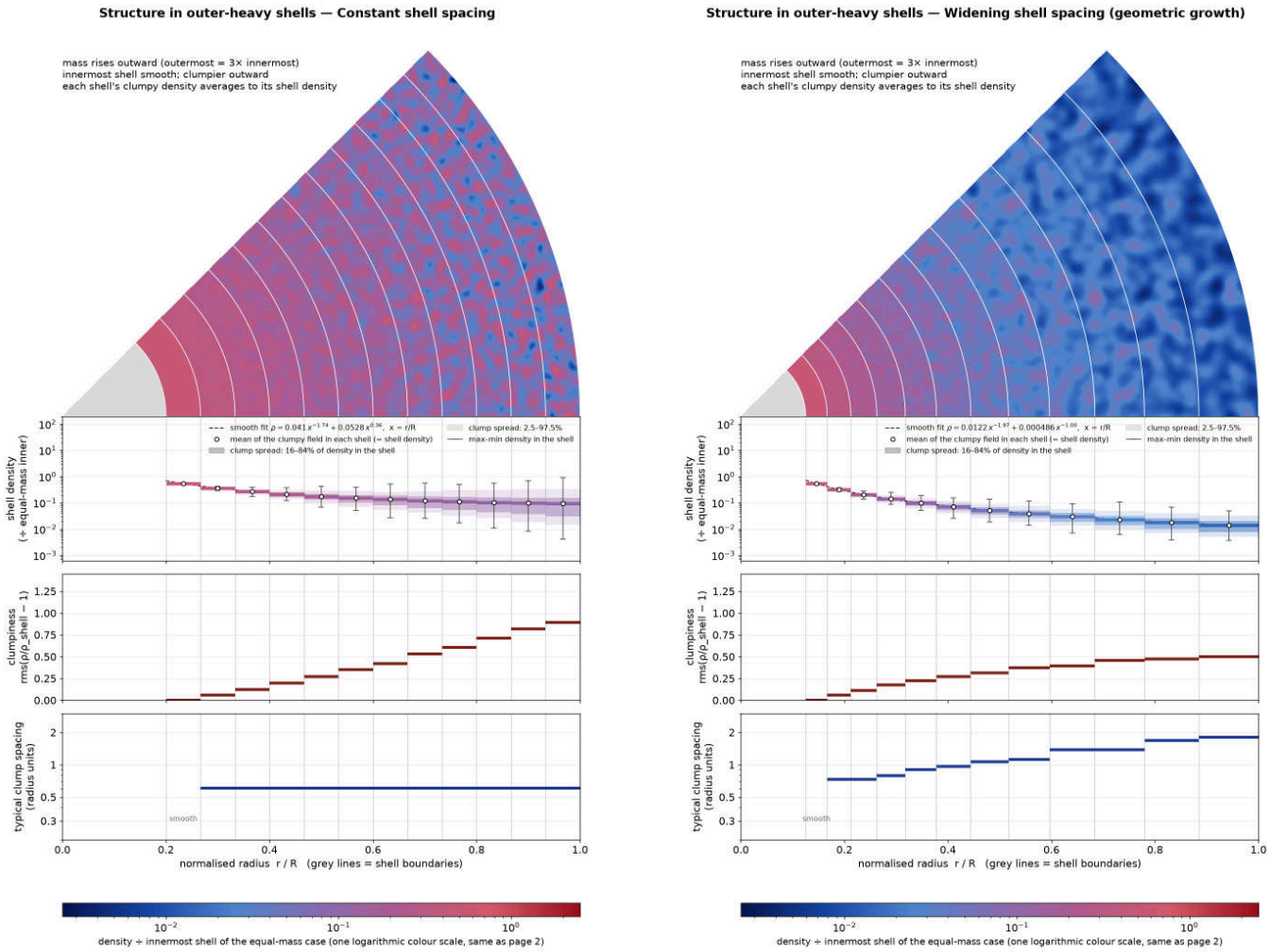


(a) narrowing spacing



(b) the three density profiles

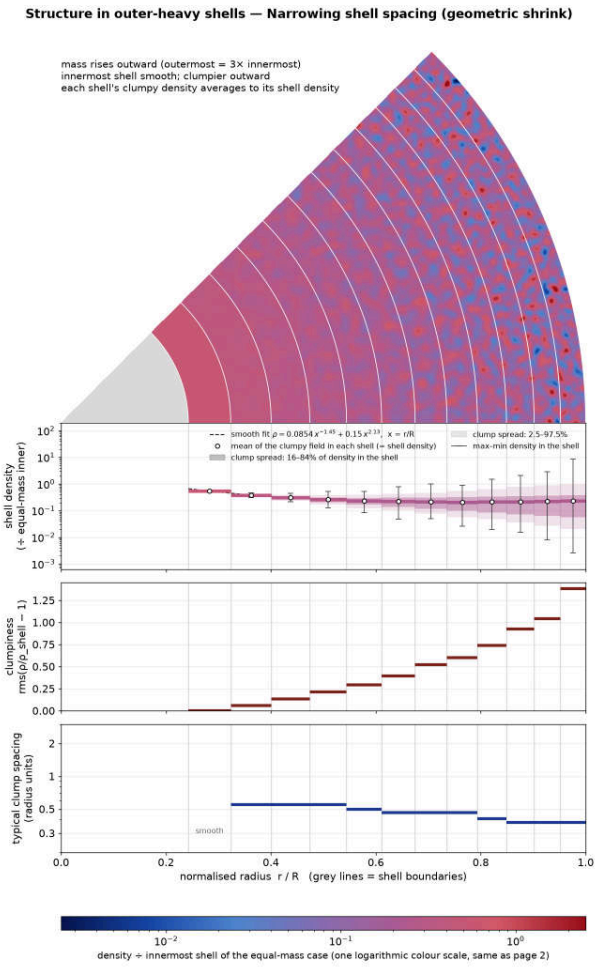
FIG. 68: (a) Shells with mass rising outward, narrowing spacing; panels as in Fig. 63. The density bottoms out near $0.8R$ and rises toward the edge, so the envelope turns slightly upward. (b) Shell density against radius for the three spacings with mass rising outward (thick) and equal mass (dotted); grey: constant; blue: widening; red: narrowing.



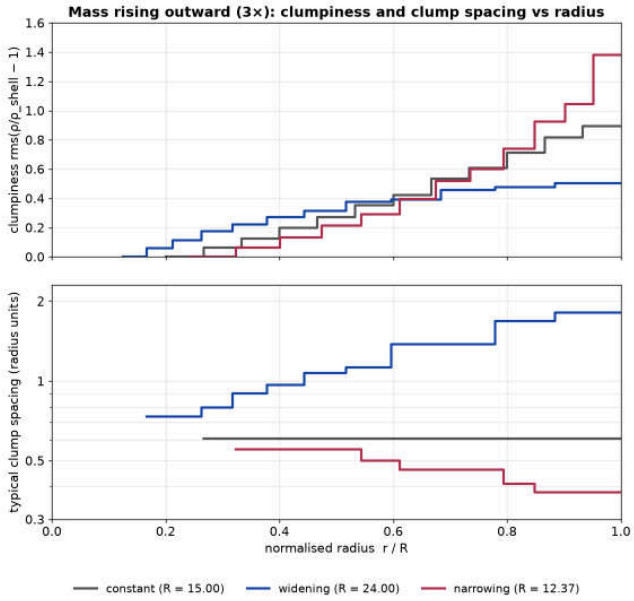
(a) constant spacing

(b) widening spacing

FIG. 69: Structure in shells with mass rising outward, (a) constant and (b) widening spacing. Panels, bands, whiskers and colour scale as in Fig. 65.



(a) narrowing spacing



(b) clumpiness and clump spacing compared

FIG. 70: (a) Structure in shells with mass rising outward, narrowing spacing; panels as in Fig. 65. The outermost shell spans about 3,300 times from its emptiest to its densest spot. (b) Clumpiness and typical clump spacing against radius for the three spacings with mass rising outward (grey: constant; blue: widening; red: narrowing).

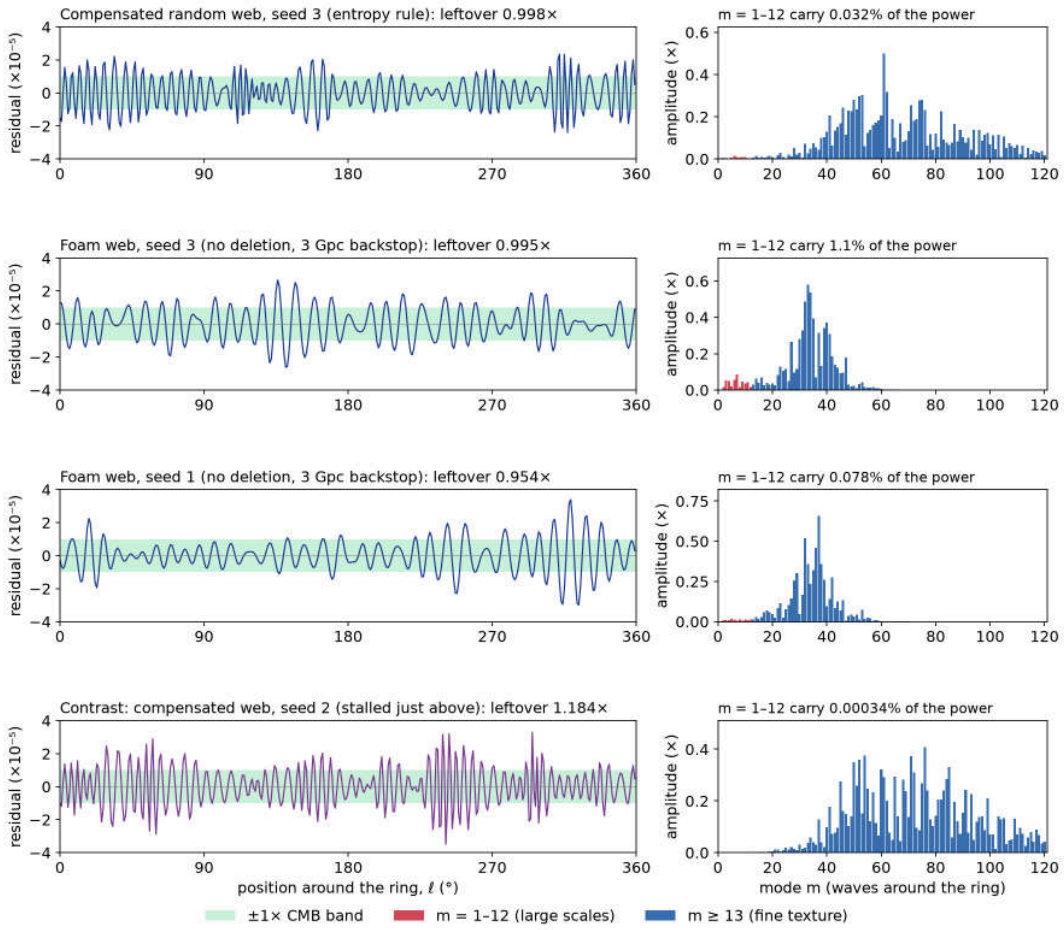


FIG. 71: The leftover after editing, shown alone (Sec. IX A). Each row is one edited random web. Left: residual around the ring in units of 10^{-5}, with the pale green ±1× band. Right: amplitude of each mode m (number of waves around the ring); red, $m = 1$–12, the large scales of the fingerprint; blue, $m \geq 13$, fine texture. The top three rows reach the CMB level (compensated web seed 3, 0.998×; foam seeds 3 and 1 with no deletions, 0.995× and 0.95×); the bottom row (compensated seed 2, 1.18×) stalled just above and is shown for contrast. In every row the large scales are gone (the red bars are barely visible) and what is left is irregular fine ripple spread over many modes, with no dominant low mode.

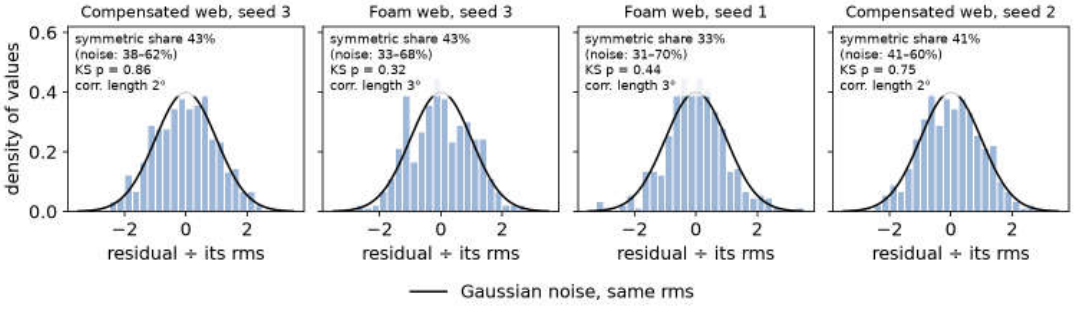


FIG. 72: Is the leftover noise-like? Histograms of the residual values of Fig. 71, divided by their rms, against a Gaussian with the same rms (black line). Each panel lists the symmetric share (the part of the leftover power that is mirror-symmetric about the CoE axis) with, in brackets, the 5–95% range for 2,000 Gaussian-noise rings with the same mode amplitudes and random phases; the Kolmogorov–Smirnov p-value against a Gaussian (above 0.05 means consistent with one); and the correlation length (where the autocorrelation first crosses zero). All four look like Gaussian noise, and none is mirror-symmetric.

XII. CAVEATS

Missing physics. The kernel includes only the Newtonian tidal and velocity effect. For bodies between us and the ring, the time-varying potentials crossed by CMB photons (integrated Sachs–Wolfe [22] and Rees–Sciama [23] effects), gravitational lensing of the ring, and the potential difference between the ring and us (Sachs–Wolfe term [22]) are left out. For 10^{52}–10^{53} kg bodies a few Gpc away they are probably not small, so the inside results describe what the tidal model alone says.

Overfitting. Each body adds four parameters, while the ring offers 360 samples dominated by about a dozen modes. Above about five bodies many different layouts fit almost equally well, and successful fits push the error into modes above 12. Reaching 1× with 8–12 bodies is thus partly a statement about the number of free parameters.

Non-uniqueness. Below about 5 bodies the answer is nearly unique (a backbone of four masses near 20 Gpc at $\ell \approx 75°/315°$ and $162°/228°$); beyond that the examples shown are illustrations, not unique answers. The smooth field is the minimum-norm solution; many other fields also work.

Ring-plane only. Only the in-plane ring is constrained, so nothing is known about structure far above or below the galactic plane; a full-sky fit would need more structure.

Point bodies near the ring. In the web runs every void and cluster is a point mass for the kernel and a Gpc-scale stand-in for a whole region; near the ring this exaggerates the short waves, which is why the starting leftovers of filled webs are so high and why a ring rule is needed. The entropy rule is itself a choice of seven measures, and the toy energies, pair forces and shell structure rule are illustrations, not dynamics.

Extrapolation. The comparison with observed structure extrapolates the 2M++ spread from 80 Mpc to 1–4 Gpc, a factor of 12–50 in radius, and assumes Gaussian tails.

Faster-than-light recession. The model expects the outer parts of the shell to recede faster than light. Spreading the shell's outer edge (72.5 Gpc) over an age of 13.8 Gyr gives an average edge speed of about $17c$ in Newtonian terms; a Sedov–Taylor scaling ($R \propto t^{2/5}$) puts today's edge speed at two fifths of that, about $7c$. This is expected for expanding space: it is a recession velocity, not motion through space, so it does not conflict with relativity, but the Newtonian kernel used here does not treat it. The next stage of model development is a full relativistic Lemaître–Tolman–Bondi (LTB) treatment, in which the energy function $E(r)$ must satisfy $2E/c^2 > -1$, grow as $E \propto r^2$ near the centre, avoid shell crossing, and match the local Hubble constant H_0 where we sit.

XIII. CONCLUSIONS

The CoE shell model leaves a specific fingerprint in the CMB ring: a 237× pattern dominated by three to five waves around the sky. Treated as an inverse problem, this fingerprint can be cancelled to the flat target by a small number of deliberately placed bodies: 11 outside our visible sphere, 4 to 8 inside it, and 8 when no placement wall is imposed. With the CoE core fully open, 10 outside bodies suffice, all within a few Gpc of the core's near edge; the deep core contributes almost nothing because the $m = 3$–5 pull fades rapidly with distance. Both masses and voids are needed outside; inside, masses alone suffice. Requiring that the layout not be a mirror image costs about one body.

The solutions are not unique, and above a handful of bodies the fits are as much a measure of freedom as of physics. Sequential growth that never revisits earlier choices fails by a wide margin. Coordination can, however, be added afterwards and modestly: a realistic random web, with every void balanced by its own clusters, reaches the CMB level (789× to 0.998×) when 12% of its bodies are moved or resized by a median of 0.6 Gpc, without deletions, without forming a ring and without becoming more ordered. What it leaves is what the model predicts: the large-scale pattern is cancelled ($m = 1$–12 down to 0.02×), and the rest is chaotic, asymmetric fine-scale ripple at the 10^{-5} limit with the statistics of Gaussian noise, as in the two foam webs that reach the target.

The smooth-field solutions show what the cancellation would demand of the real universe: a coherent pattern of about 1% rms and 5–7% peaks on gigaparsec scales, arranged in the same three-to-five-lobe shape as the fingerprint. That is about an 8σ fluctuation at 1 Gpc even against the most generous extrapolation of observed structure. A fuller treatment would add the line-of-sight effects listed above, fit the whole sky rather than the ring alone, and replace the Newtonian kernel with a relativistic LTB model that handles the faster-than-light recession the model expects.

Spontaneous Pattern Formation: Forcing Functions, Scale Selection and Common Mechanisms

Draft

Abstract

Spatial patterns arise spontaneously in fluids, chemical media, magnets, mixtures, growing crystals and colonies. For twenty examples this draft shows the pattern and gives its forcing function or control parameter in standard form, its threshold and its selected scale. The examples are classified by how the scale is selected and tagged by what sustains the pattern. All share a uniform state, one dimensionless drive, a wavenumber-dependent growth rate and two competing effects that fix the scale.

1 Introduction

A uniform or disordered medium often organizes itself into cells, rolls, spots, stripes, fingers or bands with a well-defined spacing. The geometry is not written in by the boundaries, and where growth starts from a seed (a dendrite, a colony), the seed starts the growth but does not set the length scale. This draft shows a set of such patterns, gives for each the forcing function or control parameter in its standard form, and emphasizes what they have in common: a uniform state, a dimensionless drive, a growth rate that depends on wavenumber, and two competing effects that fix the scale.

Section 2 gives the common framework, Section 3 the classes; Table 1 collects the forcing functions.

2 A common framework

Growth rate versus wavenumber. Every example starts from a uniform state and one dimensionless drive (Ra, Ma, Γ, B/B_c, Ca, quench depth, …). A perturbation $\propto e^{i\mathbf{k}\cdot\mathbf{x}+\sigma(k)t}$ grows if $\sigma(k) > 0$, and the shape of $\sigma(k)$ sorts the examples [1]: a band at finite k_c that opens at a threshold (class A), near which

$$\sigma(k) \approx \tau_0^{-1}\big[\epsilon - \xi_0^2(k-k_c)^2\big], \qquad \epsilon = R/R_c - 1;$$

a long-wave band pinned at $\sigma(0) = 0$ by a conservation law (spinodal decomposition, class C); or a band $0 < k < k_c$ with no threshold, cut off by surface tension (driven interfaces, class D).

A scale from two competing effects. The selected length always comes from one effect that penalizes long waves and one that penalizes short ones: confinement against dissipation (convection rolls of width $\approx d$); gravity against capillarity (Faraday, Rosensweig and Rayleigh–Taylor all give the capillary length $2\pi\sqrt{\sigma/g\Delta\rho}$ [2]); diffusion against capillarity (the Mullins–Sekerka length $2\pi\sqrt{d_0\ell}$ [3]); local activation against lateral inhibition (Turing [1]); and wall or line tension against long-range dipolar energy (magnetic domains, ferrofluid labyrinths [4, 2]). For equilibrium modulated phases this competition is written as a free energy per mode [4, 2],

$$F(q) = \tfrac{1}{2}\big[b\,q^2 - \mu^2 G(q)\big], \qquad G(q) \approx G(0) - g_1|q| \;\Rightarrow\; q^* \neq 0.$$

Universality near onset. Just above a finite-k threshold the slowly varying amplitude A of the pattern obeys a universal Ginzburg–Landau equation [1, 5],

$$\tau_0\,\partial_t A = \epsilon A + \xi_0^2\,\partial_x^2 A - g|A|^2 A,$$

so the amplitude grows as $\sqrt{\epsilon}$, and only a band of wavenumbers inside the unstable band is stable (the Busse balloon in convection). The model equation of this class is Swift–Hohenberg, $\partial_t\psi = \epsilon\psi - (k_c^2 + \nabla^2)^2\psi - \psi^3$ [6]. Where up–down symmetry is broken, hexagons appear through a subcritical, hysteretic onset: Bénard–Marangoni cells, Rosensweig spikes [2], thin-film Rayleigh–Taylor drops [7] and low-frequency granular squares [8]. Localized states (oscillons, ferrofluid "solitons") live in the hysteresis windows.



Figure 73: Additional experimental BZ target and spiral patterns in a thin layer.

Why driven and equilibrium patterns look alike. For a stationary finite-k instability the lowest-order amplitude equation is a gradient flow, $\partial_t A = -\delta\mathcal{F}/\delta A^*$ [1]. Near onset, convection rolls and Turing stripes therefore behave as if they were minimizing a functional with the same q-dependence as an equilibrium modulated phase [4], which is why stripes, hexagons, labyrinths and dislocation climb look the same in convection and in garnet films. Far from onset they differ: only gradient systems have a unique preferred wavenumber; driven systems show a band of states, persistent dynamics and chaos [1].

Interfaces. Interfacial patterns (fingers, dendrites, billows) do not saturate at small amplitude. Their final scales, such as tip radius and finger width, are selected by surface tension acting as a singular perturbation, and dendrites need crystalline anisotropy to have any steady solution at all [1, 3, 7].

Fronts. A front into an unstable state moves at the linear spreading speed v^* (pulled) or faster (pushed); for the Fisher–KPP equation $\partial_t u = D\partial_x^2 u + f(u)$, $v^* = 2\sqrt{Df'(0)}$, and the pattern left behind inherits its wavenumber from the front [7]. Examples: thin-film Rayleigh–Taylor drops, BZ waves, bacterial colonies, Liesegang bands.

Coarsening. When there is no equilibrium length (spinodal decomposition, foams, granular clusters), a single length grows as a power law, $L \propto t^{1/3}$ for conserved diffusive dynamics, and statistics collapse when scaled by L [9, 7]. Arresting coarsening with a long-range interaction, as chain connectivity does in block copolymers, turns class C into class B [2].

What sustains the pattern. This is a separate question from how the scale is selected. Open driven systems need a flux; their full dynamics has no Lyapunov functional [1]. Closed systems relax and their patterns are transients: a closed BZ dish runs down within about 100 periods [1], Liesegang bands freeze kinetically, and spinodal domains persist for long times only because coarsening is slow. True equilibrium patterns persist with no input [2].

3 Classification

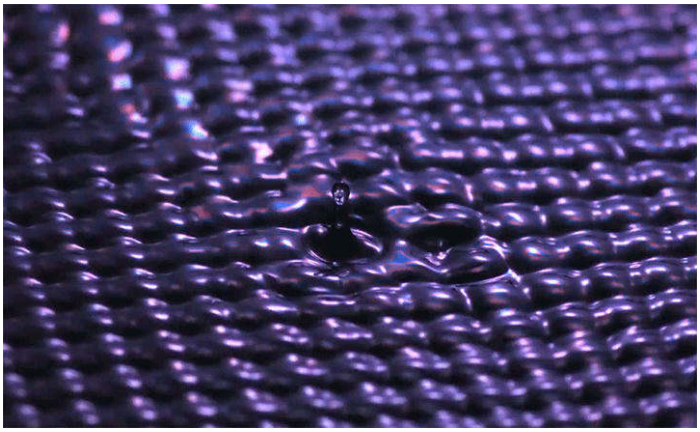


Figure 74: Additional experimental Faraday surface showing organized peaks under vertical oscillation.

Classes. Each pattern is labelled by *how its length scale is selected* and *what keeps it alive.*

A Bulk finite-wavelength instability: a uniform state becomes unstable above a threshold to a mode of finite wavenumber k_c (Cross & Hohenberg type I [1]).

B Equilibrium modulated phase: a short-range cost and a long-range gain give a free-energy minimum at finite period [4, 2].

C Coarsening: phase separation or ripening with no equilibrium scale; a single length grows as $L \propto t^n$ [9].

D Interfacial instability: a driven sharp interface is unstable at long wavelengths, with surface tension cutting off short ones (Mullins–Sekerka, Saffman–Taylor, Rayleigh–Taylor, Kelvin–Helmholtz) [3].

E Fronts and waves: a front invades an unstable, excitable or supersaturated state and leaves structure behind it [7].

Tags: **D**riven (open), **R**elaxing (closed transient), **E**quilibrium. Chladni figures are a forced linear response, not a self-organized pattern, and serve as the reference case.

The classes follow the instability types of Cross and Hohenberg [1] (I_s, I_o for A, II for C, III_o for excitable media in E), the interface and front literature [3, 7] and the modulated-phase picture [4, 2]. The tag replaces the test "does the pattern collapse when the drive stops", which fails for closed transients such as a BZ dish or a Liesegang tube. Each section heading gives the class and tag.

4 Reference case: Chladni figures (forced mode, D)

A plate driven near one of its eigenfrequencies vibrates in a standing mode. Coarse sand is thrown off the moving regions and gathers along the nodal lines; fine powder instead collects at the antinodes, carried there by air streaming [8]. The figure is the forced linear response of the plate, selected by its shape and edges, with no threshold and no symmetry breaking, so it is the contrast case for everything that follows [10].

$$D\nabla^4 w + \rho h\, \partial_t^2 w = f(x,y)\cos\omega t, \qquad D = \frac{Eh^3}{12(1-\nu_P^2)}.$$

Threshold and scale: a figure appears when $\omega \approx \omega_n = k_n^2\sqrt{D/\rho h}$; the nodal spacing scales as $\omega^{-1/2}$. *Class:* reference (forced mode), tag D; it shows the eigenmode structure that true instabilities inherit in finite containers.

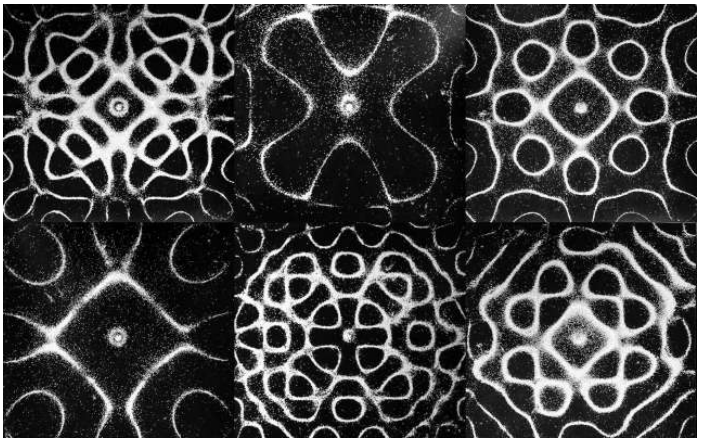


Figure 75: Experimental Chladni figures. Sand accumulates along the nodal lines of standing-wave modes on a vibrated plate.

5 Class A: bulk finite-wavelength instabilities

5.1 Faraday waves (A, D)

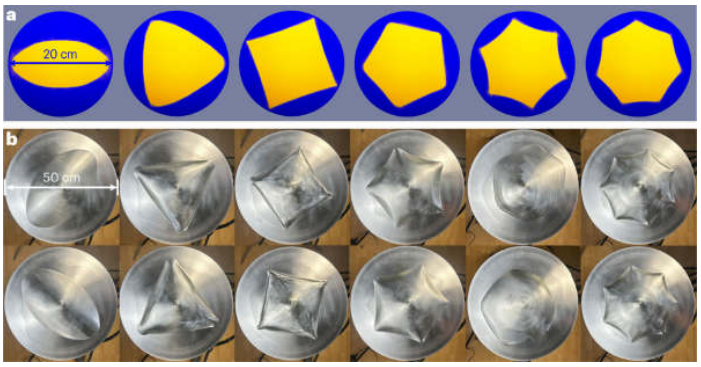


Figure 76: Experimental polygonal Faraday-wave patterns (elliptical through hexagonal) on vertically oscillated liquid surfaces. X. Liu and X. Wang, *Nat. Phys.* **20**, 287 (2024), Fig. 3, doi:10.1038/s41567-023-02294-y.

Vertical shaking modulates the effective gravity and parametrically pumps surface waves at half the drive frequency. Above a threshold set by damping, standing waves fill the dish at once: squares near onset in low-viscosity fluids, hexagons at high viscosity, and quasipatterns with two-frequency forcing [1, 11]; shaped containers give polygons [12]. For drive frequency 2ω the surface mode $a(t)$ obeys a damped Mathieu equation [1],

$$\ddot{a} + \gamma\dot{a} + \omega_s^2(q)\big[1 + 2\Delta\cos 2\omega t\big]a = 0, \qquad \omega_s^2 = q\tanh(qh)\Big(g + \frac{\sigma q^2}{\rho}\Big).$$

Threshold and scale: $\Delta_c = \gamma/\omega_0$; the wavenumber is fixed by the subharmonic resonance $\omega_s(q_0) = \omega$. *Class:* A, D; the parametric cousin of the granular patterns below.

5.2 Rayleigh–Bénard convection (A, D)

A fluid layer heated from below conducts heat until buoyancy beats viscosity and thermal diffusion; then rolls or hexagonal cells about one layer depth wide appear everywhere at once. Beyond onset the rolls are stable only inside the Busse balloon, and at Prandtl number ≈ 1 they give way to spiral-defect chaos [1, 13].

$$\mathrm{Ra} = \frac{g\alpha\Delta T\, d^3}{\nu\kappa}.$$

Threshold and scale: $\mathrm{Ra}_c = 1707.76$, $k_c d = 3.117$ for rigid plates (657.5 and $\pi/\sqrt{2}$ for free–free surfaces); away from onset a band of wavenumbers is stable, not a unique one [1]. *Class:* A, D; the parent of the Lorenz equations (Section 11).

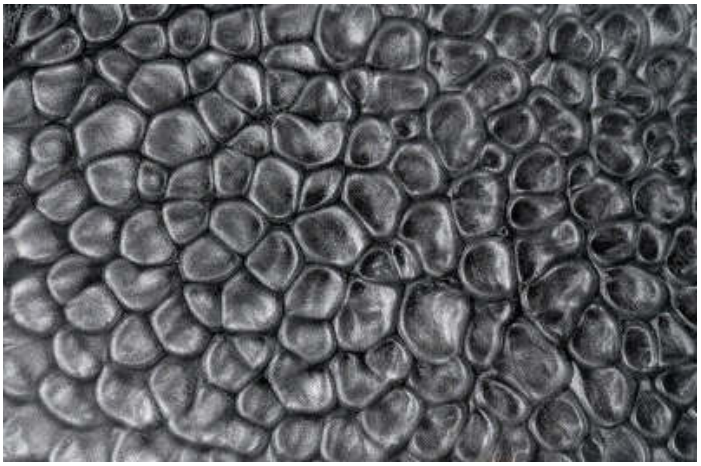


Figure 77: Experimental Rayleigh–Bénard convection cells visualized with aluminum powder in a thin layer of oil heated from below (Scott Camazine).

5.3 Bénard–Marangoni convection (A, D)

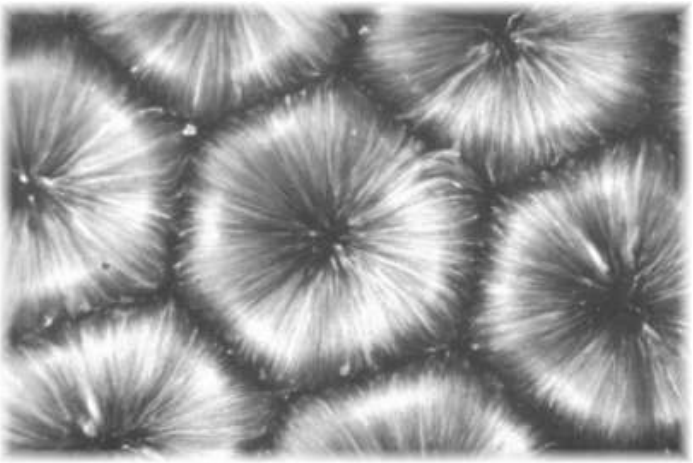


Figure 78: Additional experimental visualization of organized Rayleigh–Bénard convection cells.

In a thin layer with a free surface, surface-tension gradients drive the flow: a warm spot has lower tension, the surroundings pull liquid away and upwelling brings more heat [14]. The free surface breaks the up–down symmetry, so hexagonal cells appear through a weakly subcritical onset and change to squares far above it [1, 15].

$$\mathrm{Ma} = \frac{\Delta T\, d\, |\partial\sigma/\partial T|}{\rho\nu\kappa}.$$

Threshold and scale: $\mathrm{Ma}_c \approx 80$ (conducting bottom, insulating surface); cells $\approx 3d$ across. *Class:* A, D; with buoyancy, the two thresholds combine roughly as $\mathrm{Ra}/\mathrm{Ra}_{c0} + \mathrm{Ma}/\mathrm{Ma}_{c0} \approx 1$ [1].

5.4 Taylor–Couette vortices (A, D)

Fluid between a rotating inner cylinder and a fixed outer one is centrifugally unstable, because angular momentum falls outward (Rayleigh's criterion); viscosity sets a finite threshold. Stacked counter-rotating toroidal vortices, each about one gap high, fill the annulus, followed by wavy, modulated and turbulent vortex states [1, 16].

$$R_i = \frac{a(b-a)\Omega_i}{\nu} \qquad \Big(\text{or} \quad \mathrm{Ta} = \frac{\Omega^2 R d^3}{\nu^2}\Big).$$

Threshold and scale: narrow gap $\mathrm{Ta}_c \approx 1708$, axial period $\approx 2d$; wavy vortices at $R_i/R_{ic} \approx 1.1$–1.2 depending on radius ratio [1]. *Class:* A, D; the rotating analogue of convection rolls.

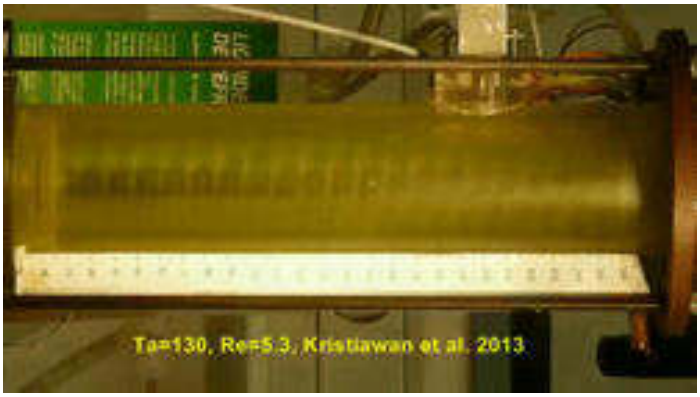


Figure 79: Experimental Taylor–Couette flow showing stacked vortices between rotating cylinders. The image's embedded label credits Kristiawan *et al.* (2013); not independently verified.

5.5 Electroconvection in nematic liquid crystals (A, D)

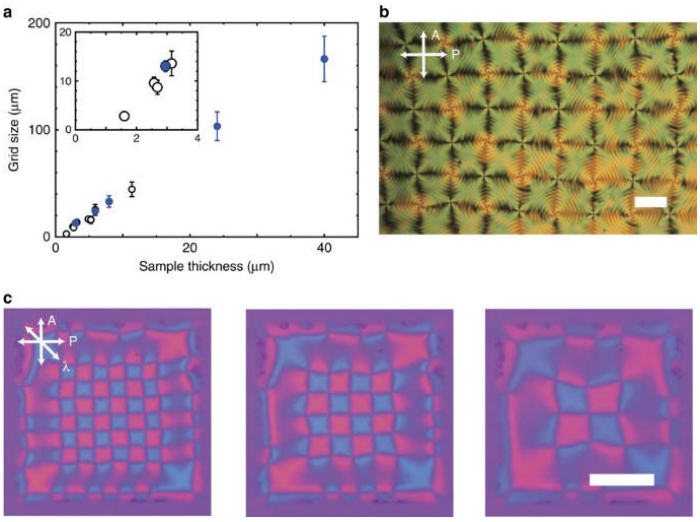


Figure 80: Electroconvection pattern in a nematic liquid crystal. The right-hand panel shows the self-organized roll or defect pattern that appears above threshold under an applied electric field. Y. Sasaki *et al.*, *Nat. Commun.* **7**, 13238 (2016), doi:10.1038/ncomms13238, CC BY 4.0.

In a thin nematic cell under an ac voltage, a bend fluctuation of the director deflects the current, the field pushes on the resulting space charge, and the flow increases the bend (Carr–Helfrich feedback). Above threshold, rolls with spacing of order the cell thickness d appear: oblique rolls below the Lifshitz frequency, normal rolls above it, and a finer dielectric regime above the charge-relaxation cutoff [1, 17, 18].

$$\varepsilon = \frac{V^2}{V_c^2} - 1.$$

Threshold and scale: V_c is nearly independent of d; rolls $\sim d$ ($d = 10$–$100\,\mu$m). *Class:* A, D; the anisotropic analogue of Rayleigh–Bénard, with large aspect ratios and fast time scales.

5.6 Turing patterns (A, D)

Two reacting species form stationary spots or stripes when an activator is short-ranged and an inhibitor long-ranged [19]. This was first realized in the CIMA reaction in an open gel reactor, where starch slows the activator; spots and stripes about 0.2 mm apart appear whatever the reactor size [20, 21].

$$\partial_t u = D_u \nabla^2 u + f(u,v), \quad \partial_t v = D_v \nabla^2 v + g(u,v); \qquad D_v f_u + D_u g_v > 2\sqrt{D_u D_v \det J}, \;\; k_c^2 = \sqrt{\frac{\det J}{D_u D_v}}.$$

Threshold and scale: the diffusivity ratio D_v/D_u must exceed a kinetic threshold; the period is intrinsic. *Class:* A, D; distinct from BZ waves (class E), which need no difference in diffusivities.

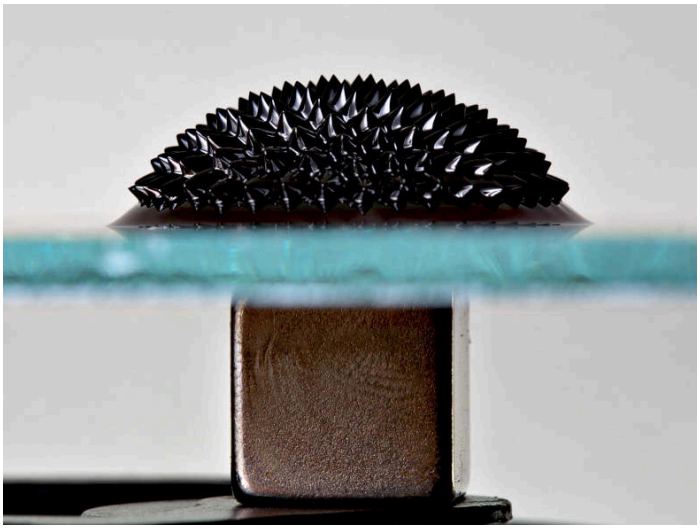


Figure 81: Experimental ferrofluid peaks formed by the Rosensweig (normal-field) instability. Side view with the magnet beneath a glass plate. Photo: Gregory F. Maxwell, Wikimedia Commons, CC BY-SA 3.0.

5.7 Langmuir circulation and cloud streets (A, D)



Figure 82: Satellite image of cloud streets: wind-aligned convective rolls in the atmospheric boundary layer, with the same roll geometry as Langmuir circulation.

Wind over waves organizes the upper ocean into wind-aligned counter-rotating roll pairs; floating material collects in lines (windrows) over the convergences [22]. The Stokes drift $\mathbf{u}_s$ of the waves tilts and stretches wind-driven vorticity into downwind rolls (the Craik–Leibovich CL2 instability [23]):

$$\partial_t \boldsymbol{\omega} = \nabla \times \big[(\mathbf{u} + \mathbf{u}_s) \times \boldsymbol{\omega}\big] + \nu \nabla^2 \boldsymbol{\omega}, \qquad \mathrm{La}_t = (u_*/U_s)^{1/2}.$$

Threshold and scale: windrows appear above about $3\,\mathrm{m\,s^{-1}}$ of wind, 2–3 mixed-layer depths apart [24]; open-ocean $\mathrm{La}_t \approx 0.3$. Cloud streets (figure) share the geometry but are convective rolls [25]. *Class:* A, D; the vortex-force analogue of convection rolls. Not to be confused with Langmuir *monolayers*, which form class-B stripe phases [2].

5.8 Vibrated granular layers and oscillons (A, D)

A vibrated layer of grains leaves the plate each cycle when $\Gamma > 1$ and, above a second threshold, forms standing waves at half the drive frequency. Below a crossover frequency f^* the patterns are subcritical squares, above it supercritical stripes; oscillons, localized peaks that alternate with

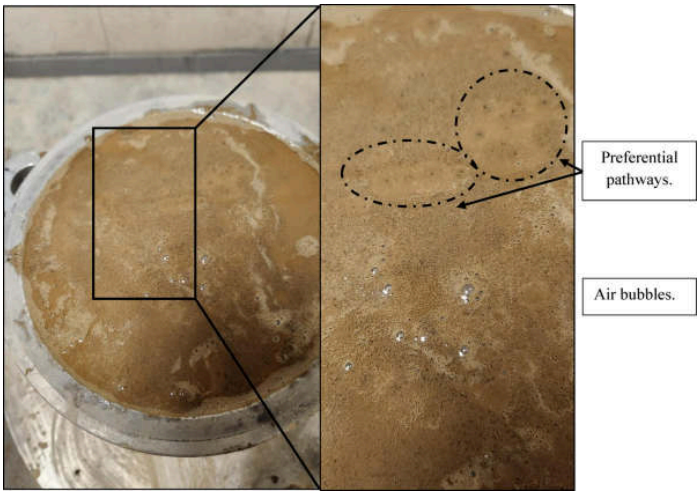


Figure 83: Experimental view of a vibrated granular bed showing preferential pathways and surface organization under vibration.

craters, live near f^*, and hexagons appear for $\Gamma > 4$ [26, 27, 8].

$$\Gamma = \frac{A\omega^2}{g}.$$

Threshold and scale: $\Gamma \approx 2.4$; $f^* \approx 45\,\mathrm{Hz}$ with $f^* \propto d^{-1/2}$; the wavelength grows with layer depth and falls with f [8]. *Class:* A, D; parametric like Faraday waves, with inelastic collisions in place of viscosity.

6 Class B: equilibrium modulated phases

6.1 Rosensweig instability (B, E)

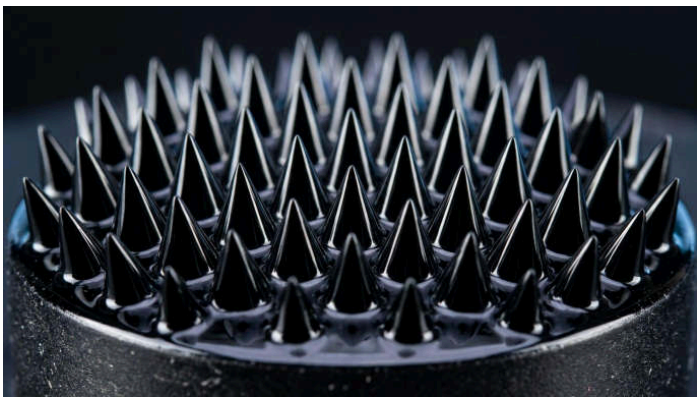


Figure 84: Close-up experimental view of the ordered spike lattice produced by the Rosensweig instability.

A ferrofluid in a vertical field forms a hexagonal lattice of spikes: a crest concentrates field lines and gains magnetic energy, while gravity and surface tension resist [28]. The lattice appears abruptly (subcritical), changes to squares at higher field, and persists with no energy input [2, 29].

$$\mu_0 M_c^2 = 2\Big(1 + \frac{1}{r_p}\Big)\sqrt{g\Delta\rho\,\sigma}, \qquad r_p = \frac{\sqrt{\mu_c \mu_t}}{\mu_0}.$$

Threshold and scale: onset spacing $\lambda = 2\pi\sqrt{\sigma/g\Delta\rho}$, about 1 cm, the same as the Rayleigh–Taylor onset wavelength [2]. *Class:* B, E; its onset dynamics is a class-A-like subcritical hexagon instability, the bridge between the classes.

6.2 Magnetic domain patterns (B, E)

Thin films with perpendicular anisotropy break into up and down stripes or labyrinths: domain walls cost energy, but alternating domains reduce the stray-field energy [30, 4]. A perpendicular field drives first-order transitions from stripes to bubbles to saturation [2]. In the Fe/Ni/Cu films of the figure the width falls sevenfold over less than one monolayer near the spin-reorientation transition, where the wall energy vanishes [31].

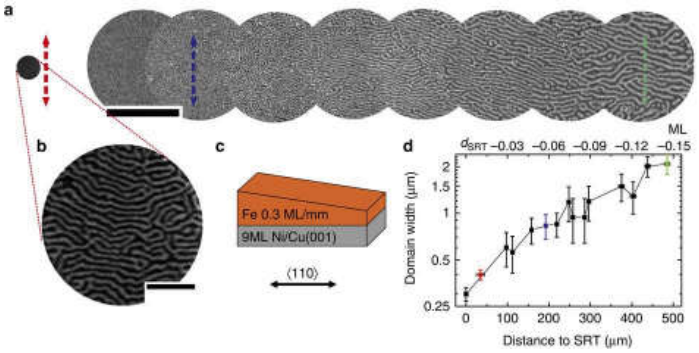


Figure 85: Experimental magnetic domain patterns in a thin ferromagnetic film (labyrinthine stripes observed by real-time microscopy). Distinct from the free-surface peaks of a ferrofluid. M. Kronseder *et al.*, *Nat. Commun.* **6**, 6832 (2015), Fig. 1, doi:10.1038/ncomms7832, CC BY 4.0.

$$\text{thick films: } w \propto \sqrt{\ell_c t}, \quad \ell_c = \frac{\sigma_w}{\mu_0 M_s^2}; \qquad \text{ultrathin films: } d^* = a\,\exp\!\Big(\frac{b\gamma}{\mu_0 M^2}\Big).$$

Threshold and scale: none (an equilibrium state); the period is set by the wall-to-dipolar energy ratio and tuned reversibly by field and temperature. *Class:* B, E; the same stripe and bubble phases occur in ferrofluid labyrinths, Langmuir monolayers and block copolymers [4].

7 Class C: coarsening

7.1 Soap films and foams (C, R)

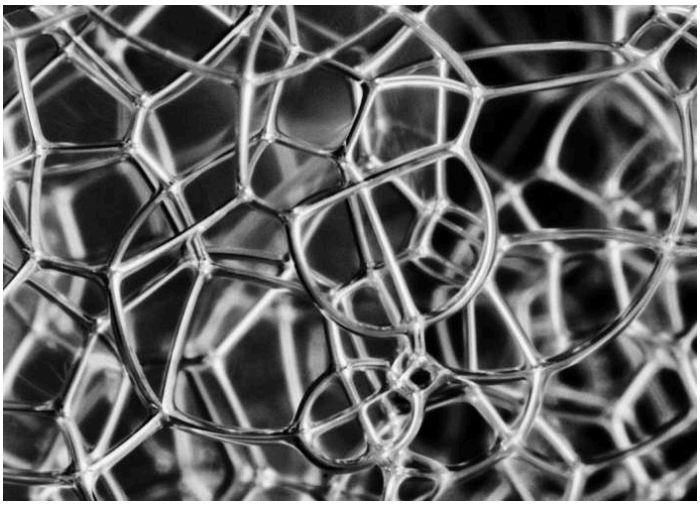


Figure 86: Experimental soap foam showing the cellular network and Plateau borders set locally by surface tension; the network coarsens as gas diffuses between cells.

Films meet three at a time at 120° and their Plateau borders four at a time at 109.5°, a local equilibrium of surface tension. The foam as a whole is metastable: gas diffuses from small, high-pressure bubbles into large ones and the network coarsens [32, 8].

$$\Delta p = 4\sigma H, \qquad \frac{dA_n}{dt} = \frac{\pi}{3} k_D (n-6).$$

Threshold and scale: none; the mean cell area grows $\propto t$ in 2D. *Class:* C, R; the same scaling as spinodal domains and grain growth.

8 Class D: interfacial instabilities

8.1 Kelvin–Helmholtz billows (D, D)



Figure 87: Kelvin–Helmholtz clouds: atmospheric manifestation of the shear-driven interfacial instability.

Shear across a density interface rolls it into billows, seen as rows of breaking-wave clouds. Gravity stabilizes long waves; for a continuous shear layer the billow spacing is set by the layer thickness δ, and stratification suppresses billows when the Richardson number exceeds $1/4$ [33, 34, 35].

$$\sigma^2 = \frac{\rho_1\rho_2 k^2 U^2}{(\rho_1+\rho_2)^2} - gk\frac{\rho_2-\rho_1}{\rho_1+\rho_2}, \qquad \mathrm{Ri} = \frac{N^2}{(\partial_z U)^2} < \frac{1}{4}.$$

Threshold and scale: $\lambda \approx 7\delta$; billows then pair and merge. *Class:* D, D; the archetype of open-flow, convectively unstable systems [1].

8.2 Saffman–Taylor viscous fingering (D, D)

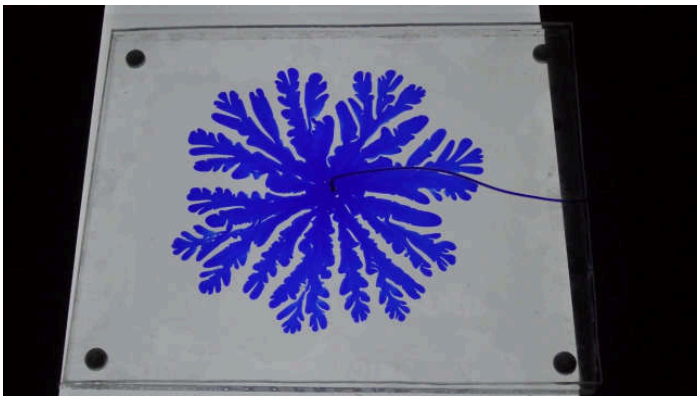


Figure 88: Experimental Saffman–Taylor viscous fingering: dyed water injected into glycerol in a Hele-Shaw cell.

A less viscous fluid pushed into a more viscous one in a Hele-Shaw cell advances in fingers that split at their tips [36]. All long waves are unstable; surface tension cuts off short ones, and in a channel a single finger narrows toward half the channel width as Ca grows [37].

$$\sigma(k) = \frac{k}{\mu_1+\mu_2}\Big[(\mu_2-\mu_1)U - \frac{b^2\gamma}{12}k^2\Big], \qquad \lambda_m = \frac{\pi b}{\sqrt{\mathrm{Ca}}}.$$

Threshold and scale: none (any $\mu_2 > \mu_1$); finger spacing $\propto b\,\mathrm{Ca}^{-1/2}$. *Class:* D, D; the hydrodynamic analogue of solidification [1].

8.3 Rayleigh–Taylor fingers (D, R)

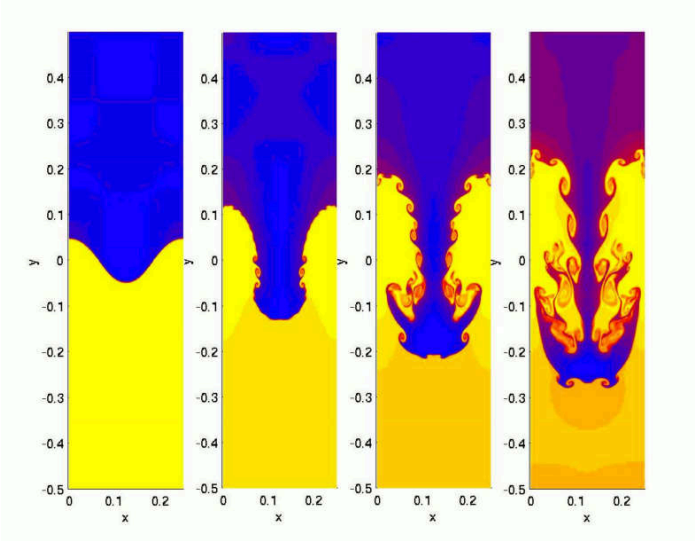


Figure 89: Numerical evolution of the Rayleigh–Taylor instability showing the development of descending fingers and rising bubbles. Simulation: Shengtai Li and Hui Li (Los Alamos National Laboratory), Wikimedia Commons.

Heavy fluid above light fluid releases potential energy through falling spikes and rising bubbles; capillarity or viscosity sets the short-wave cutoff [38]. A hanging film breaks into hexagonal drops that spread from the edges as a front [7].

$$\sigma^2 = Agk - \frac{\gamma k^3}{\rho_h + \rho_l}, \qquad A = \frac{\rho_h - \rho_l}{\rho_h + \rho_l}.$$

Threshold and scale: none; $\lambda_c = 2\pi\sqrt{\gamma/\Delta\rho g}$ and fastest $\lambda_m = \sqrt{3}\lambda_c$, the same capillary length as the Rosensweig spacing [2]. *Class:* D, R (a one-shot release of potential energy).

8.4 Dendritic solidification (D, R / D)



Figure 90: Manganese-oxide "dendrites" on a rock surface: diffusion-limited branching grown from solution, shown for its dendritic morphology.

A seeded crystal growing into an undercooled melt sheds latent heat; a bump reaches colder liquid and grows faster (Mullins–Sekerka), while the Gibbs–Thomson effect stabilizes short waves [39, 3]. The tip radius is selected by anisotropic surface tension acting as a singular perturbation (solvability): with isotropic surface tension there is no steady needle at all [1]. Sidebranches are tip noise amplified on the flanks [7].

$$\sigma = vk\Big[1 - \tfrac{1}{2}(1+\beta)\, d_0 \ell k^2\Big], \quad \ell = \frac{2D}{v}; \qquad \rho^2 v = \frac{2Dd_0}{\sigma^*},\ \sigma^* \approx 0.025.$$

Threshold and scale: always unstable; $\lambda_s = 2\pi\sqrt{d_0\ell}$ in the symmetric model ($\beta = 1$), comparable to the measured tip radius [3, 40]. *Class:* D; tag R for free growth, D for directional solidification. A front into a metastable melt, not class E.

8.5 Bacterial colonies (D on an E front, D)



Figure 91: Experimental bacterial colony patterns on Petri dishes showing branching and labyrinthine morphologies.

An expanding colony on nutrient-poor hard agar branches into a DLA-like fractal of dimension ≈ 1.7; richer or softer agar gives dense branching, compact disks or rings [41, 42, 43]. Branching needs a *pushed* front (nonlinear cell diffusion or cooperative growth) and slow cell spreading compared with nutrient diffusion; the front then has a Mullins–Sekerka-type instability, whereas a simple Fisher front stays flat [7].

$$\partial_t \beta = \delta\, \nabla \cdot (\beta^k \nabla \beta) + \alpha\beta^m, \qquad \partial_t \alpha = \nabla^2 \alpha - \alpha\beta^m.$$

Threshold and scale: small δ and $k > 0$ or $m > 1$; branch width $\sim$ nutrient diffusion length. *Class:* D on an E front, D; it links fronts to fingering and dendrites.

9 Class E: fronts and waves

9.1 Belousov–Zhabotinsky waves (E, R in a dish / D in an open reactor)



Figure 92: Experimental Belousov–Zhabotinsky spiral wave in a Petri dish (ferroin-catalyzed reaction).

In a thin unstirred layer of the BZ reagent, oxidation waves run across an excitable medium as blue rings on red; breaking a front leaves a pair of counter-rotating spirals [44]. Targets and spirals are defects of the uniform oscillation (type III_o) [1]. A closed dish runs down within about 100 periods; an open, fed reactor sustains the waves [1]. With the Oregonator kinetics [1],

$$\partial_t u_1 = \eta^{-1}\Big[u_1(1-u_1) - b u_2 \frac{u_1 - a}{u_1 + a}\Big] + D_1 \nabla^2 u_1, \qquad \partial_t u_2 = u_1 - u_2 + D_2 \nabla^2 u_2.$$

Threshold and scale: excitability (acid and bromate concentrations); wave speed $c \propto \sqrt{Dk}$ (Luther's law), spiral pitch $\approx$ period $\times$ speed, a few millimetres. *Class:* E; the medium is linearly stable and is invaded by trigger waves.

9.2 River–ocean density fronts (E, D)



Figure 93: Surface photograph of a river-plume front. Turbid freshwater on one side of a sharp boundary meets clearer seawater on the other. The interface is the visible signature of incomplete mixing maintained by the density difference.

Buoyant river water spreads over the sea as a thin layer; surface convergence sharpens the density contrast (frontogenesis) until instability or mixing arrests it, and the front is seen as a sharp colour line with foam [45, 46]. Rotation organizes the plume into a bulge and coastal current on the deformation radius.

$$R_d = \frac{\sqrt{g'h}}{f}, \qquad \frac{D_g}{Dt}\,|\nabla_h b|^2 = 2\,\mathbf{Q}\cdot\nabla_h b, \quad Q_i = -\frac{\partial u_{gj}}{\partial x_i}\frac{\partial b}{\partial x_j}.$$

Threshold and scale: no wavelength-selecting instability; plume width $\sim R_d$ (a few km), front width set where frontogenesis is arrested [47]. *Class:* E, D (discharge, tides); a sharpened front that can host class-D billows.

9.3 Liesegang rings (E + C, R)

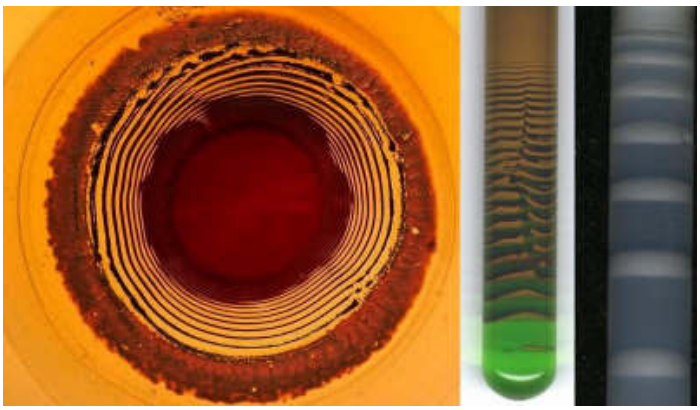


Figure 94: Experimental Liesegang rings: concentric precipitation bands in a gel and layered bands in a test tube.

Two reactants meeting by diffusion in a gel deposit discrete precipitation bands rather than a continuous front; each band depletes its neighbourhood, so the next forms farther away. Behind the front the product may phase-separate, linking the bands to spinodal decomposition [48]. The bands

are kinetically frozen in a closed system, not an equilibrium.

$$\frac{x_{n+1}}{x_n} \to 1 + p, \qquad x_n \propto \sqrt{t_n}.$$

Threshold and scale: a nucleation (supersaturation) threshold; spacing grows geometrically, more densely for a stronger outer electrolyte. *Class:* E (with C behind the front), R.

10 Self-similarity



Figure 95: Romanesco broccoli: each bud is a cone of smaller spiralling buds, repeated over about four generations. It is a natural object that is geometrically self-similar over a limited range of scales. Photo: Jon Sullivan, Wikimedia Commons, public domain.

Branched patterns (dendrites, viscous fingers in porous media, bacterial colonies) are statistically self-similar over one to three decades, with fractal dimensions near the DLA value 1.71 [42, 41, 49]. Coarsening patterns (class C) are dynamically self-similar: a single length $L(t)$ grows and the structure factor collapses [9],

$$S(k,t) = L(t)^d \, f\big(kL(t)\big), \qquad L \propto t^{1/3} \text{ (conserved)}, \; t^{1/2} \text{ (non-conserved)}.$$

Classes A and B select one wavelength and are not self-similar.

11 Chaos

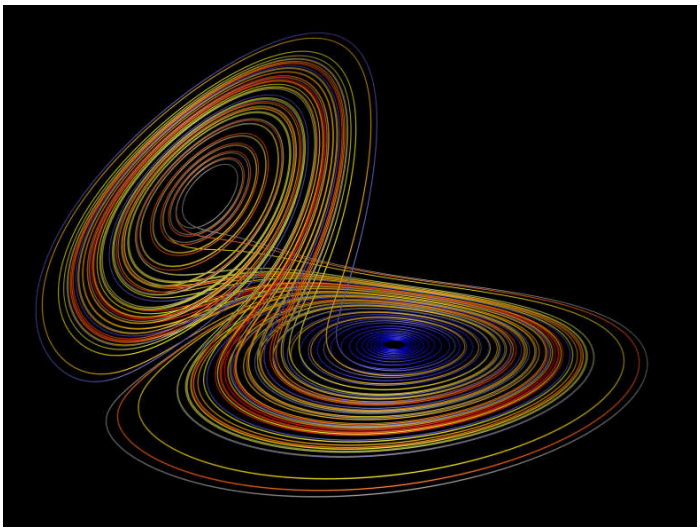


Figure 96: The Lorenz attractor. Trajectories remain bounded yet never repeat, illustrating sensitive dependence on initial conditions in a system derived from thermal convection.

Class-A patterns become chaotic as the drive increases. In small containers few modes are active: period doubling with Feigenbaum's ratio $\delta \approx 4.669$ in a small convection cell [50, 51], and chaos after two frequencies in Taylor–Couette flow [52]. The Lorenz equations, a truncation of Rayleigh–Bénard convection, and the logistic map are the standard models [53]:

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z; \qquad x_{n+1} = r x_n (1 - x_n).$$

Large containers give spatiotemporal chaos (spiral-defect chaos, defect turbulence) [1, 13], and the single selected wavelength is lost.

12 Summary: what the patterns share

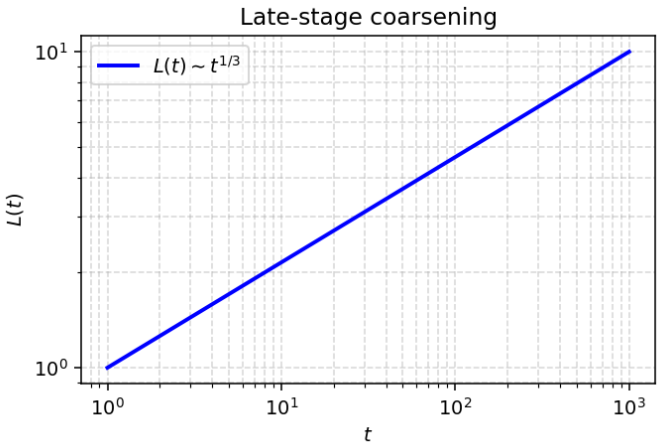


Figure 97: Late-stage coarsening: the characteristic length grows as $L(t) \sim t^{1/3}$ for curvature-driven, conserved order-parameter dynamics. Statistics collapse when lengths are scaled by $L(t)$.

Every example starts from a uniform state and one dimensionless drive, and its length scale comes from a balance between an effect that suppresses long waves and one that suppresses short waves; the capillary length, a diffusion length or the layer depth recurs as the result. The shape of $\sigma(k)$, or of $F(q)$, sorts the examples into classes A–E. Near onset, classes A and B share gradient amplitude dynamics, hence the same stripes, hexagons, labyrinths and defects in rolls, Turing patterns, domains and ferrofluids. Far from onset, driven patterns develop bands of states and chaos, interfaces are selected by small surface tension, and fronts and coarsening add speed and time as variables. What sustains a pattern (flux, slow relaxation, or nothing) is a separate label.

13 Wrinkled thin shells: supernova remnants and the CoE onion

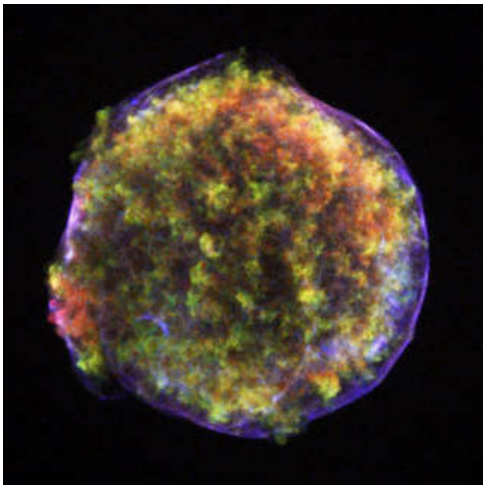


Figure 98: Tycho's supernova remnant in X-rays. The outer shock is smooth; the shell of ejecta inside it is broken into wrinkles and clumps. NASA/CXC/Rutgers/J. Warren & J. Hughes et al., public domain, Chandra via Wikimedia Commons.

A thin dense shell swept up by a blast wave and decelerating against the gas around it is unstable to wrinkling [54, 55]. In the shell frame the deceleration acts as gravity pointing outward, so the dense shell sits on top of lighter gas: Rayleigh–Taylor with

$$g_{\text{eff}} = -\ddot{R}, \qquad \sigma = \sqrt{A\,k\,g_{\text{eff}}}, \qquad A = \frac{\rho_{\text{shell}} - \rho_{\text{in}}}{\rho_{\text{shell}} + \rho_{\text{in}}} \approx 1.$$

In the Vishniac thin-shell instability the ram pressure pushes along the tilted shell and drives material from troughs into crests, so the wrinkle also modulates the surface density [54]. The shell thickness Δ cuts off short waves and the shell size cuts off long ones:

$$\lambda \approx 3\Delta, \qquad \Delta \lesssim \lambda \lesssim \pi R, \qquad \ell_{\max} \approx \frac{2\pi R}{\Delta}.$$

Threshold and scale: needs strong compression ($\rho_{\text{shell}}/\rho_0 \gtrsim 10$, nearly isothermal shell); spacing set by thickness. *Class:* D, R; the same interface physics as Rayleigh–Taylor fingers and the same "thickness sets spacing" rule as Marangoni cells ($\approx 3d$) and Kelvin–Helmholtz billows ($\approx 7\times$ layer). Laser-driven shells reproduce the remnant clumps [55].

The CoE onion. In the center-of-expansion (CoE) picture the visible universe is countless thin concentric shells moving out from one center. For a point mass M at the CoE each shell obeys $\ddot{r} = -GM/r^2$, so every shell is a decelerating thin shell with $g_{\text{eff}} = GM/r^2$, the forcing above. The sequence (Figs. 103–105 for the wrinkle growth, Figs. 109–113 for clump and void):

1. Smooth thin shells: no preferred spot on the sky.
2. A slight wrinkle seeds the pattern, crests about 3Δ apart.
3. Crests collect material (overdense), troughs thin out (underdense).
4. Once seeded, gravity keeps growing the contrast even if the shell instability itself dies away (the dominoes keep falling). Overdensities cross $\delta_c = 1.686$ and collapse into clumps; underdensities reach $\delta_v = -2.81$ and become voids. Growing linearly from recombination, this needs seeds of about 1.5×10^{-3} (cluster) and -2.5×10^{-3} (void) [56].

Stacking many thin shells and letting their thickness go to zero gives a continuous 3D field: crests of neighbouring shells line up into walls and filaments with clumps at their joins, and troughs line up into voids. If the seed spacing survives, clump and void spacing traces $\approx 3\Delta$.

Inserted figures — wrinkled thin shells. Michaut et al. (2012) followed the Vishniac instability numerically in a thin, dense, radiatively cooled supernova-remnant shell. Fig. 99 shows how the perturbation is set up; Fig. 100 shows the density of the rippled shell at the start and the sectors used to track how mass moves from troughs into crests. Fig. 101 follows the shell in time.

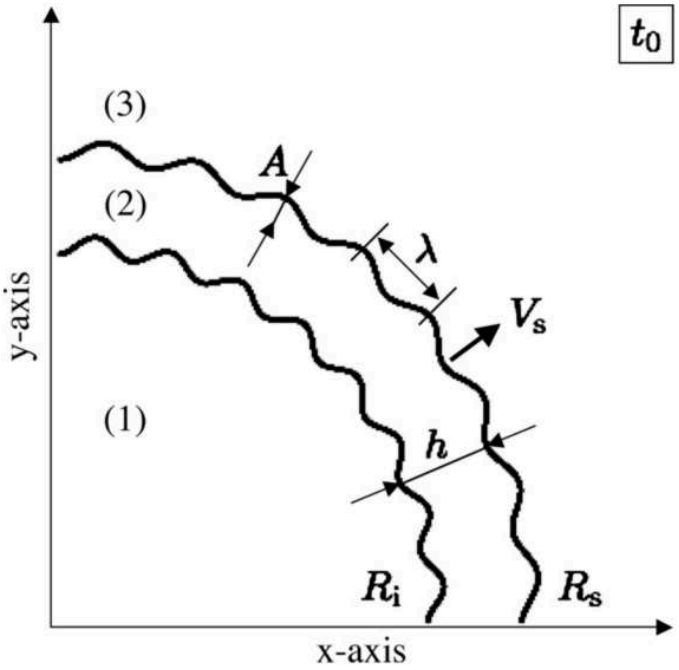


Figure 99: Set-up of the perturbation: the thin dense shell (2), of thickness $h = R_s - R_i$, lies between the hot cavity (1) and the ambient gas (3) and moves out at V_s; both faces carry a sinusoidal ripple of amplitude A and wavelength λ. This is the seed ripple of step 2 below, imposed on a shell whose thickness sets the scale of the instability. Reproduced from Michaut et al. (2012), ApJ 759, 78, doi:10.1088/0004-637X/759/2/78.

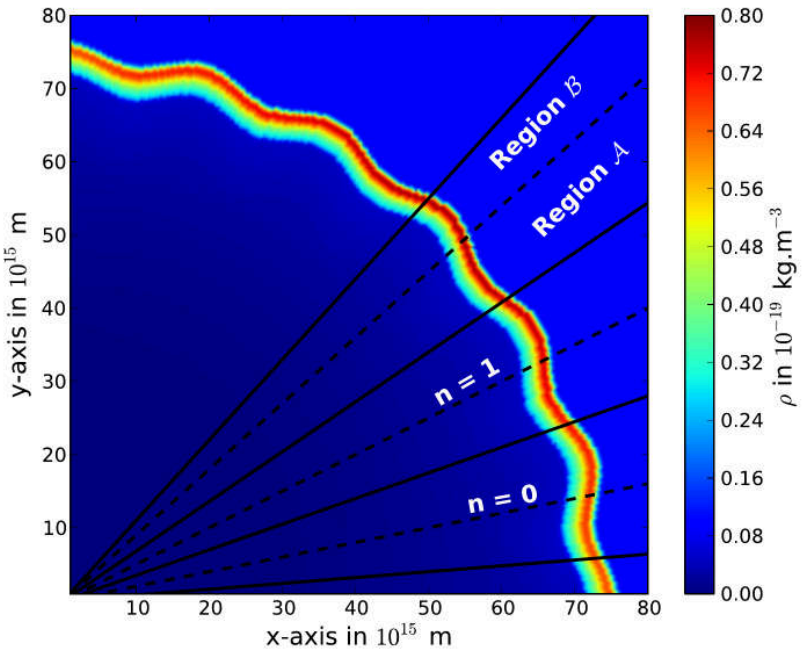


Figure 100: Density of the rippled shell at the start, with the angular sectors used to track mass: Region $\mathcal{A}$ (initially a bulge) and Region $\mathcal{B}$ (initially a hollow), repeated each wavelength ($n = 0, 1, \ldots$). Comparing the mass in $\mathcal{A}$ and $\mathcal{B}$ measures how much material moves from troughs into crests. Reproduced from Michaut et al. (2012), ApJ 759, 78, doi:10.1088/0004-637X/759/2/78.

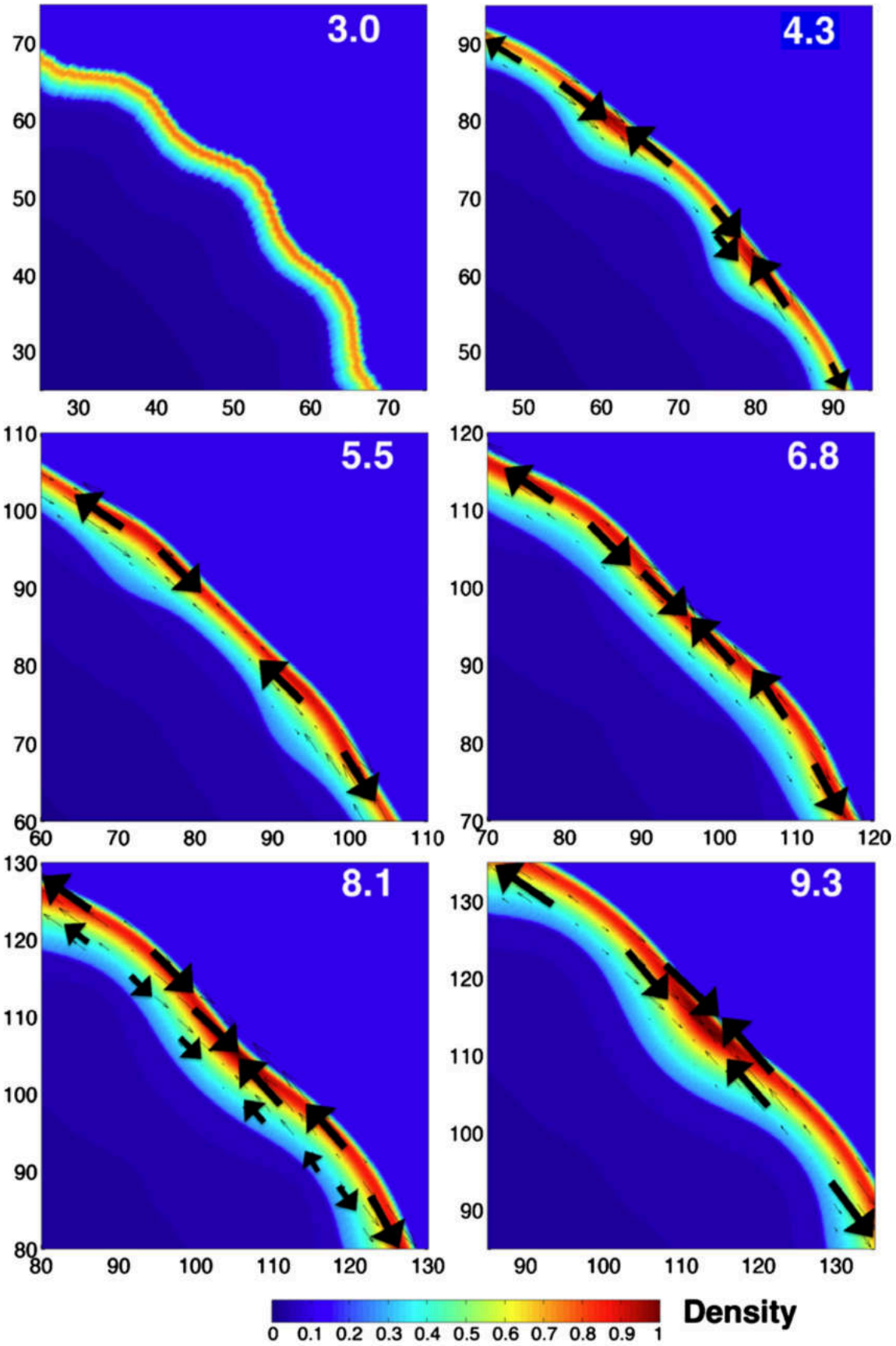


Figure 101: Density in the perturbed shell at $t = 3.0$, 4.3, 5.5, 6.8, 8.1 and 9.3 kyr (left to right, top to bottom); arrows show the flow along the shell. Material slides along the tilted shell and piles up in dense knots, and the flow reverses from one snapshot to the next: the Vishniac overstable oscillation. In this run the shell thickens and the oscillation is eventually damped rather than fragmenting the shell. Reproduced from Michaut et al. (2012), ApJ 759, 78, doi:10.1088/0004-637X/759/2/78.

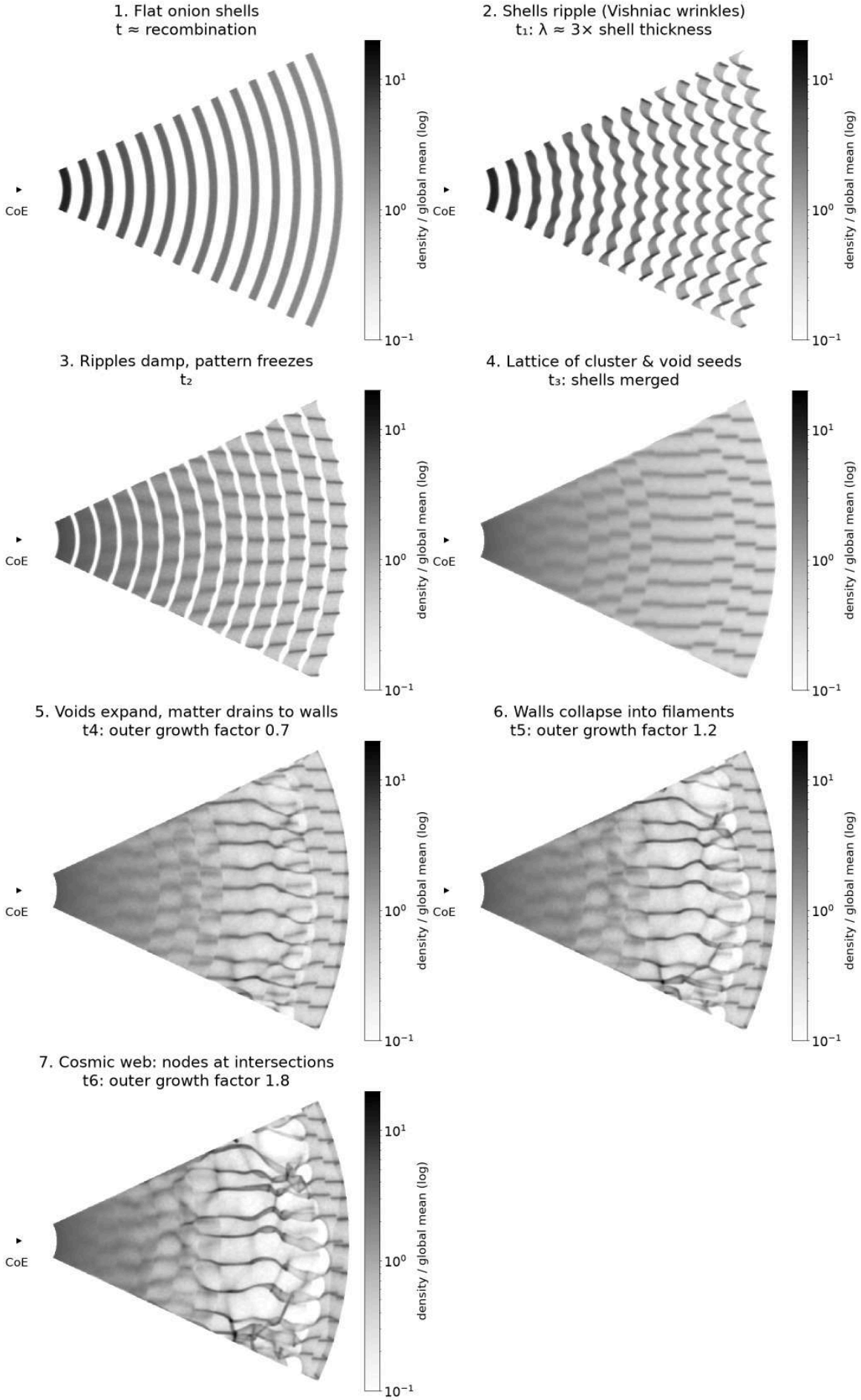


Figure 102: Wedge evolution storyboard (7 frames), equal mass per shell (density falling outward, 2D $\sigma \propto 1/r$) with structure forming outside-in (outer, oldest shells evolve first); the CoE is at the left of each wedge. Flat concentric shells ripple, the ripples fade and leave a density pattern, seeds of voids and clusters form, and gravity gathers them into filaments and a cosmic-web-like pattern. Illustrative toy model (Zel'dovich approximation, schematic units).

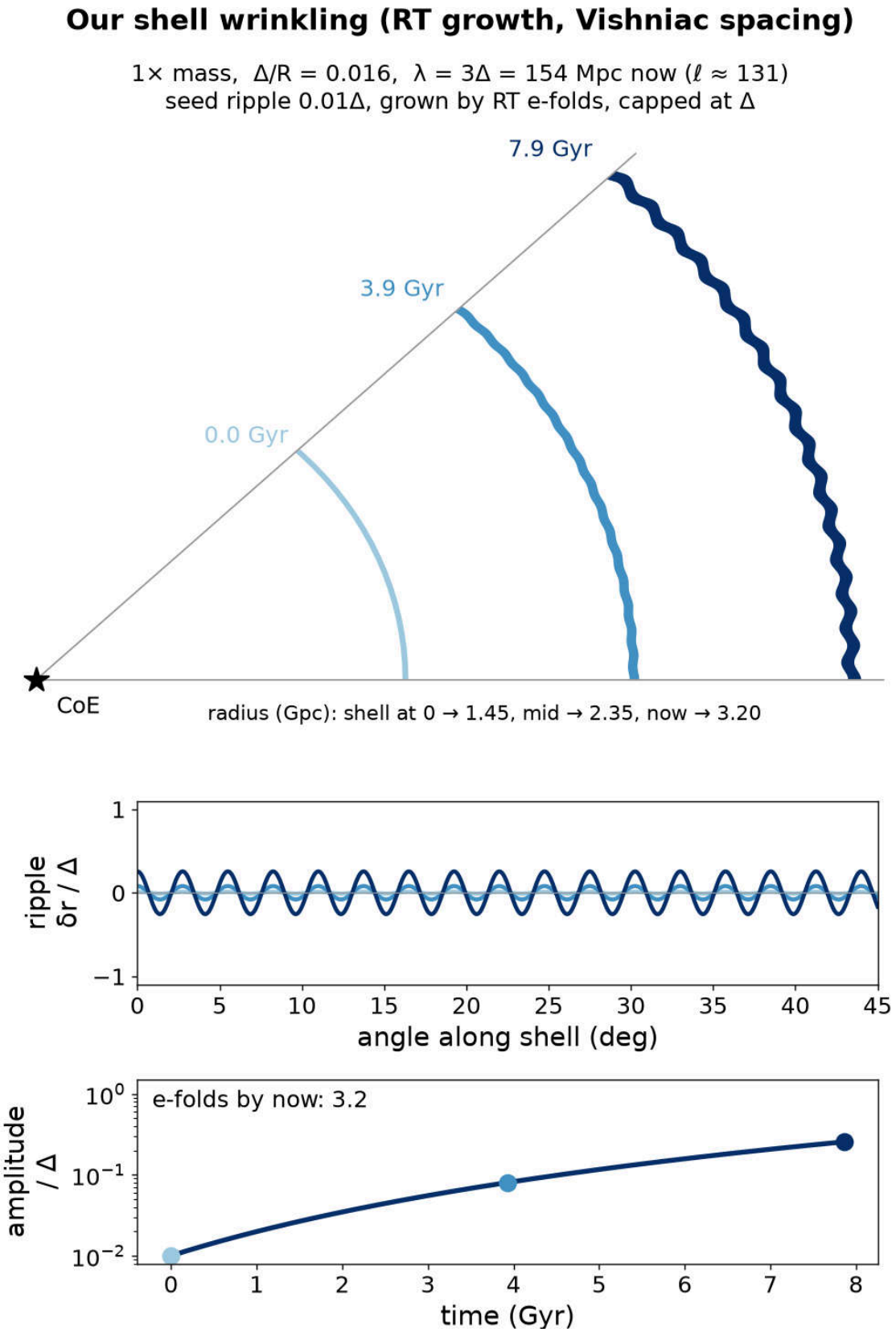


Figure 103: Our shell wrinkling, 1× mass, $\Delta/R = 0.016$: a 0.01Δ seed ripple at Vishniac spacing $\lambda = 3\Delta$ grown by Rayleigh–Taylor e-folds in $g_{\rm eff} = GM/r^2$ (capped at Δ). Shell drawn at start, mid and now.

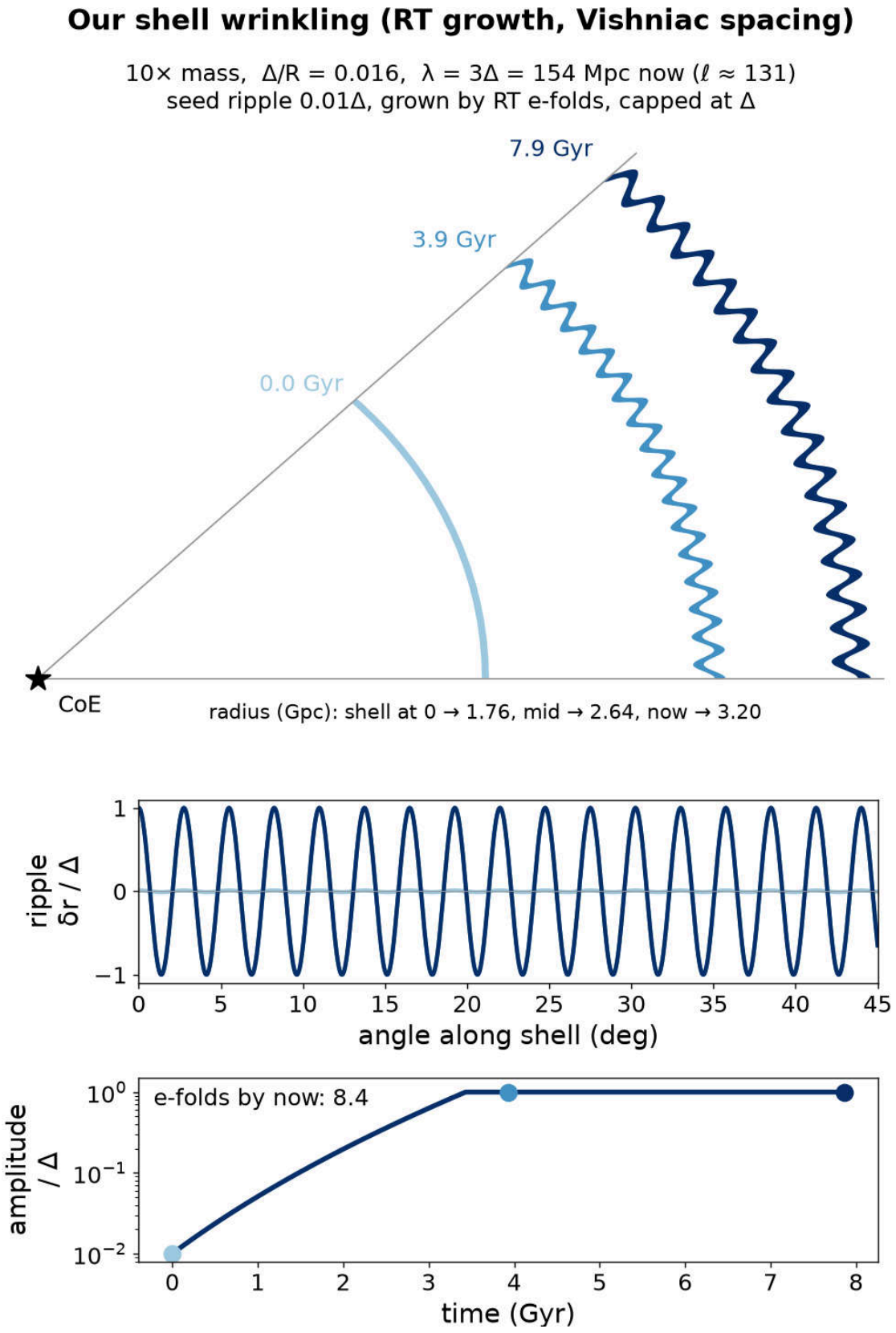


Figure 104: As Fig. 103 for 10× mass: stronger deceleration, faster growth; the ripple saturates at Δ by mid-run.

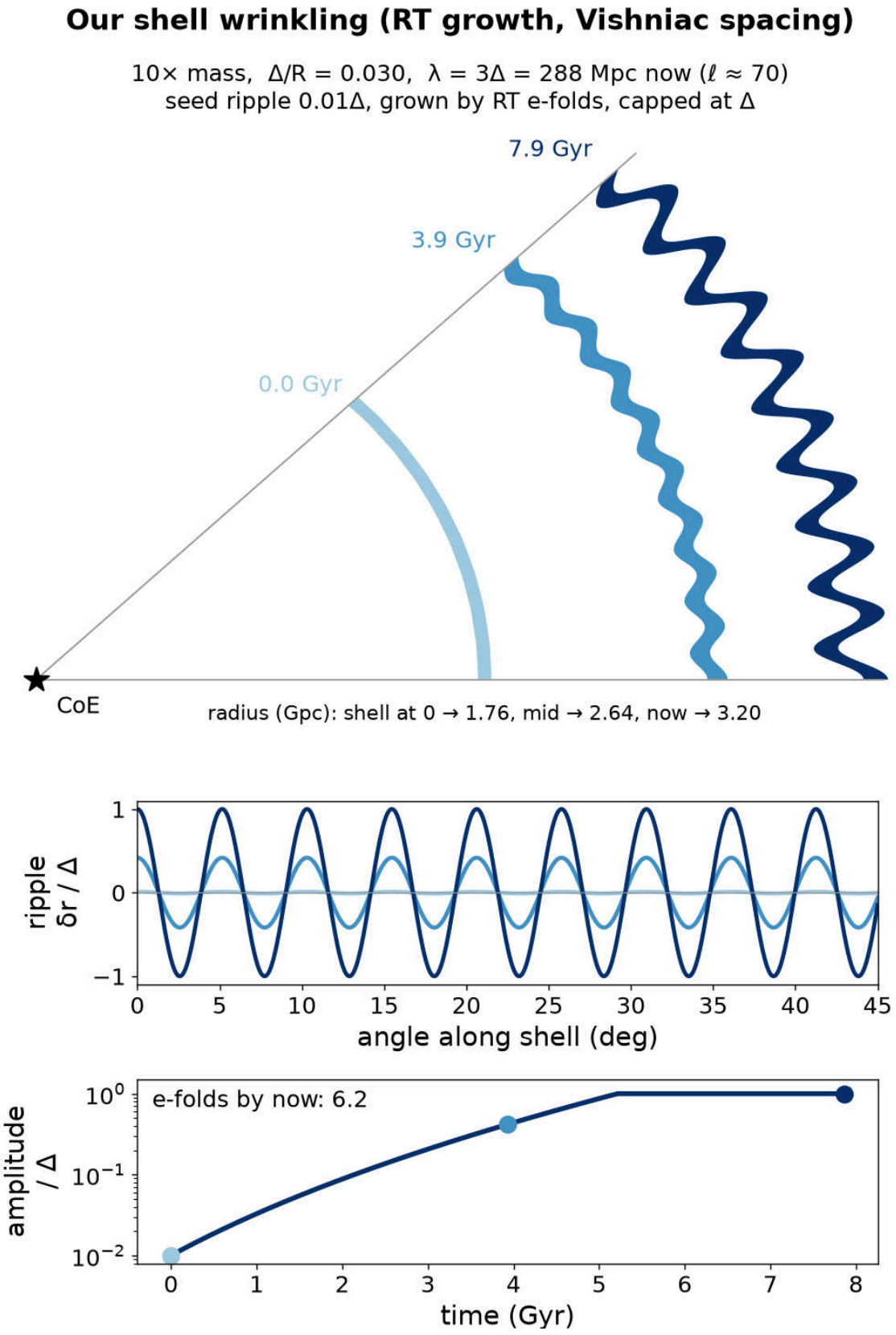


Figure 105: As Fig. 104 with a thicker shell, $\Delta/R = 0.03$: wider spacing (lower ℓ).

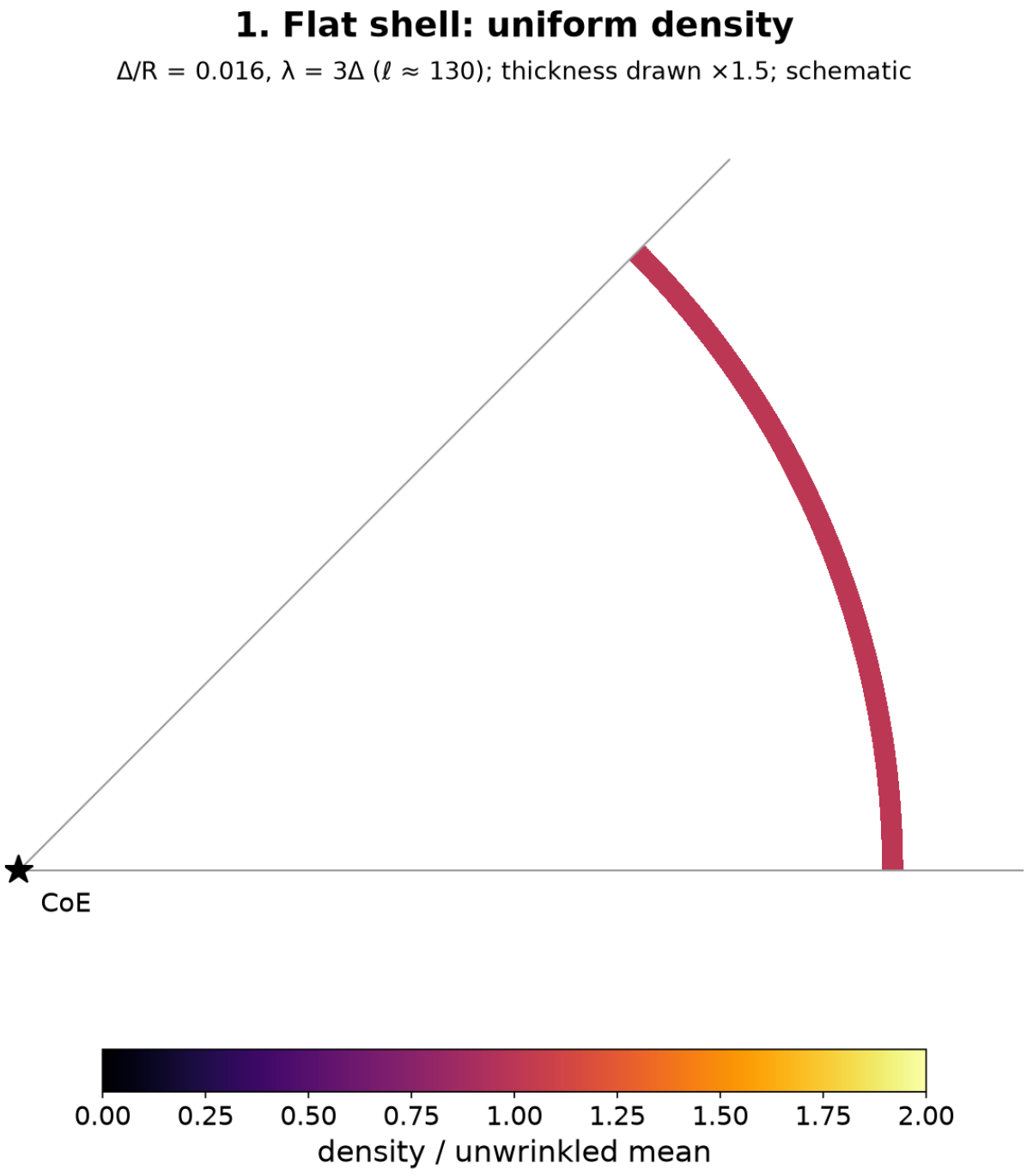


Figure 106: Flat shell: uniform density. Colour: density relative to the unwrinkled mean. Schematic.

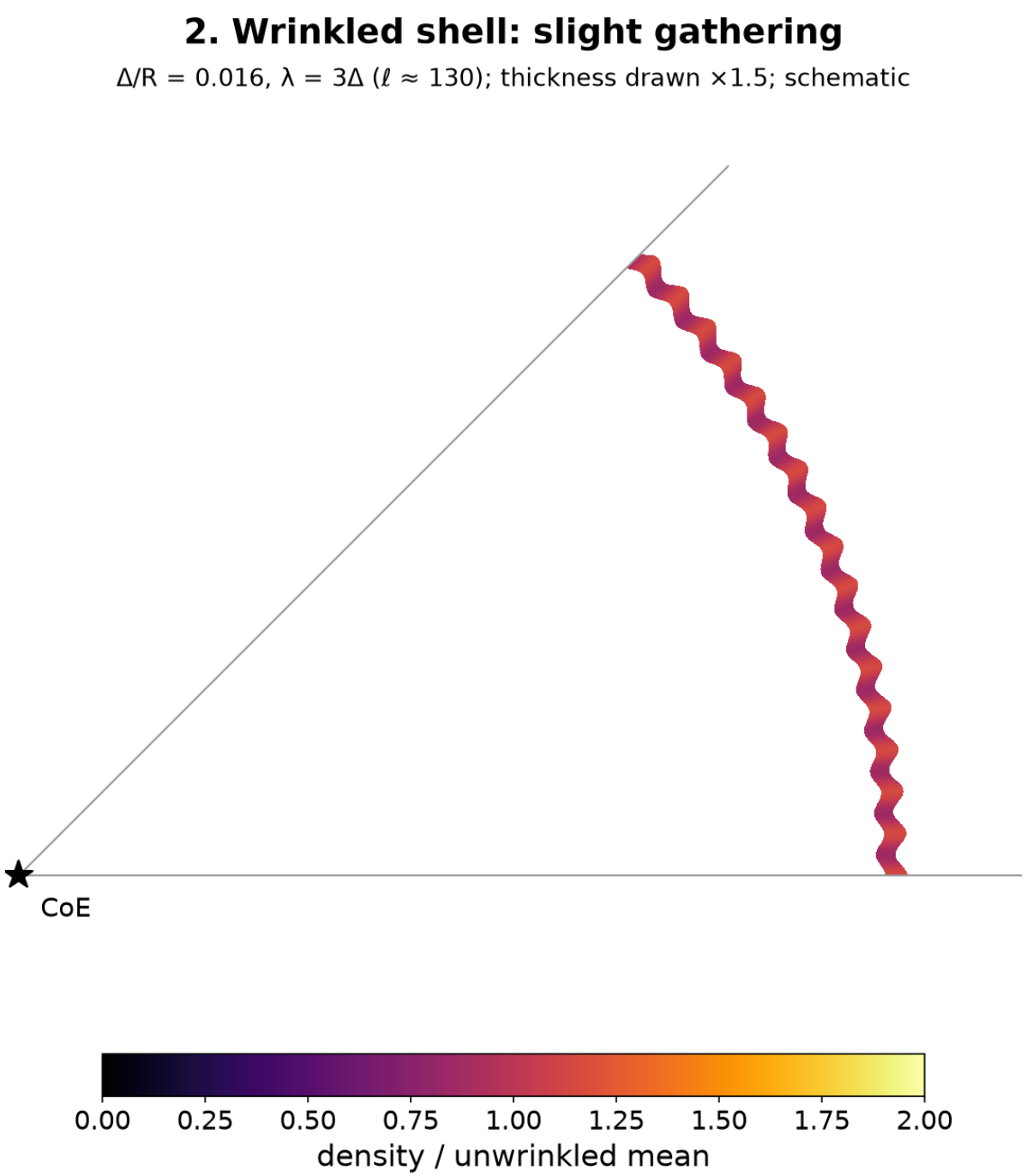


Figure 107: Wrinkled shell: slight gathering at the crests. Colour: density relative to the unwrinkled mean. Schematic.

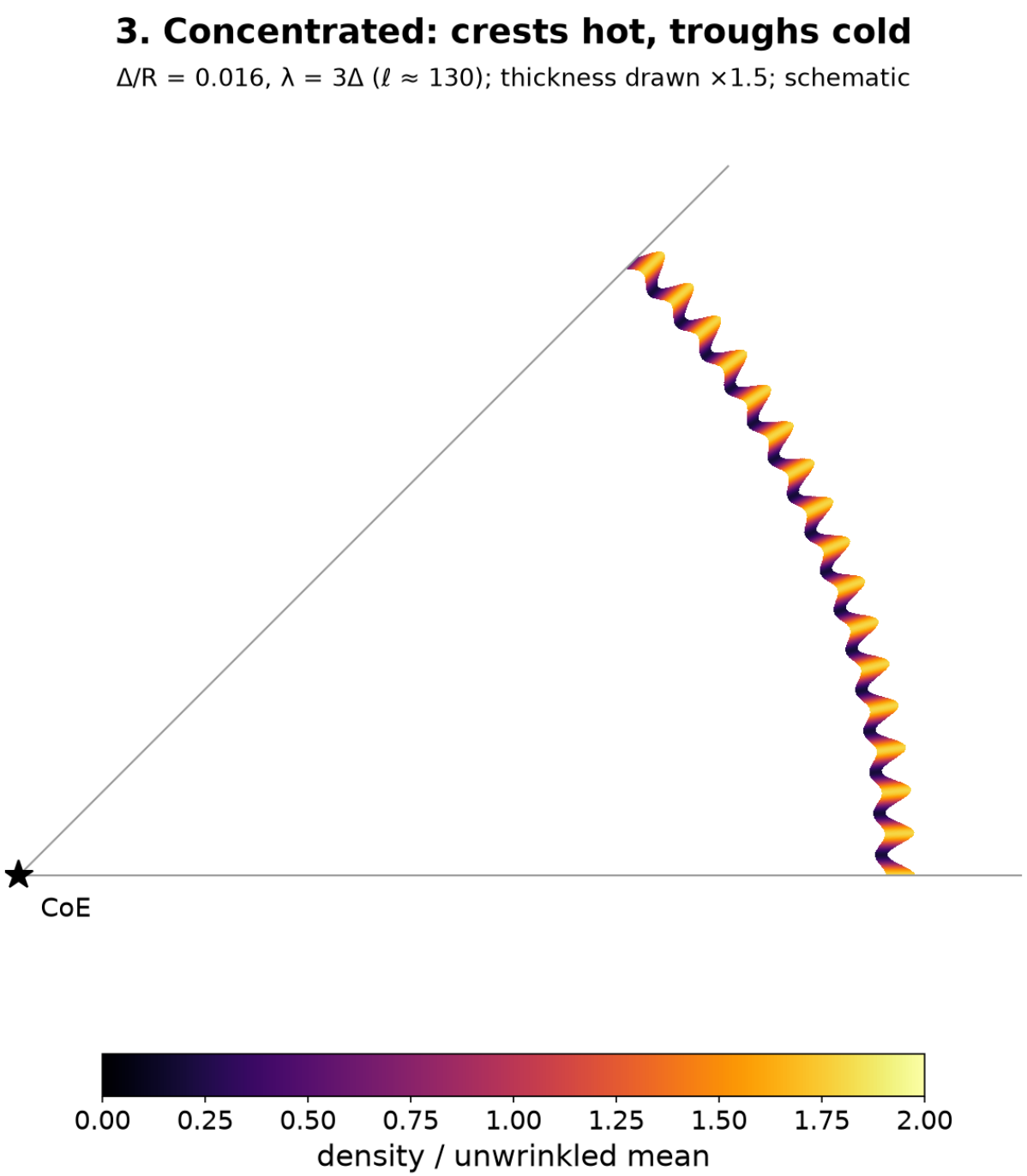


Figure 108: Concentrated: crests dense (hot), troughs thinned (cold). Colour: density relative to the unwrinkled mean. Schematic.

Where the mass gathers. Figs. 106–108 show the same shell as a heat map of mass density: as the wrinkle grows, material moves into the crests (hot) and out of the troughs (cold).

13.1 Potential geometry of clumps and voids

Relative to the mean, an overdensity is a potential well and an underdensity a potential hill (Fig. 109). Two overdensities merge into one wider well, two underdensities into one wider hill. A clump next to a void forms a potential dipole: the steepest slope runs between them, so matter flows down the gradient out of the void and into the clump, deepening both.

13.2 From flat density to clump and void

Figs. 110–113 follow one patch of a thin shell. A flat shell carries a seed of about 10^{-3}; the crest becomes a shallow well and the trough a shallow hill; the dipole drains the trough into the crest; linear growth ($\delta \propto a$) carries the $+1.5 \times 10^{-3}$ seed to $\delta_c = 1.686$ (clump) and the -2.5×10^{-3} seed to $\delta_v = -2.81$ (void) by today. The growth continues after the shell instability has stopped.

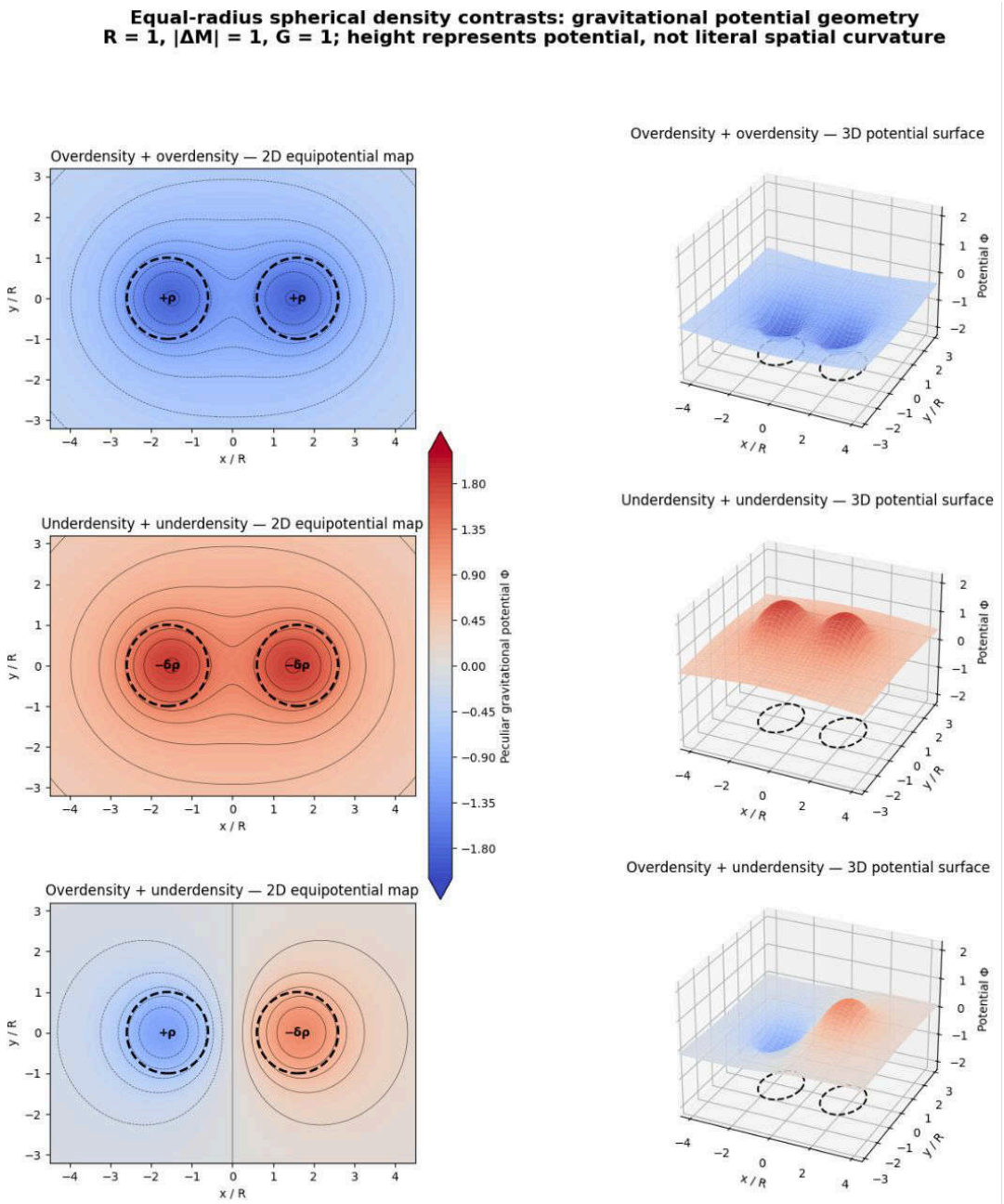


Figure 109: Equal-radius spherical density contrasts ($R = 1$, $|\Delta M| = 1$, $G = 1$): 2D equipotential maps (left) and 3D potential surfaces (right; height is potential, not spatial curvature) for over+over, under+under and over+under pairs.

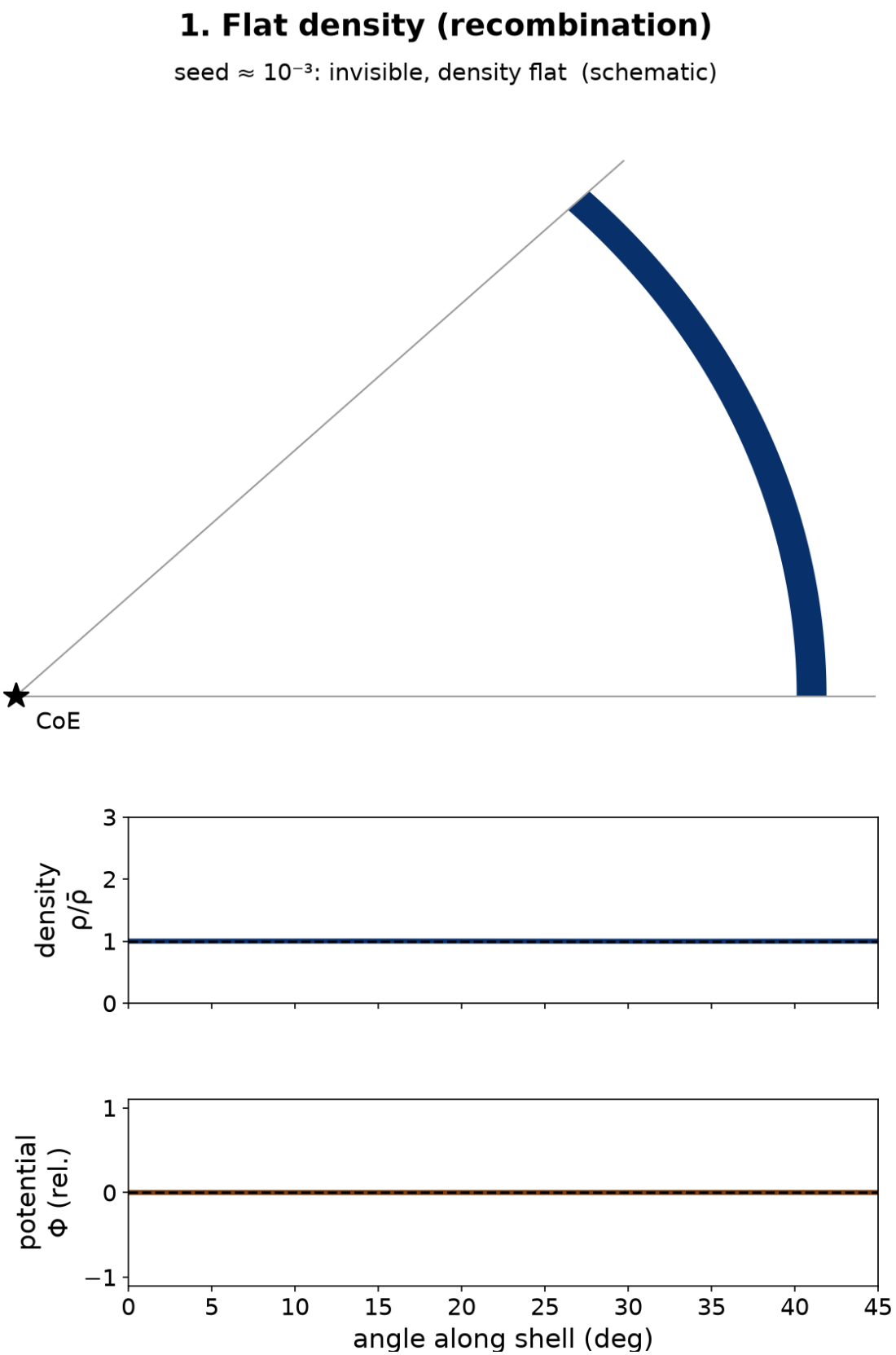


Figure 110: Flat density at recombination: the seed is too small to see. Graphs: density along the shell and potential. Schematic.

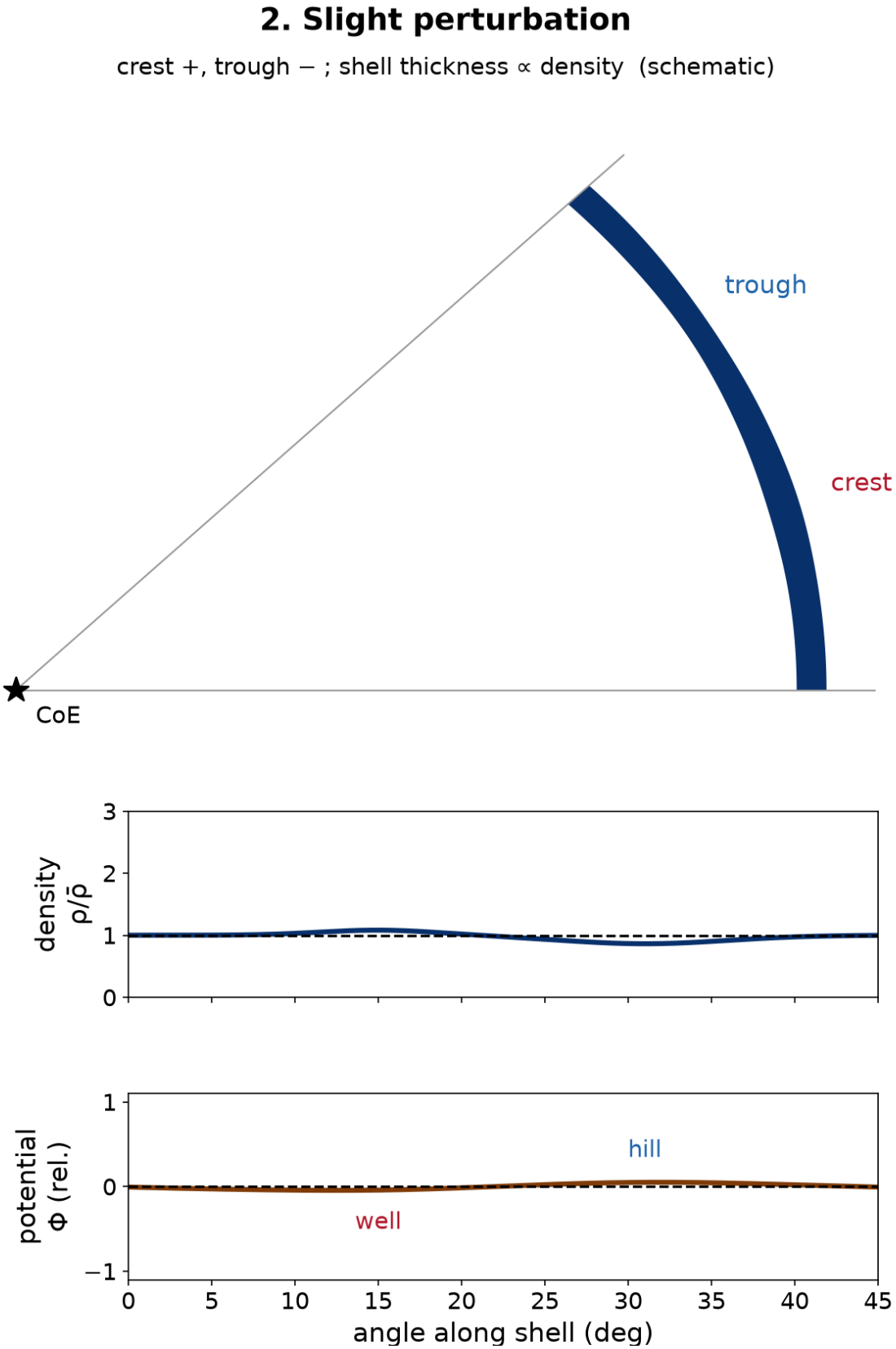


Figure 111: Slight perturbation: crest above, trough below the mean; a shallow well and hill appear. Schematic.

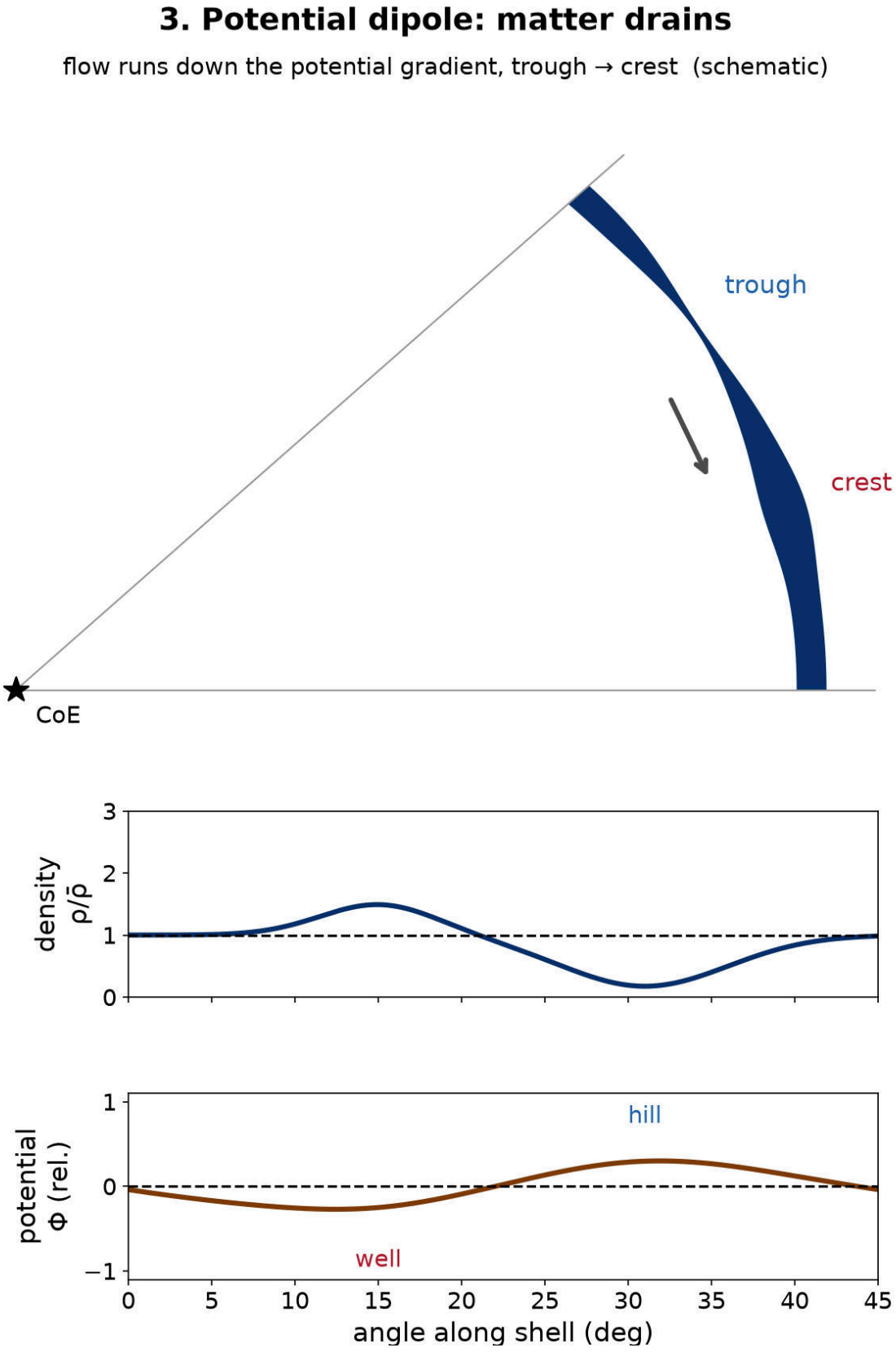


Figure 112: Potential dipole: matter drains down the gradient from trough to crest (arrow). Schematic.

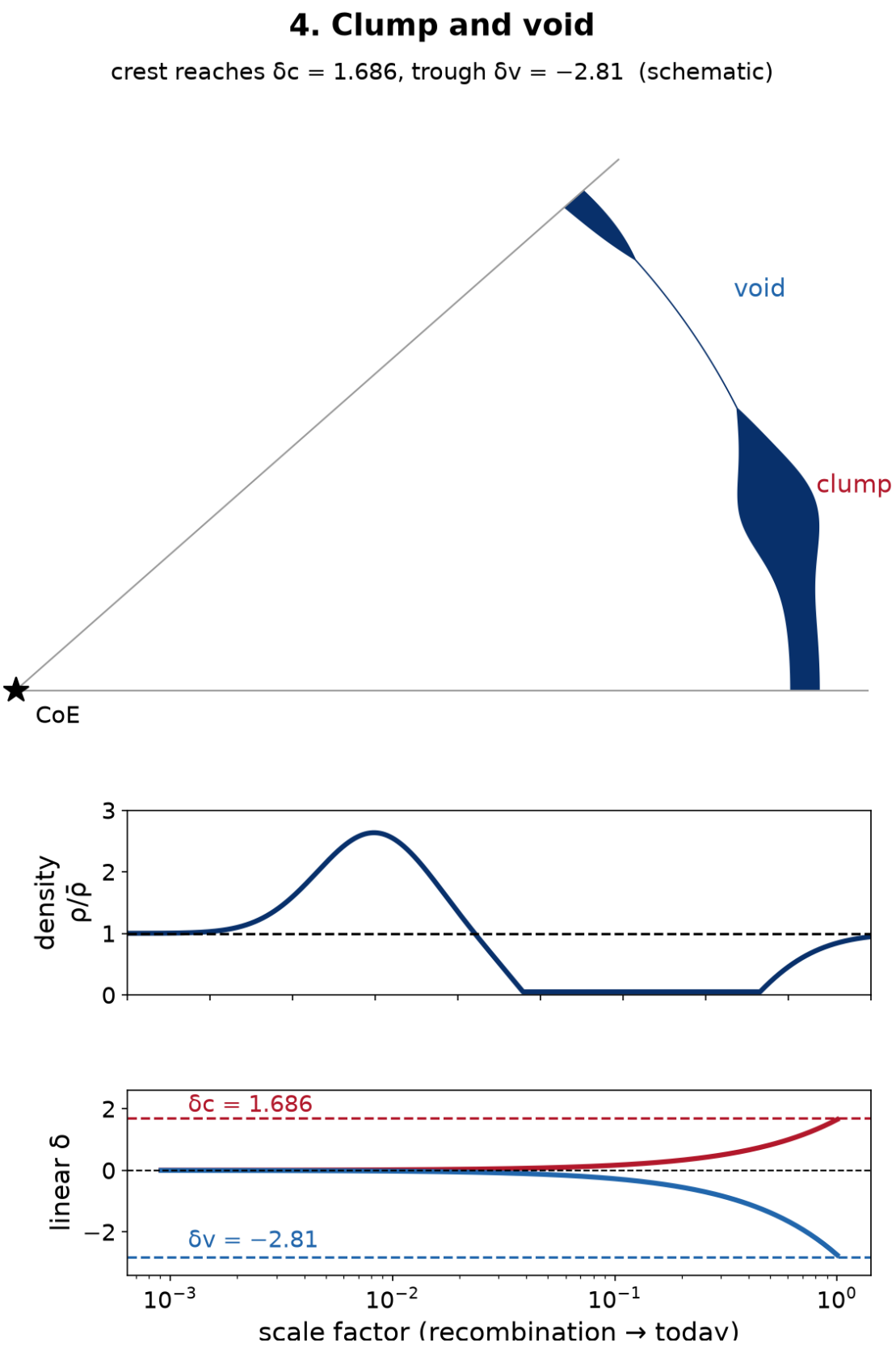


Figure 113: Clump and void: the crest reaches $\delta_c = 1.686$, the trough $\delta_v = -2.81$; bottom graph shows linear growth of the seeds. Schematic.

A Summary Table

Table 1: Phenomena, class·tag, key control parameter, forcing function, and geometry.

Phenomenon	Class	Key control	Forcing function	Geometry	How the geometry changes with the control parameter / forcing
Chladni	ref.·D	drive frequency ω	$D\nabla^4 w + \rho h\,\partial_t^2 w = f\cos\omega t$, $\omega_n = k_n^2\sqrt{D/\rho h}$	nodal lines	Higher frequency excites higher modes with more nodal lines; spacing $\propto \omega^{-1/2}$. Coarse sand marks nodes, fine powder antinodes. Amplitude only sharpens the lines.
Faraday waves	A·D	forcing Δ	$\ddot{a} + \gamma\dot{a} + \omega_s^2(q)[1 + 2\Delta\cos 2\omega t]a = 0$, $\omega_s(q_0) = \omega$, $\Delta_c = \gamma/\omega_0$	standing waves	Wavelength from the subharmonic resonance ($\lambda \propto f^{-2/3}$ for capillary waves). Squares near onset (low viscosity), hexagons (high viscosity); quasipatterns with two-frequency forcing; larger Δ gives defects and chaos.
Rayleigh–Bénard	A·D	Rayleigh number	$\mathrm{Ra} = \dfrac{g\alpha\Delta T\, d^3}{\nu\kappa}$, $\mathrm{Ra}_c = 1707.76$, $k_c d = 3.117$	rolls / hexagons	Rolls of width $\approx d$ at onset, then a stable band (Busse balloon) bounded by cross-roll, zigzag, Eckhaus and skewed-varicose instabilities; hexagons with broken up–down symmetry; spiral-defect chaos near $\mathrm{Pr} \approx 1$.
Bénard–Marangoni	A·D	Marangoni number	$\mathrm{Ma} = \dfrac{\Delta T\, d\,\lvert\partial\sigma/\partial T\rvert}{\rho\nu\kappa}$, $\mathrm{Ma}_c \approx 80$	hexagons	Hexagons $\approx 3d$ across from a weakly subcritical onset; thinner layers give smaller cells; squares far above onset.
Taylor–Couette	A·D	R_i or Ta	$R_i = \dfrac{a(b-a)\Omega_i}{\nu}$, narrow gap $\mathrm{Ta}_c \approx 1708$	toroidal rolls	Axial period $\approx 2d$ at onset; wavy vortices at $R_i/R_{ic} \approx 1.1$–1.2, then modulated waves and turbulent vortices; counter-rotation gives spirals.
Electro-convection	A·D	ε, frequency	$\varepsilon = V^2/V_c^2 - 1$	rolls	Rolls $\sim d$; oblique below the Lifshitz frequency, normal above; finer dielectric rolls above the cutoff; larger ε gives defect turbulence and dynamic scattering.
Turing (CIMA)	A·D	$D_{\mathrm{inh}}/D_{\mathrm{act}}$	$D_v f_u + D_u g_v > 2\sqrt{D_u D_v \det J}$, $k_c^2 = \sqrt{\det J/D_u D_v}$	spots / stripes	Intrinsic period, independent of reactor size; feed concentrations move the pattern between hexagonal spots, stripes and mixed states.
Langmuir circulation	A·D	Stokes drift, La_t	$\partial_t\boldsymbol{\omega} = \nabla\times[(\mathbf{u}+\mathbf{u}_s)\times\boldsymbol{\omega}] + \nu\nabla^2\boldsymbol{\omega}$	roll pairs	Windrows 2–3 mixed-layer depths apart; smaller La_t gives stronger, more regular cells; Y-junctions as the wind changes.
Granular layers	A·D	Γ, f, depth	$\Gamma = A\omega^2/g$, onset $\Gamma \approx 2.4$	squares / stripes / oscillons	Squares subcritical for $f < f^* \approx 45\,\mathrm{Hz}$ ($f^* \propto d^{-1/2}$), stripes supercritical above; oscillons near f^*; hexagons for $\Gamma > 4$, chaos for $\Gamma > 7$. Wavelength grows with depth, falls with f.
Rosensweig	B·E	B/B_c	$\mu_0 M_c^2 = 2(1+1/r_p)\sqrt{g\Delta\rho\sigma}$, $\lambda = 2\pi\sqrt{\sigma/g\Delta\rho}$	hexagonal spikes	Spacing near the capillary length, weakly field dependent. Subcritical hexagons, then squares, both hysteretic; taller spikes at higher field.
Magnetic domains	B·E	Bond number, t, H, T	thick: $w \propto \sqrt{\ell_c t}$; ultrathin: $d^* = a\exp(b\gamma/\mu_0 M^2)$	stripes / bubbles	Ultrathin: width $\propto \exp(b\gamma/\mu_0 M^2)$, falling steeply as the anisotropy (wall energy) $\to 0$. A perpendicular field takes stripes to bubbles to saturation by first-order transitions; history dependent.

Classes (how the scale is selected). **A** bulk finite-wavelength instability above a threshold. **B** equilibrium modulated phase: short-range cost vs. long-range gain gives a free-energy minimum at finite period. **C** coarsening: no equilibrium scale, $L \propto t^n$. **D** interfacial instability: long waves unstable, surface tension cuts off short ones. **E** fronts and waves invading an unstable, excitable or supersaturated state. ref. = forced linear mode (Chladni). **Tags** (what sustains it): **D** driven (open), **R** relaxing (closed transient), **E** equilibrium.

Table 1 (continued)

Phenomenon	Class	Key control	Forcing function	Geometry	How the geometry changes with the control parameter / forcing
Spinodal	C·R	quench depth $F''(\phi_0)$	$\partial_t\phi = \nabla\cdot M\nabla(F' - \varepsilon^2\nabla^2\phi)$, $k_m = \sqrt{-F''/2}/\varepsilon$	bicontinuous	Transient; coarsens toward full demixing. Initial scale shrinks for deeper quenches; then $L \propto t^{1/3}$ (diffusive) or $\propto t$ (viscous); off-critical mixtures form droplets.
Soap foam	C·R	gas, liquid fraction	$\Delta p = 4\sigma H$, $dA_n/dt = \frac{\pi}{3}k_D(n-6)$	cellular network	120° junctions fixed (local equilibrium); mean area $\propto t$ in 2D; wetter foams unjam near 36% liquid.
Kelvin–Helmholtz	D·D	velocity jump, Ri	$\sigma^2 = \frac{\rho_1\rho_2 k^2U^2}{(\rho_1+\rho_2)^2} - gk\frac{\rho_2-\rho_1}{\rho_1+\rho_2}$, $\mathrm{Ri} < \frac{1}{4}$	billows	Wavelength $\approx 7\times$ shear-layer thickness; no billows for Ri > 1/4; downstream pairing doubles the wavelength.
Saffman–Taylor	D·D	capillary number	$\sigma(k) = \frac{k}{\mu_1+\mu_2}\left[(\mu_2-\mu_1)U - \frac{b^2\gamma}{12}k^2\right]$	fingers	$\lambda_m = \pi b/\sqrt{\mathrm{Ca}}$: faster injection, narrower fingers, more tip splitting; channel finger tends to half width; radial cells give dense branching.
Rayleigh–Taylor	D·R	Atwood number	$\sigma^2 = Agk - \frac{\gamma k^3}{\rho_h+\rho_l}$	bubbles / spikes	$\lambda_m = \sqrt{3}\,\lambda_c$, $\lambda_c = 2\pi\sqrt{\gamma/\Delta\rho g}$; larger A more asymmetric; mixing zone $h \approx \alpha A g t^2$; thin films give hexagonal drops spreading as a front.
Dendrites	D·R/D	undercooling Δ	$\sigma = vk[1 - \frac{1}{2}(1+\beta)d_0\ell k^2]$, $\rho^2 v = 2Dd_0/\sigma^*$	dendrites	Tip radius $\rho \propto V^{-1/2}$ via solvability (σ^* depends on anisotropy); finer sidebranches at larger undercooling; anisotropy sets the arm symmetry; directional growth goes planar → cells → dendrites.
Bacterial colonies	D on E·D	nutrient, agar	$\partial_t\beta = \delta\nabla\cdot(\beta^k\nabla\beta) + \alpha\beta^m$, $\partial_t\alpha = \nabla^2\alpha - \alpha\beta^m$	branching fronts	Low nutrient/hard agar: DLA-like branches; more nutrient: dense branching; soft agar: compact disks or rings. Branch width ∼ nutrient diffusion length.
BZ waves	E·R/D	excitability (acid, bromate)	Oregonator: $\partial_t u_1 = \eta^{-1}\left[u_1(1-u_1) - bu_2\frac{u_1-a}{u_1+a}\right] + D_1\nabla^2 u_1$, $\partial_t u_2 = u_1 - u_2 + D_2\nabla^2 u_2$	targets / spirals	Speed $c \propto \sqrt{Dk}$ (Luther); higher excitability gives tighter spirals, then meandering and breakup; a closed dish runs down in ∼ 100 periods.
River–ocean fronts	E·D	discharge, R_d	$R_d = \sqrt{g'h}/f$, $\frac{D_g}{Dt}\lvert\nabla_h b\rvert^2 = 2\mathbf{Q}\cdot\nabla_h b$	density front	Plume and bulge scale with R_d; larger discharge or buoyancy gives a wider bulge; wind and tides broaden or detach the front; front width set where frontogenesis is arrested.
Liesegang rings	E·R	concentrations	$x_{n+1}/x_n \to 1+p$, $x_n \propto \sqrt{t_n}$	bands / rings	Spacing grows geometrically; stronger outer electrolyte lowers p (Matalon–Packter); bands in tubes, rings in plates, shells in 3D; kinetically frozen.
Thin-shell wrinkling	D·R	deceleration g_{eff}, Δ/R	$\sigma = \sqrt{Akg_{\mathrm{eff}}}$, $g_{\mathrm{eff}} = -\ddot{R}$, $\lambda \approx 3\Delta$	wrinkles / clumps	Thinner shells give finer wrinkles ($\ell_{\max} \approx 2\pi R/\Delta$); stronger deceleration grows them faster; crests become clumps, troughs voids.

Classes (how the scale is selected). **A** bulk finite-wavelength instability above a threshold. **B** equilibrium modulated phase: short-range cost vs. long-range gain gives a free-energy minimum at finite period. **C** coarsening: no equilibrium scale, $L \propto t^n$. **D** interfacial instability: long waves unstable, surface tension cuts off short ones. **E** fronts and waves invading an unstable, excitable or supersaturated state. ref. = forced linear mode (Chladni). **Tags** (what sustains it): **D** driven (open), **R** relaxing (closed transient), **E** equilibrium.

Does everything have a centre? A philosophical note

Chris Alameddin

1. The thesis

Every physical object or system that has ever been observed has a centre. Depending on what is being described, that centre may be geometric (the centroid of a shape), dynamical (the centre of mass), or kinematic (the centre-of-momentum frame, in which the total momentum vanishes). A galaxy, a cluster, a star, a supernova remnant, a planetary system and a laboratory sample all admit one or more of these. Even an irregular or diffuse distribution of matter has a well-defined first moment,

$$\mathbf{x}_{\mathrm{c}} = \frac{\int \rho(\mathbf{x})\, \mathbf{x}\, d^3x}{\int \rho(\mathbf{x})\, d^3x}, \tag{1}$$

provided the total mass is finite. The integral fails to define a centre only when the distribution is infinite in extent or exactly uniform without bound.

The second half of the thesis is empirical. Over many discussions, the author has asked for counter-examples, including from several AI systems. Every example proposed of something *without* a centre has turned out to be a mathematical abstraction, and none of them has ever been observed by any scientist as a physical object.

2. The candidate counter-examples

The usual candidates fall into a few classes:

- **Infinite uniform space, an infinite plane or an infinite line.** These are idealisations. No finite measurement can establish that something extends without bound, and any observed region of them is finite and so has a centroid.
- **The surface of a sphere, taken as an intrinsic two-dimensional manifold.** No point on the surface is special. But every physical sphere (a balloon, a planet, a soap film) is embedded in three-dimensional space and has a centre there. A centreless sphere exists only if you deliberately forget the embedding.
- **Fractal limits.** A physical fractal-like structure (a coastline, a turbulent cascade, galaxy clustering on intermediate scales) is cut off at small and large scales. The ideal infinite-scale fractal is never observed.
- **The homogeneous, isotropic FLRW universe.** Under the cosmological principle every point is equivalent, so there is no centre. This is the case that matters here, and it is taken up next.

In each case the absence of a centre comes from an idealising step: infinite extent, exact uniformity, or ignoring the embedding. It does not come from a measurement.

3. "No centre" in cosmology is an assumption

The standard model's statement that the universe has no centre is not an observational result. It follows from the cosmological principle, which assumes spatial homogeneity and isotropy and thereby selects the FLRW metrics [5, 1]. Two statements are often merged and should be kept apart:

1. "The CMB is very nearly isotropic about us." This is data.

2. "Therefore there is no centre anywhere." This is not a deduction from (1). Isotropy about one point plus the Copernican assumption that we are not special implies homogeneity, but that assumption is a philosophical preference, not a measurement [1, 2].

Spherically symmetric inhomogeneous models such as Lemaître–Tolman–Bondi solutions, and the earlier kinematic cosmology of Milne [4], show that a universe with a centre is mathematically consistent with general relativity and with many of the same observations. Ellis and collaborators have stressed that what we see on large scales depends on how local structure is averaged and fitted to a smooth model [2, 3]. Whether a centre exists is therefore an empirical question that the cosmological principle settles by assumption.

4. Connection to the CoE model

The papers in this collection take the opposite starting point. If the universe began as an expansion from a point, a finite expansion necessarily has a centre, the centre of expansion (CoE). We sit a distance D from it. Nothing in this requires local speeds above c: bulk recession of a distant shell faster than c is expansion, not local motion.

On this view the thesis of Section 1 applies to the universe itself. A universe without a centre would be the one physical system that behaves like the mathematical abstractions of Section 2, and would be the only such system ever claimed.

5. What could test it

A centre is not just a philosophical position. An off-centre observer should see direction-dependent signatures:

- a dependence of the local Hubble rate on direction relative to the CoE, measured in the papers as $A/A_{\rm all}(\theta)$, the per-angle expansion rate divided by the full-sky mean, compared with cluster-scaling anisotropy results such as those of Migkas et al. [6];
- dipoles that do not line up with the kinematic CMB dipole, such as the quasar number-count dipole reported by Secrest et al. [7];
- a spread in the inferred H_0 with direction and distance, which bears on the Hubble tension.

These signals may also have conventional explanations: survey systematics, local bulk flows, or ordinary structure. A null result at sufficient precision would push the CoE far away or remove it. A consistent, aligned set of anisotropies would be evidence that the universe, like everything else observed, has a centre.

6. Balance

None of this proves that the universe has a centre. The argument is an inductive one: every observed system has a centre, and every centreless example is an abstraction. Induction can fail, and the universe may be unique. But the burden is symmetric. "No centre" should be treated as a hypothesis to be tested, on the same footing as the CoE, and not as a settled result.

References

References are grouped by paper. Each group keeps that paper's own numbering, so the bracketed citations in the original pages refer to the matching group below.

1. CoE cosmology: Apparent acceleration from an off-centre finite geometry

[1] A. G. Riess et al., Astron. J. 116, 1009 (1998). doi:10.1086/300499

[2] S. Perlmutter et al., Astrophys. J. 517, 565 (1999). doi:10.1086/307221

[3] G. F. R. Ellis and W. R. Stoeger, Class. Quantum Grav. 4, 1697 (1987). doi:10.1088/0264-9381/4/6/025

[4] E. Hubble, Proc. Natl. Acad. Sci. USA 15, 168 (1929). doi:10.1073/pnas.15.3.168

[5] K. Migkas et al., Astron. Astrophys. 636, A15 (2020). doi:10.1051/0004-6361/201936602, arXiv:2004.03305

[6] L. I. Sedov, Prikl. Mat. Mekh. [J. Appl. Math. Mech.] 10, 241 (1946). *[no online link found]*

[7] G. I. Taylor, Proc. R. Soc. Lond. A 201, 159 (1950). doi:10.1098/rspa.1950.0049

[8] G. Lemaître, Ann. Soc. Sci. Bruxelles A53, 51 (1933); English translation: Gen. Rel. Grav. 29, 641 (1997). doi:10.1023/A:1018855621348 (translation)

[9] R. C. Tolman, Proc. Natl. Acad. Sci. USA 20, 169 (1934). doi:10.1073/pnas.20.3.169

[10] H. Bondi, Mon. Not. R. Astron. Soc. 107, 410 (1947). doi:10.1093/mnras/107.5-6.410

[11] Planck Collaboration, Astron. Astrophys. 641, A7 (2020). doi:10.1051/0004-6361/201935201

[12] N. J. Secrest, S. von Hausegger, M. Rameez, R. Mohayaee, S. Sarkar, and J. Colin, Astrophys. J. Lett. 908, L51 (2021). doi:10.3847/2041-8213/abdd40, arXiv:2009.14826

The supplement on the exact relativistic (LTB) calculation cites no additional references.

2. Reading the CMB fingerprint of the CoE shell model

[1] J. Clampitt and B. Jain, Lensing measurements of the mass distribution in SDSS voids, MNRAS 454, 3357 (2015). doi:10.1093/mnras/stv2215, arXiv:1404.1834

[2] J. Tinker et al., Toward a halo mass function for precision cosmology: The limits of universality, ApJ 688, 709 (2008). doi:10.1086/591439, arXiv:0803.2706

[3] E. S. Rykoff et al., redMaPPer. I. algorithm and SDSS DR8 catalog, ApJ 785, 104 (2014). doi:10.1088/0004-637X/785/2/104, arXiv:1303.3562

[4] M. Simet et al., Weak lensing measurement of the mass–richness relation of SDSS redMaPPer clusters, MNRAS 466, 3103 (2017). doi:10.1093/mnras/stw3250, arXiv:1603.06953

[5] L. J. Liivamägi, E. Tempel, and E. Saar, SDSS DR7 superclusters. The catalogues, A&A 539, A80 (2012). doi:10.1051/0004-6361/201016288, arXiv:1012.1989

[6] Q. Mao et al., A cosmic void catalog of SDSS DR12 BOSS galaxies, ApJ 835, 161 (2017). doi:10.3847/1538-4357/835/2/161, arXiv:1602.02771

[7] N. Hamaus, P. M. Sutter, and B. D. Wandelt, Universal density profile for cosmic voids, PRL 112, 251302 (2014). doi:10.1103/PhysRevLett.112.251302, arXiv:1403.5499

[8] I. Szapudi et al., Detection of a supervoid aligned with the cold spot of the cosmic microwave background, MNRAS 450, 288 (2015). doi:10.1093/mnras/stv488, arXiv:1405.1566

[9] A. Kovács and J. García-Bellido, Cosmic troublemakers: the cold spot, the eridanus supervoid, and the great walls, MNRAS 462, 1882 (2016). doi:10.1093/mnras/stw1752, arXiv:1511.09008

[10] R. C. Keenan, A. J. Barger, and L. L. Cowie, Evidence for a ∼300 megaparsec scale under-density in the local galaxy distribution, ApJ 775, 62 (2013). doi:10.1088/0004-637X/775/1/62, arXiv:1304.2884

[11] J. R. Whitbourn and T. Shanks, The local hole revealed by galaxy counts and redshifts, MNRAS 437, 2146 (2014). doi:10.1093/mnras/stt2024, arXiv:1307.4405

[12] W. D. Kenworthy, D. Scolnic, and A. Riess, The local perspective on the Hubble tension: Local structure does not impact measurement of the Hubble constant, ApJ 875, 145 (2019). doi:10.3847/1538-4357/ab0ebf, arXiv:1901.08681

[13] J. Carrick, S. J. Turnbull, G. Lavaux, and M. J. Hudson, Cosmological parameters from the comparison of peculiar velocities with predictions from the 2M++ density field, MNRAS 450, 317 (2015). doi:10.1093/mnras/stv547, arXiv:1504.04627

[14] M. I. Scrimgeour et al., The WiggleZ dark energy survey: the transition to large-scale cosmic homogeneity, MNRAS 425, 116 (2012). doi:10.1111/j.1365-2966.2012.21402.x, arXiv:1205.6812

[15] N. J. Secrest, S. von Hausegger, M. Rameez, R. Mohayaee, S. Sarkar, and J. Colin, A test of the cosmological principle with quasars, ApJL 908, L51 (2021). doi:10.3847/2041-8213/abdd40, arXiv:2009.14826

[16] K. Migkas, G. Schellenberger, T. H. Reiprich, F. Pacaud, M. E. Ramos-Ceja, and L. Lovisari, Probing cosmic isotropy with a new X-ray galaxy cluster sample through the L_X–T scaling relation, A&A 636, A15 (2020). doi:10.1051/0004-6361/201936602, arXiv:2004.03305

[17] D. J. Eisenstein and W. Hu, Baryonic features in the matter transfer function, ApJ 496, 605 (1998). doi:10.1086/305424, arXiv:astro-ph/9709112

[18] J. Einasto, M. Einasto, S. Gottlöber, V. Müller, V. Saar, A. A. Starobinsky, E. Tago, D. Tucker, H. Andernach, and P. Frisch, A 120-Mpc periodicity in the three-dimensional distribution of galaxy superclusters, Nature 385, 139 (1997). doi:10.1038/385139a0, arXiv:astro-ph/9701018

[19] J. Einasto, M. Einasto, P. Frisch, S. Gottlöber, V. Müller, V. Saar, A. A. Starobinsky, E. Tago, D. Tucker, and H. Andernach, The supercluster–void network II. An oscillating cluster correlation function, MNRAS 289, 801 (1997). doi:10.1093/mnras/289.4.801, arXiv:astro-ph/9704127

[20] E. Jennings, Y. Li, and W. Hu, The abundance of voids and the excursion set formalism, MNRAS 434, 2167 (2013). doi:10.1093/mnras/stt1169, arXiv:1304.6087

[21] R. K. Sheth and R. van de Weygaert, A hierarchy of voids: much ado about nothing, MNRAS 350, 517 (2004). doi:10.1111/j.1365-2966.2004.07661.x, arXiv:astro-ph/0311260

[22] R. K. Sachs and A. M. Wolfe, Perturbations of a cosmological model and angular variations of the microwave background, ApJ 147, 73 (1967). doi:10.1086/148982

[23] M. J. Rees and D. W. Sciama, Large-scale density inhomogeneities in the universe, Nature 217, 511 (1968). doi:10.1038/217511a0

3. Pattern formation (v5)

[1] M. C. Cross, P. C. Hohenberg, Pattern formation outside of equilibrium, Rev. Mod. Phys. 65, 851 (1993). doi:10.1103/RevModPhys.65.851

[2] D. Andelman, R. E. Rosensweig, Modulated phases: review and recent results, J. Phys. Chem. B 113, 3785 (2009). doi:10.1021/jp807770n

[3] J. S. Langer, Instabilities and pattern formation in crystal growth, Rev. Mod. Phys. 52, 1 (1980). doi:10.1103/RevModPhys.52.1

[4] M. Seul, D. Andelman, Domain shapes and patterns: the phenomenology of modulated phases, Science 267, 476 (1995). doi:10.1126/science.267.5197.476

[5] P. C. Hohenberg, A. P. Krekhov, An introduction to the Ginzburg–Landau theory of phase transitions and nonequilibrium patterns, Phys. Rep. 572, 1 (2015). doi:10.1016/j.physrep.2015.01.001, arXiv:1410.7285

[6] J. Swift, P. C. Hohenberg, Hydrodynamic fluctuations at the convective instability, Phys. Rev. A 15, 319 (1977). doi:10.1103/PhysRevA.15.319

[7] W. van Saarloos, Front propagation into unstable states, Phys. Rep. 386, 29 (2003). doi:10.1016/j.physrep.2003.08.001, arXiv:cond-mat/0308540

[8] I. S. Aranson, L. S. Tsimring, Patterns and collective behavior in granular media: theoretical concepts, Rev. Mod. Phys. 78, 641 (2006). doi:10.1103/RevModPhys.78.641, arXiv:cond-mat/0507419

[9] A. J. Bray, Theory of phase-ordering kinetics, Adv. Phys. 43, 357 (1994). doi:10.1080/00018739400101505, arXiv:cond-mat/9501089

[10] T. D. Rossing, Chladni's law for vibrating plates, Am. J. Phys. 50, 271 (1982). doi:10.1119/1.12866

[11] W. S. Edwards, S. Fauve, Patterns and quasi-patterns in the Faraday experiment, J. Fluid Mech. 278, 123 (1994). doi:10.1017/S0022112094003642

[12] X. Liu, X. Wang, Polygonal patterns of Faraday water waves analogous to collective excitations in Bose–Einstein condensates, Nat. Phys. 20, 287 (2024). doi:10.1038/s41567-023-02294-y

[13] S. W. Morris, E. Bodenschatz, D. S. Cannell, G. Ahlers, Spiral defect chaos in large aspect ratio Rayleigh–Bénard convection, Phys. Rev. Lett. 71, 2026 (1993). doi:10.1103/PhysRevLett.71.2026

[14] J. R. A. Pearson, On convection cells induced by surface tension, J. Fluid Mech. 4, 489 (1958). doi:10.1017/S0022112058000616

[15] K. Eckert, M. Bestehorn, A. Thess, Square cells in surface-tension-driven Bénard convection: experiment and theory, J. Fluid Mech. 356, 155 (1998). doi:10.1017/S0022112097007842

[16] C. D. Andereck, S. S. Liu, H. L. Swinney, Flow regimes in a circular Couette system with independently rotating cylinders, J. Fluid Mech. 164, 155 (1986). doi:10.1017/S0022112086002513

[17] L. Kramer, W. Pesch, Convection instabilities in nematic liquid crystals, Ann. Rev. Fluid Mech. 27, 515 (1995). doi:10.1146/annurev.fl.27.010195.002503

[18] Y. Sasaki, V. S. R. Jampani, C. Tanaka, N. Sakurai, S. Sakane, K. V. Le, F. Araoka, H. Orihara, Large-scale self-organization of reconfigurable topological defect networks in nematic liquid crystals, Nat. Commun. 7, 13238 (2016). doi:10.1038/ncomms13238

[19] A. M. Turing, The chemical basis of morphogenesis, Phil. Trans. R. Soc. B 237, 37 (1952). doi:10.1098/rstb.1952.0012

[20] V. Castets, E. Dulos, J. Boissonade, P. De Kepper, Experimental evidence of a sustained standing Turing-type nonequilibrium chemical pattern, Phys. Rev. Lett. 64, 2953 (1990). doi:10.1103/PhysRevLett.64.2953

[21] Q. Ouyang, H. L. Swinney, Transition from a uniform state to hexagonal and striped Turing patterns, Nature 352, 610 (1991). doi:10.1038/352610a0

[22] I. Langmuir, Surface motion of water induced by wind, Science 87, 119 (1938). doi:10.1126/science.87.2250.119

[23] A. D. D. Craik, S. Leibovich, A rational model for Langmuir circulations, J. Fluid Mech. 73, 401 (1976). doi:10.1017/S0022112076001420

[24] S. A. Thorpe, Langmuir circulation, Ann. Rev. Fluid Mech. 36, 55 (2004). doi:10.1146/annurev.fluid.36.052203.071431

[25] D. Etling, R. A. Brown, Roll vortices in the planetary boundary layer: a review, Boundary-Layer Meteorol. 65, 215 (1993). doi:10.1007/BF00705527

[26] F. Melo, P. B. Umbanhowar, H. L. Swinney, Hexagons, kinks, and disorder in oscillated granular layers, Phys. Rev. Lett. 75, 3838 (1995). doi:10.1103/PhysRevLett.75.3838

[27] P. B. Umbanhowar, F. Melo, H. L. Swinney, Localized excitations in a vertically vibrated granular layer, Nature 382, 793 (1996). doi:10.1038/382793a0

[28] M. D. Cowley, R. E. Rosensweig, The interfacial stability of a ferromagnetic fluid, J. Fluid Mech. 30, 671 (1967). doi:10.1017/S0022112067001697

[29] H. Knieling, R. Richter, I. Rehberg, G. Matthies, A. Lange, Growth of surface undulations at the Rosensweig instability, Phys. Rev. E 76, 066301 (2007). doi:10.1103/PhysRevE.76.066301, arXiv:0801.0140

[30] C. Kittel, Theory of the structure of ferromagnetic domains in films and small particles, Phys. Rev. 70, 965 (1946). doi:10.1103/PhysRev.70.965

[31] M. Kronseder, T. N. G. Meier, M. Zimmermann, M. Buchner, M. Vogel, C. H. Back, Real-time observation of domain fluctuations in a two-dimensional magnetic model system, Nat. Commun. 6, 6832 (2015). doi:10.1038/ncomms7832

[32] J. A. Glazier, S. P. Gross, J. Stavans, Dynamics of two-dimensional soap froths, Phys. Rev. A 36, 306 (1987). doi:10.1103/PhysRevA.36.306

[33] J. W. Miles, On the stability of heterogeneous shear flows, J. Fluid Mech. 10, 496 (1961). doi:10.1017/S0022112061000305

[34] L. N. Howard, Note on a paper of John W. Miles, J. Fluid Mech. 10, 509 (1961). doi:10.1017/S0022112061000317

[35] A. Michalke, On the inviscid instability of the hyperbolic-tangent velocity profile, J. Fluid Mech. 19, 543 (1964). doi:10.1017/S0022112064000908

[36] P. G. Saffman, G. I. Taylor, The penetration of a fluid into a porous medium or Hele-Shaw cell containing a more viscous liquid, Proc. R. Soc. Lond. A 245, 312 (1958). doi:10.1098/rspa.1958.0085

[37] G. M. Homsy, Viscous fingering in porous media, Ann. Rev. Fluid Mech. 19, 271 (1987). doi:10.1146/annurev.fl.19.010187.001415

[38] D. H. Sharp, An overview of Rayleigh–Taylor instability, Physica D 12, 3 (1984). doi:10.1016/0167-2789(84)90510-4

[39] W. W. Mullins, R. F. Sekerka, Stability of a planar interface during solidification of a dilute binary

alloy, J. Appl. Phys. 35, 444 (1964). doi:10.1063/1.1713333
[40] S.-C. Huang, M. E. Glicksman, Fundamentals of dendritic solidification I. Steady-state tip growth, Acta Metall. 29, 701 (1981). doi:10.1016/0001-6160(81)90115-2
[41] M. Matsushita, H. Fujikawa, Diffusion-limited growth in bacterial colony formation, Physica A 168, 498 (1990). doi:10.1016/0378-4371(90)90402-E
[42] T. A. Witten, L. M. Sander, Diffusion-limited aggregation, a kinetic critical phenomenon, Phys. Rev. Lett. 47, 1400 (1981). doi:10.1103/PhysRevLett.47.1400
[43] E. Ben-Jacob, O. Schochet, A. Tenenbaum, I. Cohen, A. Czirók, T. Vicsek, Generic modelling of cooperative growth patterns in bacterial colonies, Nature 368, 46 (1994). doi:10.1038/368046a0
[44] A. N. Zaikin, A. M. Zhabotinsky, Concentration wave propagation in two-dimensional liquid-phase self-oscillating system, Nature 225, 535 (1970). doi:10.1038/225535b0
[45] J. O'Donnell, S. G. Ackleson, E. R. Levine, On the spatial scales of a river plume, J. Geophys. Res. Oceans 113, C04017 (2008). doi:10.1029/2007JC004440
[46] A. R. Horner-Devine, R. D. Hetland, D. G. MacDonald, Mixing and Transport in Coastal River Plumes, Ann. Rev. Fluid Mech. 47, 569 (2015). doi:10.1146/annurev-fluid-010313-141408
[47] J. C. McWilliams, Oceanic Frontogenesis, Ann. Rev. Mar. Sci. 13, 227 (2021). doi:10.1146/annurev-marine-032320-120725
[48] T. Antal, M. Droz, J. Magnin, Z. Rácz, Formation of Liesegang patterns: a spinodal decomposition scenario, Phys. Rev. Lett. 83, 2880 (1999). doi:10.1103/PhysRevLett.83.2880
[49] K. J. Måløy, J. Feder, T. Jøssang, Viscous fingering fractals in porous media, Phys. Rev. Lett. 55, 2688 (1985). doi:10.1103/PhysRevLett.55.2688
[50] A. Libchaber, C. Laroche, S. Fauve, Period doubling cascade in mercury, a quantitative measurement, J. Physique Lett. 43, 211 (1982). doi:10.1051/jphyslet:01982004307021100
[51] M. J. Feigenbaum, Quantitative universality for a class of nonlinear transformations, J. Stat. Phys. 19, 25 (1978). doi:10.1007/BF01020332
[52] J. P. Gollub, H. L. Swinney, Onset of turbulence in a rotating fluid, Phys. Rev. Lett. 35, 927 (1975). doi:10.1103/PhysRevLett.35.927
[53] E. N. Lorenz, Deterministic nonperiodic flow, J. Atmos. Sci. 20, 130 (1963). doi:10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2
[54] E. T. Vishniac, The dynamic and gravitational instabilities of spherical shocks, Astrophys. J. 274, 152 (1983). doi:10.1086/161433
[55] C. Michaut, C. Cavet, S. E. Bouquet, F. Roy, H. C. Nguyen, Numerical study of the Vishniac instability in supernova remnants, Astrophys. J. 759, 78 (2012). doi:10.1088/0004-637X/759/2/78
[56] R. K. Sheth, R. van de Weygaert, A hierarchy of voids: much ado about nothing, Mon. Not. R. Astron. Soc. 350, 517 (2004). doi:10.1111/j.1365-2966.2004.07661.x

Inserted Figs. 99–101 are reproduced from Michaut et al. (2012), Astrophys. J. 759, 78, doi:10.1088/0004-637X/759/2/78 (ref. [55] above). Fig. 102 is original to this collection.

4. Does everything have a centre? A philosophical note

[1] G. F. R. Ellis, "The homogeneity of the universe," *Gen. Rel. Grav.* **11**, 281 (1979). doi:10.1007/BF00759562

[2] G. F. R. Ellis, "Cosmology and local physics," *New Astron. Rev.* **46**, 645 (2002). doi:10.1016/S1387-6473(02)00234-8

[3] G. F. R. Ellis and W. R. Stoeger, "The 'fitting problem' in cosmology," *Class. Quantum Grav.* **4**, 1697 (1987). doi:10.1088/0264-9381/4/6/025

[4] E. A. Milne, *Relativity, Gravitation and World-Structure* (Oxford, 1935). archive.org/details/dli.ernet.7948

[5] W. Rindler, *Relativity: Special, General, and Cosmological*, 2nd ed. (Oxford, 2006). doi:10.1093/oso/9780198567318.001.0001

[6] K. Migkas et al., "Probing cosmic isotropy with a new X-ray galaxy cluster sample through the L_X–T scaling relation," *A&A* **636**, A15 (2020). doi:10.1051/0004-6361/201936602, arXiv:2004.03305

[7] N. J. Secrest et al., "A test of the cosmological principle with quasars," *ApJL* **908**, L51 (2021). doi:10.3847/2041-8213/abdd40, arXiv:2009.14826

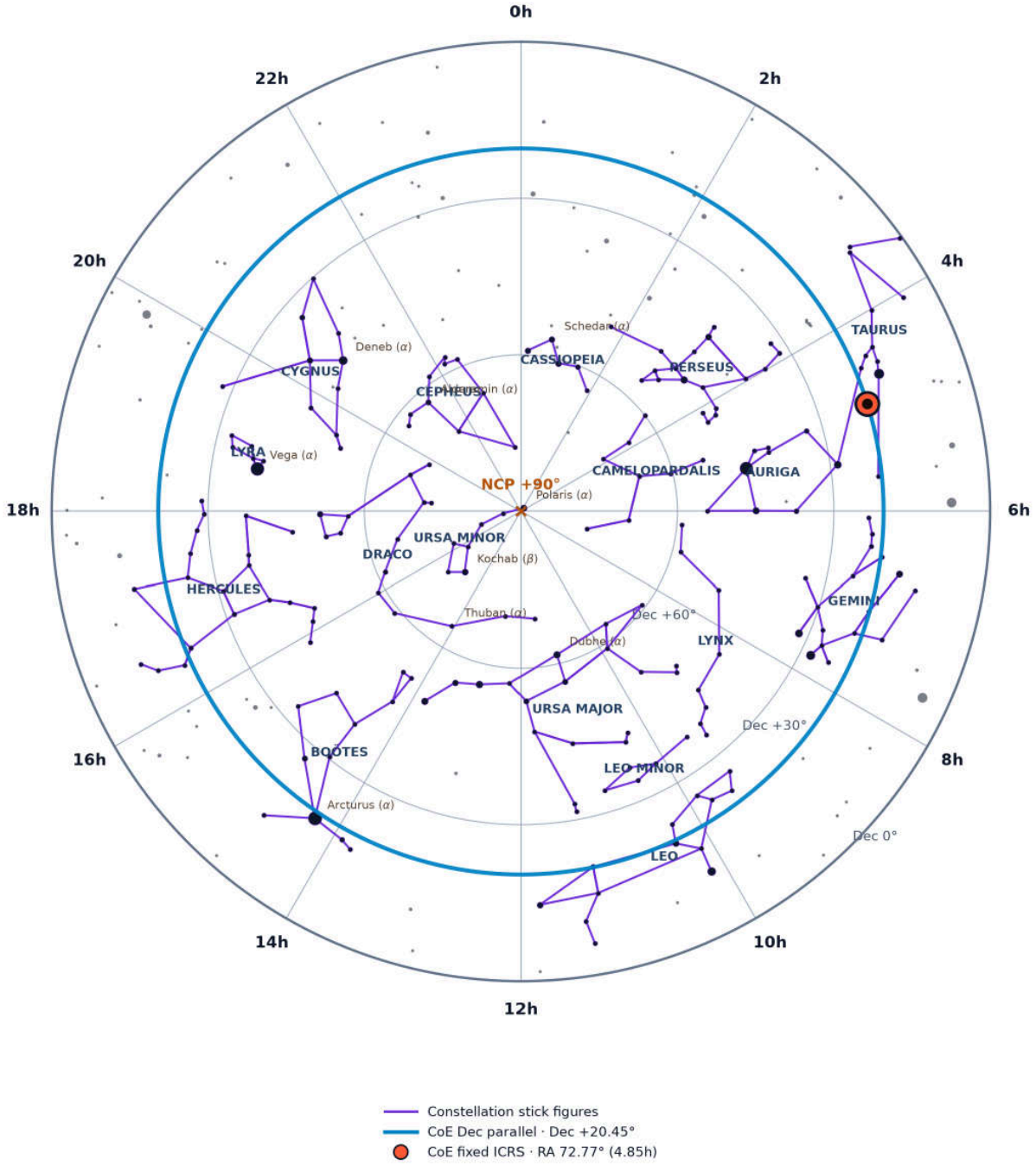


Figure 114: CoE on the northern celestial hemisphere (NCP-centered): fixed ICRS direction from Galactic $(l, b) = (180°, -15°)$; nearest bright star ≈ Aldebaran ($\sim 5.3°$). Uses the earlier rounded axis.